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# PHYSICAL WORLD, UNITS AND MEASUREMENTS

# 1

## MCQs with One Correct Answer

**1.** If x = at + bt$^2$, where x is the distance travelled by the body in kilometers while t is the time in seconds, then the unit of b is

(a) km/s (b) kms
(c) km/s$^2$ (d) kms$^2$

**2.** A metal sample carrying a current along X-axis with density $J_x$ is subjected to a magnetic field $B_z$ (along z-axis). The electric field $E_y$ developed along Y-axis is directly proportional to $J_x$ as well as $B_z$. The constant of proportionality has SI unit.

(a) $\frac{m^2}{A}$ (b) $\frac{m^3}{As}$
(c) $\frac{m^2}{As}$ (d) $\frac{As}{m^3}$

**3.** The refractive index of water measured by the relation $\mu = \frac{\text{real depth}}{\text{apparent depth}}$ is found to have values of 1.34, 1.38, 1.32 and 1.36; the mean value of refractive index with percentage error is

(a) $1.35 \pm 1.48\,\%$ (b) $1.35 \pm 0\,\%$
(c) $1.36 \pm 6\,\%$ (d) $1.36 \pm 0\,\%$

**4.** Write the dimensions of a × b in the relation $E = \frac{b - x^2}{at}$, where E is the energy, x is the displacement and t is time

(a) ML$^2$T (b) M$^{-1}$L$^2$T$^1$
(c) ML$^2$T$^{-2}$ (d) MLT$^{-2}$

**5.** In the relation $P = \frac{\alpha}{\beta} e^{-\frac{\alpha z}{k\theta}}$ where $P$ is pressure, $Z$ is distance, $k$ is Boltzmann constants and θ is the temperature. The dimensional formula of β will be

(a) $[M^0L^2T^0]$ (b) $[M^1L^2T^1]$
(c) $[M^1L^0T^{-1}]$ (d) $[M^0L^2T^{-1}]$

**6.** In a new system of units, the fundamental quantities mass, length and time are replaced by acceleration '$a$', density 'ρ' and frequency '$f$'. The dimensional formula for force in this system is

(a) $[\rho a^4 f]$ (b) $[\rho a^4 f^{-6}]$
(c) $[\rho^{-1} a^{-4} f^{\,6}]$ (d) $[\rho^{-1} a^{-4} f^{-1}]$

**7.** A formula is given as $P = \frac{b}{a}\sqrt{1 + \frac{k.\theta.t^3}{m.a}}$

where $P$ = pressure; $k$ = Boltzmann's constant; θ = temperature; $t$ = time; '$a$' and '$b$' are constants.

Dimensional formula of '$b$' is same as

(a) Force
(b) Linear momentum
(c) Angular momentum
(d) Torque

**8.** The pair of physical quantities that has the different dimensions is :

(a) Reynolds number and coefficient of friction
(b) Curie and frequency of a light wave
(c) Latent heat and gravitational potential
(d) Planck's constant and torque

**9.** Force F is given in terms of time t and distance x by F = A sin (Ct) + B cos (Dx). Then, dimensions of $\frac{A}{B}$ and $\frac{C}{D}$ are

(a) $[M^0 L^0 T^0], [M^0 L^0 T^{-1}]$
(b) $[M L T^{-2}], [M^0 L^{-1} T^0]$
(c) $[M^0 L^0 T^0], [M^0 L T^{-1}]$
(d) $[M^0 L^1 T^{-1}], [M^0 L^0 T^0]$

**10.** The respective number of significant figures for the numbers 23.023, 0.0003 and $2.1 \times 10^{-3}$ are

(a) 5, 1, 2 (b) 5, 1, 5
(c) 5, 5, 2 (d) 4, 4, 2

**11.** $N$ divisions on the main scale of a vernier calliper coincide with $(N+1)$ divisions of the vernier scale. If each division of main scale is '$a$' units, then the least count of the instrument is

(a) $a$ (b) $\frac{a}{N}$
(c) $\frac{N}{N+1} \times a$ (d) $\frac{a}{N+1}$

**12.** Given that $K$ = energy, $V$ = velocity, $T$ = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension?

(a) $[KV^{-2}T^{-2}]$ (b) $[K^2V^2T^{-2}]$
(c) $[K^2V^{-2}T^{-2}]$ (d) $[KV^2T^2]$

**13.** In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ?

(a) $[M^{-3} L^{-2} T^8 A^4]$ (b) $[M^{-1}L^{-2}T^4 A^2]$
(c) $[M^{-2} L^0 T^{-4} A^{-2}]$ (d) $[M^{-2}L^{-2}T^6 A^3]$

**14.** The relative error in the determination of the surface area of a sphere is $\alpha$. Then the relative error in the determination of its volume is

(a) $\frac{2}{3}\alpha$ (b) $\frac{2}{3}\alpha$
(c) $\frac{3}{2}\alpha$ (d) $\alpha$

**15.** In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half- a degree (= 0.5°), then the least count of the instrument is:

(a) half minute (b) one degree
(c) half degree (d) one minute

**16.** In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

(a) $A^{-1}TML^3$ (b) $AT^2 M^{-1}L^{-1}$
(c) $AT^{-3}ML^{3/2}$ (d) $A^2T^3 M^{-1}L^{-2}$

**17.** From the following combinations of physical constants (expressed through their usual symbols) the only combination, that would have the same value in different systems of units, is:

(a) $\frac{ch}{2\pi\varepsilon_o^2}$

(b) $\frac{e^2}{2\pi\varepsilon_o G m_e^2}$ ($m_e$ = mass of electron)

(c) $\frac{\mu_o\varepsilon_o}{c^2}\frac{G}{he^2}$

(d) $\frac{2\pi\sqrt{\mu_o\varepsilon_o}}{ce^2}\frac{h}{G}$

**18.** A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm, The average of these four reading is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as :

(a) $(5.5375 \pm 0.0739)$ mm
(b) $(5.5375 \pm 0.0740)$ mm
(c) $(5.538 \pm 0.074)$ mm
(d) $(5.54 \pm 0.07)$ mm

**19.** A quantity $x$ is given by $(IFv^2/WL^4)$ in terms of moment of inertia $I$, force $F$, velocity $v$, work $W$ and Length $L$. The dimensional formula for $x$ is same as that of :

(a) planck's constant
(b) force constant
(c) energy density
(d) coefficient of viscosity

**20.** The period of revolution ($T$) of a planet moving round the sun in a circular orbit depends upon the radius (r) of the orbit, mass (M) of the sun and the gravitation constant (G). Then T is proportional to

(a) $r^{1/2}$ (b) r

(c) $r^{3/2}$ (d) $r^2$

## Numeric Value Answer

**21.** The current voltage relation of a diode is given by $I = (e^{1000\ V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring $\pm 0.01$ V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA?

**22.** A physical quantity P is described by the relation $P = a^{1/2}\, b^2\, c^3\, d^{-4}$

If the relative errors in the measurement of *a, b, c* and *d* respectively, are 2%, 1%, 3% and 5%, then the percentage error in P will be :

**23.** The density of a material in SI unit is 128 kg $m^{-3}$. In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is:

**24.** If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count (in cm) of the screw gauge is:

**25.** Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then percentage error in the value of resistance of the wire is

**26.** If $3.8 \times 10^{-6}$ is added to $4.2 \times 10^{-5}$ giving the regard to significant figures then the result will be $x \times 10^{-5}$. Find the value of $x$.

**27.** The mass of a liquid flowing per second per unit area of cross section of a tube is proportional to $P^x$ and $v^y$, where P is the pressure difference and v is the velocity. Then $x \div y$ is

**28.** The specific resistance $\rho$ of a circular wire of radius r, resistance R and length $\ell$ is given by $\rho = \dfrac{\pi r^2 R}{l}$. Given, $r = 0.24 \pm 0.02$ cm, $R = 30 \pm 1W$ and $l = 4.80 \pm 0.01$ cm. The percentage error in $\rho$ is nearly

**29.** To determine the Young's modulus of a wire, the formula is

$$Y = \frac{F}{A} \times \frac{L}{\Delta L}$$

: where L = length, A = area of cross-section of the wire, $\Delta L$ = change in length of the wire when stretched with a force F. The conversion factor to change it from CGS to MKS system is

**30.** The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\dfrac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1s resolution. The percentage accuracy in the determination of g is:

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (b) | **7** | (b) | **10** | (a) | **13** | (a) | **16** | (d) | **19** | (c) | **22** | (32) | **25** | (6) |
| **2** | (b) | **5** | (a) | **8** | (d) | **11** | (d) | **14** | (c) | **17** | (b) | **20** | (c) | **23** | (40) | **26** | (4.6) |
| **3** | (a) | **6** | (b) | **9** | (c) | **12** | (a) | **15** | (d) | **18** | (d) | **21** | (0.2) | **24** | (0.001) | **27** | (–1) |

| | |
|---|---|
| **28** | (20) |
| **29** | (0.1) |
| **30** | (3) |

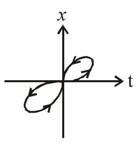
# MOTION IN A STRAIGHT LINE

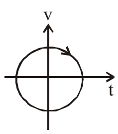
# 2

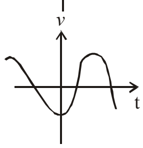
## MCQs with One Correct Answer

**1.** The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, $\alpha$ and $\beta$ are positive constants. The velocity of the particle will
(a) be independent of $\alpha$ and $\beta$
(b) drop to zero when $\alpha = \beta$
(c) go on decreasing with time
(d) go on increasing with time

**2.** Which of the following graph cannot possibly represent one dimensional motion of a particle?

(a) 


(b) 

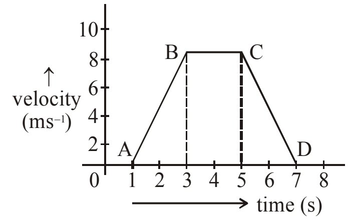

(c) 


(d) All of the above

**3.** A body moving with a uniform acceleration crosses a distance of 65 m in the 5 th second and 105 m in 9th second. How far will it go in 20 s?
(a) 2040 m (b) 240 m
(c) 2400 m (d) 2004 m

**4.** When two bodies move uniformly towards each other, the distance decreases by 6 $ms^{-1}$. If both bodies move in the same directions with the same speed (as above), the distance between them increases by 4 $ms^{-1}$. Then the speed of the two bodies are
(a) 3 $ms^{-1}$ and 3 $ms^{-1}$
(b) 4 $ms^{-1}$ and 2 $ms^{-1}$
(c) 5 $ms^{-1}$ and 1 $ms^{-1}$
(d) 7 $ms^{-1}$ and 3 $ms^{-1}$

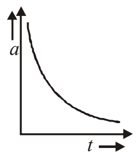
**5.** For the velocity time graph shown in the figure below the distance covered by the body in the last two seconds of its motion is what fraction of the total distance travelled by it in all the seven seconds?

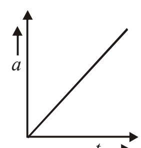


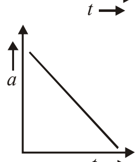

(a) $\frac{1}{2}$
(b) $\frac{1}{4}$
(c) $\frac{2}{3}$
(d) $\frac{1}{3}$

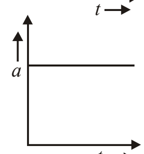
**6.** A particle moves for 20 seconds with velocity 3 m/s and then with velocity 4 m/s for another 20 seconds and finally moves with velocity 5 m/s for next 20 seconds. What is the average velocity of the particle ?
(a) 3 m/s (b) 4 m/s
(c) 5 m/s (d) Zero

**7.** The distance travelled by a body moving along a line in time $t$ is proportional to $t^3$.
The acceleration-time ($a$, $t$) graph for the motion of the body will be

(a) 


(b) 


(c) 


(d) 

**8.** A goods train accelerating uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity $u$ and the guard's room passes with velocity $v$. The middle wagon of the train passes the pole with a velocity.

(a) $\frac{u+v}{2}$ (b) $\frac{1}{2}\sqrt{u^2+v^2}$

(c) $\sqrt{uv}$ (d) $\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

**9.** A juggler keeps on moving four balls in the air throwing the balls after intervals. When one ball leaves his hand (speed = 20 ms$^{-1}$) the position of other balls (height in m) will be
(Take g = 10 ms$^{-2}$)

(a) 10, 20, 10 (b) 15, 20, 15
(c) 5, 15, 20 (d) 5, 10, 20

**10.** A car, starting from rest, accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is 15 $S$, then

(a) $S = \frac{1}{6}ft^2$ (b) $S = ft$

(c) $S = \frac{1}{4}ft^2$ (d) $S = \frac{1}{72}ft^2$

**11.** A particle located at $x = 0$ at time $t = 0$, starts moving along with the positive $x$-direction with a velocity '$v$' that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as

(a) $t^2$ (b) $t$ (c) $t^{1/2}$ (d) $t^3$

**12.** A person climbs up a stalled escalator in 60 s. If standing on the same but escalator running with constant velocity he takes 40 s. How much time is taken by the person to walk up the moving escalator?

(a) 37 s (b) 27 s (c) 24 s (d) 45 s

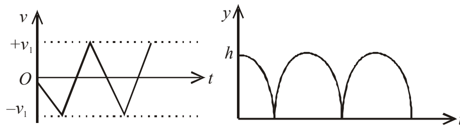
**13.** From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H, u and n is:

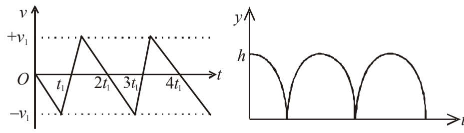
(a) $2gH = n^2u^2$ (b) $gH = (n-2)^2 u^2 d$
(c) $2gH = nu^2 (n-2)$ (d) $gH = (n-2)u^2$

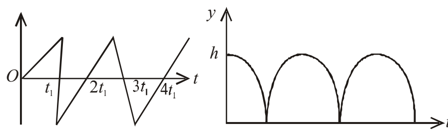
**14.** If a body looses half of its velocity on penetrating 3 cm in a wooden block, then how much will it penetrate more before coming to rest?

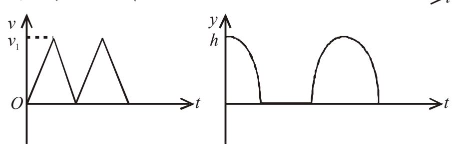
(a) 1 cm (b) 2 cm
(c) 3 cm (d) 4 cm.

**15.** Consider a rubber ball freely falling from a height $h = 4.9$ m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic.
Then the velocity as a function of time and the height as a function of time will be :

(a) 


(b) 


(c) 


(d) 


**16.** The position of a particle as a function of time $t$, is given by
$x(t) = at + bt^2 - ct^3$
where, $a$, $b$ and $c$ are constants. When the particle attains zero acceleration, then its velocity will be:

(a) $a + \frac{b^2}{4c}$ (b) $a + \frac{b^2}{3c}$

(c) $a + \frac{b^2}{c}$ (d) $a + \frac{b^2}{2c}$

**17.** A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s$^2$ and the car has acceleration 4 m/s$^2$. The car will catch up with the bus after a time of :

(a) $\sqrt{110}$ s (b) $\sqrt{120}$ s

(c) $10\sqrt{2}$ s (d) 15 s

**18.** A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60º with ground level. But he finds the aeroplane right vertically above his position. If $v$ is the speed of sound, speed of the plane is:

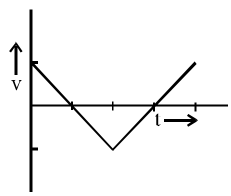
(a) $\frac{\sqrt{3}}{2}v$ (b) $\frac{2v}{\sqrt{3}}$

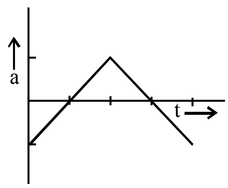
(c) $v$ (d) $\frac{v}{2}$

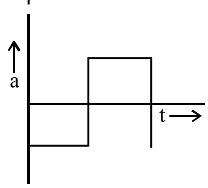
**19.** A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/h. The ratio of times taken by the passenger train to completely cross the freight train when: (i) they are moving in same direction, and (ii) in the opposite directions is:

(a) $\frac{11}{5}$ (b) $\frac{5}{2}$ (c) $\frac{3}{2}$ (d) $\frac{25}{11}$

**20.** The graph shown in figure shows the velocity v versus time t for a body. Which of the graphs represents the corresponding acceleration versus time graphs?




(a)



(b)
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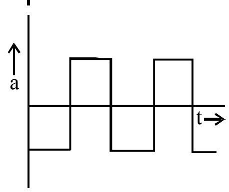
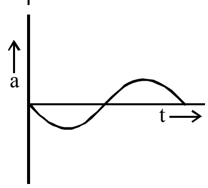


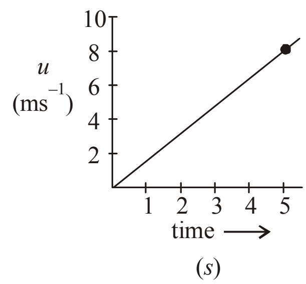

(d)



**Numeric Value Answer**

**21.** A parachutist after bailing out falls 50 m without friction. When parachute opens, it decelerates at 2 m/s$^2$. He reaches the ground with a speed of 3 m/s. At what height (in m), did he bail out ?

**22.** An automobile travelling with a speed of 60 km/h, can brake to stop within a distance of 20m. If the car is going twice as fast i.e., 120 km/h, the stopping distance (in m) will be

**23.** The speed verses time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval $t = 0$ to $t = 5$ s will be __________.




**24.** The distance x covered by a particle in one dimensional motion varies with time $t$ as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on $x$ as $x^{-n}$, where $n$ is an integer, the value of $n$ is ______.

**25.** A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}$ s before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms$^{-2}$) near the surface on that planet is _______.

**26.** An object, moving with a speed of 6.25 m/s, is decelerated at a rate given by

$\frac{dv}{dt} = -2.5\sqrt{v}$ where $v$ is the instantaneous speed. The time (in second) taken by the object, to come to rest, would be:

**27.** A cat, on seeing a rat at a distant of $d = 5$ m, starts with velocity $u = 5$ ms$^{-1}$ and moves with acceleration $\alpha = 2.5$ ms$^{-2}$ in order to catch it, while the rate with acceleration $\beta$ starts from rest. For what value of $\beta$ will be the cat overtake the rat ? (in $ms^{-2}$)

**28.** A particle is moving in a straight line with initial velocity and uniform acceleration a. If the sum of the distance travelled in $t^{th}$ and $(t + 1)^{th}$ seconds is 100 cm, then its velocity after $t$ seconds, in cm/s, is

**29.** A body is thrown vertically upwards with velocity u. The distance travelled by it in the fifth and the sixth seconds are equal. The velocity u (in m/s) is given by ($g = 9.8\ m/s^2$)

**30.** If you throw a ball vertically upward with an initial velocity of 50 m/s, approximately how long (in second) would it take for the ball to return to your hand? Assume air resistance is negligible.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (c) | **7** | (b) | **10** | (d) | **13** | (c) | **16** | (b) | **19** | (a) | **22** | (80) | **25** | (8) | **28** | (50) |
| **2** | (d) | **5** | (b) | **8** | (d) | **11** | (a) | **14** | (a) | **17** | (c) | **20** | (b) | **23** | (20) | **26** | (2) | **29** | (49) |
| **3** | (c) | **6** | (b) | **9** | (b) | **12** | (c) | **15** | (b) | **18** | (d) | **21** | (293) | **24** | (3) | **27** | (5) | **30** | (10) |

# MOTION IN A PLANE

3

## MCQs with One Correct Answer

1. If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 7\hat{i} + 24\hat{j}$, the vector having the same magnitude as B and parallel to A is
   (a) $5\hat{i} + 20\hat{j}$ (b) $15\hat{i} + 10\hat{j}$
   (c) $20\hat{i} + 15\hat{j}$ (d) $15\hat{i} + 20\hat{j}$

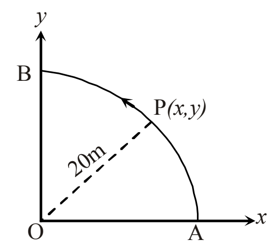
2. Two balls are projected at an angle θ and $(90^\circ - \theta)$ to the horizontal with the same speed. The ratio of their maximum vertical heights is
   (a) 1 : 1 (b) $\tan\theta : 1$
   (c) $1 : \tan\theta$ (d) $\tan^2\theta : 1$

3. A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height $H_1$. When it is projected with velocity u at an angle $\left(\frac{\pi}{2} - \theta\right)$ with the horizontal, it reaches maximum height $H_2$. The relation between the horizontal range R of the projectile, heights $H_1$ and $H_2$ is
   (a) $R = 4\sqrt{H_1H_2}$ (b) $R = 4(H_1 - H_2)$
   (c) $R = 4(H_1 + H_2)$ (d) $R = \frac{H_1^2}{H_2^2}$

4. The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$
   The angle of projection is given by
   (a) $\tan\theta = \frac{1}{\sqrt{3}}$ (b) $\tan\theta = \sqrt{3}$
   (c) $\frac{\pi}{2}$ (d) zero.

5. A particle moves along a circle of radius $\left(\frac{20}{\pi}\right)$m with constant tangential acceleration. It the velocity of particle is 80 m/sec at end of second revolution after motion has begun, the tangential acceleration is
   (a) $40\,\pi$ m/sec$^2$ (b) 40 m/sec$^2$
   (c) $640\,\pi$ m/sec$^2$ (d) $160\,\pi$ m/sec$^2$

6. A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and $t$ is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 2$ s is nearly.




   (a) 13m/s$^2$ (b) 12 m/s$^2$
   (c) 7.2 ms$^2$ (d) 14m/s$^2$

7. The vectors $\vec{A}$ and $\vec{B}$ are such that
   $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$
   The angle between the two vectors is
   (a) 60° (b) 75°
   (c) 45° (d) 90°

**8.** Two balls are projected simultaneously in the same vertical plane from the same point with velocities $v_1$ and $v_2$ with angle $\theta_1$ and $\theta_2$ respectively with the horizontal. If $v_1 \cos\theta_1 = v_2 \cos\theta_2$, the path of one ball as seen from the position of other ball is :

(a) parabola
(b) horizontal straight line
(c) vertical straight line
(d) straight line making 45° with the vertical

**9.** A projectile with same projection velocity can have the same range '$R$' for two angles of projection. If '$T_1$' and '$T_2$' be time of flights in the two cases, then the product of the two time of flights is directly proportional to

(a) $R$ (b) $\frac{1}{R}$
(c) $\frac{1}{R^2}$ (d) $R^2$

**10.** A bomber plane moves horizontally with a speed of 500 m/s and a bomb released from it, strikes the ground in 10 sec. Angle with the ground at which it strikes the ground will be ($g = 10\ m/s^2$)

(a) $\tan^{-1}\left(\frac{1}{5}\right)$ (b) $\tan\left(\frac{1}{5}\right)$
(c) $\tan^{-1}(1)$ (d) $\tan^{-1}(5)$

**11.** Starting from the origin at time $t = 0$, with initial velocity $5\hat{j}$ ms$^{-1}$, a particle moves in the $x$–$y$ plane with a constant acceleration of $(10\hat{i} + 4\hat{j})$ ms$^{-2}$. At time $t$, its coordiantes are (20 m, $y_0$ m). The values of $t$ and $y_0$ are, respectively :

(a) 2 s and 18 m (b) 4 s and 52 m
(c) 2 s and 24 m (d) 5 s and 25 m

**12.** The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$. What is the magnitude of the acceleration at $t = 1$?

(a) 40 (b) 25
(c) 100 (d) 50

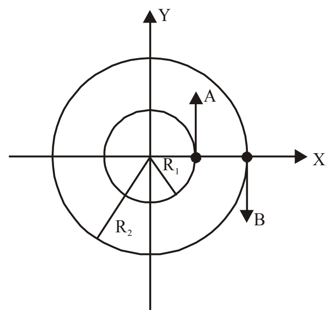
**13.** Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A} + \vec{B})$ is 'n' times the magnitude of $(\vec{A} - \vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is:

(a) $\cos^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (b) $\cos^{-1}\left[\frac{n-1}{n+1}\right]$
(c) $\sin^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (d) $\sin^{-1}\left[\frac{n-1}{n+1}\right]$

**14.** Ship A is sailing towards north-east with velocity km/hr where points east and , north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in:

(a) 4.2 hrs. (b) 2.6 hrs.
(c) 3.2 hrs. (d) 2.2 hrs.

**15.** Two particles A, B are moving on two concentric circles of radii $R_1$ and $R_2$ with equal angular speed $\omega$. At t = 0, their positions and direction of motion are shown in the figure :




The relative velocity $v_{\vec{A}} - v_{\vec{B}}$ and $t = \frac{\pi}{2\omega}$ is given by:

(a) $\omega(R_1 + R_2)\hat{i}$ (b) $-\omega(R_1 + R_2)\hat{i}$
(c) $\omega(R_2 - R_1)\hat{i}$ (d) $\omega(R_1 - R_2)\hat{i}$

**16.** A particle is moving with velocity $\vec{v} = k(y\hat{i} + x\hat{j})$, where k is a constant. The general equation for its path is

(a) $y = x^2$ + constant
(b) $y^2 = x$ + constant
(c) $xy$ = constant
(d) $y^2 = x^2$ + constant

**17.** A particle moves such that its position vector $\vec{r}\ (t) = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ where ω is a constant and $t$ is time. Then which of the following statements is true for the velocity $\vec{v}\ (t)$ and acceleration $\vec{a}\ (t)$ of the particle:

(a) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin

(b) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$

(c) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$

(d) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin

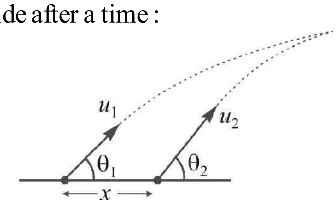
**18.** The position of a projectile launched from the origin at $t = 0$ is given by $\vec{r} = \left(40\hat{i} + 50\hat{j}\right)$m at $t$ = 2s. If the projectile was launched at an angle θ from the horizontal, then θ is

(take g = 10 $ms^{-2}$)

(a) $\tan^{-1}\frac{2}{3}$ (b) $\tan^{-1}\frac{3}{2}$

(c) $\tan^{-1}\frac{7}{4}$ (d) $\tan^{-1}\frac{4}{5}$

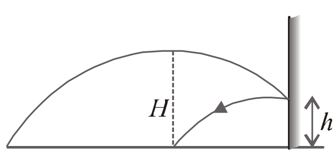
**19.** Two particles are projected simultaneously from the level ground as shown in figure. They may collide after a time :




(a) $\frac{x\sin\theta_2}{u_1}$ (b) $\frac{x\cos\theta_2}{u_2}$

(c) $\frac{x\sin\theta_2}{u_1\sin(\theta_2-\theta_1)}$ (d) $\frac{2x\sin\theta_1}{u_2\sin(\theta_2-\theta_1)}$

**20.** A stone is projected from a horizontal plane. It attains maximum height $H$ and strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assuming the collision to be elastic, the height of the point on the wall where ball will strike is:




(a) $\frac{H}{4}$ (b) $\frac{H}{2}$

(c) $\frac{3H}{4}$ (d) $\frac{7H}{8}$

## Numeric Value Answer

**21.** The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to the vector $\vec{A}$ and its magnitude is equal to half the magnitude of vector $\vec{B}$. The angle (in degree) between $\vec{A}$ and $\vec{B}$ is

**22.** A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity (in $ms^{-1}$) of the body is (g = 10 $ms^{-2}$)

**23.** A particle describes uniform circular motion in a circle of radius 2 m, with the angular speed of 2 rad $s^{-1}$. The magnitude of the change in its velocity in $\frac{\pi}{2}$ s is _____m $s^{-1}$.

**24.** A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is :

**25.** If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then the value of α is

**26.** A particle moves from the point $\left(2.0\hat{i} + 4.0\hat{j}\right)$m, at t = 0, with an initial velocity $\left(5.0\hat{i} + 4.0\hat{j}\right)ms^{-1}$. It is acted upon by a constant force which produces a constant acceleration $\left(4.0\hat{i} + 4.0\hat{j}\right)ms^{-2}$. What is the distance (in m) of the particle from the origin at time 2s?

**27.** A particle starts from the origin at $t = 0$ with an initial velocity of $3.0\hat{i}$ m/s and moves in the $x$-$y$ plane with a constant acceleration $(6.0\hat{i} + 4.0\hat{j})$ m/s². The $x$-coordinate of the particle at the instant when its $y$-coordinate is 32 m is D meters. The value of D is:

**28.** A particle is moving along the $x$-*axis* with its coordinate with time '$t$' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the $y$-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1$ s, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then $v$ (in m/s) is____

**29.** A force $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k})$ N acts at a point $(4\hat{i} + 3\hat{j} - \hat{k})$ m. Then the magnitude of torque about the point $(\hat{i} + 2\hat{j} + \hat{k})$ m will be $\sqrt{x}$ N-m. The value of $x$ is ______.

**30.** The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$. The angle θ (in degrees) that the resultant of 2 $\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is _______.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (b) | **7** | (d) | **10** | (a) | **13** | (a) | **16** | (d) | **19** | (c) | **22** | (10) | **25** | (–0.5) | **28** | (580) |
| **2** | (d) | **5** | (b) | **8** | (c) | **11** | (a) | **14** | (b) | **17** | (d) | **20** | (c) | **23** | (8) | **26** | (20√2) | **29** | (195) |
| **3** | (a) | **6** | (d) | **9** | (a) | **12** | (d) | **15** | (c) | **18** | (c) | **21** | (150) | **24** | (7√2) | **27** | (60) | **30** | (90) |

# 4 LAWS OF MOTION

## MCQs with One Correct Answer

**1.** A particle of mass m is moving in a straight line with momentum p. Starting at time t = 0, a force F = $kt$ acts in the same direction on the moving particle during time interval T so that its momentum changes from p to 3p. Here $k$ is a constant. The value of T is :

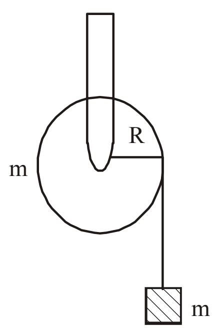
(a) $2\sqrt{\frac{k}{p}}$ (b) $2\sqrt{\frac{p}{k}}$ (c) $\sqrt{\frac{2k}{p}}$ (d) $\sqrt{\frac{2p}{k}}$

**2.** A rocket with a lift-off mass $3.5 \times 10^4$ kg is blasted upwards with an initial acceleration of $10 m/s^2$. Then the initial thrust of the blast is

(a) $3.5 \times 10^5$ N (b) $7.0 \times 10^5$ N

(c) $14.0 \times 10^5$ N (d) $1.75 \times 10^5$ N

**3.** A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall or release?



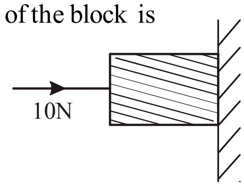

(a) $\frac{2g}{3}$

(b) $\frac{g}{2}$

(c) $\frac{5g}{6}$

(d) g

**4.** A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is




(a) 20 N

(b) 50 N

(c) 100 N

(d) 2 N

**5.** A body of mass 2kg slides down with an acceleration of $3 m/s^2$ on a rough inclined plane having a slope of 30°. The external force required to take the same body up the plane with the same acceleration will be: ($g = 10 m/s^2$)

(a) 4N (b) 14N

(c) 6N (d) 20N

**6.** A block of mass m = 10 kg rests on a horizontal table. The coefficient of friction between the block and the table is 0.05. When hit by a bullet of mass 50 g moving with speed $v$, that gets embedded in it, the block moves and comes to stop after moving a distance of 2 m on the table. If a freely falling object were to acquire speed $\frac{v}{10}$ after being dropped from height H, then neglecting energy losses and taking $g = 10 ms^{-2}$, the value of H is close to:

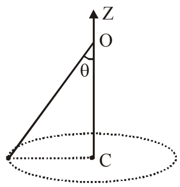
(a) 0.05 km (b) 0.02 km

(c) 0.03 km (d) 0.04 km

**7.** A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45°at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10\ ms^{-2}$)

(a) 200 N (b) 140 N
(c) 70 N (d) 100 N

**8.** A conical pendulum of length 1 m makes an angle $\theta = 45°$ w.r.t. Z-axis and moves in a circle in the XY plane.The radius of the circle is 0.4 m and its centre is vertically below O. The speed of the pendulum, in its circular path, will be :
(Take $g = 10\ ms^{-2}$)




(a) 0.4 m/s
(b) 4 m/s
(c) 0.2 m/s
(d) 2 m/s

**9.** A particle of mass 0.3 kg subject to a force $F = -kx$ with $k = 15$ N/m . What will be its initial acceleration if it is released from a point 20 cm away from the origin ?

(a) 15 $m/s^2$ (b) 3 $m/s^2$
(c) 10 $m/s^2$ (d) 5 $m/s^2$

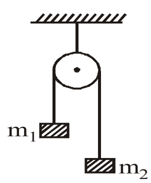
**10.** When forces $F_1$, $F_2$, $F_3$ are acting on a particle of mass $m$ such that $F_2$ and $F_3$ are mutually perpendicular, then the particle remains stationary. If the force $F_1$ is now removed then the acceleration of the particle is

(a) $F_1/m$ (b) $F_2F_3\ /mF_1$
(c) $(F_2 - F_3)/m$ (d) $F_2\ /m.$

**11.** A lift is moving down with acceleration $a$. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively

(a) $g, g$ (b) $g - a, g - a$
(c) $g - a, g$ (d) $a, g$

**12.** Two blocks $m_1 = 5$ gm and $m_2 = 10$ gm are hung vertically over a light frictionless pulley as shown here. What is the velocity of separation of the masses after 1 second when they are left free? [take $g = 10\ m/s^2$]



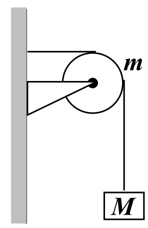

(a) 20/3 m/s
(b) 10/3 m/s
(c) 5/3 m/s
(d) 2/3 m/s

**13.** A block of mass $m$ is connected to another block of mass $M$ by a spring (massless) of spring constant $k$. The block are kept on a smooth horizontal plane. Initially the blocks are at rest and the spring is unstretched. Then a constant force $F$ starts acting on the block of mass $M$ to pull it. Find the force on the block of mass $m$.

(a) $\dfrac{MF}{(m+M)}$ (b) $\dfrac{mF}{M}$

(c) $\dfrac{(M+m)F}{m}$ (d) $\dfrac{mF}{(m+M)}$

**14.** A string of negligible mass going over a clamped pulley of mass $m$ supports a block of mass $M$ as shown in the figure. The force on the pulley by the clamp is given by




(a) $\sqrt{2}$ Mg
(b) $\sqrt{2}$ mg
(c) $\sqrt{(M+m)^2 + m^2}\ g$
(d) $\sqrt{(M+m)^2 + M^2}\ g$

**15.** A car is moving along a straight horizontal road with a speed $v_0$. If the coefficient of friction between the tyres and the road is $\mu$, The shortest distance in which the car can be stopped is

(a) $\dfrac{v_0^2}{2\mu g}$ (b) $\dfrac{v_0}{\mu g}$

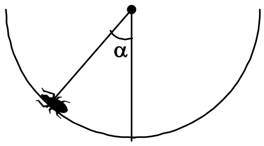
(c) $\left(\dfrac{v_0}{\mu g}\right)^2$ (d) $\dfrac{v_0}{\mu}$

**16.** A uniform metal chain is placed on a rough table such that one end of chain hangs down over the edge of the table. When one-third of its length hangs down over the edge, the chain starts sliding. Then the value of coefficient of static friction is

(a) $\dfrac{3}{4}$ (b) $\dfrac{1}{4}$

(c) $\dfrac{2}{3}$ (d) $\dfrac{1}{2}$

**17.** An insect crawls up a hemispherical surface very slowly (see fig.). The coefficient of friction between the insect and the surface is 1/3. If the line joining the center of the hemispherical surface to the insect makes an angle $\alpha$ with the vertical, the maximum possible value of $\alpha$ is given by




(a) $\cot \alpha = 3$
(b) $\tan \alpha = 3$
(c) $\sec \alpha = 3$
(d) $\operatorname{cosec} \alpha = 3$

**18.** A ball of mass 0.2 kg is thrown vertically upwards by applying a force by hand. If the hand moves 0.2 m while applying the force and the ball goes upto 2 m height further, find the magnitude of the force. (Consider $g = 10\ m/s^2$).

(a) 4 N (b) 16 N
(c) 20 N (d) 22 N

**19.** A person with his hands in his pockets is skating on ice at the velocity of 10 m/s and describes a circle of radius 50 m. What is his inclination with vertical?

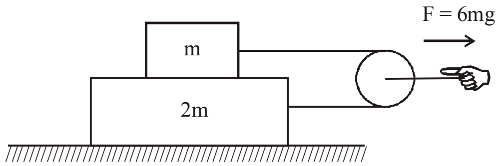
(a) $\tan^{-1}\left(\frac{1}{10}\right)$ (b) $\tan^{-1}\left(\frac{3}{5}\right)$
(c) $\tan^{-1}(1)$ (d) $\tan^{-1}\left(\frac{1}{5}\right)$

**20.** A monkey is decending from the branch of a tree with constant acceleration. If the breaking strength is 75% of the weight of the monkey, the minimum acceleration with which monkey can slide down without breaking the branch is

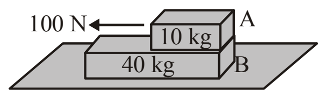
(a) g (b) $\frac{3g}{4}$
(c) $\frac{g}{4}$ (d) $\frac{g}{2}$

## Numeric Value Answer

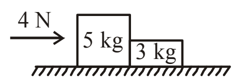
**21.** A block of mass m is placed on top of a block of mass 2m which in turn is placed on fixed horizontal surface. The coefficient of friction between all surfaces is $\mu = 1$. A massless string is connected to each mass and wraps halfway around a massless and frictionless pulley, as shown. The pulley is pulled by horizontal force of magnitude F = 6mg towards right as shown. If the magnitude of acceleration of pulley is $\frac{X}{2}$ $m/s^2$, find the value of $\sqrt{X}$. (Take $g = 10\ m/s^2$)




**22.** A 40 kg slab rests on a frictionless floor as shown in the figure. A 10 kg block rests on the top of the slab. The static coefficient of friction between the block and slab is 0.60 while the coefficient of kinetic friction is 0.40. The 10 kg block is acted upon by a horizontal force 100 N. If $g = 9.8\ m/s^2$, the resulting acceleration (in $m/s^2$) of the slab will be




**23.** Two blocks of masses 5 kg and 3 kg are placed in contact on a horizontal frictionless surface as shown in the figure. A force of 4N is applied on mass 5 kg. The acceleration (in $m/s^2$) of the mass 3 kg will be




**24.** The coefficient of friction between a body and the surface of an inclined plane at 45° is 0.5. If $g = 9.8\ m/s^2$, the acceleration of the body in downwards in $m/s^2$ is

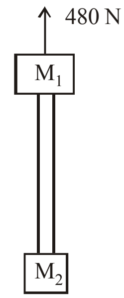
**25.** A body of mass 0.4 kg is whirled in a vertical circle making 2 rev/sec. If the radius of the circle is 1.2 m, then tension (in N) in the string when the body is at the top of the circle, is

**26.** A block starts moving up an inclined plane of inclination 30° with an initial velocity of $v_0$. It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$. The nearest integer to $I$ is_________.

**27.** The minimum velocity (in $ms^{-1}$) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is

**28.** The minimum force required to start pushing a body up rough (frictional coefficient $\mu$) inclined plane is $F_1$ while the minimum force needed to prevent it from sliding down is $F_2$. If the inclined plane makes an angle $\theta$ from the horizontal such that $\tan\theta = 2\mu$ then the ratio $\frac{F_1}{F_2}$ is

**29.** Two blocks of mass $M_1 = 20$ kg and $M_2 = 12$ kg are connected by a metal rod of mass 8 kg. The system is pulled vertically up by applying a force of 480 N as shown. The tension (in N) at the mid-point of the rod is :




**30.** A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 49 N, when the lift is stationary. If the lift moves downward with an acceleration of 5 m/$s^2$, the reading (in N) of the spring balance will be

| ANSWER KEY | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (d) | **7** | (d) | **10** | (a) | **13** | (d) | **16** | (d) | **19** | (d) | **22** | (0.98) | **25** | (71.8) | **28** | (3) |
| **2** | (b) | **5** | (d) | **8** | (d) | **11** | (c) | **14** | (d) | **17** | (a) | **20** | (c) | **23** | (0.5) | **26** | (346) | **29** | (192) |
| **3** | (b) | **6** | (d) | **9** | (c) | **12** | (a) | **15** | (a) | **18** | (d) | **21** | (5) | **24** | (3.47) | **27** | (30) | **30** | (24.5) |

# WORK, ENERGY AND POWER

**5**

## MCQs with One Correct Answer

**1.** A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement $x$ is proportional to

(a) $x$ (b) $e^x$
(c) $x^2$ (d) $\log_e x$

**2.** A body is moved along a straight line by a machine delivering constant power. The distance moved by the body in time '$t$' is proportional to

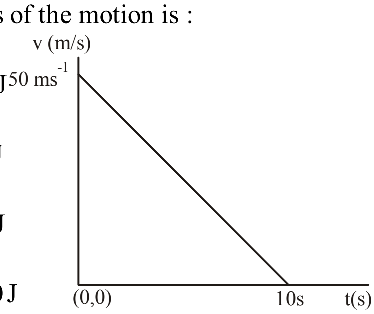
(a) $t^{3/4}$ (b) $t^{3/2}$ (c) $t^{1/4}$ (d) $t^{1/2}$

**3.** A ball is let to fall from a height $h_0$. There are $n$ collisions with the earth. If the velocity of rebound after $n$ collisions is $v_n$ and the ball rises to a height $h_n$ then coefficient of restitution $e$ is given by

(a) $e^n = \sqrt{\dfrac{h_n}{h_0}}$ (b) $e^n = \sqrt{\dfrac{h_0}{h_n}}$

(c) $ne = \sqrt{\dfrac{h_n}{h_0}}$ (d) $\sqrt{ne} = \sqrt{\dfrac{h_n}{h_0}}$

**4.** Velocity–time graph for a body of mass 10 kg is shown in figure. Work–done on the body in first two seconds of the motion is :




(a) –9300 J
(b) 12000 J
(c) –4500 J
(d) –12000 J

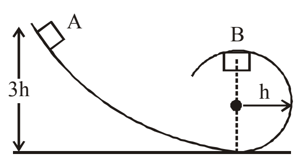
**5.** A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain (hanging portion) on the table ?

(a) 12 J (b) 3.6 J
(c) 7.2 J (d) 1200 J

**6.** The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \dfrac{a}{x^{12}} - \dfrac{b}{x^6}$ where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is $D = [U(x = \infty) - U_{\text{at equilibrium}}]$, D is

(a) $\dfrac{b^2}{4a}$ (b) $\dfrac{b^2}{2a}$ (c) $\dfrac{b^2}{12a}$ (d) $\dfrac{b^2}{6a}$

**7.** In the figure shown, a particle of mass m is released from the position A on a smooth track. When the particle reaches at B, then normal reaction on it by the track is

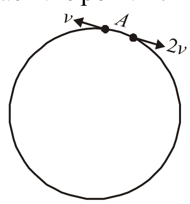



(a) mg
(b) 2mg
(c) $\dfrac{2}{3}$mg
(d) $\dfrac{m^2 g}{h}$

**8.** A 10 H.P. motor pumps out water from a well of depth 20 m and fills a water tank of volume 22380 litres at a height of 10 m from the ground. The running time of the motor to fill the empty water tank is (g = 10ms$^{-2}$)

(a) 5 minutes (b) 10 minutes
(c) 15 minutes (d) 20 minutes

**9.** Two small particles of equal masses start moving in opposite directions from a point *A* in a horizontal circular orbit. Their tangential velocities are $v$ and $2v$, respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at *A*, these two particles will again reach the point *A*?




(a) 4

(b) 3

(c) 2

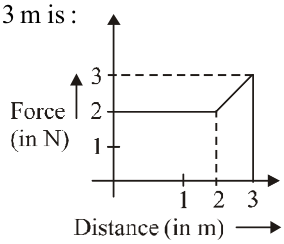
(d) 1

**10.** A spring of spring constant $5 \times 10^3$ N/m is stretched initially by 5cm from the unstretched position. Then the work required to stretch it further by another 5 cm is

(a) 12.50 N-m (b) 18.75 N-m

(c) 25.00 N-m (d) 6.25 N-m

**11.** A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is :




(a) 4 J

(b) 2.5 J

(c) 6.5 J

(d) 5 J

**12.** A bullet looses $\left(\frac{1}{n}\right)^{th}$ of its velocity passing through one plank. The number of such planks that are required to stop the bullet can be:

(a) $\frac{n^2}{2n-1}$ (b) $\frac{2n^2}{n-1}$

(c) infinite (d) n

**13.** At time $t = 0$ a particle starts moving along the $x$-axis. If its kinetic energy increases uniformly with time '$t$', the net force acting on it must be proportional to

(a) constant (b) $t$

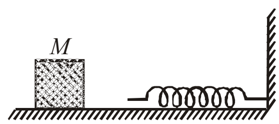
(c) $\frac{1}{\sqrt{t}}$ (d) $\sqrt{t}$

**14.** A wedge of mass M = 4m lies on a frictionless plane. A particle of mass $m$ approaches the wedge with speed $v$. There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by:

(a) $\frac{v^2}{g}$ (b) $\frac{2v^2}{7g}$

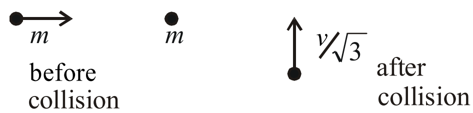
(c) $\frac{2v^2}{5g}$ (d) $\frac{v^2}{2g}$

**15.** The block of mass $M$ moving on the frictionless horizontal surface collides with the spring of spring constant $k$ and compresses it by length $L$. The maximum momentum of the block after collision is




(a) $\frac{kL^2}{2M}$ (b) $\sqrt{Mk}\,L$ (c) $\frac{ML^2}{k}$ (d) zero

**16.** A mass 'm' moves with a velocity '$v$' and collides inelastically with another identical mass. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision.




(a) $\sqrt{3}v$ (b) $v$

(c) $\frac{v}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}v$

**17.** A particle is moving in a circle of radius r under the action of a force F = $\alpha r^2$ which is directed towards centre of the circle. Total mechanical energy (kinetic energy + potential energy) of the particle is (take potential energy = 0 for r = 0) :

(a) $\frac{1}{2}\alpha r^3$ (b) $\frac{5}{6}\alpha r^3$

(c) $\frac{4}{3}\alpha r^3$ (d) $\alpha r^3$

**18.** In a collinear collision, a particle with an initial speed $v_0$ strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is:

(a) $\frac{v_0}{4}$ (b) $\sqrt{2}v_0$

(c) $\frac{v_0}{2}$ (d) $\frac{v_0}{\sqrt{2}}$

**19.** A body of mass 3 kg is under a constant force which causes a displacement s in metre in it, given by the relation $s = \frac{1}{3}t^3$, where t is in second. Work done by the force in 2 second is

(a) $\frac{3}{8}$J (b) 24 J

(c) $\frac{19}{5}$J (d) $\frac{5}{19}$J

**20.** A running man has half the kinetic energy of that of a boy of half of his mass. The man speeds up by 1m/s so as to have same K.E. as that of the boy. The original speed of the man will be

(a) $\sqrt{2}\ m/s$ (b) $\left(\sqrt{2}-1\right)m/s$

(c) $\frac{1}{\left(\sqrt{2}-1\right)}m/s$ (d) $\frac{1}{\sqrt{2}}m/s$

## Numeric Value Answer

**21.** A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the $y$ direction with speed **v**. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to :

**22.** Four smooth steel balls of equal mass at rest are free to move along a straight line without friction. The first ball is given a velocity of 0.4 m/s. It collides head on with the second elastically, the second one similarly with the third and so on. The velocity (in m/s) of the last ball is

**23.** A ball collides elastically with another ball of the same mass. The collision is oblique and initially one of the balls was at rest. After the collision, both the balls move with same speed. What will be the angle (in degree) between the velocities of the balls after the collision ?

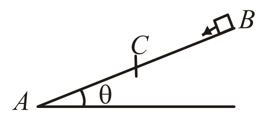
**24.** A car of weight W is on an inclined road that rises by 100 m over a distance of 1 km and applies a constant frictional force $\frac{W}{20}$ on the car. While moving uphill on the road at a speed of 10 ms$^{-1}$, the car needs power P. If it needs power $\frac{P}{2}$ while moving downhill at speed $v$ then value of $v$ (in ms$^{-1}$) is:

**25.** The potential energy (in joule) of a body of mass 2 kg moving in the x – y plane is given by

U = 6x + 8y, where x and y are in metre.

If the body is at rest at point (6m, 4m) at time t = 0, it will cross y-axis at time t (in second) equal to

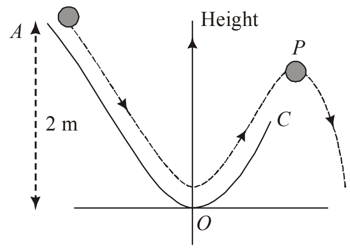
**26.**




A small block starts slipping down from a point $B$ on an inclined plane $AB$, which is making an angle θ with the horizontal section $BC$ is smooth and the remaining section $CA$ is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point $A$) of the inclined plane. If $BC = 2AC$, the coefficient of friction is given by $\mu = k\tan\theta$. The value of $k$ is ______.

**27.** A cricket ball of mass 0.15 kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force $F$ on the ball and moves horizontally a distance of 0.2 m while launching the ball, the value of $F$ (in N) is ($g$ = 10 ms$^{-2}$) ______.

**28.** A particle ($m$ = 1 kg) slides down a frictionless track (AOC) starting from rest at a point $A$ (height 2 m). After reaching $C$, the particle continues to move freely in air as a projectile. When it reaching its highest point $P$ (height 1 m), the kinetic energy of the particle (in $J$) is: (Figure drawn is schematic and not to scale; take $g = 10\ ms^{-2}$) ______.




**29.** A body of mass 2 kg is driven by an engine delivering a constant power of 1 J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m) ______.

**30.** Two bodies of the same mass are moving with the same speed, but in different directions in a plane. They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. The angle between the initial velocities of the two bodies (in degree) is ______.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (c) | **7** | (a) | **10** | (b) | **13** | (c) | **16** | (d) | **19** | (b) | **22** | (0.4) | **25** | (2) | **28** | (10.00) |
| **2** | (b) | **5** | (b) | **8** | (c) | **11** | (c) | **14** | (c) | **17** | (b) | **20** | (c) | **23** | (45) | **26** | (3) | **29** | (18) |
| **3** | (a) | **6** | (a) | **9** | (c) | **12** | (a) | **15** | (b) | **18** | (b) | **21** | (56) | **24** | (15) | **27** | (150.00) | **30** | (120) |

# 6

# SYSTEM OF PARTICLES AND ROTATIONAL MOTION

## MCQs with One Correct Answer

1. The centre of mass of three particles of masses 1 kg, 2 kg and 3 kg is at (3, 3, 3) with reference to a fixed coordinate system. Where should a fourth particle of mass 4 kg be placed so that the centre of mass of the system of all particles shifts to a point (1, 1, 1)?

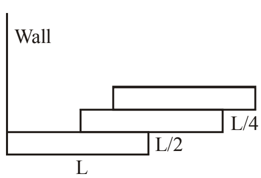
   (a) $(-1,-1,-1)$ (b) $(-2,-2,-2)$
   (c) $(2,2,2)$ (d) $(1,1,1)$

2. A loop of radius r and mass m rotating with an angular velocity $\omega_0$ is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip?

   (a) $\frac{r\omega_0}{4}$ (b) $\frac{r\omega_0}{3}$
   (c) $\frac{r\omega_0}{2}$ (d) $r\omega_0$

3. Three bricks each of length L and mass M are arranged as shown from the wall. The distance of the centre of mass of the system from the wall is

   (a) L/4
   (b) L/2
   (c) (3/2)L
   (d) (11/12)L




4. A particle of mass 2 kg is on a smooth horizontal table and moves in a circular path of radius 0.6 m. The height of the table from the ground is 0.8 m. If the angular speed of the particle is 12 rad $s^{-1}$, the magnitude of its angular momentum about a point on the ground right under the centre of the circle is :

   (a) 14.4 kg $m^2s^{-1}$ (b) 8.64 kg $m^2s^{-1}$
   (c) 20.16 kg $m^2s^{-1}$ (d) 11.52 kg $m^2s^{-1}$

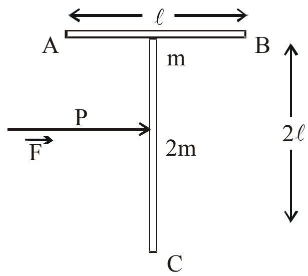
5. The moment of inertia of a rod about an axis through its centre and perpendicular to it is $\frac{1}{12}ML^2$ (where, M is the mass and L is the length of the rod). The rod is bent in the middle so that the two halves make an angle of 60°. The moment of inertia of the bent rod about the same axis would be

   (a) $\frac{1}{48}ML^2$ (b) $\frac{1}{12}ML^2$
   (c) $\frac{1}{24}ML^2$ (d) $\frac{ML^2}{8\sqrt{3}}$

6. A solid cylinder rolls up an inclined plane of angle of inclination 30°. At the bottom of the inclined plane, the C.M. of the cylinder has a speed of 5 m/s. How long will it take to return to the bottom?

   (a) 1.53 sec (b) 9.23 sec
   (c) 11.11 sec (d) 15.55 sec

**7.** A 'T' shaped object with dimensions shown in the figure, is lying on a smooth floor. A force '$\vec{F}$' is applied at the point P parallel to AB, such that the object has only the translational motion without rotation. Find the location of P with respect to C.



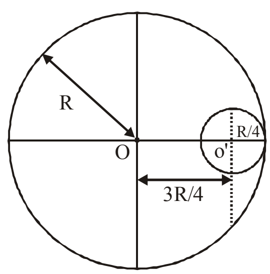

(a) $\frac{3}{2}\ell$

(b) $\frac{2}{3}\ell$

(c) $\ell$

(d) $\frac{4}{3}\ell$

**8.** A circular hole of radius $\frac{R}{4}$ is made in a thin uniform disc having mass M and radius R, as shown in figure. The moment of inertia of the remaining portion of the disc about an axis passing through the point O and perpendicular to the plane of the disc is :




(a) $\frac{219\,MR^2}{256}$

(b) $\frac{237\,MR^2}{512}$

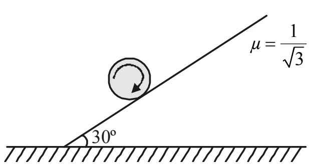
(c) $\frac{19\,MR^2}{512}$

(d) $\frac{197\,MR^2}{256}$

**9.** A uniform solid cylindrical roller of mass 'm' is being pulled on a horizontal surface with force F parallel to the surface and applied at its centre. If the acceleration of the cylinder is 'a' and it is rolling without slipping then the value of 'F' is :

(a) ma
(b) $\frac{5}{3}$ma
(c) $\frac{3}{2}$ma
(d) 2ma

**10.** A disc is rotated about its axis with a certain angular velocity and lowered gently on a rough inclined plane as shown in fig., then



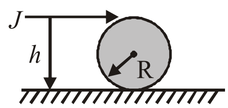

(a) it will rotate at the position where it was placed and then will move downwards
(b) it will go downwards just after it is lowered
(c) it will go downwards first and then climb up
(d) it will climb upwards and then move downwards

**11.** The moment of inertia of a body about a given axis is 1.2 kg m$^2$. Initially, the body is at rest. In order to produce a rotational kinetic energy of 1500 joule, an angular acceleration of 25 radian/sec$^2$ must be applied about that axis for a duration of

(a) 4 seconds
(b) 2 seconds
(c) 8 seconds
(d) 10 seconds

**12.** A solid sphere of mass $M$ and radius $R$ is placed on a rough horizontal surface. It is struck by a horizontal cuestick at a height $h$ above the surface.

The value of $h$ so that the sphere performs pure rolling motion immediately after it has been struck is

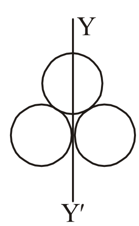



(a) $\frac{2R}{5}$
(b) $\frac{5R}{2}$
(c) $\frac{7R}{5}$
(d) $\frac{9R}{5}$

**13.** A hot solid sphere is rotating about a diameter at an angular velocity ω. If it cools so that its radius reduces to $\frac{1}{n}$ of its original value, its angular velocity becomes

(a) $\frac{\omega}{n}$
(b) $\frac{\omega}{n^2}$
(c) $\omega n$
(d) $n^2\omega$

**14.** Three rings each of mass M and radius R are arranged as shown in the figure. The moment of inertia of the system about YY′ will be



(a) $\frac{9}{2}MR^2$ (b) $\frac{3}{2}MR^2$

(c) $5MR^2$ (d) $\frac{7}{2}MR^2$

**15.** A torque of 30 N-m is applied on a 5 kg wheel whose moment of inertia is 2 kg–m$^2$ for 10 sec. The angle covered by the wheel in 10 sec will be

(a) 750 rad (b) 1500 rad

(c) 3000 rad (d) 6000 rad

**16.** A mass $m$ hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass $m$ and radius $R$. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass $m$, if the string does not slip on the pulley, is:

(a) $g$ (b) $\frac{2}{3}g$

(c) $\frac{g}{3}$ (d) $\frac{3}{2}g$

**17.** A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights $h_{sph}$ and $h_{cyl}$ on the incline. The ratio $\frac{h_{sph}}{h_{cyl}}$ is given by :

(a) $\frac{2}{\sqrt{5}}$ (b) 1

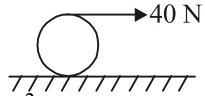
(c) $\frac{14}{15}$ (d) $\frac{4}{5}$

**18.** A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is:

(a) $\frac{3F}{2mR}$ (b) $\frac{F}{3mR}$

(c) $\frac{F}{2mR}$ (d) $\frac{2F}{3mR}$

**19.** A string is wound around a hollow cylinder of mass 5 kg and radius 0.5 m. If the string is now pulled with a horizontal force of 40 N, and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string)




(a) 20 rad/s$^2$ (b) 16 rad/s$^2$

(c) 12 rad/s$^2$ (d) 10 rad/s$^2$

**20.** A circular disc of radius $R$ is removed from a bigger circular disc of radius $2R$ such that the circumferences of the discs coincide. The centre of mass of the new disc is $\alpha/R$ form the centre of the bigger disc. The value of $\alpha$ is

(a) 1/4 (b) 1/3

(c) 1/2 (d) 1/6

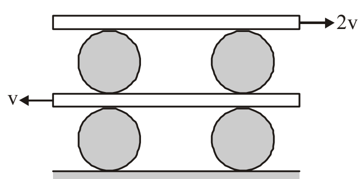
## Numeric Value Answer

**21.** A wheel rotates with a constant acceleration of 2.0 radian/sec$^2$. If the wheel starts from rest, the number of revolutions it makes in the first ten seconds will be approximately

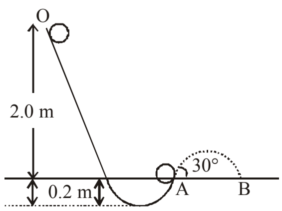
**22.** A circular thin disc of mass 2 kg has a diameter 0.2 m. Calculate its moment of inertia about an axis passing through the edge tangential to its axis and perpendicular to the plane of the disc (in kg-m$^2$)

**23.** A stone of mass $m$, tied to the end of a string, is whirled around in a horizontal circle (neglect the force due to gravity). The length of the string is reduced gradually keeping the angular momentum of the stone about the centre of the circle constant. Then, the tension in the string is given by $T = Ar^n$, where $A$ is a constant, $r$ is the instantaneous radius of the circle. The value of n is equal to

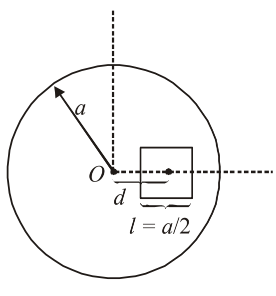
**24.** A system of uniform cylinders and plates is shown in fig. All the cylinders are identical and there is no slipping at any contact. The velocity of lower and upper plates is V and 2V, respectively as shown in fig. Then the ratio of angular speeds of the upper cylinders to lower cylinders is



**25.** A tennis ball (treated as hollow spherical shell) starting from O rolls down a hill. At point A the ball becomes air borne leaving at an angle of 30° with the horizontal. The ball strikes the ground at B. What is the value of the distance AB (in m)? (Moment of inertia of a spherical shell of mass $m$ and radius $R$ about its diameter $= \frac{2}{3}mR^2$)




**26.** A square shaped hole of side $l = \frac{a}{2}$ is carved out at a distance $d = \frac{a}{2}$ from the centre '$O$' of a uniform circular disk of radius $a$. If the distance of the centre of mass of the remaining portion from $O$ is $-\frac{a}{X}$, value of $X$ (to the nearest integer) is ___________.

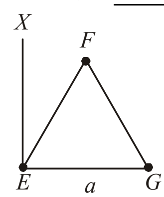



**27.** A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be :

**28.** A thin rod of mass 0.9 kg and length 1 m is suspended, at rest, from one end so that it can freely oscillate in the vertical plane. A particle of move 0.1 kg moving in a straight line with velocity 80 m/s hits the rod at its bottom most point and sticks to it (see figure). The angular speed (in rad/s) of the rod immediately after the collision will be ______________.

**29.** A person of 80 kg mass is standing on the rim of a circular platform of mass 200 kg rotating about its axis at 5 revolutions per minute (rpm). The person now starts moving towards the centre of the platform. What will be the rotational speed (in rpm) of the platform when the person reaches its centre __________.

**30.** An massless equilateral triangle $EFG$ of side '$a$' (As shown in figure) has three particles of mass $m$ situated at its vertices. The moment of inertia of the system about the line $EX$ perpendicular to $EG$ in the plane of $EFG$ is $\frac{N}{20}ma^2$ where $N$ is an integer. The value of $N$ is _________.




**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (a) | **7** | (d) | **10** | (a) | **13** | (d) | **16** | (b) | **19** | (b) | **22** | (0.03) | **25** | (2.08) | **28** | (20) |
| **2** | (c) | **5** | (b) | **8** | (b) | **11** | (b) | **14** | (d) | **17** | (c) | **20** | (b) | **23** | (–3) | **26** | (23.00) | **29** | (9.00) |
| **3** | (d) | **6** | (a) | **9** | (c) | **12** | (c) | **15** | (a) | **18** | (d) | **21** | (16) | **24** | (3) | **27** | (2) | **30** | (25) |

# 7

# GRAVITATION

## MCQs with One Correct Answer

**1.** Suppose the gravitational force varies inversely as the $n^{th}$ power of distance. Then the time period of a planet in circular orbit of radius 'R' around the sun will be proportional to

(a) $R^n$ (b) $R^{\left(\frac{n-1}{2}\right)}$

(c) $R^{\left(\frac{n+1}{2}\right)}$ (d) $R^{\left(\frac{n-2}{2}\right)}$

**2.** A straight rod of length $L$ extends from $x = a$ to $x = L + a$. Find the gravitational force it exerts on a point mass $m$ at $x = 0$ if the linear density of rod $\mu = A + Bx^2$.

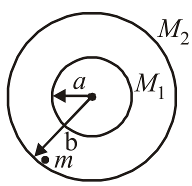
(a) $Gm\left[\frac{A}{a} + BL\right]$

(b) $Gm\left[A\left(\frac{1}{a} - \frac{1}{a+L}\right) + BL\right]$

(c) $Gm\left[BL + \frac{A}{a+L}\right]$

(d) $Gm\left[BL - \frac{A}{a}\right]$

**3.** Two concentric uniform shells of mass $M_1$ and $M_2$ are as shown in the figure. A particle of mass $m$ is located just within the shell $M_2$ on its inner surface. Gravitational force on '$m$' due to $M_1$ and $M_2$ will be




(a) zero

(b) $\frac{GM_1 m}{b^2}$

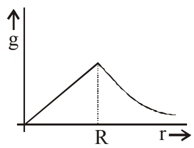
(c) $\frac{G(M_1 + M_2)m}{b^2}$

(d) None of these

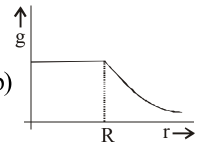
**4.** The mass density of a spherical body is given by $\rho(r) = \frac{k}{r}$ for $r \le R$ and $\rho(r) = 0$ for $r > R$, where $r$ is the distance from the centre.

The correct graph that describes qualitatively the acceleration due to gravity, g of a test particle as a function of r is :

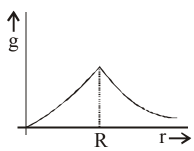
(a)



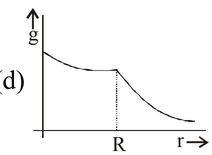

(b)




(c)




(d)

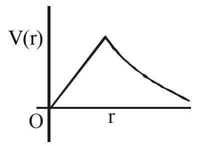


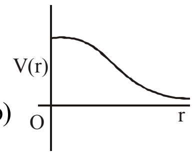

**5.** The change in potential energy, when a body of mass m is raised to a height nR from the earth's surface is (R = radius of earth)

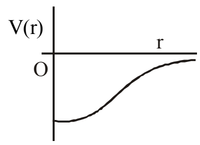
(a) $mgR\left(\frac{n}{n-1}\right)$ (b) nmgR

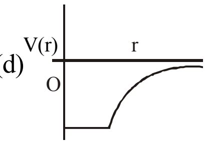
(c) $mgR\left(\frac{n^2}{n^2+1}\right)$ (d) $mgR\left(\frac{n}{n+1}\right)$

**6.** Which of the following most closely depicts the correct variation of the gravitational potential V(r) due to a large planet of radius R and uniform mass density ? (figures are not drawn to scale)

(a) 


(b) 


(c) 

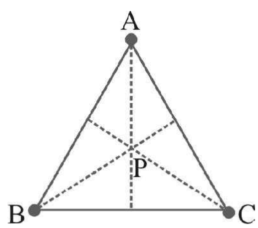

(d) 


**7.** A satellite of mass m revolves around the earth of radius R at a height 'x' from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is

(a) $\frac{gR^2}{R+x}$ (b) $\frac{gR}{R-x}$

(c) gx (d) $\left(\frac{gR^2}{R+x}\right)^{1/2}$

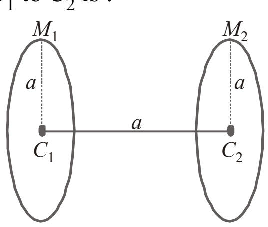
**8.** Three equal masses (each m) are placed at the corners of an equilateral triangle of side '$a$'. Then the escape velocity of an object from the circumcentre $P$ of triangle is :




(a) $\sqrt{\frac{2\sqrt{3}\,Gm}{a}}$ (b) $\sqrt{\frac{\sqrt{3}\,Gm}{a}}$

(c) $\sqrt{\frac{6\sqrt{3}\,Gm}{a}}$ (d) $\sqrt{\frac{3\sqrt{3}\,Gm}{a}}$

**9.** Two rings each of radius '$a$' are coaxial and the distance between their centres is $a$. The masses of the rings are $M_1$ and $M_2$. The work done in transporting a particle of a small mass $m$ from centre $C_1$ to $C_2$ is :




(a) $\frac{Gm(M_2-M_1)}{a}$

(b) $\frac{Gm(M_2-M_1)}{a\sqrt{2}}(\sqrt{2}+1)$

(c) $\frac{Gm(M_2-M_1)}{a\sqrt{2}}(\sqrt{2}-1)$

(d) $\frac{Gm(M_2-M_1)}{\sqrt{2}}a$

**10.** The depth $d$ at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value at the surface of the earth, is [R = radius of the earth]

(a) $\frac{R}{n}$ (b) $R\left(\frac{n-1}{n}\right)$

(c) $\frac{R}{n^2}$ (d) $R\left(\frac{n}{n+1}\right)$

**11.** A particle of mass $M$ is situated at the centre of a spherical shell of same mass and radius $a$. The gravitational potential at a point situated at $\frac{a}{4}$ distance from the centre, will be:

(a) $-\frac{5GM}{a}$ (b) $-\frac{2GM}{a}$

(c) $-\frac{GM}{a}$ (d) $-\frac{4GM}{a}$

**12.** If the gravitational force between two objects were proportional to 1/R (and not as $1/R^2$) where R is separation between them, then a particle in circular orbit under such a force would have its orbital speed v proportional to

(a) $1/R^2$ (b) $R^0$

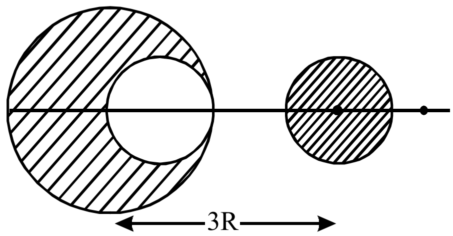
(c) $R^1$ (d) 1/R

**13.** A satellite is moving with a constant speed '$V$' in a circular orbit about the earth. An object of mass '$m$' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of its ejection, the kinetic energy of the object is

(a) $\frac{1}{2}mV^2$ (b) $mV^2$

(c) $\frac{3}{2}mV^2$ (d) $2mV^2$

**14.** From a sphere of mass M and radius R, a smaller sphere of radius $\frac{R}{2}$ is carved out such that the cavity made in the original sphere is between its centre and the periphery (See figure). For the configuration in the figure where the distance between the centre of the original sphere and the removed sphere is 3R, the gravitational force between the two sphere is:




(a) $\frac{41\,GM^2}{3600\,R^2}$ (b) $\frac{41\,GM^2}{450\,R^2}$

(c) $\frac{59\,GM^2}{450\,R^2}$ (d) $\frac{GM^2}{225\,R^2}$

**15.** Two particles of equal mass '*m*' go around a circle of radius *R* under the action of their mutual gravitational attraction. The speed of each particle with respect to their centre of mass is

(a) $\sqrt{\frac{Gm}{4R}}$ (b) $\sqrt{\frac{Gm}{3R}}$

(c) $\sqrt{\frac{Gm}{2R}}$ (d) $\sqrt{\frac{Gm}{R}}$

**16.** The change in the value of '*g*' at a height '*h*' above the surface of the earth is the same as at a depth '*d*' below the surface of earth. When both '*d*' and '*h*' are much smaller than the radius of earth, then which one of the following is correct?

(a) $d=\frac{3h}{2}$ (b) $d=\frac{h}{2}$

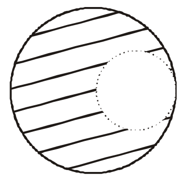
(c) $d=h$ (d) $d=2\,h$

**17.** A body of mass m is moving in a circular orbit of radius *R* about a planet of mass *M*. At some instant, it splits into two equal masses. The first mass moves in a circular orbit of radius $\frac{R}{2}$, and the other mass, in a circular orbit of radius $\frac{3R}{2}$. The difference between the final and initial total energies is:

(a) $-\frac{GMm}{2R}$ (b) $+\frac{GMm}{6R}$

(c) $-\frac{GMm}{6R}$ (d) $\frac{GMm}{2R}$

**18.** From a solid sphere of mass M and radius R, a spherical portion of radius R/2 is removed, as shown in the figure. Taking gravitational potential V = 0 at r = ∞, the potential at the centre of the cavity thus formed is :
(*G = gravitational constant*)



(a) $\frac{-2GM}{3R}$ (b) $\frac{-2GM}{R}$

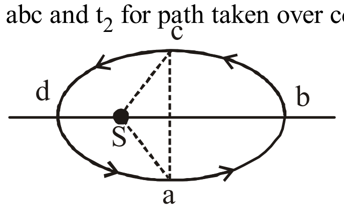
(c) $\frac{-GM}{2R}$ (d) $\frac{-GM}{R}$

**19.** Two bodies of masses *m* and 4 *m* are placed at a distance *r*. The gravitational potential at a point on the line joining them where the gravitational field is zero is:

(a) $-\frac{4Gm}{r}$ (b) $-\frac{6Gm}{r}$

(c) $-\frac{9Gm}{r}$ (d) zero

**20.** Figure shows elliptical path abcd of a planet around the sun S such that the area of triangle csa is $\frac{1}{4}$ the area of the ellipse. (See figure) With db as the semimajor axis, and ca as the semiminor axis. If $t_1$ is the time taken for planet to go over path abc and $t_2$ for path taken over cda then:




(a) $t_1 = 4t_2$ (b) $t_1 = 2t_2$

(c) $t_1 = 3t_2$ (d) $t_1 = t_2$

## Numeric Value Answer

**21.** If the distance of earth is halved from the sun, then the no. of days in a year will be

**22.** Mass M is divided into two parts xM and (1 – x )M. For a given separation, the value of x for which the gravitational attraction between the two pieces becomes maximum is

**23.** The mass of the earth is 81 times that of the moon and the radius of the earth is 3.5 times that of the moon. The ratio of the acceleration due to gravity at the surface of the moon to that at the surface of the earth is

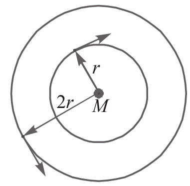
**24.** The escape velocity for a rocket from earth is 11.2 km/sec. Its value on a planet where acceleration due to gravity is double that on the earth and diameter of the planet is twice that of earth will be in km/sec

**25.** The earth is assumed to be a sphere of radius R. A platform is arranged at a height R from the surface of the earth. The escape velocity of a body from this platform is fv, v is its escape velocity from the surface of the earth. The value of f is

**26.** The gravitational potential difference between the surface of a planet and a point 10 m above is 4.0 J/kg. The gravitational field (in N/kg) in this region, assumed uniform is:

**27.** A geo-stationary satellite orbits around the earth in a circular orbit of radius 36,000km. Then, the time period (in hr) of a spy satellite orbiting a few hundred km above the earth's surface ($R_{earth}$ = 6,400km) will approximately be

**28.** Two satellites of masses $m$ and $2m$ are revolving around a planet of mass $M$ with different speeds in orbits of radii $r$ and $2r$ respectively. The ratio of minimum and maximum forces on the planet due to satellites is




**29.** Take the mean distance of the moon and the sun from the earth to be $0.4 \times 10^6$ km and $150 \times 10^6$ km respectively. Their masses are $8 \times 10^{22}$ kg and $2 \times 10^{30}$ kg respectively. The radius of the earth is 6400 km. Let $\Delta F_1$ be the difference in the forces exerted by the moon at the nearest and farthest points on the earth and $\Delta F_2$ be the difference in the force exerted by the sun at the nearest and farthest points on the earth. Then, the number closest to $\frac{\Delta F_1}{\Delta F_2}$ is:

**30.** The value of acceleration due to gravity is $g_1$ at a height $h = \frac{R}{2}$ ($R$ = radius of the earth) from the surface of the earth. It is again equal to $g_1$ and a depth $d$ below the surface of the earth. The ratio $\left(\frac{d}{R}\right)$ equals :

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (b) | **7** | (d) | **10** | (b) | **13** | (b) | **16** | (d) | **19** | (c) | **22** | (0.5) | **25** | (0.70) | **28** | (0.33) |
| **2** | (b) | **5** | (d) | **8** | (c) | **11** | (a) | **14** | (a) | **17** | (c) | **20** | (c) | **23** | (0.15) | **26** | (0.40) | **29** | (2) |
| **3** | (b) | **6** | (c) | **9** | (c) | **12** | (b) | **15** | (a) | **18** | (d) | **21** | (129) | **24** | (22.4) | **27** | (2) | **30** | (0.56) |

# 8

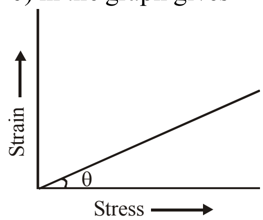
# MECHANICAL PROPERTIES OF SOLIDS

## MCQs with One Correct Answer

1. The upper end of a wire of diameter 12mm and length 1m is clamped and its other end is twisted through an angle of 30°. The angle of shear is
(a) 18° (b) 0.18°
(c) 36° (d) 0.36°

2. What will be the density of ocean water at a depth where the pressure is 80 atm? [Given that its density at the surface is $1.03 \times 10^3$ kg/m$^3$, compressibility of water = $45.8 \times 10^{-11}$/Pa. Given: 1 atm = $1.013 \times 10^5$ Pa.]
(a) $1.03 \times 10^3$ kg/m$^3$ (b) $5.03 \times 10^3$ kg/m$^3$
(c) $8.03 \times 10^3$ kg/m$^3$ (d) $9.03 \times 10^3$ kg/m$^3$

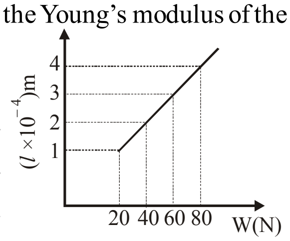
3. The value of tan (90° – θ) in the graph gives
(a) Young's modulus of elasticity
(b) compressibility
(c) shear strain
(d) tensile strength




4. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area $A$ and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by $\Delta x$ on applying force $F$, how much force is needed to stretch wire 2 by the same amount?
(a) $4F$ (b) $6F$
(c) $9F$ (d) $F$

5. A beam of metal supported at the two edges is loaded at the centre. The depression at the centre is proportional to
(a) $Y^2$ (b) Y
(c) 1/Y (d) $1/Y^2$

6. The adjacent graph shows the extension ($\Delta l$) of a wire of length 1m suspended from the top of a roof at one end with a load $W$ connected to the other end. If the cross-sectional area of the wire is $10^{-6}$m$^2$, calculate the Young's modulus of the material of the wire.
(a) $2 \times 10^{11}$ N/m$^2$
(b) $2 \times 10^{-11}$ N/m$^2$
(c) $2 \times 10^{-12}$ N/m$^2$
(d) $2 \times 10^{-13}$ N/m$^2$




7. The force required to stretch a steel wire of 1 cm$^2$ cross-section to 1.1 times its length would be [$Y = 2 \times 10^{11}$ Nm$^{-2}$]
(a) $2 \times 10^6$ N (b) $2 \times 10^3$ N
(c) $2 \times 10^{-6}$ N (d) $2 \times 10^{-7}$ N

8. A steel ring of radius $r$ and cross-section area 'A' is fitted on to a wooden disc of radius R(R > r). If Young's modulus be E, then the force with which the steel ring is expanded is
(a) $AE\dfrac{R}{r}$ (b) $AE\left(\dfrac{R-r}{r}\right)$
(c) $\dfrac{E}{A}\left(\dfrac{R-r}{A}\right)$ (d) $\dfrac{Er}{AR}$

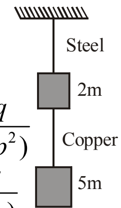
9. One end of a uniform wire of length L and of weight W is attached rigidly to a point in the roof and $W_1$ weight is suspended from lower end. If A is area of cross-section of the wire, the stress in the wire at a height $\dfrac{L}{4}$ from the upper end is
(a) $\dfrac{W_1 + W}{A}$ (b) $\dfrac{W_1 + 3W/4}{A}$
(c) $\dfrac{W_1 + W/4}{A}$ (d) $\dfrac{4W_1 + 3W}{A}$

**10.** The length of a metal wire is $\ell_1$ when the tension in it is $T_1$ and is $\ell_2$ when the tension is $T_2$. The original length of the wire is

(a) $\dfrac{\ell_1+\ell_2}{2}$ (b) $\dfrac{\ell_1 T_2+\ell_2 T_1}{T_1+T_2}$

(c) $\dfrac{\ell_1 T_2-\ell_2 T_1}{T_2-T_1}$ (d) $\sqrt{T_1 T_2 \ell_1 \ell_2}$

**11.** If the ratio of diameters, lengths and Young's modulus of steel and copper wires shown in the figure are $p$, $q$ and $s$ respectively, then the corresponding ratio of increase in their lengths would be




(a) $\dfrac{7q}{(5sp)}$ (b) $\dfrac{5q}{(7sp^2)}$

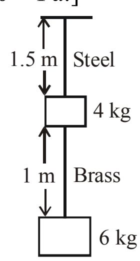
(c) $\dfrac{7q}{(5sp^2)}$ (d) $\dfrac{2q}{(5sp)}$

**12.** A 14.5 kg mass, fastened to the end of a steel wire of unstretched length 1m, is whirled in a vertical circle with an angular velocity of 2 rev/s at the bottom of the circle. The cross-sectional area of the wire is 0.065 $cm^2$. The elongation of the wire when the mass is at the lowest point of its path is [$Y_{steel} = 2 \times 10^{11}$ N/$m^2$]

(a) 9.67 mm (b) 6.67 mm

(c) 1.87 mm (d) 0.12 mm

**13.** The elongation of the steel and brass wire in the adjacent figure are respectively. [Unloaded length of steel wire is 1.5 m and of brass wire is 1m, diameter of each wire = 0.25 cm. Young's modulus of steel is $2 \times 10^{11}$ Pa and that of brass is $0.91 \times 10^{11}$ Pa.]




(a) $1.49 \times 10^{-4}$ m, $1.3 \times 10^{-4}$ m

(b) $2.94 \times 10^{-4}$ m, $2.3 \times 10^{-4}$ m

(c) $5.12 \times 10^{-4}$ m, $3.2 \times 10^{-4}$ m

(d) $1.12 \times 10^{-4}$ m, $6.2 \times 10^{-4}$ m

**14.** The edge of an aluminium cube is 10 cm long. One face of the cube is firmly fixed to a vertical wall. A mass of 100 kg is then attached to the opposite face of the cube. The shear modulus of aluminium is 25 G Pa. What is the vertical deflection of this face?
(1Pa = 1N/$m^2$) (g = 10 m/$s^2$).

(a) $4 \times 10^{-7}$ m (b) $3 \times 10^{-7}$ m

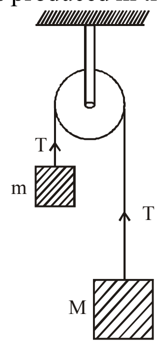
(c) $2 \times 10^{-7}$ m (d) $1 \times 10^{-7}$ m

**15.** Four identical hollow cylindrical columns of mild steel support a big structure of mass 50,000 kg. The inner and outer radii of each column are 30 cm and 60 cm respectively. Assuming the load distribution to be uniform, the compressional strain in each column is [The Young's modulus of steel is $2 \times 10^{11}$ Pa.]

(a) $2.31 \times 10^{-7}$ (b) $4.12 \times 10^{-7}$

(c) $7.21 \times 10^{-7}$ (d) $9.93 \times 10^{-7}$

**16.** Two blocks of masses m and M are connected by means of a metal wire of cross-sectional area A passing over a frictionless fixed pulley as shown in the figure. The system is then released. If M = 2 m, then the stress produced in the wire is :




(a) $\dfrac{2mg}{3A}$

(b) $\dfrac{4mg}{3A}$

(c) $\dfrac{mg}{A}$

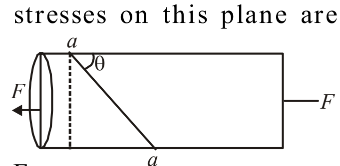
(d) $\dfrac{3mg}{4A}$

**17.** Two strips of metal are riveted together at their ends by four rivets, each of diameter 6 mm. What is the maximum tension that can be exerted by the riveted strip if the shearing stress on the rivet is not to exceed $6.9 \times 10^7$ Pa? Assume that each rivet is to carry one quarter of the load.

(a) 7800 N (b) 7000 N

(c) 9000 N (d) 1000 N

**18.** Consider a long steel bar under a tensile stress due to forces $F$ acting at the edges along the length of the bar (figure). Consider a plane making an angle θ with the length. The tensile and shearing stresses on this plane are respectively



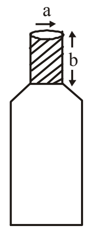

(a) $\dfrac{F}{A}\cos^2\theta, \dfrac{F}{2A}\sin 2\theta$

(b) $\dfrac{F}{5A}\sin^2\theta, \dfrac{F}{2A}\sin 4\theta$

(c) $\dfrac{F}{9A}\sin^3\theta, \dfrac{F}{3A}\sin 5\theta$

(d) None of these

**19.** A bottle has an opening of radius a and length b. A cork of length b and radius $(a + \Delta a)$ where $(\Delta a << a)$ is compressed to fit into the opening completely (see figure). If the bulk modulus of cork is B and frictional coefficient between the bottle and cork is $\mu$ then the force needed to push the cork into the bottle is :




(a) $(\pi \mu B b)\, a$

(b) $(2\pi\mu Bb)\, \Delta a$

(c) $(\pi\mu B b)\, \Delta a$

(d) $(4\,\pi\mu B b)\, \Delta a$

**20.** A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's modulii of copper and steel are respectively $1.0 \times 10^{11}$ $Nm^{-2}$ and $2.0 \times 10^{11}$ $Nm^{-2}$, the total extension of the composite wire is :

(a) 1.75 mm (b) 2.0 mm
(c) 1.50 mm (d) 1.25 mm

## Numeric Value Answer

**21.** A 2 m long rod of radius 1 cm which is fixed from one end is given a twist of 0.8 radians. The shear strain developed will be

**22.** If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is $1 \times 10^3$ $kg/m^3$ and $g = 10 m/s^2$, then the volume elasticity in $N/m^2$ will be

**23.** A uniform cube is subjected to volume compression. If each side is decreased by 1%, then bulk strain is

**24.** What is the bulk modulus (in Pa) of water for the given data : Initial volume = 100 litre, pressure increase = 100 atmosphere, final volume = 100.5 litre (1 atmosphere = $1.013 \times 10^5$ Pa)

**25.** The breaking stress of the material of a wire is $6 \times 10^6$ $Nm^{-2}$. Then density $\rho$ of the material is $3 \times 10^3$ $kg\ m^{-3}$. If the wire is to break under its own weight, the length of the wire (in m) made of that material should be (take $g = 10\ ms^{-2}$)

**26.** A body of mass m = 10 kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad $s^{-1}$) with which it can be rotated about its other end in space station is (Breaking stress of wire = $4.8 \times 10^7$ $Nm^{-2}$ and area of cross-section of the wire = $10^{-2}$ $cm^2$) is _______ .

**27.** A steel wire can sustain 100 kg weight without breaking. If the wire is cut into two equal parts, each part can sustain a weight (in kg) of

**28.** Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is:

**29.** Young's moduli of two wires A and B are in the ratio 7 : 4. Wire A is 2 m long and has radius R. Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R (in mm) is close to :

**30.** The elastic limit of brass is 379 MPa. What should be the minimum diameter (in mm) of a brass rod if it is to support a 400 N load without exceeding its elastic limit?

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (c) | **7** | (a) | **10** | (c) | **13** | (a) | **16** | (b) | **19** | (d) | **22** | $(2\times10^9)$ | **25** | (200) | **28** | (1.41) |
| **2** | (a) | **5** | (c) | **8** | (b) | **11** | (c) | **14** | (a) | **17** | (a) | **20** | (d) | **23** | (0.03) | **26** | (4) | **29** | (1.75) |
| **3** | (a) | **6** | (a) | **9** | (b) | **12** | (c) | **15** | (c) | **18** | (a) | **21** | (0.04) | **24** | $(2.026\times10^9)$ | **27** | (100) | **30** | (1.15) |

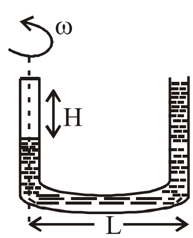
# 9 MECHANICAL PROPERTIES OF FLUIDS

**MCQs with One Correct Answer**

1. The total weight of a piece of wood is 6 kg. In the floating state in water its $\frac{1}{3}$ part remains inside the water. On this floating piece of wood what maximum weight is to be put such that the whole of the piece of wood is to be drowned in the water?
(a) 15 kg (b) 14 kg
(c) 10 kg (d) 12 kg

2. A hydraulic automobile lift is designed to lift cars with a maximum mass of 3000 kg. The area of cross - section of the piston carrying the load is 425 cm$^2$. What maximum pressure would the smaller piston have to bear?
(a) $15.82 \times 10^5$ Pa (b) $6.92 \times 10^5$ Pa
(c) $2.63 \times 10^5$ Pa (d) $1.12 \times 10^5$ Pa

3. A U-shaped tube contains a liquid of density $\rho$ and it is rotated about the left dotted line as shown in the figure. Find the difference in the levels of liquid column.




(a) $\frac{\omega^2 L^2}{2g}$
(b) $\frac{\omega^2 L^2}{2\sqrt{2}g}$
(c) $\frac{2\omega^2 L^2}{g}$
(d) $\frac{2\sqrt{2}\omega^2 L^2}{g}$

4. Air of density 1.2 kg m$^{-3}$ is blowing across the horizontal wings of an aeroplane in such a way that its speeds above and below the wings are 150 ms$^{-1}$ and 100 ms$^{-1}$, respectively. The pressure difference between the two sides of the wings, is :
(a) 60 Nm$^{-2}$ (b) 180 Nm$^{-2}$
(c) 7500 Nm$^{-2}$ (d) 12500 Nm$^{-2}$

5. If it takes 5 minutes to fill a 15 litre bucket from a water tap of diameter $\frac{2}{\sqrt{\pi}}$ cm then the Reynolds number for the flow is (density of water = $10^3$ kg/m$^3$) and viscosity of water = $10^{-3}$ Pa s) close to :
(a) 1100 (b) 11000
(c) 550 (d) 5500

6. Water flows into a large tank with flat bottom at the rate of $10^{-4}$ m$^3$ s$^{-(1)}$ Water is also leaking out of a hole of area 1 cm$^2$ at its bottom. If the height of the water in the tank remains steady, then this height is:
(a) 5.1 cm (b) 7 cm
(c) 4 cm (d) 9 cm

7. A spherical drop of radius R is divided into eight equal droplets. If surface tension is T, then the work done in this process is
(a) $2\pi R^2 T$ (b) $3\pi R^2 T$
(c) $4\pi R^2 T$ (d) $2\pi R T^2$

8. A U-shaped wire is dipped in a soap solution and removed. The thin soap film formed between the wire and the light slider supports a weight of

$1.5 \times 10^{-2}$ N (which includes the small weight of the slider). The length of the slider is 30 cm. What is the surface tension of the film?

(a) $2.5 \times 10^{-2}\,\text{Nm}^{-1}$ (b) $5.5 \times 10^{-2}\,\text{Nm}^{-1}$

(c) $9.5 \times 10^{-2}\,\text{Nm}^{-1}$ (d) $11.5 \times 10^{-2}\,\text{Nm}^{-1}$

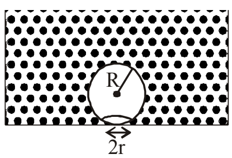
**9.** Water rises in a capillary tube to a certain height such that the upward force due to surface tension is balanced by $7.5 \times 10^{-4}$N force due to the weight of the liquid. If the surface tension of water is $6 \times 10^{-2}\text{Nm}^{-1}$, the inner circumference of the capillary must be

(a) $1.25 \times 10^{-2}$m (b) $0.50 \times 10^{-2}$m

(c) $6.5 \times 10^{-2}$m (d) $12.5 \times 10^{-2}$m

**10.** On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be sphere of radius R and making a circular contact of radius r with the bottom of the vessel. If r ≪ R and the surface tension of water is T, value of r just before bubbles detach is:

(density of water is $\rho_w$)




(a) $R^2\sqrt{\dfrac{\rho_w g}{3T}}$ (b) $R^2\sqrt{\dfrac{2\rho_w g}{3T}}$

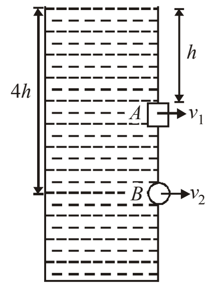
(c) $R^2\sqrt{\dfrac{\rho_w g}{T}}$ (d) $R^2\sqrt{\dfrac{3\rho_w g}{T}}$

**11.** A U tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are in level with 10.0 cm of water in one arm and 12.5 cm of spirit in the other, the relative density of spirit is

(a) 0.8 (b) 1.32

(c) 2.38 (d) 3.52

**12.** A square hole of side length $\ell$ is made at a depth of h and a circular hole of radius r is made at a depth of 4h from the surface of water in a water tank kept on a horizontal surface (See figure). If $\ell << h, r << h$ and the rate of water flow from the two holes is the same, then r is equal to




(a) $\dfrac{\ell}{\sqrt{2\pi}}$

(b) $\dfrac{\ell}{\sqrt{3\pi}}$

(c) $\dfrac{\ell}{3\pi}$

(d) $\dfrac{\ell}{2\pi}$

**13.** If the terminal speed of a sphere of gold (density = 19.5 kg/m$^3$) is 0.2 m/s in a viscous liquid (density = 1.5 kg/m$^3$), find the terminal speed of a sphere of silver (density = 10.5 kg/m$^3$) of the same size in the same liquid

(a) 0.4 m/s (b) 0.133 m/s

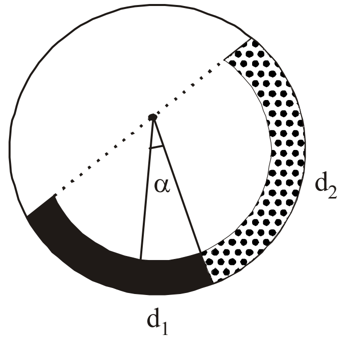
(c) 0.1 m/s (d) 0.2 m/s

**14.** Two tubes of radii $r_1$ and $r_2$, and lengths $l_1$ and $l_2$, respectively, are connected in series and a liquid flows through each of them in streamline conditions. $P_1$ and $P_2$ are pressure differences across the two tubes. If $P_2$ is $4P_1$ and $l_2$ is $\dfrac{l_1}{4}$, then the radius $r_2$ will be equal to :

(a) $r_1$ (b) $2r_1$

(c) $4r_1$ (d) $\dfrac{r_1}{2}$

**15.** There is a circular tube in a vertical plane. Two liquids which do not mix and of densities $d_1$ and $d_2$ are filled in the tube. Each liquid subtends 90º angle at centre. Radius joining their interface makes an angle $\alpha$ with vertical. Ratio $\dfrac{d_1}{d_2}$ is:




(a) $\dfrac{1+\sin\alpha}{1-\sin\alpha}$ (b) $\dfrac{1+\cos\alpha}{1-\cos\alpha}$

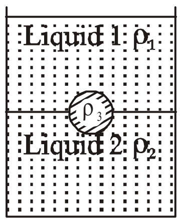
(c) $\dfrac{1+\tan\alpha}{1-\tan\alpha}$ (d) $\dfrac{1+\sin\alpha}{1-\cos\alpha}$

**16.** A spherical solid ball of volume $V$ is made of a material of density $\rho_1$. It is falling through a liquid of density $\rho_1$ ($\rho_2 < \rho_1$). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed $v$, i.e., $F_{viscous} = -kv^2$ ($k > 0$). The terminal speed of the ball is

(a) $\sqrt{\dfrac{Vg(\rho_1-\rho_2)}{k}}$  (b) $\dfrac{Vg\rho_1}{k}$

(c) $\sqrt{\dfrac{Vg\rho_1}{k}}$  (d) $\dfrac{Vg(\rho_1-\rho_2)}{k}$

**17.** A jar is filled with two non-mixing liquids 1 and 2 having densities $\rho_1$ and, $\rho_2$ respectively. A solid ball, made of a material of density $\rho_3$, is dropped in the jar. It comes to equilibrium in the position shown in the figure.Which of the following is true for $\rho_1$, $\rho_2$ and $\rho_3$?




(a) $\rho_3 < \rho_1 < \rho_2$  (b) $\rho_1 > \rho_3 > \rho_2$

(c) $\rho_1 < \rho_2 < \rho_3$  (d) $\rho_1 < \rho_3 < \rho_2$

**18.** A thin uniform tube is bent into a circle of radius $r$ in the vertical plane. Equal volumes of two immiscible liquids, whose densities are $\rho_1$ and $\rho_1$ ($\rho_1 > \rho_2$) fill half the circle. The angle $\theta$ between the radius vector passing through the common interface and the vertical is

(a) $\theta = \tan^{-1}\left[\dfrac{\pi}{2}\left(\dfrac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)\right]$

(b) $\theta = \tan^{-1}\dfrac{\pi}{2}\left(\dfrac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)$

(c) $\theta = \tan^{-1}\pi\left(\dfrac{\rho_1}{\rho_2}\right)$

(d) None of above

**19.** A body of density $\rho'$ is dropped from rest at a height h into a lake of density $\rho$ where $\rho > \rho'$ neglecting all dissipative forces, calculate the maximum depth to which the body sinks.

(a) $\dfrac{h}{\rho-\rho'}$  (b) $\dfrac{h\rho'}{\rho}$

(c) $\dfrac{h\rho'}{\rho-\rho'}$  (d) $\dfrac{h\rho}{\rho-\rho'}$

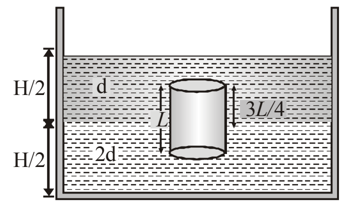
**20.** A homogeneous solid cylinder of length L (L < H/2) cross-sectional area A/5 is immersed such that it floats with its axis vertical at the liquid-liquid interface with length L/4 in the denser liquid as shown in the fig. The lower density liquid is open to atmosphere having pressure $P_0$. Then density of solid (material of cylinder) D is given by

(a) $\dfrac{5}{4}d$

(b) $\dfrac{4}{5}d$

(c) d

(d) $\dfrac{d}{5}$




## Numeric Value Answer

**21.** A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H. Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice.

[Take atmospheric pressure = $1.0 \times 10^5$ N/m$^2$, density of water = 1000 kg/m$^3$ and g = 10 m/s$^2$. Neglect any effect of surface tension.]

**22.** When a ball is released from rest in a very long column of viscous liquid, its downward acceleration is 'a' (just after release). Its acceleration when it has acquired two third of the maximum velocity is a/X. Find the value of X.

**23.** An isolated and charged spherical soap bubble has a radius r and the pressure inside is atmospheric. If T is the surface tension of soap solution, then charge on drop is $X\,\pi r\sqrt{2rT\varepsilon_0}$ find the value of X.

**24.** A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm. If the entire arrangement is put in a freely falling elevator the length (in m) of water column in the capillary tube will be

**25.** A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in $ms^{-1}$) through a small hole on the side wall of the cylinder near its bottom is

**26.** Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8g $cm^{-3}$, the angle remains the same. If density of the material of the sphere is 1.6 g $cm^{-3}$, the dielectric constant of the liquid is

**27.** When a long glass capillary tube of radius 0.015 cm is dipped in a liquid, the liquid rises to a height of 15 cm within it. If the contact angle between the liquid and glass to close to 0°, the surface tension of the liquid, in milliNewton $m^{-1}$, is [$\rho_{(liquid)} = 900\ kgm^{-3}$, $g = 10\ ms^{-2}$] (Give answer in closest integer) __________.

**28.** An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density $10^3$ kg/$m^3$. The pressure inside the bubble is 1100 $Nm^{-2}$ greater than the atmospheric pressure. At what depth (in m) is the bubble below the surface of the liquid? ($g = 9.8\ ms^{-2}$)

**29.** An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional 46 cm. What will be length (in cm) of the air column above mercury in the tube now?
(Atmospheric pressure = 76 cm of Hg)

**30.** The velocity of water in a river is 18 km/h near the surface. If the river is 5 m deep, find the shearing stress (in N/$m^2$) between the horizontal layers of water. The co-efficient of viscosity of water = $10^{-2}$ poise.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (c) | **7** | (c) | **10** | (b) | **13** | (c) | **16** | (a) | **19** | (c) | **22** | (3) | **25** | (20) | **28** | (0.1) |
| **2** | (b) | **5** | (d) | **8** | (a) | **11** | (a) | **14** | (d) | **17** | (d) | **20** | (a) | **23** | (8) | **26** | (2) | **29** | (16) |
| **3** | (a) | **6** | (a) | **9** | (a) | **12** | (a) | **15** | (c) | **18** | (d) | **21** | (6) | **24** | (20) | **27** | (101) | **30** | ($10^{-2}$) |

# 10

# THERMAL PROPERTIES OF MATTER

## MCQs with One Correct Answer

**1.** Two rods, one of aluminum and the other made of steel, having initial length $\ell_1$ and $\ell_2$ are connected together to form a single rod of length $\ell_1 + \ell_2$. The coefficients of linear expansion for aluminum and steel are $\alpha_a$ and $\alpha_s$ respectively. If the length of each rod increases by the same amount when their temperature are raised by $t^{\circ}$C, then find the ratio $\ell_1/(\ell_1 + \ell_2)$.

(a) $\alpha_s/\alpha_a$ (b) $\alpha_a/\alpha_s$
(c) $\alpha_s/(\alpha_a+\alpha_s)$ (d) $\alpha_a/(\alpha_a+\alpha_s)$

**2.** If a graph is plotted taking the temperature in Fahrenheit along Y-axis and the corresponding temperature in Celsius along the X-axis, it will be a straight line

(a) having a + ve intercept on Y-axis
(b) having a + ve intercept on X-axis
(c) passing through the origin
(d) having a – ve intercepts on both the axis

**3.** A steel rail of length 5 m and area of cross-section 40 cm$^2$ is prevented from expanding along its length while the temperature rises by 10°C. If coefficient of linear expansion and Young's modulus of steel are $1.2 \times 10^{-5}$ K$^{-1}$ and $2 \times 10^{11}$ Nm$^{-2}$ respectively, the force developed in the rail is approximately:

(a) $2 \times 10^7$ N (b) $1 \times 10^5$ N
(c) $2 \times 10^9$ N (d) $3 \times 10^{-5}$ N

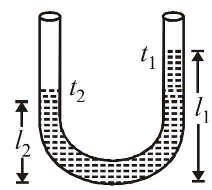
**4.** A glass flask of volume one litre at 0°C is filled full with mercury at this temperature. The flask and mercury are now heated to 100°C. How much mercury will spill out, if coefficient of volume expansion of mercury is $1.82 \times 10^{-4}$/ºC and linear expansion of glass is $0.1 \times 10^{-4}$ /ºC respectively?

(a) 21.2 cc (b) 15.2 cc
(c) 1.52 cc (d) 2.12 cc

**5.** The coefficient of linear expansion of crystal in one direction is $\alpha_1$ and that in every direction perpendicular to it is $\alpha_2$ . The coefficient of cubical expansion is

(a) $\alpha_1 + \alpha_2$ (b) $\alpha_1 + 2\alpha_2$
(c) $2\alpha_1 + \alpha_2$ (d) $\alpha_1 + \alpha_2/2$

**6.** In a vertical U-tube containing a liquid, the two arms are maintained at different temperatures $t_1$ and $t_2$. The liquid columns in the two arms have heights $l_1$ and $l_2$ respectively. The coefficient of volume expansion of the liquid is equal to




(a) $\dfrac{l_1 - l_2}{l_2 t_1 - l_1 t_2}$

(b) $\dfrac{l_1 - l_2}{l_1 t_1 - l_2 t_2}$

(c) $\dfrac{l_1 + l_2}{l_2 t_1 + l_1 t_2}$

(d) $\dfrac{l_1 + l_2}{l_1 t_1 + l_2 t_2}$

**7.** In an experiment a sphere of aluminium of mass 0.20 kg is heated upto 150°C. Immediately, it is put into water of volume 150 cc at 27°C kept in a calorimeter of water equivalent to 0.025 kg. Final temperature of the system is 40°C. The specific heat of aluminium is :
(take 4.2 joule=1 calorie)

(a) 378 J/kg – °C (b) 315 J/kg – °C
(c) 476 J/kg – °C (d) 434 J/kg – °C

**8.** A black body at 1227°C emits radiations with maximum intensity at a wavelength of 5000Å. If the temperature of the body is increased by 1000°C, the maximum intensity will be observed at

(a) 5000Å (b) 6000Å

(c) 3000Å (d) 4000Å

**9.** Two rods of same length transfer a given amount of heat in 12 second, when they are joined as shown in figure (i). But when they are joined as shown in figure (ii), then they will transfer same heat in same conditions in

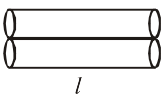



Fig. (i)

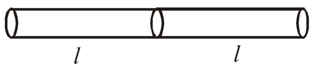



Fig. (ii)

(a) 24 s (b) 13 s

(c) 15 s (d) 48 s

**10.** Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature $T_0$, while Box B contains one mole of helium at temperature $\left(\frac{7}{3}\right)T_0$. The boxes are then put into thermal contact with each other, and heat flows between them until the gases reach a common final temperature (ignore the heat capacity of boxes). Then, the final temperature of the gases, $T_f$ in terms of $T_0$ is

(a) $T_f = \frac{3}{7}T_0$ (b) $T_f = \frac{7}{3}T_0$

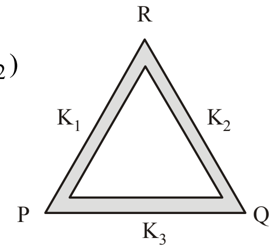
(c) $T_f = \frac{3}{2}T_0$ (d) $T_f = \frac{5}{2}T_0$

**11.** A bullet of mass 10gm moving with a speed of 20 m/s hits an ice block of mass 990gm kept on a frictionless floor and gets stuck in it. How much ice will melt if 50% of the lost KE goes to ice ? (Initial temperature of the ice block = 0°C, J = 4.2 J/cal and latent heat of ice = 80 cal/g)

(a) 0.001 gm (b) 0.002 gm

(c) 0.003 gm (d) None of these

**12.** Three rods of same dimensions are arranged as shown in figure. They have thermal conductivities $K_1$, $K_2$ and $K_3$ . The points P and Q are maintained at different temperatures for the heat to flow at the same rate along PRQ and PQ then which of the following option is correct?




(a) $K_3 = \frac{1}{2}(K_1 + K_2)$

(b) $K_3 = K_1 + K_2$

(c) $K_3 = \frac{K_1K_2}{K_1 + K_2}$

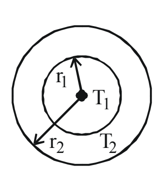
(d) $K_3 = 2(K_1 + K_2)$

**13.** A copper sphere cools from 62°C to 50°C in 10 minutes and to 42°C in the next 10 minutes. Calculate the temperature of the surroundings.

(a) 28°C (b) 26°C

(c) 32°C (d) 62°C

**14.** The figure shows a system of two concentric spheres of radii $r_1$ and $r_2$ kept at temperatures $T_1$ and $T_2$, respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to




(a) $\ell n\left(\frac{r_2}{r_1}\right)$

(b) $\frac{(r_2 - r_1)}{(r_1 r_2)}$

(c) $(r_2 - r_1)$

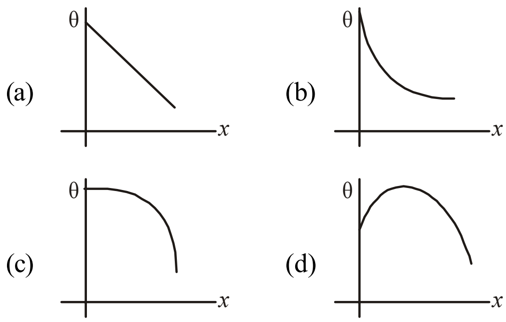
(d) $\frac{r_1 r_2}{(r_2 - r_1)}$

**15.** A thermometer graduated according to a linear scale reads a value $x_0$ when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer in the contact with the object reads $x_0/2$?

(a) 25 (b) 60

(c) 40 (d) 35

**16.** A long metallic bar is carrying heat from one of its ends to the other end under steady–state. The variation of temperature $\theta$ along the length $x$ of the bar from its hot end is best described by which of the following figures?




**17.** 500 g of water and 100 g of ice at 0°C are in a calorimeter whose water equivalent is 40 g. 10 g of steam at 100°C is added to it. Then water in the calorimeter is : (Latent heat of ice = 80 cal/g, Latent heat of steam = 540 cal/ g)

(a) 580 g (b) 590 g
(c) 600 g (d) 610 g

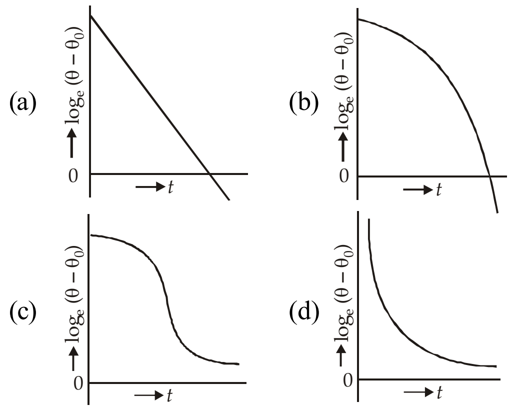
**18.** Two rods A and B of identical dimensions are at temperature 30°C. If A is heated upto 180°C and B upto T°C, then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4 : 3, then the value of T is :

(a) 230°C (b) 270°C
(c) 200°C (d) 250°C

**19.** A large cylindrical rod of length $L$ is made by joining two identical rods of copper and steel of length $\left(\frac{L}{2}\right)$ each. The rods are completely insulated from the surroundings. If the free end of copper rod is maintained at 100°C and that of steel at 0°C then the temperature of junction is (Thermal conductivity of copper is 9 times that of steel)

(a) 90°C (b) 50°C
(c) 10°C (d) 67°C

**20.** A liquid in a beaker has temperature $\theta(t)$ at time $t$ and $\theta_0$ is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e(\theta - \theta_0)$ and $t$ is :

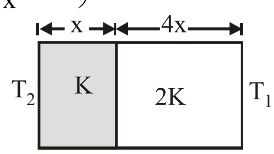



## Numeric Value Answer

**21.** The coefficient of apparent expansion of mercury in a glass vessel is $153 \times 10^{-6}$/°C and in a steel vessel is $144 \times 10^{-6}$/°C. If $\alpha$ for steel is $12 \times 10^{-6}$/°C, then that of glass (in /°C) is

**22.** A pendulum clock loses 12 s a day if the temperature is 40°C and gains 4 s a day if the temperature is 20° C. The temperature (in °C) at which the clock will show correct time is

**23.** The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficient of thermal conductivity K and 2K and thickness $x$ and $4x$ respectively are $T_2$ and $T_1$ ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)f$ with f equal to




**24.** A body cools from 50.0°C to 48°C in 5s. How long (in s) will it take to cool from 40.0°C to 39°C? Assume the temperature of surroundings to be 30.0°C and Newton's law of cooling to be valid.

**25.** A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with $V_m$ volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}$ °C$^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}$ °C$^{-1}$, where $\gamma$ is the coefficient of volume expansion, then $V_m$ (in cc) is close to __________.

**26.** M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/ g and heat of fusion of ice is 80 cal/g], the value of M (in g) is ________.

**27.** According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surroundings, and n is equal to

**28.** If the temperature of the sun were to increase from $T$ to $2T$ and its radius from $R$ to $2R$, then the ratio of the radiant energy received on earth to what it was previously will be

**29.** Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is

**30.** At 40°C, a brass wire of 1 mm radius is hung from the ceiling. A small mass, M is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of M (in kg) is close to: (Coefficient of linear expansion and Young's modulus of brass are $10^{-5}$/°C and $10^{11}$ N/m$^2$, respectively; $g = 10$ ms$^{-2}$)

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (b) | **7** | (d) | **10** | (c) | **13** | (b) | **16** | (a) | **19** | (a) | **22** | (25) | **25** | (20) | **28** | (64) |
| **2** | (a) | **5** | (b) | **8** | (c) | **11** | (c) | **14** | (d) | **17** | (b) | **20** | (a) | **23** | (0.33) | **26** | (40) | **29** | (1) |
| **3** | (b) | **6** | (a) | **9** | (d) | **12** | (c) | **15** | (a) | **18** | (a) | **21** | $(9\times10^{-6})$ | **24** | (10) | **27** | (1) | **30** | (6.28) |

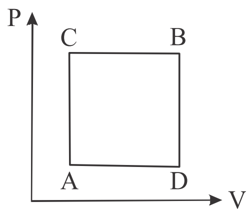
# 11

# THERMODYNAMICS

## MCQs with One Correct Answer

**1.** Which of the following is not a thermodynamical function
(a) Enthalpy (b) Work done
(c) Gibb's energy (d) Internal energy

**2.** A gas can be taken from A to B via two different processes ACB and ADB.

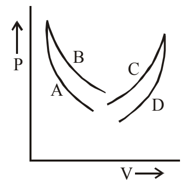



When path ACB is used 60 J of heat flows into the system and 30J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat Flow into the system in path ADB is :
(a) 40 J (b) 80 J
(c) 100 J (d) 20 J

**3.** Unit mass of a liquid with volume $V_1$ is completely changed into a gas of volume $V_2$ at a constant external pressure $P$ and temperature $T$. If the latent heat of evaporation for the given mass is $L$, then the increase in the internal energy of the system is
(a) Zero (b) $P(V_2 - V_1)$
(c) $L - P(V_2 - V_1)$ (d) $L$

**4.** The specific heat capacity of a monoatomic gas for the process $TV^2$ = constant is (where $R$ is gas constant)
(a) R (b) 2R
(c) $\frac{R}{3}$ (d) $\frac{R}{4}$

**5.** Four curves A, B, C and D are drawn in the figure for a given amount of a gas. The curves which represent adiabatic and isothermal changes are
(a) C and D respectively
(b) D and C respectively
(c) A and B respectively
(d) B and A respectively




**6.** One mole of an ideal gas at an initial temperature of $T$ $K$ does 6R joules of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is 5/3, the final temperature of gas will be
(a) $(T-4)K$ (b) $(T+2.4)K$
(c) $(T-2.4)K$ (d) $(T+4)K$

**7.** A thermally insulted vessel contains 150 g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closed to :
(Latent heat of vaporization of water = $2.10 \times 10^6$ J kg$^{-1}$ and Latent heat of Fusion of water = $3.36 \times 10^5$ J kg$^{-1}$)
(a) 150 g (b) 20 g
(c) 130 g (d) 35 g

**8.** Two Carnot engines A and B are operated in series. The engine A receives heat from the source at temperature $T_1$ and rejects the heat to the sink at temperature T. The second engine B receives the heat at temperature T and rejects to its sink at temperature $T_2$. For what value of T the efficiencies of the two engines are equal?

(a) $\frac{T_1+T_2}{2}$ (b) $\frac{T_1-T_2}{2}$

(c) $T_1T_2$ (d) $\sqrt{T_1T_2}$

**9.** An ideal heat engine works between temperatures $T_1 = 500$ K and $T_2 = 375$ K. If the engine absorbs 600J of heat from the source, then the amount of heat released to the sink is:

(a) 450 J (b) 600 J

(c) 45 J (d) 500 J

**10.** In a Carnot engine, the temperature of reservoir is 927°C and that of sink is 27°C. If the work done by the engine when it transfers heat from reservoir to sink is $12.6 \times 10^6$J, the quantity of heat absorbed by the engine from the reservoir is

(a) $16.8 \times 10^6$ J (b) $4 \times 10^6$ J

(c) $7.6 \times 10^6$ J (d) $4.2 \times 10^6$ J

**11.** A reversible engine converts one-sixth of the heat input into work. When the temperature of the sink is reduced by 62ºC, the efficiency of the engine is doubled. The temperatures of the source and sink are

(a) 99ºC, 37ºC (b) 80ºC, 37ºC

(c) 95ºC, 37ºC (d) 90ºC, 37ºC

**12.** Adiabatic modulus of elasticity of gas is $2.1 \times 10^5$ N/m$^2$. What will be its iosthermal modulus of elasticity? $\left(\frac{C_p}{C_v} = 1.4\right)$

(a) $1.8 \times 10^5$ N/m$^2$ (b) $1.5 \times 10^5$ N/m$^2$

(c) $1.4 \times 10^5$ N/m$^2$ (d) $1.2 \times 10^5$ N/m$^2$.

**13.** In an adiabatic process, the pressure is increased by $\frac{2}{3}$%. If $\gamma = \frac{3}{2}$, then the volume decreases by nearly

(a) $\frac{4}{9}$% (b) $\frac{2}{3}$%

(c) 1% (d) $\frac{9}{4}$%

**14.** Two moles of helium gas ($\gamma = 5/3$) are initially at temperature 27°C and occupy a volume of 20 litres. The gas is first expanded at constant pressure until the volume is doubled. Then, it undergoes and adiabatic change until the temperature returns to the initial value. What is the final volume of the gas?

(a) 112.4 lit. (b) 115.2 lit

(c) 120 lit (d) 125 lit

**15.** The relation between $U$, $P$ and $V$ for an ideal gas in an adiabatic process is given by relation $U = a + bPV$. Find the value of adiabatic exponent ($\gamma$) of this gas.

(a) $\frac{b+1}{b}$ (b) $\frac{b+1}{a}$

(c) $\frac{a+1}{b}$ (d) $\frac{a}{a+b}$

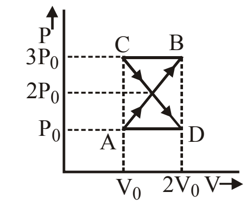
**16.** A thermodynamic system undergoes cyclic process ABCDA as shown in fig. The work done by the system in the cycle is :

(a) $P_0V_0$

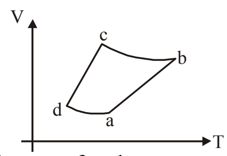
(b) $2P_0V_0$

(c) $\frac{P_0V_0}{2}$

(d) Zero

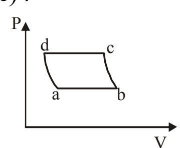



**17.** An ideal gas goes through a reversible cycle a→b→c→d has the V - T diagram shown below. Process d→a and b→c are adiabatic.

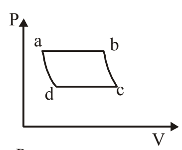



The corresponding P - V diagram for the process is (all figures are schematic and not drawn to scale) :

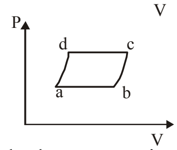
(a)



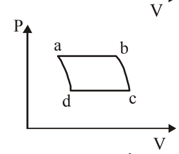

(b)




(c)




(d)




**18.** An ideal monatomic gas with pressure $P$, volume $V$ and temperature $T$ is expanded isothermally to a volume $2V$ and a final pressure $P_i$. If the same gas is expanded adiabatically to a volume $2V$, the final pressure is $P_a$. The ratio $\frac{P_a}{P_i}$ is

(a) $2^{-1/3}$ (b) $2^{1/3}$

(c) $2^{2/3}$ (d) $2^{-2/3}$

19. Three samples of the same gas A, B and C $\left(\gamma = \frac{3}{2}\right)$ have initially equal volume. Now the volume of each sample is double. The process is adiabatic for A, Isobaric for B and isothermal for C. If the finanl pressures are equal for all the three samples, the ratio of their initial pressure is

(a) $2\sqrt{2}:2:1$ (b) $2\sqrt{2}:1:2$

(c) $\sqrt{2}:1:2$ (d) $2:1:\sqrt{2}$

20. A Carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased?

(a) 840 K (b) 280 K

(c) 560 K (d) 373 K

## Numeric Value Answer

21. An ideal gas at 27ºC is compressed adiabatically to $\frac{8}{27}$ of its original volume. The rise in temperature (in °C) is $\left(\gamma = \frac{5}{3}\right)$

22. During an adiabatic process of an ideal gas, if P is proportional to $\frac{1}{V^{1.5}}$, then the ratio of specific heat capacities at constant pressure to that at constant volume for the gas is

23. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio $C_P/C_V$ for the gas is

24. A Carnot freezer takes heat from water at 0°C inside it and rejects it to the room at a temperature of 27°C. The latent heat of ice is $336 \times 10^3$ J kg$^{-1}$. If 5 kg of water at 0°C is converted into ice at 0°C by the freezer, then the energy consumed (in J) by the freezer is close to :

25. A Carnot engine whose efficiency is 50% has an exhaust temperature of 500 K. If the efficiency is to be 60% with the same intake temperature, the exhaust temperature must be (in K)

26. An engine takes in 5 mole of air at 20°C and 1 atm, and compresses it adiabaticaly to 1/10$^{th}$ of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be $X$ kJ. The value of $X$ to the nearest integer is ________.

27. Starting at temperature 300 K, one mole of an ideal diatomic gas ($\gamma$ = 1.4) is first compressed adiabatically from volume $V_1$ to $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in °K) is (to the nearest integer) ______.

28. A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is _______.

29. Two Carnot engines A and B are operated in series. The first one, A receives heat at $T_1$ (= 600 K) and rejects to a reservoir at temperature $T_2$. The second engine B receives heat rejected by the first engine and in turn, rejects to a heat reservoir at $T_3$ (= 400 K). Calculate the temperature $T_2$ (in K) if the work outputs of the two engines are equal.

30. A heat engine is involved with exchange of heat of 1915 J, – 40 J, +125 J and – $Q$ J, during one cycle achieving an efficiency of 50.0%. The value of $Q$ (in J) is :

| ANSWER KEY | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (a) | **7** | (b) | **10** | (a) | **13** | (a) | **16** | (d) | **19** | (b) | **22** | (1.5) | **25** | (400) | **28** | (600) |
| **2** | (a) | **5** | (c) | **8** | (d) | **11** | (a) | **14** | (a) | **17** | (b) | **20** | (d) | **23** | (1.5) | **26** | (46) | **29** | (500) |
| **3** | (c) | **6** | (a) | **9** | (a) | **12** | (b) | **15** | (a) | **18** | (d) | **21** | (402) | **24** | ($1.67 \times 10^5$) | **27** | (1818) | **30** | (980) |

12

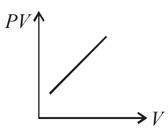
# KINETIC THEORY

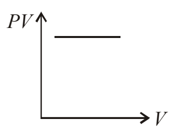
## MCQs with One Correct Answer

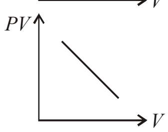
**1.** The pressure is $P$, volume $V$ and temperature $T$ of a gas in the jar $A$ and the other gas in the jar $B$ is at pressure $2P$, volume $V/4$ and temperature $2T$, then the ratio of the number of molecules in the jar $A$ and $B$ will be

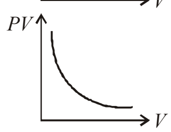
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 4 : 1

**2.** Which one the following graphs represents the behaviour of an ideal gas at constant temperature?

(a)



(b)



(c)



(d)



**3.** One mole of an ideal gas undergoes a process

$$P = \frac{P_0}{1+\left(\frac{V_0}{V}\right)^2}$$

Here $P_0$ and $V_0$ are constant. Change in temperature of the gas when volume is changed from $V = V_0$ to $V = 2V_0$

(a) $\frac{-2P_0V_0}{5R}$ (b) $\frac{11P_0V_0}{10R}$
(c) $\frac{-5P_0V_0}{4R}$ (d) $P_0V_0$

**4.** Work done by a system under isothermal change from a volume $V_1$ to $V_2$ for a gas which obeys Vander Waal's equation

$$(V-\beta n)\left(P+\frac{\alpha n^2}{V^2}\right) = nRT \text{ is}$$

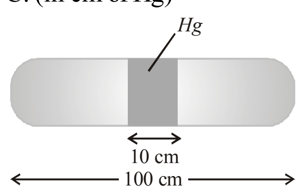
(a) $nRT \log_e\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) + \alpha n^2\left(\frac{V_1 - V_2}{V_1V_2}\right)$

(b) $nRT \log_{10}\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) - \alpha n^2\left(\frac{V_1 - V_2}{V_1V_2}\right)$

(c) $nRT \log_e\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) + \beta n^2\left(\frac{V_1 - V_2}{V_1V_2}\right)$

(d) $nRT \log_e\left(\frac{V_1 - n\beta}{V_2 - n\beta}\right) + \alpha n^2\left(\frac{V_1V_2}{V_1 - V_2}\right)$

**5.** A horizontal uniform glass tube of 100 cm, length sealed at both ends contain 10 cm mercury column in the middle. The temperature and pressure of air on either side of mercury column are respectively 31°C and 76 cm of mercury. If the air column at one end is kept at 0°C and the other end at 273°C, find the pressure of mercury of air which is at 0°C. (in cm of Hg)




(a) 194.32 cm
(b) 181.5 cm
(c) 173.2 cm
(d) 102.4 cm

**6.** A gaseous mixture consists of 16 g of helium and 16 g of oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is

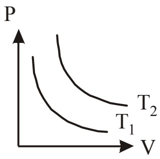
(a) 1.62 (b) 1.59
(c) 1.54 (d) 1.4

**7.** Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to:

(a) 3.67 (b) 1.83
(c) 1.09 (d) 4.67

**8.** The adjoining figure shows graph of pressure and volume of a gas at two tempertures $T_1$ and $T_2$. Which of the following inferences is correct?

(a) $T_1 > T_2$
(b) $T_1 = T_2$
(c) $T_1 < T_2$
(d) None of these




**9.** A jar has a mixture of hydrogen and oxygen gases in the ratio of 1 : 5. The ratio of mean kinetic energies of hydrogen and oxygen molecules is

(a) 1 : 16 (b) 1 : 4
(c) 1 : 5 (d) 1 : 1

**10.** In the isothermal expansion of 10g of gas from volume V to 2V the work done by the gas is 575J. What is the change in root mean square speed of the molecules of the gas at the end of the process?

(a) 398m/s (b) 520m/s
(c) 0 (d) 532m/s

**11.** The kinetic theory of gases states that the average squared velocity of molecules varies linearly with the mean molecular weight of the gas. If the root mean square (rms) velocity of oxygen molecules at a certain temperature is 0.5 km/sec. The rms velocity for hydrogen molecules at the same temperature will be :

(a) 2 km/sec (b) 4 km/sec
(c) 8 km/sec (d) 16 km/sec

**12.** The root mean square speed of hydrogen molecules at 300 *K* is 1930 *m/s*. Then the root mean square speed of oxygen molecules at 900 *K* will be.

(a) $1930\sqrt{3}$ m/s (b) 836 m/s
(c) 643 m/s (d) $\frac{1930}{\sqrt{3}}$ m/s

**13.** A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0°C. If it were to go up straight without colliding with any other molecules, how high it would rise? Assume that the height attained is much less than radius of the earth. ($k_B$ is Boltzmann constant).

(a) 0 (b) $\frac{273k_B}{2Mg}$
(c) $\frac{546k_B}{3Mg}$ (d) $\frac{819k_B}{2Mg}$

**14.** Four mole of hydrogen, two mole of helium and one mole of water vapour form an ideal gas mixture. What is the molar specific heat at constant pressure of mixture?

(a) $\frac{16}{7}R$ (b) $\frac{7}{16}R$
(c) R (d) $\frac{23}{7}R$

**15.** One mole of an ideal monatomic gas undergoes a process described by the equaton $PV^3$ = constant. The heat capacity of the gas during this process is

(a) $R$ (b) $\frac{3}{2}R$
(c) $\frac{5}{2}R$ (d) $2R$

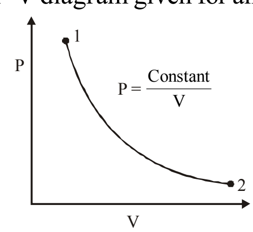
**16.** The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to :
[Boltzmann Constant $k_B = 1.38 \times 10^{-23}$ J/K
Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg
Radius of Earth : $6.4 \times 10^6$ m
Gravitational acceleration on Earth = 10 $ms^{-2}$]

(a) 800 K (b) $3 \times 10^5$ K
(c) $10^4$ K (d) 650 K

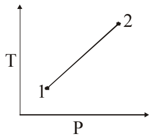
**17.** A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0°C. If it were to go up straight without colliding with any other molecules, how high it would rise? Assume that the height attained is much less than radius of the earth. ($k_B$ is Boltzmann constant).

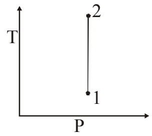
(a) 0 (b) $\frac{273k_B}{2Mg}$
(c) $\frac{546k_B}{3Mg}$ (d) $\frac{819k_B}{2Mg}$

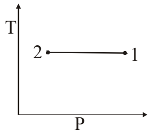
**18.** For the P-V diagram given for an ideal gas,

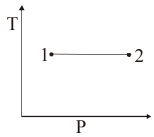


Which one correctly represents the T-P diagram?

(a) 


(b) 


(c) 


(d) 


**19.** At room temperature a diatomic gas is found to have an r.m.s. speed of 1930 $ms^{-1}$. The gas is:
(a) $H_2$ (b) $Cl_2$ (c) $O_2$ (d) $F_2$

**20.** Three perfect gases at absolute temperatures $T_1$, $T_2$ and $T_3$ are mixed. The masses of molecules are $m_1$, $m_2$ and $m_3$ and the number of molecules are $n_1$, $n_2$ and $n_3$ respectively. Assuming no loss of energy, the final temperature of the mixture is :

(a) $\dfrac{n_1T_1 + n_2T_2 + n_3T_3}{n_1 + n_2 + n_3}$

(b) $\dfrac{n_1T_1^2 + n_2T_2^2 + n_3T_3^2}{n_1T_1 + n_2T_2 + n_3T_3}$

(c) $\dfrac{n_1^2T_1^2 + n_2^2T_2^2 + n_3^2T_3^2}{n_1T_1 + n_2T_2 + n_3T_3}$

(d) $\dfrac{(T_1 + T_2 + T_3)}{3}$

## Numeric Value Answer

**21.** A cylinder rolls without slipping down an inclined plane, the number of degrees of freedom it has is

**22.** Internal energy of $n_1$ mol of hydrogen of temperature T is equal to the internal energy of $n_2$ mol of helium at temperature 2T. The ratio $\dfrac{n_1}{n_2}$ is :

**23.** One mole of ideal monatomic gas ($\gamma = 5/3$) is mixed with one mole of diatomic gas ($\gamma = 7/5$). What is $\gamma$ for the mixture? $\gamma$ denotes the ratio of specific heat at constant pressure, to that at constant volume

**24.** Using equipartition of energy, the specific heat (in J $kg^{-1}$ $K^{-1}$) of aluminium at room temperature can be estimated to be (atomic weight of aluminium = 27)

**25.** The temperature of an open room of volume 30 $m^3$ increases from 17°C to 27°C due to sunshine. The atmospheric pressure in the room remains $1 \times 10^5$ Pa. If $n_i$ and $n_f$ are the number of molecules in the room before and after heating, then $n_f - n_i$ will be:

**26.** Initially a gas of diatomic molecules is contained in a cylinder of volume $V_1$ at a pressure $P_1$ and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$ is given by $P_2$. The ratio $P_2/P_1$ is ______.

**27.** The change in the magnitude of the volume of an ideal gas when a small additional pressure $\Delta P$ is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity $\Delta T$ at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm. respectively. If $|\Delta T| = C|\Delta P|$, then value of $C$ in (K/atm.) is __________.

**28.** Nitrogen gas is at 300°C temperature. The temperature (in K) at which the rms speed of a $H_2$ molecule would be equal to the rms speed of a nitrogen molecule, is _____________. (Molar mass of $N_2$ gas 28 g);

**29.** A mixture of 2 moles of helium gas (atomic mass = 4u), and 1 mole of argon gas (atomic mass = 40u) is kept at 300 K in a container. The ratio of their rms speeds

$\left[\dfrac{V_{rms}(\text{helium})}{V_{rms}(\text{argon})}\right]$ is close to :

**30.** A closed vessel contains 0.1 mole of a monatomic ideal gas at 200 K. If 0.05 mole of the same gas at 400 K is added to it, the final equilibrium temperature (in K) of the gas in the vessel will be close to _________.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (a) | **7** | (c) | **10** | (c) | **13** | (d) | **16** | (c) | **19** | (a) | **22** | (1.2) | **25** | ($-2.5\times10^{25}$) | **28** | (41) |
| **2** | (b) | **5** | (d) | **8** | (c) | **11** | (a) | **14** | (d) | **17** | (d) | **20** | (a) | **23** | (1.5) | **26** | (5) | **29** | (3.16) |
| **3** | (b) | **6** | (a) | **9** | (d) | **12** | (b) | **15** | (a) | **18** | (c) | **21** | (2) | **24** | (925) | **27** | (150) | **30** | (266.67) |

13

# OSCILLATIONS

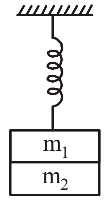
## MCQs with One Correct Answer

**1.** The motion of a particle is given by $x = A \sin\omega t + B \cos\omega t$. The motion of the particle is
(a) not simple harmonic
(b) simple harmonic with amplitude (A–B)/2
(c) simple harmonic with amplitude (A + B)/2
(d) simple harmonic with amplitude $\sqrt{A^2 + B^2}$

**2.** Two particles P and Q start from origin and execute simple harmonic motion along X-axis with same amplitude but with periods 3 seconds and 6 seconds respectively. The ratio of the velocities of P and Q when they meet at mean position is
(a) 1 : 2 (b) 2 : 1
(c) 2 : 3 (d) 3 : 2

**3.** A boy is executing Simple Harmonic Motion. At a displacement x its potential energy is $E_1$ and at a displacement y its potential energy is $E_2$. If the potential energy is E at displacement (x + y) then:
(a) $\sqrt{E} = \sqrt{E_1} - \sqrt{E_2}$
(b) $\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$
(c) $E = E_1 + E_2$
(d) $E = E_1 - E_2$

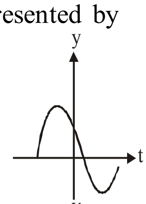
**4.** Two masses $m_1$ and $m_2$ are suspended together by a massless spring of force constant k, as shown in figure. When the masses are in equilibrium, mass $m_1$ is removed without disturbing the system. The angular frequency of oscillation of mass $m_2$ is

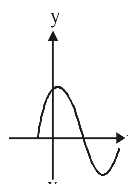


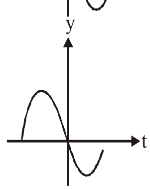

(a) $\sqrt{\frac{k}{m_2}}$ (b) $\sqrt{\frac{k}{m_1}}$
(c) $\sqrt{\frac{km_1}{m_2^2}}$ (d) $\sqrt{\frac{km_2}{m_1^2}}$

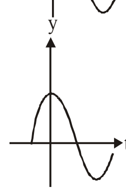
**5.** In damped oscillations, the amplitude of oscillations is reduced to one-third of its inital value $a_0$ at the end of 100 oscillations. When the oscillator completes 200 oscillations, its amplitude must be
(a) $a_0/2$ (b) $a_0/4$
(c) $a_0/6$ (d) $a_0/9$

**6.** The displacement $y(t) = A \sin(\omega t + \phi)$ of a particle executing S.H.M. for $\phi = \frac{2\pi}{3}$ is correctly represented by

(a)



(b)



(c)



(d)



**7.** A particle of mass 1 kg is placed on a platform and the platform executes S.H.M. along vertical line, along with particle. The amplitude of oscillation is 5 cm and at topmost position particle is just weightless. Maximum speed of particle is [in m/s]

(a) $\frac{3}{2\sqrt{5}}$ (b) $\frac{1}{\sqrt{2}}$

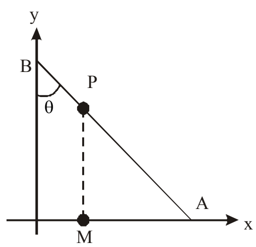
(c) $\frac{\pi}{2\sqrt{7}}$ (d) $\frac{1}{2\sqrt{7}}$

**8.** Two particles are performing simple harmonic motion in a straight line about the same equilibrium point. The amplitude and time period for both particles are same and equal to A and T, respectively. At time t=0 one particle has displacement A while the other one has displacement $\frac{-A}{2}$ and they are moving towards each other. If they cross each other at time t, then t is:

(a) $\frac{5T}{6}$ (b) $\frac{T}{3}$

(c) $\frac{T}{4}$ (d) $\frac{T}{6}$

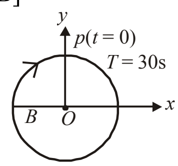
**9.** A rod of length $\ell$ is in motion such that its ends A and B are moving along x-axis and y-axis respectively. It is given that $\frac{d\theta}{dt} = 2$ rad/sec always. P is a fixed point on the rod.




Let M be the projection of P on x-axis. For the time interval in which θ changes from 0 to $\frac{\pi}{2}$, choose the correct statement.

(a) The speed of M is always directed towards right
(b) M executes S.H.M.
(c) M moves with constant speed
(d) M moves with constant velocity

**10.** Figure shows the circular motion of a particle. The radius of the circle, the period, sense of revolution and the initial position are indicated on the figure. The simple harmonic motion of the $x$-projection of the radius vector of the rotating particle $P$ is [radius = B]




(a) $x(t) = B\sin\left(\frac{2\pi t}{30}\right)$

(b) $x(t) = B\cos\left(\frac{\pi t}{15}\right)$

(c) $x(t) = B\sin\left(\frac{\pi t}{15} + \frac{\pi}{3}\right)$

(d) $x(t) = B\cos\left(\frac{\pi t}{15} + \frac{\pi}{3}\right)$

**11.** A point particle of mass 0.1 kg is executing S.H.M. of amplitude of 0.1 m. When the particle passes through the mean position, its kinetic energy is $8 \times 10^{-3}$ joule. Obtain the equation of motion of this particle if this initial phase of oscillation is 45º.

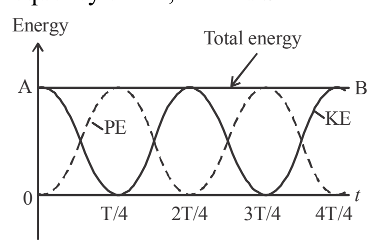
(a) $y = 0.1\sin\left(4t + \frac{\pi}{4}\right)$

(b) $y = 0.2\sin\left(4t + \frac{\pi}{4}\right)$

(c) $y = 0.1\sin\left(2t + \frac{\pi}{4}\right)$

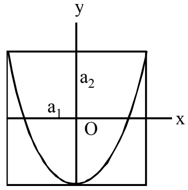
(d) $y = 0.2\sin\left(2t + \frac{\pi}{4}\right)$

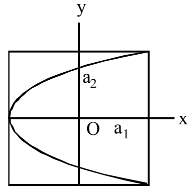
**12.** What do you conclude from the graph about the frequency of KE, PE and SHM ?

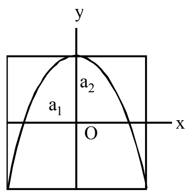


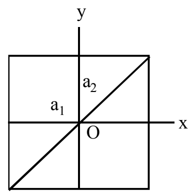

(a) Frequency of KE and PE is double the frequency of SHM
(b) Frequency of KE and PE is four times the frequency SHM.
(c) Frequency of PE is double the frequency of K.E.
(d) Frequency of KE and PE is equal to the frequency of SHM.

**13.** A particle which is simultaneously subjected to two perpendicular simple harmonic motions represented by; $x = a_1 \cos \omega t$ and $y = a_2 \cos 2\omega t$ traces a curve given by:

(a) 


(b) 


(c) 


(d) 


**14.** A flat horizontal board moves up and down (vertically) in SHM with amplitude A. Then the shortest permissible time period of the vibration such that an object placed on the board may not loose contact with the board is

(a) $2\pi\sqrt{\dfrac{g}{A}}$ (b) $2\pi\sqrt{\dfrac{A}{g}}$

(c) $2\pi\sqrt{\dfrac{2A}{g}}$ (d) $\dfrac{\pi}{2}\sqrt{\dfrac{A}{g}}$

**15.** The bob of a simple pendulum executes simple harmonic motion in water with a period t, while the period of oscillation of the bob is $t_0$ in air. Nglecting frictional force of water and given that the density of the bob is $(4/3) \times 1000$ kg/m$^3$. What relationship between $t$ and $t_0$ is true

(a) $t = 2t_0$ (b) $t = t_0/2$

(c) $t = t_0$ (d) $t = 4t_0$

**16.** A particle of mass m is attached to a spring (of spring constant $k$) and has a natural angular frequency $\omega_0$. An external force $F(t)$ proportional to $\cos \omega t (\omega \neq \omega_0)$ is applied to the oscillator. The displacement of the oscillator will be proportional to

(a) $\dfrac{1}{|m(\omega_0^2 + \omega^2)|}$ (b) $\dfrac{1}{|m(\omega_0^2 - \omega^2)|}$

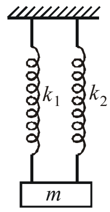
(c) $\dfrac{m}{|\omega_0^2 - \omega^2|}$ (d) $\dfrac{m}{|(\omega_0^2 + \omega^2)|}$

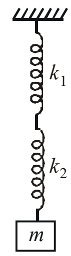
**17.** A pendulum with time period of 1s is losing energy. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s$^{-1}$) is:

(a) $\dfrac{1}{2}$ (b) $\dfrac{1}{30}\ln 3$

(c) 2 (d) $\dfrac{1}{15}\ln 3$

**18.** What is the ratio of the frequencies in the following arrangement of springs?

(i) 


(ii) 


(a) $\dfrac{k_1 + k_2}{\sqrt{k_1 k_2}}$ (b) $\dfrac{k_1 k_2}{\sqrt{k_1 - k_2}}$

(c) $\dfrac{k_1 k_2 - k_1}{\sqrt{k_1 + k_2}}$ (d) $\dfrac{(k_2 - k_1) k_2}{\sqrt{k_1 + k_2}}$

**19.** A particle of mass m oscillates with a potential energy $U = U_0 + \alpha x^2$, where $U_0$ and $\alpha$ are constants and x is the displacement of particle from equilibrium position. The time period of oscillation is

(a) $2\pi\sqrt{\dfrac{m}{\alpha}}$ (b) $2\pi\sqrt{\dfrac{m}{2\alpha}}$

(c) $\pi\sqrt{\dfrac{2m}{\alpha}}$ (d) $2\pi\sqrt{\dfrac{m}{\alpha^2}}$

**20.** A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to $T_M$. If the Young's modulus of the material of the wire is Y then $\dfrac{1}{Y}$ is equal to: ($g$ = *gravitational acceleration)*

(a) $\left[1 - \left(\dfrac{T_M}{T}\right)^2\right]\dfrac{2A}{Mg}$

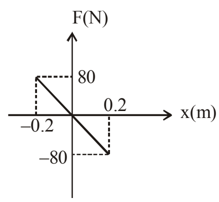
(b) $\left[1 - \left(\dfrac{T}{T_M}\right)^2\right]\dfrac{2A}{Mg}$

(c) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{A}{Mg}$

(d) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{Mg}{2A}$

## Numeric Value Answer

**21.** A body of mass 0.01 kg executes simple harmonic motion about x = 0 under the influence of a force as shown in figure. The time period (in s) of S.H.M. is




**22.** In an engine the piston undergoes vertical simple harmonic motion with amplitude 7 cm. A washer rests on top of the piston and moves with it. The motor speed is slowly increased. The frequency (in Hz) of the piston at which the washer no longer stays in contact with the piston, is close to :

**23.** A forced oscillation is acted upon by a force $F = F_0 \sin \omega t$. The amplitude of oscillation is given by $55/\sqrt{2\omega^2 - 36\omega + 9}$. The resonant angular frequency is

**24.** A particle undergoing simple harmonic motion has time dependent displacement given by $x(t) = A \sin \frac{\pi t}{90}$. The ratio of kinetic to potential energy of this particle at $t = 210$s will be :

**25.** A block of mass 0.1 kg is connected to an elastic spring of spring constant 640 $Nm^{-1}$ and oscillates in a medium of constant $10^{-2}$ kg $s^{-1}$. The system dissipates its energy gradually. The time taken for its mechanical energy of vibration to drop to half of its initial value, is closest to :

**26.** The displacement of a damped harmonic oscillator is given by $x(t) = e^{-0.1t}. \cos(10\pi t + \varphi)$. Here t is in seconds.

The time taken for its amplitude of vibration to drop to half of its initial value is close to :

**27.** The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to $\alpha$ times its original magnitude, where $\alpha$ equals

**28.** A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to :

**29.** In an experiment to determine the period of a simple pendulum of length 1 m, it is attached to different spherical bobs of radii $r_1$ and $r_2$. The two spherical bobs have uniform mass distribution. If the relative difference in the periods, is found to be $5 \times 10^{-4}$ s, the difference in radii, $|r_1 - r_2|$ is best given by:

**30.** The displacement of a particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (a) | **7** | (b) | **10** | (a) | **13** | (a) | **16** | (b) | **19** | (b) | **22** | (1.9) | **25** | (3.5) | **28** | (0.37) |
| **2** | (b) | **5** | (d) | **8** | (d) | **11** | (a) | **14** | (b) | **17** | (d) | **20** | (c) | **23** | (9) | **26** | (7) | **29** | (0.1) |
| **3** | (b) | **6** | (a) | **9** | (b) | **12** | (a) | **15** | (a) | **18** | (a) | **21** | (0.0314) | **24** | (0.33) | **27** | (0.729) | **30** | ($4\sqrt{2}$) |

# WAVES

14

## MCQs with One Correct Answer

**1.** A wave travelling along the x-axis is described by the equation y(x, t) = 0.005 cos ($\alpha$ x – $\beta$t). If the wavelength and the time period of the wave are 0.08 m and 2.0s, respectively, then $\alpha$ and $\beta$ in appropriate units are

(a) $\alpha = 25.00\pi, \beta = \pi$

(b) $\alpha = \frac{0.08}{\pi}, \beta = \frac{2.0}{\pi}$

(c) $\alpha = \frac{0.04}{\pi}, \beta = \frac{1.0}{\pi}$

(d) $\alpha = 12.50\pi, \beta = \frac{\pi}{2.0}$

**2.** A tuning fork of unknown frequency makes 3 beats/sec with a standard fork of frequency 384 Hz. The beat frequency decreases when a small piece of wax is put on the prong of the first. The frequency of the fork is:

(a) 387 Hz (b) 381 Hz
(c) 384 Hz (d) 390 Hz

**3.** A car emitting sound of frequency 500 Hz speeds towards a fixed wall at 4 m/s. An observer in the car hears both the source frequency as well as the frequency of sound reflected from the wall. If he hears 10 beats per second between the two sounds, the velocity of sound in air will be

(a) 330 m/s (b) 387 m/s
(c) 404 m/s (d) 340 m/s

**4.** 26 tuning forks are placed in a series such that each tuning fork produces 4 beats with its previous tuning fork. If the frequency of last tuning fork be three times the frequency of first tuning fork then the frequency of first tuning fork will be

(a) 76 Hz (b) 75 Hz
(c) 50 Hz (d) 25 Hz

**5.** Two trains are moving towards each other with speeds of 20m/s and 15 m/s relative to the ground. The first train sounds a whistle of frequency 600 Hz. The frequency of the whistle heard by a passenger in the second train before the train meets, is (the speed of sound in air is 340 m/s)

(a) 600 Hz (b) 585 Hz
(c) 645 Hz (d) 666 Hz

**6.** A progressive sound wave of frequency 500 Hz is travelling through air with a speed of 350 $ms^{-1}$. A compression maximum appears at a place at a given instant. The minimum time interval after which the rarefraction maximum occurs at the same point, is

(a) 200 s (b) $\frac{1}{250}$ s
(c) $\frac{1}{500}$ s (d) $\frac{1}{1000}$ s

**7.** When a wave travel in a medium, the particle displacement is given by the equation y = a sin $2\pi$ (bt– cx) where a, b and c are constants. The maximum particle velocity will be twice the wave velocity if

(a) $c = \frac{1}{\pi a}$ (b) $c = \pi a$
(c) b = ac (d) $b = \frac{1}{ac}$

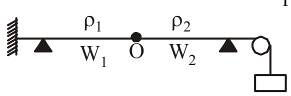
**8.** The number of possible natural oscillation of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are : (velocity of sound = 340 $ms^{-1}$)

(a) 4 (b) 5
(c) 7 (d) 6

**9.** A hollow pipe of length 0.8 m is closed at one end. At its open end a 0.5 m long uniform string is vibrating in its second harmonic and it resonates with the fundamental frequency of the pipe. If the tension in the wire is 50 N and the speed of sound is 320 ms$^{-1}$, the mass of the string is
(a) 5 grams (b) 10 grams
(c) 20 grams (d) 40 grams

**10.** A uniform tube of length 60.5 cm is held vertically with its lower end dipped in water. A sound source of frequency 500 Hz sends sound waves into the tube. When the length of tube above water is 16 cm and again when it is 50 cm, the tube resonates with the source of sound. Two lowest frequencies (in Hz), to which tube will resonate when it is taken out of water, are (approximately).
(a) 281, 562 (b) 281, 843
(c) 276, 552 (d) 272, 544

**11.** Two wires $W_1$ and $W_2$ have the same radius r and respective densities $\rho_1$ and $\rho_2$ such that $\rho_2 = 4\rho_1$. They are joined together at the point O, as shown in the figure. The combination is used as a sonometer wire and kept under tension T. The point O is midway between the two bridges. When a stationary waves is set up in the composite wire, the joint is found to be a node. The ratio of the number of antinodes formed in $W_1$ and $W_2$ is :




(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 4 : 1

**12.** Two engines pass each other moving in opposite directions with uniform speed of 30 m/s. One of them is blowing a whistle of frequency 540 Hz. Calculate the frequency heard by driver of second engine before they pass each other. Speed of sound is 330 m/sec:
(a) 450 Hz (b) 540 Hz
(c) 270 Hz (d) 648 Hz

**13.** A motor cycle starts from rest and accelerates along a straight path at 2m/s$^2$. At the starting point of the motor cycle there is a stationary electric siren. How far has the motor cycle gone when the driver hears the frequency of the siren at 94% of its value when the motor cycle was at rest? (Speed of sound = 330 ms$^{-1}$)
(a) 98 m (b) 147 m
(c) 196 m (d) 49 m

**14.** A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120 dB intensity sound ? [Given reference intensity of sound as $10^{-12}$W/m$^2$]
(a) 40 cm (b) 20 cm
(c) 10 cm (d) 30 cm

**15.** A travelling wave represented by $y = A \sin(\omega t - kx)$ is superimposed on another wave represented by $y = A \sin(\omega t + kx)$. The resultant is
(a) A wave travelling along + x direction
(b) A wave travelling along – x direction
(c) A standing wave having nodes at

$$x = \frac{n\lambda}{2}, n = 0, 1, 2....$$

(d) A standing wave having nodes at

$$x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2};\ n = 0, 1, 2....$$

**16.** A wire of length L and mass per unit length $6.0 \times 10^{-3}$ kgm$^{-1}$ is put under tension of 540 N. Two consecutive frequencies that it resonates at are: 420 Hz and 490 Hz. Then L in meters is:
(a) 2.1 m (b) 1.1 m
(c) 8.1 m (d) 5.1 m

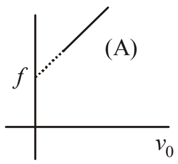
**17.** An air column in a pipe, which is closed at one end, will be in resonance wtih a vibrating tuning fork of frequency 264 Hz if the length of the column in cm is (velocity of sound = 330 m/s)
(a) 125.00 (b) 93.75
(c) 62.50 (d) 187.50

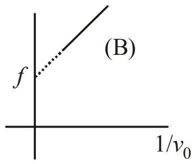
**18.** When two sound waves travel in the same direction in a medium, the displacements of a particle located at 'x' at time '$t$' is given by :

$$y_1 = 0.05 \cos(0.50\,\pi x - 100\,\pi t)$$
$$y_2 = 0.05 \cos(0.46\,\pi x - 92\,\pi t)$$

where $y_1, y_2$ and $x$ are in meters and $t$ in seconds. The speed of sound in the medium is :
(a) 92 m/s (b) 200 m/s
(c) 100 m/s (d) 332 m/s

**19.** A source of sound emits sound waves at frequency $f_0$. It is moving towards an observer with fixed speed $v_s$ ($v_s < v$, where v is the speed of sound in air). If the observer were to move towards the source with speed $v_0$, one of the following two graphs (A and B) will given the correct variation of the frequency f heard by the observer as $v_0$ is changed.




The variation of f with $v_0$ is given correctly by:

(a) graph A with slope = $\dfrac{f_0}{(v+v_s)}$

(b) graph B with slope = $\dfrac{f_0}{(v-v_s)}$

(c) graph A with slope = $\dfrac{f_0}{(v-v_s)}$

(d) graph B with slope = $\dfrac{f_0}{(v+v_s)}$

**20.** A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. Then, the lowest resonant frequency for this string is

(a) 105 Hz (b) 1.05 Hz
(c) 1050 Hz (d) 10.5 Hz

## Numeric Value Answer

**21.** Equation of a stationary and a travelling waves are as follows $y_1 = a \sin kx \cos \omega t$ and $y_2 = a \sin(\omega t - kx)$. The phase difference between two points $x_1 = \dfrac{\pi}{3k}$ and $x_2 = \dfrac{3\pi}{2k}$ is $\phi_1$ in the standing wave ($y_1$) and is $\phi_2$ in travelling wave ($y_2$) then ratio $\dfrac{\phi_1}{\phi_2}$ is

**22.** The amplitude of a wave disturbance propagating in the positive *y*-direction is given by $y = \dfrac{1}{1+x^2}$ at time $t = 0$ and $y = \dfrac{1}{\left[1+(x-1)^2\right]}$ at $t = 2$ s, where $x$ and $y$ are in meters. The shape of the wave disturbance does not change during the propagation. The velocity of wave in m/s is

**23.** An organ pipe $P_1$ closed at one end vibrating in its first overtone and another pipe $P_2$ open at both ends vibrating in third overtone are in resonance with a given tuning fork. The ratio of the length of $P_1$ to that of $P_2$ is

**24.** A string of length 0.4 m and mass $10^{-2}$ kg is clamped at its ends. The tension in the string is1.6 N. Identical wave pulses are generated at one end at regular intervals of time, $\Delta t$. The minimum value of $\Delta t$, so that a constructive interference takes place between successive pulses is

**25.** A source of sound of frequency 256Hz is moving rapidly towards a wall with a velocity of 5m/s. The speed of sound is 330 m/s. If the observer is between the wall and the source, then beats per second heard will be

**26.** A bat moving at 10 $ms^{-1}$ towards a wall sends a sound signal of 8000 Hz towards it. On reflection it hears a sound of frequency *f*. The value of *f* in Hz is close to (speed of sound = 320 $ms^{-1}$)

**27.** Two whistles A and B produce tones of frequencies 660 Hz and 596 Hz respectively. There is a listener at the mid-point of the line joining them. Now the whistle B and the listener both start moving with speed 30 m/s away from the whistle A. If the speed of sound be 330 m/s, how many beats will be heard by the listener?

**28.** A wire of density $9 \times 10^{-3}$ kg $cm^{-3}$ is stretched between two clamps 1 m apart. The resulting strain in the wire is $4.9 \times 10^{-4}$. The lowest frequency (in Hz) of the transverse vibrations in the wire is
(Young's modulus of wire $Y = 9 \times 10^{10}$ $Nm^{-2}$), (to the nearest integer), ___________.

**29.** A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is ______ Hz.

**30.** Two identical strings $X$ and $Z$ made of same material have tension $T_X$ and $T_Z$ in them. If their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio $T_X/T_Z$ is:

**ANSWER KEY**

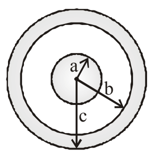
| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (c) | **7** | (a) | **10** | (d) | **13** | (a) | **16** | (a) | **19** | (c) | **22** | (0.5) | **25** | (7.7) | **28** | (35.00) |
| **2** | (a) | **5** | (d) | **8** | (d) | **11** | (b) | **14** | (a) | **17** | (b) | **20** | (a) | **23** | (0.375) | **26** | (8516) | **29** | (106) |
| **3** | (c) | **6** | (d) | **9** | (b) | **12** | (d) | **15** | (d) | **18** | (b) | **21** | (0.85) | **24** | (0.1) | **27** | (4) | **30** | (2.25) |

# ELECTRIC CHARGES AND FIELDS

15

## MCQs with One Correct Answer

1. A solid conducting sphere of radius a has a net positive charge 2Q. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge – Q.




The surface charge density on the inner and outer surfaces of the spherical shell will be

(a) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$ (b) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(c) $0, \frac{Q}{4\pi c^2}$ (d) None of the above

2. Two spheres carrying charges $+6\mu C$ and $+ 9\mu C$, separated by a distance d, experiences a force of repulsion F. when a charge of $- 3\mu C$ is given to both the sphere and kept at the same distance as before, the new force of repulsion is

(a) $3F$ (b) $F/9$

(c) $F$ (d) $F/3$

3. Two equally charged, identical metal spheres A and B repel each other with a force 'F'. The spheres are kept fixed with a distance 'r' between them. A third identical, but uncharged sphere C is brought in contact with A and then placed at the mid point of the line joining A and B. The magnitude of the net electric force on C is

(a) F (b) $\frac{3F}{4}$

(c) $\frac{F}{2}$ (d) $\frac{F}{4}$

4. Five point charges, each of value +q, are placed on five vertices of a regular hexagon of side L. The magnitude of the force on a point charge of value –q coulomb placed at the center of the hexagon is

(a) $\frac{1}{\pi\varepsilon_0}\left(\frac{q}{L}\right)^2$ (b) $\frac{2}{\pi\varepsilon_0}\left(\frac{q}{L}\right)^2$

(c) $\frac{1}{2\pi\varepsilon_0}\left(\frac{q}{L}\right)^2$ (d) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{L}\right)^2$

5. Two balls of same mass and carrying equal charge are hung from a fixed support of length $l$. At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, $x$ between the balls is proportional to :

(a) $l$ (b) $l^2$

(c) $l^{2/3}$ (d) $l^{1/3}$

6. Three charges $+Q_1$, $+Q_2$ and q are placed on a straight line such that q is somewhere in between $+Q_1$ and $+Q_2$. If this system of charges is in equilibrium, what should be he magnitude and sign of charge q?

(a) $\frac{Q_1Q_2}{\sqrt{(Q_1+Q_2)^2}}, +$ (b) $\frac{Q_1+Q_2}{2}, -$

(c) $\frac{Q_1Q_2}{\left(\sqrt{Q_1}+\sqrt{Q_2}\right)^2}, -$ (d) $\frac{Q_1+Q_2}{2}, -$

**7.** Three point charges +q, –2q and +q are placed at points (x = 0, y = a, z = 0), (x = 0, y = 0, z = 0) and (x = a, y = 0, z = 0) respectively. The magnitude and direction of the electric dipole moment vector of this charge assembly are

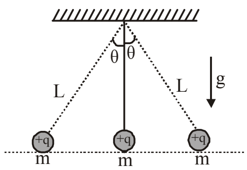
(a) $\sqrt{2}qa$ along the line joining points (x = 0, y = 0, z = 0) and (x = a, y = a, z = 0)

(b) qa along the line joining points (x = 0, y = 0, z = 0) and (x = a, y = a, z = 0)

(c) $\sqrt{2}qa$ along +ve x direction

(d) $\sqrt{2}qa$ along +ve y direction

**8.** Three identical point charge, each of mass m and charge q, hang from three strings as shown in Fig. The value of q in terms of m, L and q is




(a) $q=\sqrt{(16/5)\pi\varepsilon_0 mgL^2\sin^2\theta\tan\theta}$

(b) $q=\sqrt{(16/15)\pi\varepsilon_0 mgL^2\sin^2\theta\tan\theta}$

(c) $q=\sqrt{(15/16)\pi\varepsilon_0 mgL^2\sin^2\theta\tan\theta}$

(d) None of these

**9.** The electric field at a distance $\frac{3R}{2}$ from the centre of a charged conducting spherical shell of radius $R$ is $E$. The electric field at a distance $\frac{R}{2}$ from the centre of the sphere is

(a) $\frac{E}{2}$ (b) zero

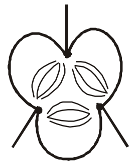
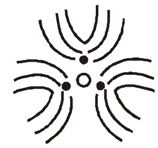
(c) E (d) $\frac{E}{2}$

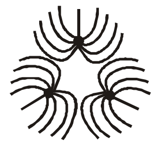
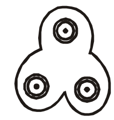
**10.** Three concentric metallic spherical shells of radii $R$, $2R$, $3R$, are given charges $Q_1$, $Q_2$, $Q_3$, respectively. It is found that the surface charge densities on the outer surfaces of the shells are equal. Then, the ratio of the charges given to the shells, $Q_1 : Q_2 : Q_3$, is

(a) 1 : 2 : 3 (b) 1 : 3 : 5

(c) 1 : 4 : 9 (d) 1 : 8 : 18

**11.** Three positive charges of equal value $q$ are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in

(a) 
(b) 

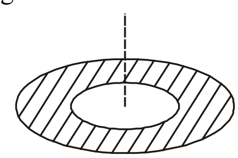
(c) 
(d) 

**12.** Two point charges + 8$q$ and –2$q$ are located at $x$ = 0 and $x$ = $L$ respectively. The location of a point on the $x$-axis at which the net electric field due to these two point charges is zero

(a) 8$L$ (b) 4$L$

(c) 2$L$ (d) $\frac{L}{4}$

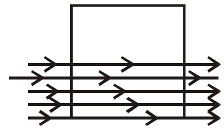
**13.** A thin disc of radius $b = 2a$ has a concentric hole of radius '$a$' in it (see figure). It carries uniform surface charge 'σ' on it. If the electric field on its axis at height '$h$' ($h << a$) from its centre is given as 'Ch' then value of 'C' is :



(a) $\frac{\sigma}{4a\epsilon_0}$ (b) $\frac{\sigma}{8a\epsilon_0}$

(c) $\frac{\sigma}{a\epsilon_0}$ (d) $\frac{\sigma}{2a\epsilon_0}$

**14.** A square surface of side L metres is in the plane of the paper. A uniform electric field $\vec{E}$ (volt /m), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The electric flux in SI units associated with the surface is



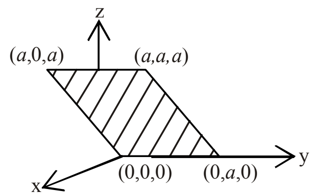

(a) $EL^2/2$

(b) zero

(c) $EL^2$

(d) $EL^2/(2\varepsilon_0)$

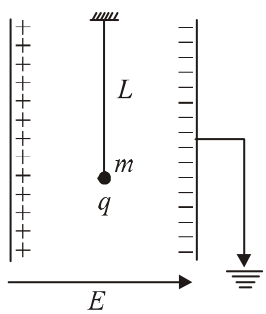
**15.** Consider an electric field $\vec{E} = E_0\hat{x}$ where $E_0$ is a constant. The flux through the shaded area (as shown in the figure) due to this field is




(a) $2E_0a_2$ (b) $\sqrt{2}E_0a^2$

(c) $E_0\,a^2$ (d) $\dfrac{E_0a^2}{\sqrt{2}}$

**16.** A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E, as shown in figure. Its bob has mass m and charge $q$. The time period of the pendulum is given by :



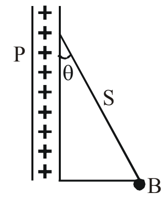

(a) $2\pi\sqrt{\dfrac{L}{\left(g+\dfrac{qE}{m}\right)}}$

(b) $2\pi\sqrt{\dfrac{L}{\sqrt{g^2-\dfrac{q^2E^2}{m^2}}}}$

(c) $2\pi\sqrt{\dfrac{L}{\left(g-\dfrac{qE}{m}\right)}}$

(d) $2\pi\sqrt{\dfrac{L}{\sqrt{g^2+\left(\dfrac{qE}{m}\right)^2}}}$

**17.** A charged ball $B$ hangs from a silk thread $S$, which makes an angle $\theta$ with a large charged conducting sheet $P$, as shown in the figure. The surface charge density $\sigma$ of the sheet is proportional to




(a) $\cot\theta$ (b) $\cos\theta$

(c) $\tan\theta$ (d) $\sin\theta$

**18.** Two point dipoles of dipole moment $\vec{p}_1$ and $\vec{p}_2$ are at a distance $x$ from each other and $\vec{p}_1 \parallel \vec{p}_2$. The force between the dipoles is :

(a) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{4p_1p_2}{x^4}$ (b) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{3p_1p_2}{x^3}$

(c) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{6p_1p_2}{x^4}$ (d) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{8p_1p_2}{x^4}$

**19.** If the electric flux entering and leaving an enclosed surface respectively is $\phi_1$ and $\phi_2$, the electric charge inside the surface will be

(a) $(\phi_2-\phi_1)\varepsilon_o$ (b) $(\phi_1-\phi_2)/\varepsilon_o$

(c) $(\phi_2-\phi_1)/\varepsilon_o$ (d) $(\phi_1-\phi_2)\varepsilon_o$

**20.** An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience

(a) a translational force only in the direction of the field

(b) a translational force only in a direction normal to the direction of the field

(c) a torque as well as a translational force

(d) a torque only

## Numeric Value Answer

**21.** An electric dipole, consisting of two opposite charges of $2 \times 10^{-6}$ C each separated by a distance 3 cm is placed in an electric field of $2 \times 10^5$ N/C. Maximum torque (in Nm) acting on the dipole is

**22.** A pendulum bob of mass $30.7 \times 10^{-6}$ kg carrying a charge $2 \times 10^{-8}$ C is at rest in a horizontal uniform electric field of 20000 V/m. The tension (in N) in the thread of the pendulum is (g = 9.8 m/s$^2$)

**23.** A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 $kVm^{-1}$. The density of liquid is $1.26 \times 10^3$ kg $m^{-3}$. The radius (in m) of the drop is (neglect buoyancy).

**24.** The electric field in a region of space is given by, $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$ where $E_0$ = 100 N/C. The flux (in $Nm^2/C$) of the field through a circular surface of radius 0.02 m parallel to the Y-Z plane is nearly:

**25.** A charge $Q$ is placed at each of the opposite corners of a square. A charge $q$ is placed at each of the other two corners. If the net electrical force on $Q$ is zero, then $Q/q$ equals:

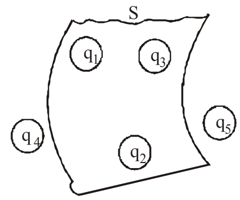
**26.** A sphere of radius R carries charge such that its volume charge density is proportional to the square of the distance from the center. What is the ratio of the magnitude of the eletric field at a distance 2R from the center to the magnitude of the electric field at a distance of R/2 from the center?

**27.** The surface charge density of a thin charged disc of radius R is $\sigma$. The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc reduces by %

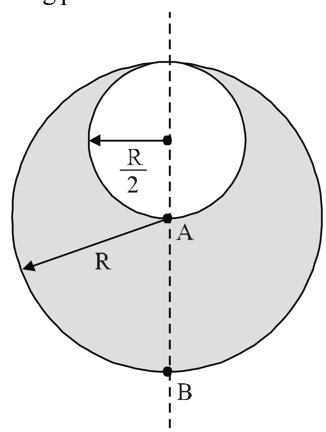
**28.** A solid sphere of radius $R$ has a charge $Q$ distributed in its volume with a charge density $\rho = \kappa r^a$, where $\kappa$ and $a$ are constants and $r$ is the distance from its centre.

If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, find the value of $a$.

**29.** Figure shows five charged lumps of plastic. The cross-section of Gaussian surface S is indicated. Assuming $q_1 = q_4 = 3.1$ nC, $q_2 = q_5 = -5.9$ nC, and $q_3 = -3.1$ nC, the net electric flux (in $Nm^2/C$) through the surface is




**30.** Consider a sphere of radius R which carries a uniform charge density $\rho$. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is:




| ANSWER KEY | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (a) | **10** | (b) | **13** | (a) | **16** | (d) | **19** | (a) | **22** | $(5 \times 10^{-4})$ | **25** | $(-2\sqrt{2})$ | **28** | (2) |
| **2** | (d) | **5** | (d) | **8** | (a) | **11** | (c) | **14** | (b) | **17** | (c) | **20** | (c) | **23** | $(7.8 \times 10^{-7})$ | **26** | (2) | **29** | (–670) |
| **3** | (a) | **6** | (c) | **9** | (b) | **12** | (c) | **15** | (c) | **18** | (c) | **21** | $(12 \times 10^{-3})$ | **24** | (0.125) | **27** | (70.7%) | **30** | (0.53) |

# ELECTROSTATIC POTENTIAL AND CAPACITANCE

16

## MCQs with One Correct Answer

**1.** Three charges 2 q, – q and – q are located at the vertices of an equilateral triangle. At the centre of the triangle

(a) the field is zero but potential is non-zero

(b) the field is non-zero, but potential is zero

(c) both field and potential are zero

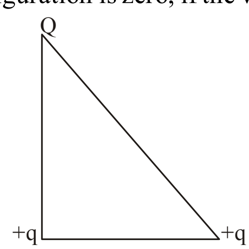
(d) both field and potential are non-zero

**2.** The 1000 small droplets of water each of radius r and charge Q, make a big drop of spherical shape. The potential of big drop is how many times the potential of one small droplet ?

(a) 1 (b) 10

(c) 100 (d) 1000

**3.** Three charges Q, + q and + q are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is :




(a) + q (b) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$

(c) $\frac{-q}{1+\sqrt{2}}$ (d) –2q

**4.** The electric potential V (in Volt) varies with $x$ (in metres) according to the relation $V = (5 + 4x^2)$. The force experienced by a negative charge of $2 \times 10^{-6}$ C located at $x = 0.5$ m is

(a) $2 \times 10^{-6}$ N (b) $4 \times 10^{-6}$ N

(c) $6 \times 10^{-6}$ N (d) $8 \times 10^{-6}$ N

**5.** The electric potential V at any point O (x, y, z all in metres) in space is given by $V = 4x^2$ volt. The electric field at the point (1 m, 0, 2 m) in volt/metre is

(a) 8 along negative X-axis

(b) 8 along positive X-axis

(c) 16 along negative X-axis

(d) 16 along positive Z-axis

**6.** Two conducting spheres of radii $R_1$ and $R_2$ having charges $Q_1$ and $Q_2$ respectively are connected to each other. There is

(a) no change in the energy of the system

(b) an increase in the energy of the system

(c) always a decrease in the energy of the system

(d) a decrease in the energy of the system unless $Q_1R_2 = Q_2R_1$

**7.** The potential to which a conductor is raised, depends on

(a) The amount of charge

(b) Geometry and size of the conductor

(c) Both (a) and (b)

(d) None of these

**8.** Calculate the area of the plates of a one farad parallel plate capacitor if separation between plates is 1 mm and plates are in vacuum

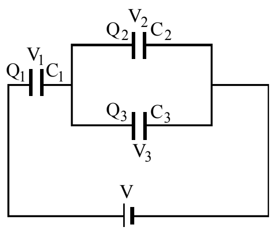
(a) $18 \times 10^8$ m$^2$ (b) $0.3 \times 10^8$ m$^2$

(c) $1.3 \times 10^8$ m$^2$ (d) $1.13 \times 10^8$ m$^2$

**9.** A parallel plate capacitor is charged and then isolated. What is the effect of increasing the plate separation on charge, potential, capacitance, respectively?

(a) Constant, decreases, decreases
(b) Increases, decreases, decreases
(c) Constant, decreases, increases
(d) Constant, increases, decreases

**10.** Three capacitors $C_1$, $C_2$ and $C_3$ are connected to a battery as shown. With symbols having their usual meanings, the correct conditions are




(a) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V$
(b) $V_1 = V_2 = V_3 = V$
(c) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(d) $Q_2 = Q_3$ and $V = V_2 + V_3$

**11.** A capacitor with capacitance 5μF is charged to 5 μC. If the plates are pulled apart to reduce the capacitance to 2 ¼F, how much work is done?

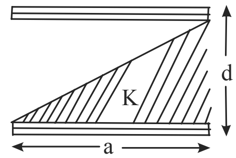
(a) $6.25 \times 10^{-6}$ J (b) $3.75 \times 10^{-6}$ J
(c) $2.16 \times 10^{-6}$ J (d) $2.55 \times 10^{-6}$ J

**12.** Four equal point charges Q each are placed in the xy plane at (0, 2), (4, 2), (4, – 2) and (0, – 2). The work required to put a fifth charge Q at the origin of the coordinate system will be:

(a) $\dfrac{Q^2}{4\pi\varepsilon_0}\left(1+\dfrac{1}{\sqrt{3}}\right)$ (b) $\dfrac{Q^2}{4\pi\varepsilon_0}\left(1+\dfrac{1}{\sqrt{5}}\right)$

(c) $\dfrac{Q^2}{2\sqrt{2}\,\pi\varepsilon_0}$ (d) $\dfrac{Q^2}{4\pi\varepsilon_0}$

**13.** A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d (d<<a). The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is:




(a) $\dfrac{K \in_0 a^2}{2d\,(K+1)}$ (b) $\dfrac{K \in_0 a^2}{d\,(K-1)}$ In K

(c) $\dfrac{K \in_0 a^2}{d}$ In K (d) $\dfrac{1}{2}\dfrac{K \in_0 a^2}{d}$

**14.** A parallel plate capacitor having capacitance 12 pF is charged by a battery to a potential difference of 10 V between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is:

(a) 692 pJ (b) 508 pJ
(c) 560 pJ (d) 600 pJ

**15.** A capacitor $C$ is fully charged with voltage $V_0$. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\dfrac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is :

(a) $\dfrac{1}{2}CV_0^2$ (b) $\dfrac{1}{3}CV_0^2$

(c) $\dfrac{1}{4}CV_0^2$ (d) $\dfrac{1}{6}CV_0^2$

**16.** Concentric metallic hollow spheres of radii $R$ and $4R$ hold charges $Q_1$ and $Q_2$ respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R) - V(4R)$ is :

(a) $\dfrac{3Q_1}{16\pi\varepsilon_0 R}$ (b) $\dfrac{3Q_2}{4\pi\varepsilon_0 R}$

(c) $\dfrac{Q_2}{4\pi\varepsilon_0 R}$ (d) $\dfrac{3Q_1}{4\pi\varepsilon_0 R}$

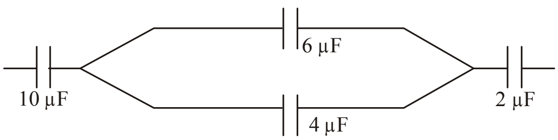
**17.** A solid conducting sphere, having a charge Q, is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V. If the shell is now given a charge of – 4 Q, the new potential difference between the same two surfaces is :

(a) –2V (b) 2V

(c) 4V (d) V

## Numeric Value Answer

**18.** In the figure shown below, the charge on the left plate of the 10 μF capacitor is –30μC. The charge (in μC) on the right plate of the 6μF capacitor is:



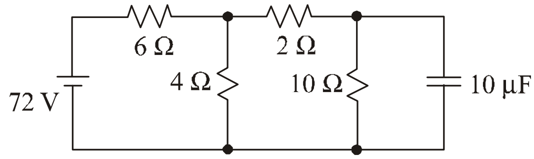

**19.** Voltage rating of a parallel plate capacitor is 500 V. Its dielectric can withstand a maximum electric field of $10^6$ V/m. The plate area is $10^{-4}$ $m^2$. What is the dielectric constant if the capacitance is 15 pF ?

(given $\in_0 = 8.86 \times 10^{-12}$ $C^2$ $m^2$)

**20.** A parallel plate capacitor has 1μF capacitance. One of its two plates is given +2μC charge and the other plate, +4μC charge. The potential difference (in volt) developed across the capacitor is :

**21.** The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in $NC^{-1}$ and $x$ is in metres. The values of constants are A = 20 SI unit and B = 10 SI unit. If the potential at $x = 1$ is $V_1$ and that at $x = -5$ is $V_2$, then $V_1 - V_2$ (in volt) is :

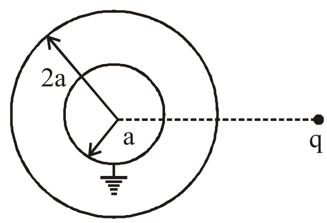
**22.** Determine the charge (in coulomb) on the capacitor in the following circuit:




**23.** The 1000 small droplets of water each of radius r and charge Q, make a big drop of spherical shape. The potential of big drop is how many times the potential of one small droplet ?

**24.** A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential (in volt) at a distance of 2 cm from the centre of the sphere is

**25.** A solid conducting sphere of radius a is surrounded by a thin uncharged concentric conducting shell of radius 2a. A point charge q is placed at a distance 4a from common centre of conducting sphere and shell. The inner sphere is then grounded. The charge on solid sphere is $\frac{q}{x}$. Find the value of $x$.




| ANSWER KEY | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (d) | **7** | (c) | **10** | (c) | **13** | (b) | **16** | (a) | **19** | (8.5) | **22** | (200) | 
| **2** | (c) | **5** | (a) | **8** | (d) | **11** | (b) | **14** | (b) | **17** | (d) | **20** | (1) | **23** | (100) |
| **3** | (b) | **6** | (d) | **9** | (d) | **12** | (b) | **15** | (d) | **18** | (18) | **21** | (180) | **24** | (10) |

| | |
|---|---|
| **25** | (4) |

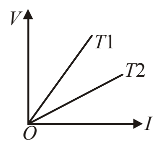
# 17 CURRENT ELECTRICITY

## MCQs with One Correct Answer

**1.** At what temperature will the resistance of a copper wire becomes three times its value at 0°C? (Temperature coefficient of resistance of copper is $4 \times 10^{-3}$/°C)

(a) 600°C (b) 500°C
(c) 450°C (d) 400°C

**2.** The voltage $V$ and current $I$ graphs for a conductor at two different temperatures $T_1$ and $T_2$ are shown in the figure. The relation between $T_1$ and $T_2$ is




(a) $T_1 > T_2$ (b) $T_1 < T_2$
(c) $T_1 = T_2$ (d) $T_1 = \frac{1}{T_2}$

**3.** A cell of emf E is connected across a resistance R. The potential difference between the terminals of the cell is found to be V volt. Then the internal resistance of the cell must be

(a) $(E-V)R$ (b) $\frac{(E-V)}{V}R$
(c) $\frac{2(E-V)R}{E}$ (d) $\frac{2(E-V)V}{R}$

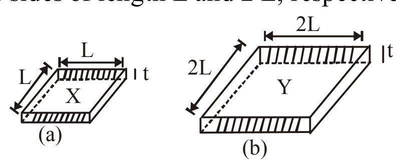
**4.** Forty electric bulbs are connected in series across a 220 V supply. After one bulb is fused the remaining 39 are connected again in series across the same supply. The illumination will be

(a) more with 40 bulbs than with 39
(b) more with 39 bulbs than with 40
(c) equal in both the cases
(d) None of these

**5.** In a Wheatstone's bridge, three resistances $P$, $Q$ and $R$ connected in the three arms and the fourth arm is formed by two resistances $S_1$ and $S_2$ connected in parallel. The condition for the bridge to be balanced will be

(a) $\frac{P}{Q} = \frac{2R}{S_1 + S_2}$ (b) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{S_1 S_2}$
(c) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{2S_1 S_2}$ (d) $\frac{P}{Q} = \frac{R}{S_1 + S_2}$

**6.** Figure shows two squares, X and Y, Cut from a sheet of metal of uniform thickness t. X and Y have sides of length L and 2 L, respectively.




The resistance $R_x$ and $R_y$ of the squares are measured between the opposite faces shaded in Fig. What is the value of $R_x/R_y$?

(a) 1/4 (b) 1/2
(c) 1 (d) 2

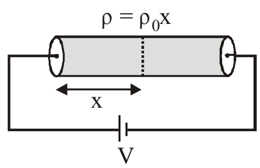
**7.** Two different conductors have same resistance at 0°C. It is found that the resistance of the first conductor at $t_1$°C is equal to the resistance of the second conductor at $t_2$°C. The ratio of the temperature coefficients of resistance of the conductors, $\frac{\alpha_1}{\alpha_2}$ is

(a) $\frac{t_1}{t_2}$ (b) $\frac{t_2 - t_1}{t_2}$
(c) $\frac{t_2 - t_1}{t_1}$ (d) $\frac{t_2}{t_1}$

8. An electrical cable of copper has just one wire of radius 9 mm. Its resistance is 5 ohm. This single copper wire of the cable is replaced by 6 different well insulated copper wires of same length in parallel, each of radius 3 mm. The total resistance of the cable will now be equal to

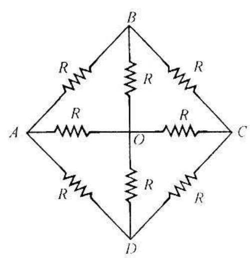
(a) 7.5 ohm (b) 45 ohm
(c) 90 ohm (d) 270 ohm

9. A cylindrical solid of length L and radius a is having varying resistivity given by $\rho = \rho_0 x$, where $\rho_0$ is a positive constant and x is measured from left end of solid. The cell shown in the figure is having emf V and negligible internal resistance. The magnitude of electric field as a function of x is best described by




(a) $\frac{2V}{L^2}x$ (b) $\frac{2V}{\rho_0 L^2}x$
(c) $\frac{V}{L^2}x$ (d) None of these

10. In the network shown, each resistance is equal to R. The equivalent resistance between adjacent corners A and D is




(a) R
(b) $\frac{2}{3}R$
(c) $\frac{3}{7}R$
(d) $\frac{8}{15}R$

11. A current source drives a current in a coil of resistance $R_1$ for a time t. The same source drives current in another coil of resistance $R_2$ for same time. If heat generated is same, find internal resistance of source. [given $R_1 > R_2$]

(a) $\frac{R_1R_2}{R_1+R_2}$ (b) $R_1+R_2$
(c) zero (d) $\sqrt{R_1R_2}$

12. The length of a wire of a potentiometer is 100 cm, and the e. m.f. of its standard cell is E volt. It is employed to measure the e.m.f. of a battery whose internal resistance is 0.5Ω. If the balance point is obtained at $\ell = 30$ cm from the positive end, the e.m.f. of the battery is

(a) $\frac{30E}{100.5}$ (b) $\frac{30E}{(100-0.5i)}$
(c) $\frac{30(E-0.5i)}{100}$ (d) $\frac{30E}{100}$

where i is the current in the potentiometer wire.

13. Two electric bulbs rated $P_1$ watt V volts and $P_2$ watt V volts are connected in parallel and V volts are applied to it. The total power will be

(a) $(P_1+P_2)\,watt$ (b) $\left(\sqrt{P_1P_2}\right)watt$
(c) $\left(\frac{P_1P_2}{P_1+P_2}\right)watt$ (d) $\left(\frac{P_1+P_2}{P_1P_2}\right)watt$

14. In an experiment of potentiometer for measuring the internal resistance of primary cell a balancing length $\ell$ is obtained on the potentiometer wire when the cell is open circuit. Now the cell is short circuited by a resistance *R*. If *R* is to be equal to the internal resistance of the cell the balancing length on the potentiometer wire will be

(a) $\ell$ (b) $2\ell$
(c) $\ell/2$ (d) $\ell/4$

15. In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28}\ m^{-3}$ and mean free time is 25 *fs* (femto second), it's approximate resistivity is: ($m_e = 9.1 \times 10^{-31}$ kg)

(a) $10^{-6}\ \Omega$ m (b) $10^{-7}\ \Omega$m
(c) $10^{-8}\ \Omega$m (d) $10^{-5}\ \Omega$m

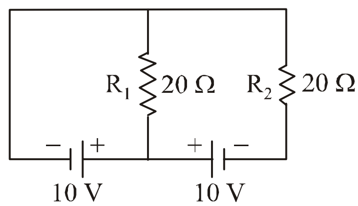
16. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm$^2$, is *v*. If the electron density in copper is $9 \times 10^{28}/m^3$ the value of *v* in mm/s close to (Take charge of electron to be = $1.6 \times 10^{-19}$C)

(a) 0.02 (b) 3
(c) 2 (d) 0.2

17. A cell of internal resistance *r* drives current through an external resistance R. The power delivered by the cell to the external resistance will be maximum when :

(a) R = 0.001 *r* (b) R = 1000 *r*
(c) R = 2*r* (d) R = *r*

**18.** In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance $R_1$ and $R_2$ respectively, are:




(a) 1, 2 (b) 2, 2

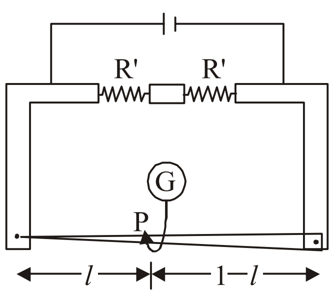
(c) 0.5, 0 (d) 0, 1

**19.** In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be:

(a) 10 A (b) 25 A

(c) 15 A (d) 20 A

**20.** In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance R with length $l$ is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP?

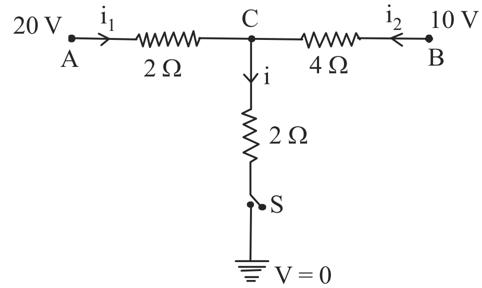



(a) 0.2 m (b) 0.3 m

(c) 0.25 m (d) 0.35 m

### Numeric Value Answer

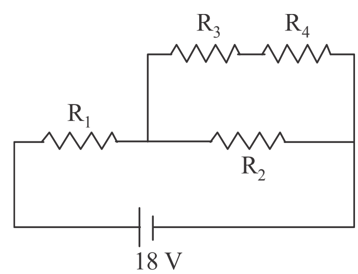
**21.** When the switch S, in the circuit shown, is closed then the value of current $i$ (in ampere) will be:




**22.** A 100 watt bulb working on 200 volt has resistance R and a 200 watt bulb working on 100 volt has resistance S. If the R/S is $\frac{8}{x}$. Find the value of $x$.

**23.** A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is:

**24.** In the given circuit the internal resistance of the 18 V cell is negligible. If $R_1 = 400\Omega$, $R_3 = 100\ \Omega$ and $R_4 = 500\ \Omega$ and the reading of an ideal voltmeter across $R_4$ is 5 V, then the value of $R_2$ (in Ω) will be:




**25.** A uniform metallic wire has a resistance of 18 Ω and is bent into an equilateral triangle. Then, the resistance (in Ω) between any two vertices of the triangle is:

**26.** A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current (in mA) which can be passed through this resistor is:

WWW.JEEBOOKS.IN

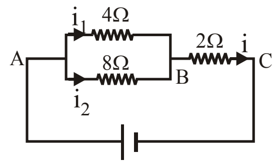
**27.** A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power (in watt) when an ideal power supply of 11 V is connected across it is:

**28.** The amount of charge Q passed in time t through a cross-section of a wire is $Q = (5t^2 + 3t + 1)$ coulomb.

The value of current (in ampere) at time t = 5 s is

**29.** A current of 1 mA flows through a copper wire. How many electrons will pass through a given point in each second?

**30.** In the circuit shown in Fig, the current in 4 Ω resistance is 1.2 A. What is the potential difference (in volt) between B and C?




| ANSWER KEY | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (b) | **7** | (d) | **10** | (d) | **13** | (a) | **16** | (a) | **19** | (d) | **22** | (1) | **25** | (4) | **28** | (53) |
| **2** | (a) | **5** | (b) | **8** | (a) | **11** | (d) | **14** | (c) | **17** | (d) | **20** | (c) | **23** | (1) | **26** | (20) | **29** | $(6.25\times10^{15})$ |
| **3** | (b) | **6** | (c) | **9** | (a) | **12** | (d) | **15** | (c) | **18** | (c) | **21** | (5) | **24** | (300) | **27** | $(11\times10^{-5})$ | **30** | (3.6) |

18

# MOVING CHARGES AND MAGNETISM

## MCQs with One Correct Answer

**1.** A beam of electrons is moving with constant velocity in a region having simultaneous perpendicular electric and magnetic fields of strength 20 $Vm^{-1}$ and 0.5 T respectively at right angles to the direction of motion of the electrons. Then the velocity of electrons must be

(a) 8 m/s (b) 20 m/s

(c) 40 m/s (d) $\frac{1}{40}$ m / s

**2.** A ring of radius $R$, made of an insulating material carries a charge $Q$ uniformly distributed on it. If the ring rotates about the axis passing through its centre and normal to plane of the ring with constant angular speed ω, then the magnitude of the magnetic moment of the ring is

(a) $Q\omega R^2$ (b) $\frac{1}{2}Q\omega R^2$

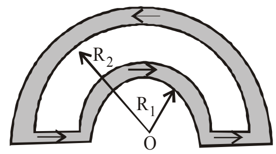
(c) $Q\omega^2 R$ (d) $\frac{1}{2}Q\omega^2 R$

**3.** If a particle of charge $10^{-12}$ coulomb moving along the $\hat{x}$- direction with a velocity $10^5$ m/s experiences a force of $10^{-10}$ newton in $\hat{y}$- direction due to magnetic field, then the minimum magnetic field is

(a) $6.25 \times 10^3$ Tesla (b) $10^{-15}$ Tesla

(c) $6.25 \times 10^{-3}$ Tesla (d) $10^{-3}$ Tesla

**4.** The magnetic induction at the centre $O$ in the figure shown is




(a) $\frac{\mu_0 i}{4}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ (b) $\frac{\mu_0 i}{4}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$

(c) $\frac{\mu_0 i}{4}(R_1 - R_2)$ (d) $\frac{\mu_0 i}{4}(R_1 + R_2)$

**5.** Two concentric circular coils of ten turns each are situated in the same plane. Their radii are 20 and 40 cm and they carry respectively 0.2 and 0.4 ampere current in opposite direction. The magnetic field in weber/$m^2$ at the centre is

(a) $\mu_0/80$ (b) $7\mu_0/80$

(c) $(5/4)\,\mu_0$ (d) zero

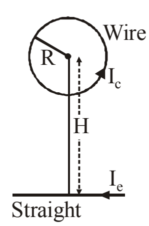
**6.** Circular loop of a wire and a long straight wire carry currents $I_c$ and $I_e$, respectively as shown in figure. Assuming that these are placed in the same plane. The magnetic fields will be zero at the centre of the loop when the separation H is

(a) $\frac{I_e R}{I_c \pi}$

(b) $\frac{I_c R}{I_e \pi}$

(c) $\frac{\pi I_c}{I_e R}$

(d) $\frac{I_e \pi}{I_c R}$



**7.** A solenoid of length 1.5 m and 4 cm diameter possesses 10 turns per cm. A current of 5A is flowing through it, the magnetic induction at axis inside the solenoid is
($\mu_0 = 4\pi \times 10^{-7}$ weber amp$^{-1}$ m$^{-1}$)
(a) $4\pi \times 10^{-5}$ gauss (b) $2\pi \times 10^{-5}$ gauss
(c) $4\pi \times 10^{-5}$ tesla (d) $2\pi \times 10^{-3}$ tesla

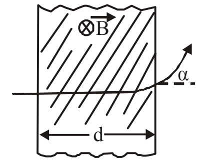
**8.** Two long conductors, separated by a distance d carry current $I_1$ and $I_2$ in the same direction. They exert a force $\vec{F}$ on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is
(a) $-\frac{2\vec{F}}{3}$ (b) $\frac{\vec{F}}{3}$
(c) $-2\vec{F}$ (d) $-\frac{\vec{F}}{3}$

**9.** A 5 cm × 12 cm coil with number of turns 600 is placed in a magnetic field of strength 0.10 Tesla. The maximum magnetic torque acting on it when a current of $10^{-5}$ A is flowing through it will be
(a) $3.6 \times 10^{-6}$ N-m
(b) $3.6 \times 10^{-6}$ dyne-cm
(c) $3.6 \times 10^{6}$ N-m
(d) $3.6 \times 10^{6}$ dyne-m

**10.** A galvanometer of 50 ohm resistance has 25 divisions. A current of $4 \times 10^{-4}$ ampere gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of
(a) 2450 Ω in series (b) 2500 Ω in series.
(c) 245 Ω in parallel (d) 2550 Ω in parallel

**11.** In a mass spectrometer used for measuring the masses of ions, the ions are initially accelerated by an electric potential V and then made to describe semicircular path of radius R using a magnetic field B. If V and B are kept constant, the ratio $\left(\frac{\text{charge on the ion}}{\text{mass of the ion}}\right)$ will be proportional to
(a) $1/R^2$ (b) $R^2$
(c) R (d) 1/R

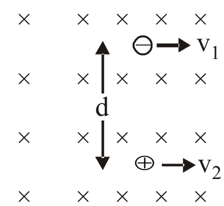
**12.** A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width '$d$'.




If α be the angle of deviation of proton from initial direction of motion (see figure), the value of sin α will be :
(a) $qV\sqrt{\frac{Bd}{2m}}$ (b) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$
(c) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$ (d) $Bd\sqrt{\frac{q}{2mV}}$

**13.** Proton, deuteron and alpha particle of same kinetic energy are moving in circular trajectories in a constant magnetic field. The radii of proton, deuteron and alpha particle are respectively $r_p$, $r_d$ and $r_\alpha$. Which one of the following relation is correct?
(a) $r_\alpha = r_p = r_d$ (b) $r_\alpha = r_p < r_d$
(c) $r_\alpha > r_d > r_p$ (d) $r_\alpha = r_d > r_p$

**14.** Two identical particles having same mass m and charges +q and – q separated by a distance d enter in a uniform magnetic field B directed perpendicular to paper inwards with speed $v_1$ and $v_2$ as shown in figure. The particles definetely will definetely not collide if
(a) $d \geq \frac{m}{Bq}(v_1 + v_2)$
(b) $d \leq \frac{m}{Bq}(v_1 + v_2)$
(c) $d > \frac{2m}{Bq}(v_1 + v_2)$
(d) $d < \frac{2m}{Bq}(v_1 + v_2)$




**15.** Two long parallel wires carry currents $i_1$ andn $i_2$ such that $i_1 > i_2$. When the currents are in the same direction, the magnetic field at a point midway between the wires is $6 \times 10^{-6}$ T. If the direction of $i_2$ is reversed, the field becomes $3 \times 10^{-5}$ T. The ratio of $\frac{i_1}{i_2}$ is
(a) $\frac{1}{2}$ (b) 2
(c) $\frac{2}{3}$ (d) $\frac{3}{2}$

**16.** An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the centre has a magnitude of

(a) $\frac{\mu_0 ne}{2r}$ (b) $\frac{\mu_0 n^2 e}{2r}$

(c) $\frac{\mu_0 ne}{2\pi r}$ (d) Zero

**17.** A solenoid of 0.4 m length with 500 turns carries a current of 3 A. A small coil of 10 turns and of radius 0.01 m carries a current of 0.4 A. The torque required to hold the coil with its axis at right angles to that of solenoid in the middle part of it, is

(a) $6\pi^2 \times 10^{-7}$ Nm (b) $3\pi^2 \times 10^{-7}$ Nm

(c) $9\pi^2 \times 10^{-7}$ Nm (d) $12\pi^2 \times 10^{-7}$ Nm

**18.** The region between $y = 0$ and $y = d$ contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$.

if $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is :

(a) $\frac{qvB}{m}\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$

(b) $\frac{qvB}{m}\left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)$

(c) $\frac{qvB}{m}\left(\frac{-\hat{j}+\hat{i}}{\sqrt{2}}\right)$

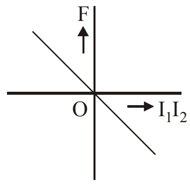
(d) $\frac{qvB}{m}\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

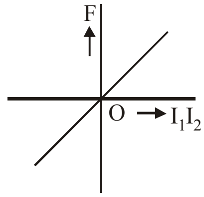
**19.** A circular coil having N turns and radius $r$ carries a current $I$. It is held in the XZ plane in a magnetic field B. The torque on the coil due to the magnetic field is :

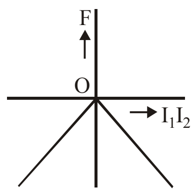
(a) $\frac{Br^2 I}{\pi N}$ (b) $B\pi r^2 IN$

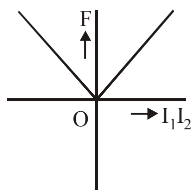
(c) $\frac{B\pi r^2 I}{N}$ (d) Zero

**20.** Two long straight parallel wires, carrying (adjustable) current $I_1$ and $I_2$, are kept at a distance d apart. If the force 'F' between the two wires is taken as 'positive' when the wires repel each other and 'negative' when the wires attract each other, the graph showing the dependence of 'F', on the product $I_1 I_2$, would be :

(a)



(b)


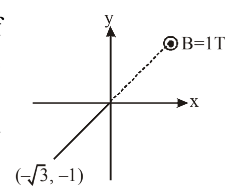

(c)



(d)



## Numeric Value Answer

**21.** An insulating rod of length $\ell$ carries a charge q distributed uniformly on it. The rod is pivoted at its mid point and is rotated at a frequency f about a fixed axis perpendicular to rod and passing through the pivot. The magnetic moment of the rod system is $\frac{1}{2a}\pi q f \ell^2$. Find the value of a.

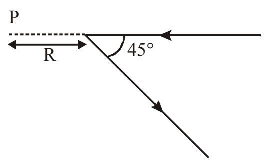
**22.** A uniform magnetic field of magnitude 1T exists in region $y \geq 0$ is along $\hat{k}$ direction as shown. A particle of charge 1C is projected from point $(-\sqrt{3}, -1)$ towards origin with speed 1 m/sec. If mass of particle is 1 kg, then co-ordinates of centre of circle in which particle moves are $\left(\frac{1}{a}, -\frac{\sqrt{b}}{2}\right)$ then find the value of (a + b).



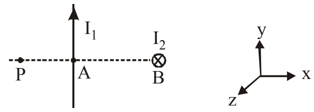
**23.** An electric current is flowing through a circular coil of radius R. The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is :

**24.** Two concentric coils each of radius equal to $2\pi$ cm are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively. The magnetic induction in Weber/m$^2$ at the centre of the coils will be $\left(\mu_0 = 4\pi \times 10^{-7} Wb/\text{A.m}\right)$

**25.** A long straight wire, carrying current I, is bent at its midpoint to form an angle of 45°. Magnitude of magnetic field at point P, distant R from point of bending is equal to $\dfrac{(\sqrt{a}-c)\mu_0 I}{b\pi R}$ then find the value of (a + b + c)




**26.** Two infinitely long linear conductors are arranged perpendicular to each other and are in mutually perpendicular planes as shown in figure. If $I_1$ = 2A along the y-axis and $I_2$ = 3A along –ve z-axis and AP = AB = 1 cm. The value of magnetic field strength $\vec{B}$ at P is $(a \times 10^{-5}\text{T})\,\hat{j} + (b \times 10^{-5}\text{T})\,\hat{k}$ then find the value of (a + b).




**27.** A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5T. If an electric field of 100V/m makes it to move in a straight path then the mass (in kg) of the particle is

(Given charge of electron = $1.6 \times 10^{-19}$C)

**28.** A beam of protons with speed $4 \times 10^5$ ms$^{-1}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch (in cm) of the resulting helical path of protons is close to : (Mass of the proton = $1.67 \times 10^{-27}$ kg, charge of the proton = $1.69 \times 10^{-19}$ C)

**29.** Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current $I$ (Ampere) in units of $\dfrac{\mu_0 I}{\pi}$ is :

**30.** A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4}$ m$^2$. If a torque of 1.5 Nm is required to keep this coil parallel to a magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is __________.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (a) | **7** | (d) | **10** | (a) | **13** | (b) | **16** | (a) | **19** | (b) | **22** | (5) | **25** | (7) |
| **2** | (b) | **5** | (d) | **8** | (a) | **11** | (a) | **14** | (c) | **17** | (a) | **20** | (a) | **23** | (27) | **26** | (7) |
| **3** | (d) | **6** | (a) | **9** | (a) | **12** | (d) | **15** | (d) | **18** | (Bonus) | **21** | (6) | **24** | $(5 \times 10^{-5})$ | **27** | $(2 \times 10^{-24})$ |

| | |
|---|---|
| **28** | (4) |
| **29** | $(500\sqrt{3})$ |
| **30** | (20) |

# MAGNETISM AND MATTER

19

## MCQs with One Correct Answer

**1.** A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm. The minimum work required to rotate it from its stable to unstable equilibrium position is :

(a) $6.4 \times 10^{-2}$ J (b) $9.2 \times 10^{-3}$ J
(c) $7.2 \times 10^{-2}$ J (d) $11.7 \times 10^{-3}$ J

**2.** A paramagnetic material has $10^{28}$ atoms/m$^3$. Its magnetic susceptibility at temperature 350 K is $2.8 \times 10^{-4}$. Its susceptibility at 300 K is:

(a) $3.267 \times 10^{-4}$ (b) $3.672 \times 10^{-4}$
(c) $3.726 \times 10^{-4}$ (d) $2.672 \times 10^{-4}$

**3.** A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of $20 \times 10^{-6}$ J/T when a magnetic intensity of $60 \times 10^3$ A/m is applied. Its magnetic susceptibility is:

(a) $3.3 \times 10^{-2}$ (b) $4.3 \times 10^{-2}$
(c) $2.3 \times 10^{-2}$ (d) $3.3 \times 10^{-4}$

**4.** A bar magnet is demagnetized by inserthing it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is:

(a) 285 A/m (b) 2600 A/m
(c) 520 A/m (d) 1200 A/m

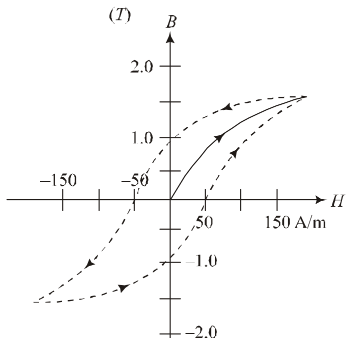
**5.** A magnetic compass needle oscillates 30 times per minute at a place where the dip is 45°, and 40 times per minute where the dip is 30°. If $B_1$ and $B_2$ are respectively the total magnetic field due to the earth and the two places, then the ratio $B_1/B_2$ is best given by :

(a) 1.8 (b) 0.7
(c) 3.6 (d) 2.2

**6.** The materials suitable for making electromagnets should have

(a) high retentivity and low coercivity
(b) low retentivity and low coercivity
(c) high retentivity and high coercivity
(d) low retentivity and high coercivity

**7.**




The figure gives experimentally measured $B$ vs. $H$ variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are:

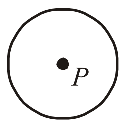
(a) 1.5 T, 50 A/m and 1.0 T
(b) 1.5 T, 50 A/m and 1.0 T
(c) 150 A/m, 1.0 T and 1.5 T
(d) 1.0 T, 50 A/m and 1.5 T

**8.** An iron rod of volume $10^{-3}$ m$^3$ and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be :

(a) $50 \times 10^2$ Am$^2$ (b) $5 \times 10^2$ Am$^2$
(c) $500 \times 10^2$ Am$^2$ (d) $0.5 \times 10^2$ Am$^2$

**9.** A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be :
(a) 1 A/m (b) 4 A/m
(c) 2.25 A/m (d) 0.75 A/m

**10.** A perfectly diamagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance. The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is :




(a) $\vec{B}$
(b) zero
(c) much large than $|\vec{B}|$ and parallel to $\vec{B}$
(d) much large than $|\vec{B}|$ but opposite to $\vec{B}$

**11.** A bar magnet having a magnetic moment of $2 \times 10^4$ JT$^{-1}$ is free to rotate in a horizontal plane. A horizontal magnetic field B = $6 \times 10^{-4}$ T exists in the space. The work done in taking the magnet slowly from a direction parallel to the field to a direction 60° from the field is
(a) 12 J (b) 6 J
(c) 2 J (d) 0.6 J

**12.** The angle of dip at a certain place is 30°. If the horizontal component of the earth's magnetic field is H, the intensity of the total magnetic field is
(a) $\frac{H}{2}$ (b) $\frac{2H}{\sqrt{3}}$
(c) $H\sqrt{2}$ (d) $H\sqrt{3}$

**13.** If the dipole moment of magnet is 0.4 amp – m$^2$ and the force acting on each pole in a uniform magnetic field of induction $3.2 \times 10^{-5}$ Weber/m$^2$ is $5.12 \times 10^{-5}$ N, the distance between the poles of the magnet is
(a) 25 cm (b) 16 cm
(c) 12.5 cm (d) 12 cm

**14.** The angle of dip at a place is 37° and the vertical component of the earth's magnetic field is $6 \times 10^{-5}$T. The earth's magnetic field at this place is (tan 37° = 3/4)
(a) $7 \times 10^{-5}$ T (b) $6 \times 10^{-5}$ T
(c) $5 \times 10^{-5}$ T (d) $10^{-4}$ T

**15.** A bar magnet of moment of inertia $9 \times 10^{-5}$ kg m$^2$ placed in a vibration magnetometer and oscillating in a uniform magnetic field $16\pi^2 \times 10^{-5}$T makes 20 oscillations in 15 s. The magnetic moment of the bar magnet is
(a) 3 Am$^2$ (b) 2 Am$^2$
(c) 5 Am$^2$ (d) 4 Am$^2$

**16.** The work done in turning a magnet of magnetic moment M by an angle of 90° from the meridian, is n times the corresponding work done to turn it through an angle of 60°. The value of n is given by
(a) 2 (b) 1
(c) 0.5 (d) 0.25

**17.** The magnetic dipole moment of a coil is $5.4 \times 10^{-6}$ joule/tesla and it is lined up with an external magnetic field whose strength is 0.80 T. Then the work done in rotating the coil (for θ = 180°) is
(a) 4.32 μJ (b) 2.16 μJ
(c) 8.6 μJ (d) None of these

**18.** A bar magnet of length 6 cm has a magnetic moment of 4 J T$^{-1}$. Find the strength of magnetic field at a distance of 200 cm from the centre of the magnet along its equatorial line.
(a) $4 \times 10^{-6}$ tesla (b) $3.5 \times 10^{-7}$ tesla
(c) $5 \times 10^{-8}$ tesla (d) $3 \times 10^{-3}$ tesla

**19.** A bar magnet has a length 8 cm. The magnetic field at a point at a distance 3 cm from the centre in the broad side-on position is found to be $4 \times 10^{-6}$ *T*. The pole strength of the magnet is
(a) $6.25 \times 10^{-2}$ Am (b) $5 \times 10^{-5}$ Am
(c) $2 \times 10^{-4}$ Am (d) $3 \times 10^{-4}$ Am

**20.** A bar magnet of length 0.2 m and pole strength 5 Am is kept in a uniform magnetic induction field of strength 15Wbm$^{-2}$ making an angle of 30° with the field. Find the couple acting on it
(a) 7.5 Nm (b) 4.5 Nm
(c) 5.5 Nm (d) 6.5 Nm

## Numeric Value Answer

**21.** A bar magnet is demagnetized by inserthing it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity (in A/m) of the bar magnet is:

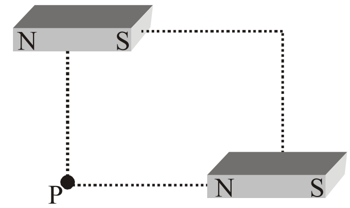
**22.** At some location on earth the horizontal component of earth's magnetic field is $18 \times 10^{-6}$ T. At this location, magnetic needle of length 0.12 m and pole strength 1.8 Am is suspended from its mid-point using a thread, it makes 45° angle with horizontal in equilibrium. To keep this needle horizontal, the vertical force (in N) that should be applied at one of its ends is:

**23.** A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of $20 \times 10^{-6}$ J/T when a magnetic intensity of $60 \times 10^{3}$ A/m is applied. Its magnetic susceptibility is:

**24.** A paramagnetic material has $10^{28}$ atoms/m$^3$. Its magnetic susceptibility at temperature 350 K is $2.8 \times 10^{-4}$. Its susceptibility at 300 K is:

**25.** If the dipole moment of magnet is 0.4 amp – m$^2$ and the force acting on each pole in a uniform magnetic field of induction $3.2 \times 10^{-5}$ Weber/m$^2$ is $5.12 \times 10^{-5}$ N, the distance (in cm) between the poles of the magnet is

**26.** Two short bar magnets of magnetic moments 1000 Am$^2$ are placed as shown at the corners of a square of side 10 cm. The net magnetic induction (in Tesla) at P is




**27.** The magnetic field of earth at the equator is approximately $4 \times 10^{-5}$ T. The radius of earth is $6.4 \times 10^{6}$ m. Then the dipole moment (in A-m$^2$) of the earth will be nearly of the order of

**28.** Two tangent galvanometers having coils of the same radius are connected in series. A current flowing in them produces deflections of 60° and 45° respectively. The ratio of the number of turns in the coils is

**29.** Two short magnets with their axes horizontal and perpendicular to the magnetic maridian are placed with their centres 40 cm east and 50 cm west of magnetic needle. If the needle remains undeflected, the ratio of their magnetic moments $M_1 : M_2$ is

**30.** A certain amount of current when flowing in a properly set tangent galvanometer, produces a deflection of 45°. If the current be reduced by a factor of $\sqrt{3}$, the deflection (in degree) would decrease by

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (b) | **7** | (d) | **10** | (b) | **13** | (a) | **16** | (a) | **19** | (a) | **22** | ($6.5\times10^{-5}$) | **25** | (25) | **28** | ($\sqrt{3}$) |
| **2** | (a) | **5** | (Bonus) | **8** | (b) | **11** | (b) | **14** | (d) | **17** | (c) | **20** | (a) | **23** | ($3.3\times10^{-4}$) | **26** | (0.1) | **29** | (0.51) |
| **3** | (d) | **6** | (b) | **9** | (d) | **12** | (b) | **15** | (d) | **18** | (c) | **21** | (2600) | **24** | ($3.266\times10^{-4}$) | **27** | ($10^{23}$) | **30** | (15) |

# 20 ELECTROMAGNETIC INDUCTION

## MCQs with One Correct Answer

**1.** A long solenoid has 500 turns. When a current of 2 ampere is passed through it, the resulting magnetic flux linked with each turn of the solenoid is $4 \times 10^{-3}$ Wb. The self- inductance of the solenoid is

(a) 2.5 henry (b) 2.0 henry
(c) 1.0 henry (d) 40 henry

**2.** A very small square loop of wire of side $\ell$ is placed inside a large square loop of wire of side L ($L \gg \ell$). The loop are coplanar and their centre coincide. The mutual inductance of the system is equal to

(a) $\frac{\mu_0}{4\pi}(\ell / L)$ (b) $\frac{\mu_0}{4\pi} 8\sqrt{2}\,(\ell^2 / L)$

(c) $\frac{\mu_0}{4\pi}(L / \ell)$ (d) $\frac{\mu_0}{4\pi} 3\sqrt{2}\,(L^2 / \ell)$

**3.** A metal rod of length $l$ moves perpendicularly across a uniform magnetic field B with a velocity v. If the resistance of the circuit of which the rod forms a part is r, then the force required to move the rod uniformly is

(a) $\frac{B^2 l^2 v}{r}$ (b) $\frac{Blv}{r}$

(c) $\frac{B^2 lv}{r}$ (d) $\frac{B^2 l^2 v^2}{r}$

**4.** A copper rod of length $l$ is rotated about one end perpendicular to the magnetic field B with constant angular velocity ω. The induced e.m.f. between the two ends is

(a) $\frac{1}{2} B\omega l^2$ (b) $\frac{3}{4} B\omega l^2$

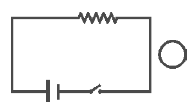
(c) $B\omega l^2$ (d) $2B\omega l^2$

**5.** A wire of length 1 m is moving at a speed of $2 ms^{-1}$ perpendicular to its length in a uniform magnetic field of 0.5 T. The ends of the wire are joined to a circuit of resistance 6Ω. The rate at which work is being done to keep the wire moving at constant speed is

(a) $\frac{1}{12} W$ (b) $\frac{1}{6} W$

(c) $\frac{1}{3} W$ (d) 1W

**6.** Consider the situation shown in figure. If the switch is closed and after some time it is opened again, the closed loop will show [Just after the closing and opening the switch]



(a) a clockwise current pulse
(b) an anticlockwise current pulse
(c) an anticlockwise current and then clockwise pulse
(d) a clockwise current and then an anticlock wise current pulse.

**7.** A coil having 100 turns and area of 0.001 metre$^2$ is free to rotate about an axis. The coil is placed perpendicular to a magnetic field of 1.0 weber/ metre$^2$. If the coil is rotate rapidly through an angle of 180°, how much charge will flow through the coil? The resistance of the coil is 10 ohm.

(a) 0.02 coulomb (b) 0.2 coulomb
(c) 3 coulomb (d) 2 coulomb

**8.** A thin circular ring of area A is perpendicular to uniform magnetic field of induction B. A small cut is made in the ring and a galvanometer is connected across the ends such that the total resistance of circuit is R. When the ring is suddenly squeezed to zero area, the charge flowing through the galvanometer is

(a) $\frac{BR}{A}$ (b) $\frac{AB}{R}$

(c) ABR (d) $B^2 A/R^2$

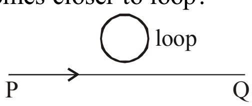
**9.** An electron moves along the line PQ as shown which lies in the same plane as a circular loop of conducting wire as shown in figure. What will be the direction of the induced current initially in the loop when electron comes closer to loop?

(a) Anticlockwise

(b) Clockwise

(c) Direction can not be predicted

(d) No current will be induced




**10.** A straight conductor of length 2m moves at a speed of 20 m/s. When the conductor makes an angle of 30° with the direction of magnetic field of induction of 0.1 wbm$^2$ then induced emf

(a) 4V (b) 3V (c) 1V (d) 2V

**11.** The self induced emf of a coil is 25 volts. When the current in it is changed at uniiform rate from 10 A to 25 A in 1s, the change in the energy of the inductance is:

(a) 740 J (b) 437.5 J (c) 540 J (d) 637.5 J

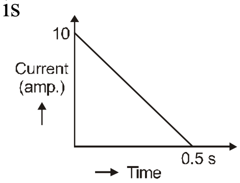
**12.** In a coil of resistance 100Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is

(a) 250 Wb

(b) 275 Wb

(c) 200 Wb

(d) 225 Wb




**13.** When current in a coil changes from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self - inductance of the coil is :

(a) 6 H (b) 0.67 H (c) 3 H (d) 1.67 H

**14.** A solid metal cube of edge length 2 cm is moving in a positive $y$-direction at a constant speed of 6 m/s. There is a uniform magnetic field of 0.1 T in the positive $z$-direction. The potential difference between the two faces of the cube perpendicular to the $x$-axis, is:

(a) 12 mV (b) 6 mV (c) 1 mV (d) 2 mV

**15.** A horizontal straight wire 20 m long extending from east to west falling with a speed of 5.0 m/s, at right angles to the horizontal component of the earth's magnetic field $0.30 \times 10^{-4}$ Wb/m$^2$. The instantaneous value of the e.m.f. induced in the wire will be

(a) 3mV (b) 4.5mV (c) 1.5mV (d) 6.0mV

**16.** There are two long co-axial solenoids of same length $l$. The inner and outer coils have radii $r_1$ and $r_2$ and number of turns per unit length $n_1$ and $n_2$, respectively. The ratio of mutual inductance to the self-inductance of the inner-coil is :

(a) $\frac{n_1}{n_2}$ (b) $\frac{n_2}{n_1} \cdot \frac{r_1}{r_2}$

(c) $\frac{n_2}{n_1} \cdot \frac{r_2^2}{r_1^2}$ (d) $\frac{n_2}{n_1}$

**17.** Two coils 'P' and 'Q' are separated by some distance. When a current of 3A flows through coil 'P', a magnetic flux of $10^{-3}$ Wb passes through 'Q'. No current is passed through 'Q'. When no current passes through 'P' and a current of 2A passes through 'Q', the flux through 'P' is:

(a) $6.67 \times 10^{-4}$ Wb (b) $3.67 \times 10^{-3}$ Wb

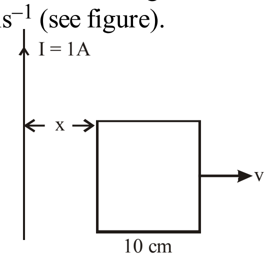
(c) $6.67 \times 10^{-3}$ Wb (d) $3.67 \times 10^{-4}$ Wb

**18.** An insulating thin rod of length $l$ has a linear charge density $\rho(x) = \rho_0 \frac{x}{l}$ on it. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is:

(a) $\pi n \rho l^3$ (b) $\frac{\pi}{3} n \rho l^3$

(c) $\frac{\pi}{4} n \rho l^3$ (d) $n \rho l^3$

**19.** A square frame of side 10 cm and a long straight wire carrying current 1 A are in the plate of the paper. Starting from close to the wire, the frame moves towards the right with a constant speed of 10 ms$^{-1}$ (see figure).



The e.m.f induced at the time the left arm of the frame is at x = 10 cm from the wire is:

(a) 2 μV (b) 1 μV
(c) 0.75 μV (d) 0.5 μV

**20.** A coil of cross-sectional area A having n turns is placed in a uniform magnetic field B. When it is rotated with an angular velocity ω, the maximum e.m.f. induced in the coil will be

(a) nBAω (b) $\frac{3}{2}$nBAω
(c) 3nBAω (d) $\frac{1}{2}$nBAω

## Numeric Value Answer

**21.** Two concentric coplanar circular loops made of wire, with resistance per unit length 10Ω/m have diameters 0.2 m and 2 m. A time varying potential difference (4 + 2.5 $t$) volt is applied to the larger loop. Calculate the current (in A) in the smaller loop.

**22.** The current in a coil of self-induction 2.0 henry is increasing according to $i = 2 \sin t^2$ ampere. Find the amount of energy (in joule) spent during the period when the current changes from 0 to 2 ampere.

**23.** The self induced emf of a coil is 25 volts. When the current in it is changed at uniform rate from 10 A to 25 A in 1s, the change in the energy (in joule) of the inductance is:

**24.** A 10 m long horizontal wire extends from North East to South West. It is falling with a speed of 5.0 $ms^{-1}$, at right angles to the horizontal component of the earth's magnetic field, of $0.3 \times 10^{-4}$ Wb/$m^2$. The value of the induced emf (in V) in wire is :

**25.** A conducting circular loop having a radius of 5.0 cm, is placed perpendicular to a magnetic field of 0.50 T. It is removed from the field in 0.50 s. Find the average emf (in V) produced in the loop during this time.

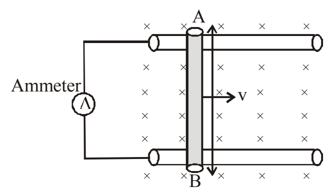
**26.** A uniform magnetic field $B$ exists in a direction perpendicular to the plane of a square frame made of copper wire. The wire has a diameter of 2 mm and a total length of 40 cm. The magnetic field changes with time at a steady rate $\frac{dB}{dt} = 0.02\,\frac{T}{s}$. Find the current (in A) induced in the frame. Resistivity of copper $= 1.7 \times 10^{-8}$ W-m.

**27.** A coil of inductance 1 H and resistance 10Ω is connected to a resistanceless battery of emf 50 V at time $t$ = 0. Calculate the ratio of the rate at which magnetic energy is stored in the coil to the rate at which energy is supplied by the battery at $t$ = 0.1 s.

**28.** A long solenoid having 200 turns per cm carries a current of 1.5 amp. At the centre of it is placed a coil of 100 turns of cross-sectional area $3.14 \times 10^{-4}$ $m^2$ having its axis parallel to the field produced by the solenoid. When the direction of current in the solenoid is reversed within 0.05 sec, the induced e.m.f. (in V) in the coil is

**29.** An air plane, with a 20 m wing spread is plying at 250 m/s straight south parallel to earth′s surface. The earth′s magnetic field has a horizontal component of $2 \times 10^{-5}$ Wb/$m^2$ and angle of dip is 60°. Calculate the induced emf (in V) between the plane tips.

**30.** If the rod is moving with a constant velocity of 12 cm/s then the power (in watt) that must be supplied by an external force in maintaining the speed will be




(Given B = 0.5 Tesla, $l$ = 15 cm, v = 12 cm/s, Resistance of rod $R_{AB}$ = 9.0 mΩ)

| ANSWER KEY | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (a) | **7** | (a) | **10** | (d) | **13** | (d) | **16** | (d) | **19** | (b) | **22** | (4) | **25** | ($7.85 \times 10^{-2}$) | **28** | (0.048) |
| **2** | (b) | **5** | (b) | **8** | (b) | **11** | (b) | **14** | (a) | **17** | (a) | **20** | (a) | **23** | (437.5) | **26** | ($9.3 \times 10^{-2}$) | **29** | (0.173) |
| **3** | (a) | **6** | (d) | **9** | (a) | **12** | (a) | **15** | (a) | **18** | (c) | **21** | (1.25) | **24** | ($1.5 \times 10^{-3}$) | **27** | (0.36) | **30** | ($9 \times 10^{-3}$) |

# 21

# ALTERNATING CURRENT

## MCQs with One Correct Answer

**1.** The current I passed in any instrument in alternating current circuit is $I = 2 \sin \omega t$ amp and potential difference applied is given by $V = 5 \cos \omega t$ volt then power loss over a complete cycle is in instrument is

(a) 2.5 watt (b) 5 watt
(c) 10 watt (d) zero

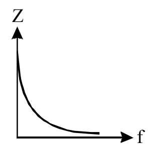
**2.** An alternating e.m.f. of angular frequency $\omega$ is applied across an inductance. The instantaneous power developed in the circuit has an angular frequency [take phase difference between emf and current is $\frac{\pi}{2}$ ]

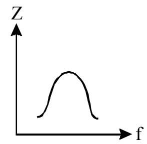
(a) $\frac{\omega}{4}$ (b) $\frac{\omega}{2}$
(c) $\omega$ (d) $2\omega$

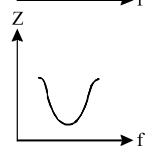
**3.** A resistance 'R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be

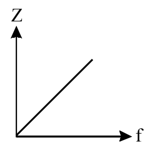
(a) $P\sqrt{\frac{R}{Z}}$ (b) $P\left(\frac{R}{Z}\right)$
(c) P (d) $P\left(\frac{R}{Z}\right)^2$

**4.** Which one of the following curves represents the variation of impedance (Z) with frequency f in series LCR circuit?

(a) 

(b) 

(c) 

(d) 


**5.** For a series RLC circuit $R = X_L = 2X_C$. The impedance of the circuit and phase difference between V and I respectively will be

(a) $\frac{\sqrt{5}R}{3}, \tan^{-1}(2)$ (b) $\frac{\sqrt{5}R}{2}, \tan^{-1}(1/2)$
(c) $\sqrt{5}X_C, \tan^{-1}(2)$ (d) $\sqrt{5}R, \tan^{-1}(1/3)$

**6.** In an a.c. circuit V and I are given by

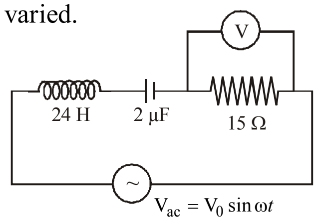
$V = 100 \sin(100\,t)$ volt
$I = 100 \sin(100\,t + \pi/3)$ mA

The power dissipated in the circuit is

(a) $10^4$ watt (b) 10 watt
(c) 2.5 watt (d) 5.0 watt

**7.** An LCR circuit as shown in the figure is connected to a voltage source $V_{ac}$ whose frequency can be varied.

The frequency, at which the voltage across the resistor is maximum, is :

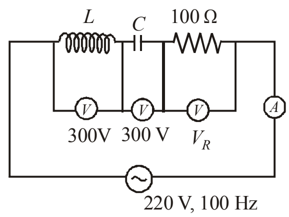



(a) 902 Hz (b) 143 Hz
(c) 23 Hz (d) 345 Hz

**8.** The primary winding of a transformer has 500 turns whereas its secondary has 5000 turns. The primary is connected to an A.C. supply of 20 V, 50 Hz. The secondary will have an output of

(a) 2 V, 5 Hz (b) 200 V, 500 Hz
(c) 2V, 50 Hz (d) 200 V, 50 Hz

**9.** In an *LCR* circuit shown in the following figure, what will be the readings of the voltmeter across the resistor and ammeter if an *a.c.* source of 220V and 100 Hz is connected to it as shown?




(a) 800 V, 8 A (b) 110 V, 1.1 A
(c) 300 V, 3 A (d) 220V, 2.2 A

**10.** The primary and secondary coil of a transformer have 50 and 1500 turns respectively. If the magnetic flux $\phi$ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where $\phi$ is in webers, t is time in seconds and $\phi_0$ is a constant, the output voltage across the secondary coil is

(a) 120 volt (b) 220 volt
(c) 30 volt (d) 90 volt

**11.** An AC circuit has $R = 100\ \Omega$, $C = 2\ \mu F$ and $L = 80$ mH, connected in series. The quality factor of the circuit is :

(a) 2 (b) 0.5
(c) 20 (d) 400

**12.** In LC circuit the inductance L = 40 mH and capacitance C = 100 μF. If a voltage V(*t*) = 10 sin(314 *t*) is applied to the circuit, the current in the circuit is given as:

(a) 0.52 cos 314 t (b) 10 cos 314 t
(c) 5.2 cos 314 t (d) 0.52 sin 314 t

**13.** A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V by the transformer. If the current in the primary of the transformer is 5A and its efficiency is 90%, the output current would be:

(a) 50 A (b) 45 A
(c) 35 A (d) 25 A

**14.** The power factor of an AC circuit having resistance (*R*) and inductance (*L*) connected in series and an angular velocity $\omega$ is

(a) $R/\omega L$ (b) $R/(R^2 + \omega^2 L^2)^{1/2}$
(c) $\omega L/R$ (d) $R/(R^2 - \omega^2 L^2)^{1/2}$

**15.** An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50Ω. The time taken for the current to rise from half of the peak value to the peak value is :

(a) 5 ms (b) 2.2 ms
(c) 7.2 ms (d) 3.3 ms

**16.** An emf of 20 V is applied at time $t = 0$ to a circuit containing in series 10 mH inductor and 5 Ω resistor. The ratio of the currents at time $t = \infty$ and at $t = 40$ s is close to:

(Take $e^2 = 7.389$)

(a) 1.06 (b) 1.15
(c) 1.46 (d) 0.84

**17.** For an RLC circuit driven with voltage of amplitude $v_m$ and frequency $\omega_0 = \dfrac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, *Q* is given by:

(a) $\dfrac{\omega_0 L}{R}$ (b) $\dfrac{\omega_0 R}{L}$

(c) $\dfrac{R}{(\omega_0 C)}$ (d) $\dfrac{CR}{\omega_0}$

**18.** A coil of inductance 300 mH and resistance 2 Ω is connected to a source of voltage 2V. The current reaches half of its steady state value in

(a) 0.1 s (b) 0.05 s
(c) 0.3 s (d) 0.15 s

**19.** A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy dissipated in the circuit in 60 s is:

(a) $5.65 \times 10^2$ J (b) $2.26 \times 10^3$ J
(c) $5.17 \times 10^2$ J (d) $3.39 \times 10^3$ J

**20.** In a series resonant LCR circuit, the voltage across $R$ is 100 volts and $R = 1\ k\Omega$ with $C = 2\mu F$. The resonant frequency $\omega$ is 200 rad/s. At resonance the voltage across $L$ is

(a) $2.5 \times 10^{-2}$ V (b) 40 V

(c) 250 V (d) $4 \times 10^{-3}$ V

## Numeric Value Answer

**21.** A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy (in joule) dissipated in the circuit in 60 s is:

**22.** A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V by the transformer. If the current in the primary of the transformer is 5A and its efficiency is 90%, the output current (in ampere) would be:

**23.** An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50 Ω. The time taken (in ms) for the current to rise from half of the peak value to the peak value is :

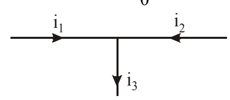
**24.** An inductor of inductance 100 mH is connected in series with a resistance, a variable capacitance and an AC source of frequency 2.0 kHz. What should be the value of the capacitance (in farad) so that maximum current may be drawn into the circuit ?

**25.** A 60 Hz AC voltage of 160 V impressed across an LR-circuit results in a current of 2 A. If the power dissipation is 200 W, calculate the maximum value of the back emf (in volt) arising in the inductance.

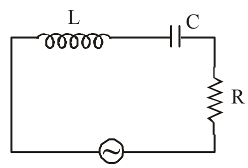
**26.** A 100 V AC source of frequency 500 Hz is connected to LCR circuit with $L = 8.1$ mH, $C = 12.5\ \mu F$ and $R = 10\Omega$, all connected in series. Find the potential (in volt) across the resistance.

**27.** A coil has a resistance of 10Ω and an inductance of 0.4 henry. It is connected to an AC source of 6.5 V, $\frac{30}{\pi} Hz.$ Find the average power (in watt) consumed in the circuit.

**28.** If $i_1 = 3 \sin \omega t$, $i_2 = 4 \cos \omega t$, and $i_3 = i_0 \sin (\omega t + 53°)$, find the value of $i_0$.




**29.** Given LCR circuit has L = 5 H, C = 80 μF, R = 40 Ω and variable frequency source of 200 V. The source frequency (in Hz) which drives the circuit at resonance is $\frac{x}{\pi}$. Find the value of $x$.




**30.** An $LCR$ series circuit with 100Ω resistance is connected to an AC source of 200 V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Calculate the current (in ampere) in the $LCR$ circuit.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (c) | **7** | (c) | **10** | (a) | **13** | (b) | **16** | (a) | **19** | (c) | **22** | (45) | **25** | (125) | **28** | (5) |
| **2** | (d) | **5** | (b) | **8** | (d) | **11** | (a) | **14** | (b) | **17** | (a) | **20** | (c) | **23** | (3.3) | **26** | (100) | **29** | (25) |
| **3** | (d) | **6** | (c) | **9** | (d) | **12** | (a) | **15** | (d) | **18** | (a) | **21** | ($5.17\times10^{2}$) | **24** | ($65\times10^{-9}$) | **27** | (0.625) | **30** | (2) |

# ELECTROMAGNETIC WAVES

22

## MCQs with One Correct Answer

**1.** If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of the electromagnetic waves, then the direction of propagation of the waves will be along

(a) $\vec{B} \times \vec{E}$ (b) $\vec{E}$

(c) $\vec{B}$ (d) $\vec{E} \times \vec{B}$

**2.** In an apparatus, the electric field was found to oscillate with an amplitude of 24 V/m. The amplitude of the oscillating magnetic field will be

(a) $6 \times 10^{-6}$ T (b) $2 \times 10^{-8}$ T

(c) $8 \times 10^{-8}$ T (d) $12 \times 10^{-6}$ T

**3.** A plane electromagnetic wave of wave intensity 10 W/m$^2$ strikes a small mirror of area 20 cm$^2$, held perpendicular to the approaching wave. The radiation force on the mirror will be

(a) $6.6 \times 10^{-11}$ N (b) $1.33 \times 10^{-11}$ N

(c) $1.33 \times 10^{-10}$ N (d) $6.6 \times 10^{-10}$ N

**4.** Given below is a list of E.M spectrum and its use. Which one does not match?

(a) U.V. rays — finger prints detection

(b) I.R.. rays — for taking photography during the fog

(c) X- rays — atomic structure

(d) Microwaves — forged document detection

**5.** A point source of electromagnetic radiation has an average power output of 800 W. The maximum value of electric field at a distance 4.0 m from the source is

(a) 64.7 V/m (b) 97.8 V/m

(c) 86.72 V/m (d) 54.77 V/m

**6.** Which of the following has/have zero average value in a plane electromagnetic wave?

(a) Both magnetic and electric fields

(b) Electric field only

(c) Magnetic field only

(d) Magnetic energy

**7.** Which of the following is correct about the electromagnetic waves?

(a) they are transverse waves

(b) they have rest mass

(c) they require medium to propagate

(d) they travel at varying speed through vaccum

**8.** In an electromagnetic wave

(a) power is transmitted along the magnetic field

(b) power is transmitted along the electric field

(c) power is equally transferred along the electric and magnetic fields

(d) power is transmitted in a direction perpendicular to both the fields

**9.** An electro magnetic wave travels along z-axis. Which of the following pairs of space and time varying fields would generate such a wave

(a) $E_x, B_y$ (b) $E_y, B_x$

(c) $E_z, B_x$ (d) $E_y, B_z$

**10.** Which of the following statements is true?

(a) The frequency of microwaves is greater than that of UV-rays

(b) The wavelength of IR rays is lesser than that of UV-rays

(c) The wavelength of microwaves is lesser than that of IR rays

(d) Gamma rays has least wavelength in the electromagnetic spectrum.

**11.** For a plane electromagnetic wave, the magnetic field at a point $x$ and time $t$ is

$$\vec{B}(x,t) = [1.2\times10^{-7}\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{k}]\text{T}$$

The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is:

(speed of light c = $3 \times 10^8$ ms$^{-1}$)

(a) $\vec{E}(x,t) = [-36\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{j}]\frac{\text{V}}{\text{m}}$

(b) $\vec{E}(x,t) = [36\sin(1\times10^3 x + 0.5\times10^{11}t)\hat{j}]\frac{\text{V}}{\text{m}}$

(c) $\vec{E}(x,t) = [36\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{k}]\frac{\text{V}}{\text{m}}$

(d) $\vec{E}(x,t) = [36\sin(1\times10^3 x + 1.5\times10^{11}t)\hat{i}]\frac{\text{V}}{\text{m}}$

**12.** The energy associated with electric field is $(U_E)$ and with magnetic fields is $(U_B)$ for an electromagnetic wave in free space. Then :

(a) $U_E = \frac{U_B}{2}$ (b) $U_E > U_B$

(c) $U_E < U_B$ (d) $U_E = U_B$

**13.** An electromagnetic wave of frequency $1 \times 10^{14}$ hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8 \times 10^{-12}$ C$^2$/N-m$^2$, then average energy density of electric field will be:

(a) $35.2 \times 10^{-10}$ J/m$^3$

(b) $35.2 \times 10^{-11}$ J/m$^3$

(c) $35.2 \times 10^{-12}$ J/m$^3$

(d) $35.2 \times 10^{-13}$ J/m$^3$

**14.** A plane electromagnetic wave in a non-magnetic dielectric medium is given by $\vec{E} = \vec{E}_0(4\times10^{-7}x - 50t)$ with distance being in meter and time in seconds. The dielectric constant of the medium is :

(a) 2.4 (b) 5.8

(c) 8.2 (d) 4.8

**15.** An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

(a) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

(b) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(c) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(d) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

**16.** Chosse the **correct** option relating wavelengths of different parts of electromagnetic wave spectrum :

(a) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{X-\text{rays}}$

(b) $\lambda_{\text{radio waves}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{x-\text{rays}}$

(c) $\lambda_{x-\text{rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{visible}}$

(d) $\lambda_{\text{visible}} > \lambda_{x-\text{rays}} > \lambda_{\text{radio waves}} > \lambda_{\text{micro waves}}$

**17.** Photons of an electromagnetic radiation has an energy11 keV each. To which region of electromagnetic spectrum does it belong ?

(a) X-ray region

(b) Ultra violet region

(c) Infrared region

(d) Visible region

**18.** The mean intensity of radiation on the surface of the Sun is about $10^8$ W/m$^2$. The rms value of the corresponding magnetic field is closest to :

(a) 1 T (b) $10^2$ T

(c) $10^{-2}$ T (d) $10^{-4}$ T

**19.** Arrange the following electromagnetic radiations per quantum in the order of increasing energy :

A : Blue light B : Yellow light

C : X-ray D : Radiowave.

(a) C, A, B, D (b) B, A, D, C

(c) D, B, A, C (d) A, B, D, C

**20.** The frequency of *X*-rays; γ-rays and ultraviolet rays are respectively *a*, *b* and *c* then

(a) $a < b; b > c$ (b) $a > b ; b > c$

(c) $a < b < c$ (d) $a = b = c$

## Numeric Value Answer

**21.** A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive *x*-direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}$ V / m. The corresponding magnetic field $\vec{B}$, at that point is x × $10^{-8}\hat{k}$T. Find the value of x.

**22.** If the magnetic field of a plane electromagnetic wave is given by (The speed of light = 3 × $10^8$ m/s)

$$B = 100 \times 10^{-6} \sin\left[2\pi \times 2 \times 10^{15}\left(t - \frac{x}{c}\right)\right]$$

then the maximum electric field (in N/C) associated with it is:

**23.** A 27 mW laser beam has a cross-sectional area of 10 $mm^2$. The magnitude of the maximum electric field (in kV/m) in this electromagnetic wave is given by :

[Given permittivity of space $\epsilon_0 = 9 \times 10^{-12}$ SI units, Speed of light c = 3 × $10^8$ m/s]

**24.** The mean intensity of radiation on the surface of the Sun is about $10^8$ W/$m^2$. The rms value of the corresponding magnetic field (in tesla) is :

**25.** The magnetic field of a plane electromagnetic wave is given by:

$$\vec{B} = B_0\hat{i}[\cos(kz - \omega t)] + B_1\hat{j}\cos(kz + \omega t)$$

Where $B_0 = 3 \times 10^{-5}$ T and $B_1 = 2 \times 10^{-6}$ T. The rms value of the force (in newton) experienced by a stationary charge Q = $10^{-4}$ C at z = 0 is :

**26.** 50 W/$m^2$ energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force (in newton) exerted on 1$m^2$ surface area will be (c = 3 × $10^8$ m/s):

**27.** A light beam travelling in the *x*-direction is described by the electric field $E_y = 300\sin\omega\left(t - \frac{x}{c}\right)$. An electron is constrained to move along the *y*-direction with a speed of $2.0 \times 10^7$ m/s. Find the maximum electric force (in newton) on the electron.

**28.** A plane electromagnetic wave of wave intensity 10 W/$m^2$ strikes a small mirror of area 20 $cm^2$, held perpendicular to the approaching wave. The radiation force (in newton) on the mirror will be

**29.** The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength (in volt $m^{-1}$) is

**30.** Light is incident normally on a completely absorbing surface with an energy flux of 25 W$cm^{-2}$. If the surface has an area of 25 $cm^2$, the momentum (in Ns) transferred to the surface in 40 min time duration will be:

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (d) | **7** | (a) | **10** | (d) | **13** | (c) | **16** | (b) | **19** | (c) | **22** | ($3\times10^4$) | **25** | (0.64) | **28** | ($1.33\times10^{-10}$) |
| **2** | (c) | **5** | (d) | **8** | (d) | **11** | (a) | **14** | (b) | **17** | (a) | **20** | (a) | **23** | (1.4) | **26** | ($20\times10^{-8}$) | **29** | (6) |
| **3** | (c) | **6** | (a) | **9** | (a) | **12** | (d) | **15** | (b) | **18** | (d) | **21** | (2.1) | **24** | ($6\times10^{-4}$) | **27** | ($4.8\times10^{-7}$) | **30** | ($5\times10^{-3}$) |

# RAY OPTICS AND OPTICAL INSTRUMENTS

23

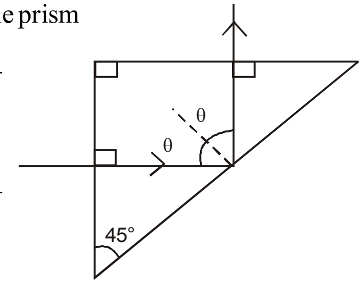
## MCQs with One Correct Answer

1. A vessel is half filled with a liquid of refractive index $\mu$. The other half of the vessel is filled with an immiscible liquid of refrative index 1.5 $\mu$. The apparent depth of the vessel is 50% of the actual depth. Then $\mu$ is
(a) 1.4 (b) 1.5
(c) 1.6 (d) 1.67

2. A convex lens is in contact with concave lens. The magnitude of the ratio of their powers is 2/3. Their equivalent focal length is 30 cm. What are their individual focal lengths (in cm)?
(a) –15, 10 (b) –10, 15
(c) 75, 50 (d) –75, 50

3. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object be placed in order to have a real image of size of the object?
(a) 60 cm (b) 30 cm
(c) 20 cm (d) 80 cm

4. A light ray is incident perpendicularly to one face of the prism shown in figure and is totally reflected if $\theta = 45°$, we conclude that the refractive index $n$ of the prism
(a) $n > \frac{1}{\sqrt{2}}$
(b) $n > \sqrt{2}$
(c) $n < \frac{1}{\sqrt{2}}$
(d) $n < \sqrt{2}$




5. The focal lengths of the objective and the eyepiece of the reflecting telescope are 225 cm and 5 cm respectively. The magnifying power of the telescope will be
(a) 49 (b) 45
(c) 35 (d) 60

6. A ray of light passes through an equilateral prism such that the angle of incidence is equal to the angle of emergence and the latter is equal to $\frac{3}{4}$ th of angle of prism. The angle of deviation is
(a) 25° (b) 30°
(c) 45° (d) 35°

7. An object is placed at a distance of 40 cm from a convex mirror of radius of curvature 20 cm. At what distance from the object a plane mirror be placed so that image in the convex mirror and plane mirror coincides?
(a) 20cm (b) 24cm
(c) 28cm (d) 32cm

8. Light propagates with speed of $2.2\times10^8$ m/s and $2.4\times10^8$ m/s in the medium P and Q respectively. The eritical angle between them is
(a) $\sin^{-1}\left(\frac{1}{11}\right)$ (b) $\sin^{-1}\left(\frac{11}{12}\right)$
(c) $\sin^{-1}\left(\frac{5}{12}\right)$ (d) $\sin^{-1}\left(\frac{5}{11}\right)$

9. A plano-convex lens fits exactly into a plano-concave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices $\mu_1$ and $\mu_2$ and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is

(a) $\frac{R}{2(\mu_1-\mu_2)}$ (b) $\frac{R}{(\mu_1-\mu_2)}$

(c) $\frac{2R}{(\mu_2-\mu_1)}$ (d) $\frac{R}{2(\mu_1+\mu_2)}$

**10.** A thin prism $P_1$ with angle 4° and made from glass of refractive index 1.54 is combined with another prism $P_2$ made of glass of refractive index 1.72 to produce dispersion without deviation. The angle of prism $P_2$ is

(a) 5.33° (b) 4°
(c) 2.6° (d) 3°

**11.** A thin lens made of glass (refractive index = 1.5) of focal length $f$ = 16 cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is $f_l$, then the ratio $f_l/f$ is closest to the integer:

(a) 1 (b) 9
(c) 5 (d) 17

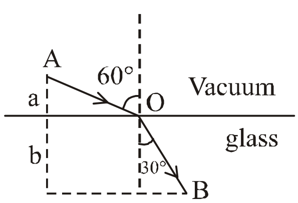
**12.** A convex lens of focal length 20 cm produces images of the same magnification 2 when an object is kept at two distances $x_1$ and $x_2$ $(x_1 > x_2)$ from the lens. The ratio of $x_1$ and $x_2$ is:

(a) 2 : 1 (b) 3 : 1
(c) 5 : 3 (d) 4 : 3

**13.** Two lenses of power –15 $D$ and +5 $D$ are in contact with each other. The focal length of the combination is

(a) + 10 cm (b) – 20 cm
(c) – 10 cm (d) + 20 cm

**14.** A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is :




(a) $\frac{2\sqrt{3}}{a}+2b$

(b) $2a+\frac{2b}{3}$

(c) $2a+\frac{2b}{\sqrt{3}}$

(d) $2a+2b$

**15.** If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to:

(a) 22mm (b) 12mm
(c) 2mm (d) 33mm

**16.** A double convex lens has power $P$ and same radii of curvature $R$ of both the surfaces. The radius of curvature of a surface of a plano-convex lens made of the same material with power 1.5 $P$ is :

(a) $2R$ (b) $\frac{R}{2}$

(c) $\frac{3R}{2}$ (d) $\frac{R}{3}$

**17.** In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be:

(a) 200 (b) 100
(c) 400 (d) 150

**18.** Light is incident from a medium into air at two possible angles of incidence (A) 20° and (B) 40°. In the medium light travels 3.0 cm in 0.2 ns. The ray will :

(a) suffer total internal reflection in both cases (A) and (B)
(b) suffer total internal reflection in case (B) only
(c) have partial reflection and partial transmission in case (B)
(d) have 100% transmission in case (A)

**19.** To get three images of a single object, one should have two plane mirrors at an angle of

(a) 60° (b) 90°
(c) 120° (d) 30°

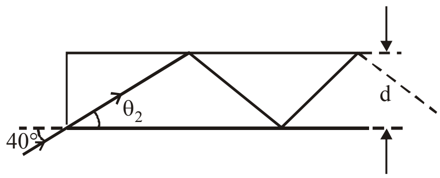
**20.** The refractive index of a glass is 1.520 for red light and 1.525 for blue light. Let $D_1$ and $D_2$ be angles of minimum deviation for red and blue light respectively in a prism of this glass. Then,

(a) $D_1 < D_2$
(b) $D_1 = D_2$
(c) $D_1$ can be less than or greater than $D_2$ depending upon the angle of prism
(d) $D_1 > D_2$

## Numeric Value Answer

**21.** A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance (in cm) of the object for which this concave mirror, by itself would produce a virtual image would be :

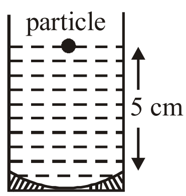
**22.** In figure, the optical fiber is $l = 2$ m long and has a diameter of $d = 20$ μm. If a ray of light is incident on one end of the fiber at angle $\theta_1 = 40°$, the number of reflections it makes before emerging from the other end is : (refractive index of fiber is 1.31 and sin 40° = 0.64)




**23.** A concave mirror for face viewing has focal length of 0.4 m. The distance (in metre) at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is:

**24.** A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure).

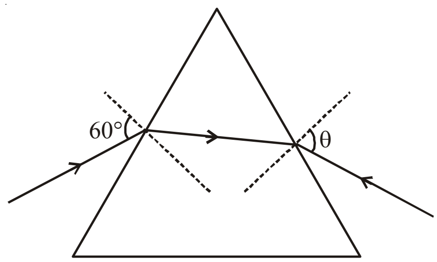
If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d (in cm) is :




(Refractive index of water = 1.33)

**25.** A ray of light falls on a glass plate of refractive index $\mu = 1.5$. What is the angle of incidence (in degree) of the ray if the angle between the reflected and refracted rays is 90°?

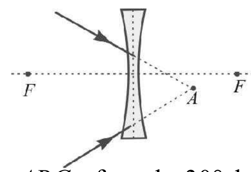
**26.** The monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle $\theta(n)$ with the normal (see the figure). For $n = \sqrt{3}$ the value of $\theta$ is 60° and $\frac{d\theta}{dn} = m$. The value of $m$ is



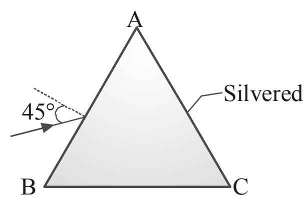

**27.** The magnifying power of a microscope with an objective of 5 mm focal length is 40. The length of its tube is 20 cm. Then the focal length (in cm) of the eye-piece is

**28.** A glass sphere of radius 5 cm has a small bubble 2 cm from its centre. The bubble is viewed along a diameter of the sphere from the side on which it lies. How far (in cm) from the surface will it appear. Refractive index of glass is 1.5.

**29.** A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed the point where the rays meet will move 5 cm closer to the mounting that holds the lens. Find focal length (in cm) of the lens.




**30.** A prism $ABC$ of angle 30° has its face $AC$ silvered. A ray of light incident at an angle of 45° at the face $AB$ retraces its path after refraction at face $AB$ and reflection at face $AC$. The refractive index of the material of the prism is




**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (b) | **7** | (b) | **10** | (d) | **13** | (c) | **16** | (d) | **19** | (b) | **22** | (57000) | **25** | (57) | **28** | (2.5) |
| **2** | (a) | **5** | (b) | **8** | (b) | **11** | (b) | **14** | (d) | **17** | (a) | **20** | (a) | **23** | (0.32) | **26** | (2) | **29** | (30) |
| **3** | (c) | **6** | (b) | **9** | (b) | **12** | (b) | **15** | (a) | **18** | (b) | **21** | (10) | **24** | (8.8) | **27** | (2.5) | **30** | ($\sqrt{2}$) |

# Wave Optics

24

## MCQs with One Correct Answer

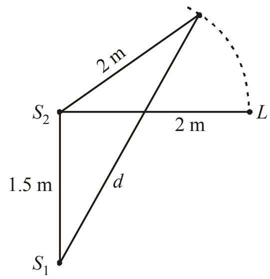
1. If two waves represented by $y_1 = 4 \sin \omega t$ and $y_2 = 3 \sin\left(\omega t + \frac{\pi}{3}\right)$ interfere at a point, then the amplitude of the resulting wave will be about
   (a) 7.99 (b) 6.08
   (c) 5.00 (d) 3.50
2. In a double slit experiment, the screen is placed at a distance of 1.25 m from the slits. When the apparatus is immersed in water ($\mu_w = 4/3$), the angular width of a fringe is found to be 0.2°. When the experiment is performed in air with same set up, the angular width of the fringe is
   (a) 0.4° (b) 0.27°
   (c) 0.35° (d) 0.15°
3. A ray of light is incident from a denser to a rarer medium. The critical angle for total internal reflection is $\theta_{iC}$ and Brewster's angle of incidence is $\theta_{iB}$, such that $\sin \theta_{iC}/\sin \theta_{iB} = \eta = 1.28$. The relative refractive index of the two media is:
   (a) 0.2 (b) 1.4
   (c) 0.8 (d) 0.12
4. In a Young's double-slit experiment, let $\beta$ be the fringe width, and $I_0$ be the intensity at the central bright fringe. At a distance x from the central bright fringe, the intensity is
   (a) $I_0 \cos\left(\frac{x}{\beta}\right)$ (b) $I_0 \cos^2\left(\frac{x}{\beta}\right)$
   (c) $I_0 \cos^2\left(\frac{\pi x}{\beta}\right)$ (d) $\left(\frac{I_0}{4}\right)\cos^2\left(\frac{\pi x}{\beta}\right)$
5. Unpolarised light is incident on a dielectric of refractive index $\sqrt{3}$. What is the angle of incidence if the reflected beam is completely polarised?
   (a) 30° (b) 45°
   (c) 60° (d) 75°
6. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angel $\phi$. If $\mu$ represents the refractive index of glass with respect to air, then the angle between the reflected and the refracted rays is
   (a) $90° + \phi$ (b) $\sin^{-1}(\mu \cos \phi)$
   (c) 90° (d) $90° - \sin^{-1}\left(\frac{\sin \phi}{\mu}\right)$
7. Unpolarized light is incident on a plane sheet on water surface. The angle of incidence for which the reflected and refracted rays are perpendicular to each other is
   $\left(\mu \text{ of water} = \frac{4}{3}\right)$
   (a) $\sin^{-1}\left(\frac{4}{3}\right)$ (b) $\tan^{-1}\left(\frac{3}{4}\right)$
   (c) $\tan^{-1}\left(\frac{4}{3}\right)$ (d) $\sin^{-1}\left(\frac{1}{3}\right)$
8. Two coherent sources of sound, $S_1$ and $S_2$, produce sound waves of the same wavelength, $\lambda = 1$ m, in phase. $S_1$ and $S_2$ are placed 1.5 m apart (see fig.). A listener, located at $L$, directly in front of $S_2$ finds that the intensity is at a minimum when he is 2 m away from $S_2$. The listener moves away from $S_1$, keeping his distance from $S_2$ fixed. The adjacent maximum of intensity is observed when the listener is at a distance $d$ from $S_1$. Then, $d$ is :

(a) 12 m

(b) 5m

(c) 2m

(d) 3m




**9.** Wavelength of light used in an optical instrument are $\lambda_1 = 4000$ Å and $\lambda_2 = 5000$ Å, then ratio of their respective resolving powers $\lambda_1 = 4000$ Å (corresponding to $\lambda_1$ and $\lambda_2$) is

(a) 16 : 25 (b) 9 : 1

(c) 4 : 5 (d) 5 : 4

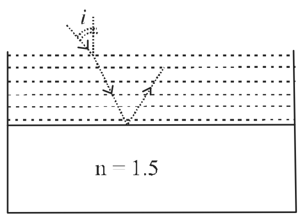
**10.** Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index $\mu$. A student finds that, irrespective of what the incident angle $i$ (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of $\mu$ is:

(a) $\sqrt{\dfrac{5}{3}}$

(b) $\dfrac{3}{\sqrt{5}}$

(c) $\dfrac{5}{\sqrt{3}}$

(d) $\dfrac{4}{3}$



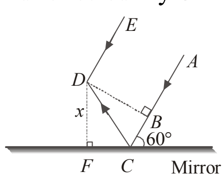

**11.** For the two parallel rays $AB$ and $DE$ shown here, $BD$ is the wavefront. For what value of wavelength of rays destructive interference takes place between ray $DE$ and reflected ray $CD$ ?

(a) $\sqrt{3}\,x$

(b) $\sqrt{2}\,x$

(c) $x$

(d) $2x$




**12.** A beam of natural light falls on a system of 5 polaroids, which arranged in succession such that the pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the incident light intensity that passes through the system is

(a) $\dfrac{1}{64}$ (b) $\dfrac{1}{32}$ (c) $\dfrac{1}{256}$ (d) $\dfrac{1}{512}$

**13.** Two polaroids are oriented with their planes perpendicular to incident light and transmission axis making an angle of 30° with each other. What fraction of incident unpolarised light is transmitted?

(a) 15.2% (b) 9.2%

(c) 11.6% (d) 37.5%

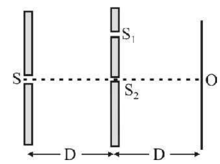
**14.** Two ideal slits $S_1$ and $S_2$ are at a distance $d$ apart, and illuminated by light of wavelength $\lambda$ passing through an ideal source slit S placed on the line through $S_2$ as shown. The distance between the planes of slits and the source slit is $D$. A screen is held at a distance $D$ from the plane of the slits. The minimum value of d for which there is darkness at $O$ is

(a) $\sqrt{\dfrac{3\lambda D}{2}}$

(b) $\sqrt{\lambda D}$

(c) $\sqrt{\dfrac{\lambda D}{2}}$

(d) $\sqrt{3\lambda D}$




**15.** A parallel beam of light of wavelength $\lambda$ is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the second minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of slit is

(a) $\pi$ (b) $2\pi$ (c) $3\pi$ (d) $4\pi$

**16.** In a Young's double slit experiment, the distance between the two identical slits is 6.1 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is:

(a) 3 (b) 6 (c) 12 (d) 24

**17.** A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 Å and diffraction bands are observed on a screen 0.5 m from the slit. The distance of the third dark band from the central bright band is :

(a) 3mm (b) 9mm

(c) 4.5mm (d) 1.5mm

**18.** In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength $\lambda$), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is

(a) $2\lambda$ (b) $\frac{2\lambda}{3}$ (c) $\frac{\lambda}{3}$ (d) $\lambda$

**19.** At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is

(a) $\frac{\pi}{2}$ radian (b) $\pi$ radian

(c) $\frac{\pi}{8}$ radian (d) $\frac{\pi}{4}$ radian

**20.** A parallel beam of monochromatic light of wavelength 5000Å is incident normally on a single narrow slit of width 0.001 mm. The light is focussed by a convex lens on a screen placed in focal plane. The first minimum will be formed for the angle of diffraction equal to

(a) 0° (b) 15° (c) 30° (d) 50°

## Numeric Value Answer

**21.** In a Young's double slit experiment, the slits are placed 0.320 mm apart. Light of wavelength $\lambda$ = 500 nm is incident on the slits. The total number of bright fringes that are observed in the angular range $-30° \leq \theta \leq 30°$ is

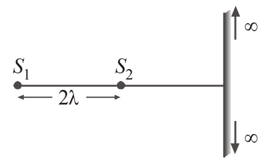
**22.** In a Young's double slit experiment, the path difference, at a certain point on the screen, betwen two interfering waves is $\frac{1}{8}$th of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is

**23.** In a double-slit experiment, green light (5303Å) falls on a double slit having a separation of 19.44 μm and a width of 4.05 μm. The number of bright fringes between the first and the second diffraction minima is :

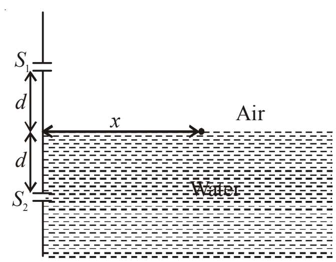
**24.** Calculate the limit of resolution (in radian) of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star.

**25.** The value of numerical aperature of the objective lens of a microscope is 1.25. If light of wavelength 5000 Å is used, the minimum separation (in μm) between two points, to be seen as distinct, will be :

**26.** There are two sources kept at distances $2\lambda$. A large screen is perpendicular to line joining the sources. Number of maximas on the screen in this case is ($\lambda$ = wavelength of light)




**27.** A Young's double slit interference arrangement with slits $S_1$ and $S_2$ is immersed in water (refractive index = $\frac{4}{3}$) as shown in the figure. The positions of maximum on the surface of water are given by $x^2 = p^2m^2\lambda^2 - d^2$, where $\lambda$ is the wavelength of light in air (refractive index = 1), $2d$ is the separation between the slits and $m$ is an integer. The value of $p$ is




**28.** Two waves of the same frequency have amplitudes 2 and 4. They interfere at a point where their phase difference is 60°. Find their resultant amplitude.

**29.** Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm, coming from a distant object, the limit of resolution of the telescope (in radian) is

**30.** In an interference pattern, at a point there observe 16th order maximum for $\lambda_1$ = 6000 Å. What order will be visible here if the source is replaced by light of wavelength $\lambda_2$ = 4800 Å?

### ANSWER KEY

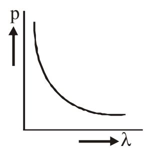
| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (c) | **7** | (c) | **10** | (b) | **13** | (d) | **16** | (c) | **19** | (b) | **22** | (0.85) | **25** | (0.24) | **28** | ($\sqrt{28}$) |
| **2** | (b) | **5** | (c) | **8** | (d) | **11** | (a) | **14** | (c) | **17** | (b) | **20** | (c) | **23** | (5) | **26** | (3) | **29** | ($3 \times 10^{-7}$) |
| **3** | (c) | **6** | (c) | **9** | (d) | **12** | (d) | **15** | (d) | **18** | (a) | **21** | (641) | **24** | ($305 \times 10^{-9}$) | **27** | (3) | **30** | (20) |

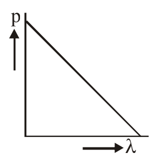
# 25

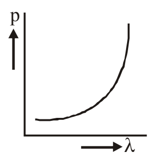
# DUAL NATURE OF RADIATION AND MATTER

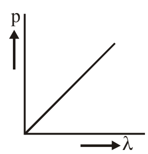
## MCQs with One Correct Answer

**1.** Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength?

(a) 


(b) 


(c) 


(d) 


**2.** When ultraviolet light of energy 6.2 eV incidents on a aluminimum surface, it emits photoelectrons. If work function for aluminium surface is 4.2 eV, then kinetic energy of emitted electrons is

(a) $3.2 \times 10^{-19}$ J (b) $3.2 \times 10^{-17}$ J
(c) $3.2 \times 10^{-16}$ J (d) $3.2 \times 10^{-11}$ J

**3.** A small photocell is placed at a distance of 4 m from a photosensitive surface. When light falls on the surface the current is 5 mA. If the distance of cell is decreased to 1 m, the current will become

(a) 10 mA (b) 40 mA
(c) 20 mA (d) 80 mA

**4.** A and B are two metals with threshold frequencies $1.8 \times 10^{14}$ Hz and $2.2 \times 10^{14}$ Hz. Two identical photons of energy 0.825 eV each are incident on them. Then photoelectrons are emitted in (Take h = $6.6 \times 10^{-34}$ Js)

(a) B alone (b) A alone
(c) neither A nor B (d) both A and B

**5.** Which is the incorrect statement of the following

(a) Photon is a particle with zero rest mass
(b) Photon is a particle with zero mementum
(c) Photon travel with velocity of light in vacuum
(d) Photon even feel the pull of gravity

**6.** An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If $\lambda_0$ is initial de-Broglie wavelength of electron, its de-Broglie wave length at time $t$ is given by:

(a) $\dfrac{\lambda_0\sqrt{2}}{\sqrt{1+\dfrac{e^2E^2t^2}{m^2v_0^2}}}$ (b) $\dfrac{\lambda_0}{\sqrt{1+\dfrac{e^2E_0^2t^2}{m^2v_0^2}}}$

(c) $\dfrac{\lambda_0}{\sqrt{1+\dfrac{e^2E^2t^2}{2m^2v_0^2}}}$ (d) $\dfrac{\lambda_0}{\sqrt{2+\dfrac{e^2E^2t^2}{m^2v_0^2}}}$

**7.** A particle 'P' is formed due to a completely inelastic collision of particles '$x$' and '$y$' having de-Broglie wavelengths '$\gamma_x$' and '$\gamma_y$' respectively. If $x$ and $y$ were moving in opposite directions, then the de-Broglie wavelength of 'P' is:

(a) $\dfrac{\gamma_x\gamma_y}{\gamma_x+\gamma_y}$ (b) $\dfrac{\gamma_x\gamma_y}{|\gamma_x-\gamma_y|}$

(c) $\gamma_x-\gamma_y$ (d) $\gamma_x+\gamma_y$

**8.** The stopping potential $V_0$ (in volt) as a function of frequency ($\nu$) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be :

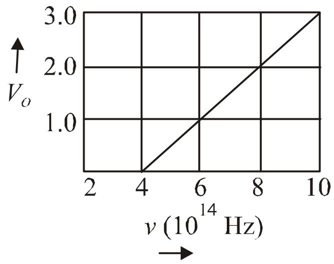
(Given : Planck's constant ($h$) = $6.63 \times 10^{-34}$ Js, electron charge $e = 1.6 \times 10^{-19}$ C)

(a) 1.82 eV

(b) 1.66 eV

(c) 1.95 eV

(d) 2.12 eV




**9.** A metal plate of area $1 \times 10^{-4}$ m$^2$ is illuminated by a radiation of intensity 16 mW/m$^2$. The work function of the metal is 5 eV. The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second and their maximum energy, respectively, will be:

[1 eV = $1.6 \times 10^{-19}$ J]

(a) $10^{14}$ and 10 eV (b) $10^{12}$ and 5 eV

(c) $10^{11}$ and 5 eV (d) $10^{10}$ and 5 eV

**10.** Two particles move at right angle to each other. Their de Broglie wavelengths are $\lambda_1$ and $\lambda_2$ respectively. The particles suffer perfectly inelastic collision. The de Broglie wavelength $\lambda$, of the final particle, is given by :

(a) $\dfrac{1}{\lambda^2} = \dfrac{1}{\lambda_1^2} + \dfrac{1}{\lambda_2^2}$ (b) $\lambda = \sqrt{\lambda_1 \lambda_2}$

(c) $\lambda = \dfrac{\lambda_2 + \lambda_2}{2}$ (d) $\dfrac{2}{\lambda} = \dfrac{1}{\lambda_1} + \dfrac{1}{\lambda_2}$

**11.** The magnetic field associated with a light wave is given at the origin by

$B = B_0 [\sin(3.14 \times 10^7)ct + \sin(6.28 \times 10^7)ct]$.

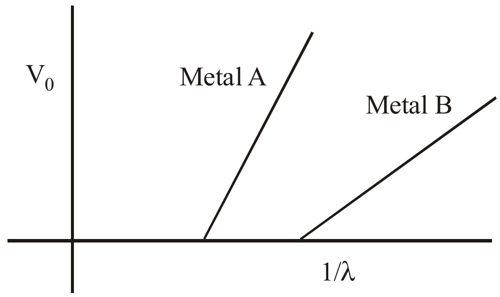
If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photoelectrons?

($c = 3 \times 10^8$ ms$^{-1}$, $h = 6.6 \times 10^{-34}$ J-s)

(a) 6.82 eV (b) 12.5 eV

(c) 8.52 eV (d) 7.72 eV

**12.** In an experiment on photoelectric effect, a student plots stopping potential $V_0$ against reciprocal of the wavelength $\lambda$ of the incident light for two different metals A and B. These are shown in the figure.




Looking at the graphs, you can most appropriately say that:

(a) Work function of metal B is greater than that of metal A

(b) For light of certain wavelength falling on both metal, maximum kinetic energy of electrons emitted from A will be greater than those emitted from B.

(c) Work function of metal A is greater than that of metal B

(d) Students data is not correct

**13.** The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5 V. The incident radiation lies in

(a) ultra-violet region (b) infra-red region

(c) visible region (d) $X$-ray region

**14.** According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal Versus the frequency, of the incident radiation gives a straight line whose slope

(a) depends both on the intensity of the radiation and the metal used

(b) depends on the intensity of the radiation

(c) depends on the nature of the metal used

(d) is the same for the all metals and independent of the intensity of the radiation

**15.** Photoelectrons are ejected from a metal when light of frequency $\upsilon$ falls on it. Pick out the wrong statement from the following.

(a) No electrons are emitted if $\upsilon$ is less than $W/h$, where $W$ is the work function of the metal

(b) The ejection of the photoelectrons is instantaneous.

(c) The maximum energy of the photoelectrons is $h\upsilon$.

(d) The maximum energy of the photoelectrons is independent of the intensity of the light.

**16.** When photons of energy $h\nu$ fall on an aluminium plate (of work function $E_0$), photoelectrons of maximum kinetic energy K are ejected. If the frequency of the radiation is doubled, the maximum kinetic energy of the ejected photoelectrons will be

(a) 2K (b) K

(c) $K + h\nu$ (d) $K + E_0$

**17.** A Laser light of wavelength 660 nm is used to weld Retina detachment. If a Laser pulse of width 60 ms and power 0.5 kW is used the approximate number of photons in the pulse are : [Take Planck's constant $h = 6.62 \times 10^{-34}$ Js]

(a) $10^{20}$ (b) $10^{18}$

(c) $10^{22}$ (d) $10^{19}$

**18.** When a metallic surface is illuminated by a light of wavelength $\lambda$, the stopping potential for the photoelectric current is 3 V. When the same surface is illuminated by light of wavelength $2\lambda$, the stopping potential is 1V. The threshold wavelength for this surface is

(a) $4\lambda$ (b) $3.5\lambda$

(c) $3\lambda$ (d) $2.75\lambda$

**19.** Photoelectric emission is observed from a metallic surface for frequencies $v_1$ and $v_2$ of the incident light rays ($v_1 > v_2$). If the maximum values of kinetic energy of the photoelectrons emitted in the two cases are in the ratio of 1 : k, then the threshold frequency of the metallic surface is

(a) $\dfrac{v_1 - v_2}{k-1}$ (b) $\dfrac{kv_1 - v_2}{k-1}$

(c) $\dfrac{kv_2 - v_1}{k-1}$ (d) $\dfrac{v_2 - v_1}{k}$

**20.** The work functions of metals A and B are in the raio 1 : 2.

If light of frequencies f and 2f are incident on the surfaces of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is (f is greater than threshold frequency of A, 2f is greater than threshold frequency of B)

(a) 1 : 1 (b) 1 : 2

(c) 1 : 3 (d) 1 : 4

## Numeric Value Answer

**21.** Surface of certain metal is first illuminated with light of wavelength $\lambda_1 = 350$ nm and then, by light of wavelength $\lambda_2 = 540$ nm. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of (2) The work function of the metal (in eV) is

(Energy of photon $= \dfrac{1240}{\lambda\,(\text{in nm})}$ eV)

**22.** The magnetic field associated with a light wave is given at the origin by

$B = B_0\,[\sin(3.14 \times 10^7)ct + \sin(6.28 \times 10^7)ct]$.

If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy (in eV) of the photoelectrons?

($c = 3 \times 10^8\ \text{ms}^{-1}$, $h = 6.6 \times 10^{-34}$ J-s)

**23.** A metal plate of area $1 \times 10^{-4}\ \text{m}^2$ is illuminated by a radiation of intensity 16 mW/m$^2$. The work function of the metal is 5 eV. The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second is

[$1\ \text{eV} = 1.6 \times 10^{-19}$ J]

**24.** If the deBroglie wavelength of an electron is equal to $10^{-3}$ times the wavelength of a photon of frequency $6 \times 10^{14}$ Hz, then the speed (in m/s) of electron is equal to :

(Speed of light $= 3 \times 10^8$ m/s)

Planck's constant $= 6.63 \times 10^{-34}$ J.s

Mass of electron $= 9.1 \times 10^{-31}$ kg)

**25.** In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential (in V) is

$$\left(\frac{hc}{e} = 1240 \text{ nm-V}\right)$$

**26.** A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50v Another particle B of mass '4m' and charge'q' is accelerated by a potential differnce of 2500V.

The ratio of de-Broglie wavelength $\dfrac{\lambda_A}{\lambda_B}$ is

**27.** A monochromatic source of light operating at 200W emits $4 \times 10^{20}$ photons per second. Find the wavelength (in nm) of the light.

**28.** A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is :

[Given Planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^8$ m/s]

**29.** Light of wavelength 180 nm ejects photoelectron from a plate of a metal whose work function is 2 eV. If a uniform magnetic field of $5 \times 10^{-5}$ T is applied parallel to plate, what would be the radius (in metre) of the path followed by electrons ejected normally from the plate with maximum energy ?

**30.** In a photoelectric effect experiment the threshold wavelength of light is 380 nm. If the wavelength of incident light is 260 nm, the maximum kinetic energy (in eV) of emitted electrons will be:

Given E (in eV) = $\dfrac{1237}{\lambda(\text{in nm})}$

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (b) | **7** | (b) | **10** | (a) | **13** | (a) | **16** | (c) | **19** | (b) | **22** | (7.7) | **25** | (1) | **28** | ($5 \times 10^{15}$) |
| **2** | (a) | **5** | (b) | **8** | (b) | **11** | (d) | **14** | (d) | **17** | (a) | **20** | (b) | **23** | ($10^{11}$) | **26** | (14.14) | **29** | (0.149) |
| **3** | (d) | **6** | (c) | **9** | (c) | **12** | (d) | **15** | (c) | **18** | (a) | **21** | (1.8) | **24** | ($1.45 \times 10^6$) | **27** | (400) | **30** | (1.5) |

26

# ATOMS

## MCQs with One Correct Answer

**1.** The significant result deduced from the Rutherford's scattering experiment is that

(a) whole of the positive charge is concentrated at the centre of atom

(b) there are neutrons inside the nucleus

(c) $\alpha$-particles are helium nuclei

(d) electrons are embedded in the atom

**2.** An $\alpha$-particle of energy 5 MeV is scattered through 180º by a fixed uranium nucleus. The distance of closest approach is of the order of

(a) $10^{-12}$ cm (b) $10^{-10}$ cm

(c) $10^{-20}$ cm (d) $10^{-15}$ cm

**3.** The energy of electron in the nth orbit of hydrogen atom is expressed as $E_n = \frac{-13.6}{n^2}$ eV. The shortest wavelength of Lyman series will be

(a) 910 Å (b) 5463 Å

(c) 1315 Å (d) None of these

**4.** Consider an electron in the $n^{th}$ orbit of a hydrogen atom in the Bohr's model. The circumference of the orbit can be expressed in terms of the de Broglie wavelength $\lambda$ of that electron as

(a) $0.529\,n\lambda$ (b) $\sqrt{n}\lambda$

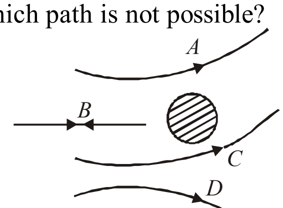
(c) $(13.6)\lambda$ (d) $n\lambda$

**5.** In Rutherford's scattering experiment, the number of $\alpha$-particles scattered at 60° is $5 \times 10^6$. The number of $\alpha$-particles scattered at 120° will be

(a) $15 \times 10^6$ (b) $\frac{3}{5} \times 10^6$

(c) $\frac{5}{9} \times 10^6$ (d) None of these

**6.** In the Rutherford experiment, $\alpha$-particles are scattered from a nucleus as shown. Out of the four paths, which path is not possible?




(a) *D*

(b) *B*

(c) *C*

(d) *A*

**7.** Electrons in a certain energy level $n = n_1$, can emit 3 spectral lines. When they are in another energy level, $n = n_2$. They can emit 6 spectral lines. The orbital speed of the electrons in the two orbits are in the ratio of

(a) 4 : 3 (b) 3 : 4

(c) 2 : 1 (d) 1 : 2

**8.** Ionization energy of hydrogen atom is 13.6eV. Hydrogen atoms in the ground state are excited by monochromatic radiation of photon energy 12.1 eV. According to Bohr's theory, the spectral lines emitted by hydrogen will be

(a) three (b) four

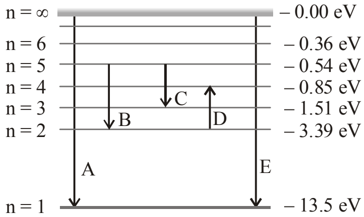
(c) one (d) fwo

**9.** The ionisation energy of hydrogen is 13.6 eV. The energy required to excite the electron from the first to the third orbit is approximately

(a) 10.2 J (b) $12.09 \times 10^{-6}$J

(c) 19.94 J (d) $19.34 \times 10^{-19}$J

**10.** The energy levels of the hydrogen spectrum is shown in figure. There are some transitions A, B, C, D and E. Transition A, B and C respectively represent




(a) first member of Lyman series, third spectral line of Balmer series and the second spectral line of Paschen series
(b) ionization potential of hydrogen, second spectral line of Balmer series, third spectral line of Paschen series
(c) series limit of Lyman series, third spectral line of Balmer series and second spectral line of Paschen series
(d) series limit of Lyman series, second spectral line of Balmer series and third spectral line of Paschen series

**11.** An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to

(a) $v^2$ (b) $\frac{1}{m}$
(c) $\frac{1}{v^2}$ (d) $\frac{1}{Ze}$

**12.** The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state?
(a) 24.2 nm (b) 11.4 nm
(c) 35.8 nm (d) 8.6 nm

**13.** In a hydrogen atom the electron makes a transition from $(n + 1)^{th}$ level to the $n^{th}$ level. If $n >> 1$, the frequency of radiation emitted is proportional to :

(a) $\frac{1}{n}$ (b) $\frac{1}{n^3}$
(c) $\frac{1}{n^2}$ (d) $\frac{1}{n^4}$

**14.** The energy required to remove the electron from a singly ionized Helium atom is 2.2 times the energy required to remove an electron from Helium atom. The total energy required to ionize the Helium atom completely is:
(a) 20 eV (b) 79 eV
(c) 109 eV (d) 34 eV

**15.** The binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of $Li^{++}$ is:
(a) 122.4 eV (b) 30.6 eV
(c) 13.6 eV (d) 3.4 eV

**16.** The electron of a hydrogen atom makes a transition from the $(n + 1)^{th}$ orbit to the $n^{th}$ orbit. For large n the wavelength of the emitted radiation is proportional to
(a) $n$ (b) $n^3$
(c) $n^4$ (d) $n^2$

**17.** The transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from :
(a) $3 \to 2$ (b) $4 \to 2$
(c) $5 \to 4$ (d) $2 \to 1$

**18.** Which of the following transitions in hydrogen atoms emit photons of highest frequency?
(a) $n = 1$ to $n = 2$ (b) $n = 2$ to $n = 6$
(c) $n = 6$ to $n = 2$ (d) $n = 2$ to $n = 1$

**19.** In a hydrogen like atom, when an electron jumps from the M-shell to the L-shell, the wavelength of emitted radiation is l. If an electron jumps from N-shell to the L-shell, the wavelength of emitted radiation will be:

(a) $\frac{27}{20}\lambda$ (b) $\frac{16}{25}\lambda$
(c) $\frac{25}{16}\lambda$ (d) $\frac{20}{27}\lambda$

**20.** Consider an electron in a hydrogen atom, revolving in its second excited state (having radius 4.65 Å). The de-Broglie wavelength of this electron is :
(a) 3.5 Å (b) 6.6 Å
(c) 12.9 Å (d) 9.7 Å

## Numeric Value Answer

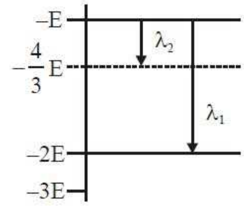
**21.** Taking the wavelength of first Balmer line in hydrogen spectrum ($n = 3$ to $n = 2$) as 660 nm, the wavelength (in nm) of the 2nd Balmer line ($n = 4$ to $n = 2$) will be;

**22.** An excited $He^+$ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number $n$, corresponding to its initial excited state is (for photon of wavelength $\lambda$, energy $E = \dfrac{1240eV}{\lambda(\text{in nm})}$

**23.** The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is

**24.** An electron in hydrogen like atom makes a transition from $n^{th}$ orbit and emits radiation corresponding to Lyman series. If de-Broglie wavelength of electron in $n^{th}$ orbit is equal to the wavelength of radiation emitted, find the value of $n$. The atomic number of atom is 11.

**25.** Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths $r = \lambda_1 / \lambda_2$, is given by




**26.** Ratio of the wavelengths of first line of Lyman series and first line of Balmer series is

**27.** As per Bohr model, the minimum energy (in eV) required to remove an electron from the ground state of doubly ionized Li atom (Z = 3) is

**28.** The ionisation energy of hydrogen atom is 13.6 eV. Following Bohr's theory, the energy (in eV) corresponding to a transition between the 3rd and the 4th orbit is

**29.** If the binding energy (in eV) of the electron in a hydrogen atom is 13.6 eV, the energy (in eV) required to remove the electron from the first excited state of $Li^{++}$ is

**30.** An electron in hydrogen atom jumps from a level $n = 4$ to $n = 1$. The momentum (in kg m/s) of the recoiled atom is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (a) | **10** | (c) | **13** | (b) | **16** | (b) | **19** | (d) | **22** | (5) | **25** | (0.34) | **28** | (0.66) |
| **2** | (a) | **5** | (c) | **8** | (a) | **11** | (b) | **14** | (b) | **17** | (c) | **20** | (d) | **23** | (823.5) | **26** | (0.18) | **29** | (30.6) |
| **3** | (a) | **6** | (c) | **9** | (d) | **12** | (b) | **15** | (b) | **18** | (d) | **21** | (488.9) | **24** | (25) | **27** | (122.4) | **30** | ($6.8 \times 10^{-27}$) |

# 27

# NUCLEI

**MCQs with One Correct Answer**

**1.** If radius of the ${}^{27}_{12}\text{Al}$ nucleus is taken to be $R_{Al}$, then the radius of ${}^{125}_{53}\text{Te}$ nucleus is nearly:

(a) $\frac{5}{3}R_{Al}$ (b) $\frac{3}{5}R_{Al}$

(c) $\left(\frac{13}{53}\right)^{1/3} R_{Al}$ (d) $\left(\frac{53}{13}\right)^{1/3} R_{Al}$

**2.** When a $U^{238}$ nucleus originally at rest, decays by emitting an alpha particle having a speed 'u', the recoil speed of the residual nucleus is

(a) $\frac{4u}{238}$ (b) $-\frac{4u}{234}$

(c) $\frac{4u}{234}$ (d) $-\frac{4u}{238}$

**3.** At any instant, the ratio of the amount of radioactive substances is 2 : 1. If their half lives be respectively 12 and 16 hours, then after two days, what will be the ratio of the substances ?

(a) 1 : 1 (b) 2 : 1

(c) 1 : 2 (d) 1 : 4

**4.** A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1. Ratio of their nuclear sizes will be

(a) $2^{1/3} : 1$ (b) $1 : 3^{1/2}$

(c) $3^{1/2} : 1$ (d) $1 : 2^{1/3}$

**5.** The nuclear radius of a nucleus with nucleon number 16 is $3 \times 10^{-15}$ m. Then, the nuclear radius of a nucleus with nucleon number 128 is

(a) $3 \times 10^{-15}$ m (b) $1.5 \times 10^{-15}$ m

(c) $6 \times 10^{-15}$ m (d) $4.5 \times 10^{-15}$ m

**6.** If $M_O$ is the mass of an oxygen isotope ${}_8O^{17}$, $M_P$ and $M_N$ are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is

(a) $(M_O - 17M_N)c^2$

(b) $(M_O - 8M_P)c^2$

(c) $(8M_P + 9M_N - M_O)c^2$

(d) $M_Oc^2$

**7.** If the nucleus ${}^{27}_{13}Al$ has nuclear radius of about 3.6 fm, then ${}^{125}_{32}Te$ would have its radius approximately as

(a) 9.6 fm (b) 12.0 fm

(c) 4.8 fm (d) 6.0 fm.

**8.** Which of the following nuclei has lowest value of the binding energy per nucleon :

(a) ${}^{4}_{2}\text{He}$ (b) ${}^{52}_{24}\text{Cr}$

(c) ${}^{152}_{62}\text{Sm}$ (d) ${}^{100}_{80}\text{Hg}$

**9.** Half lives for $\alpha$ and $\beta$ emission of a radioactive material are 16 years and 48 years respectively. When material decays giving $\alpha$ and $\beta$ emission simultaneously, time in which $3/4^{th}$ material decays is

(a) 29 years (b) 24 years

(c) 64 years (d) 12 years

**10.** The initial activity of a certain radioactive isotope was measured as 16000 counts $min^{-1}$ and its activity after 12 h was 2100 counts $min^{-1}$, its half-life, in hours, is nearest to [Given $\log_e(7.2) = 2$]

(a) 9.0 (b) 6.0
(c) 4.0 (d) 3.0

**11.** The radius $R$ of a nucleus of mass number $A$ can be estimated by the formula $R = (1.3 \times 10^{-15}) A^{1/3}$ m. It follows that the mass density of a nucleus is of the order of :

$(M_{\text{prot.}} \cong M_{\text{neut.}} \simeq 1.67 \times 10^{-27} \text{ kg})$

(a) $10^3$ kg $m^{-3}$ (b) $10^{10}$ kg $m^{-3}$
(c) $10^{24}$ kg $m^{-3}$ (d) $10^{17}$ kg $m^{-3}$

**12.** Find the Binding energy per neucleon for $^{120}_{50}Sn$. Mass of proton $m_p = 1.00783$ U, mass of neutron $m_n = 1.00867$ U and mass of tin nucleus $m_{Sn} = 119.902199$ U. (take 1U = 931 MeV)

(a) 7.5 MeV (b) 9.0 MeV
(c) 8.0 MeV (d) 8.5 MeV

**13.** In a reactor, 2 kg of $_{92}U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, N = 6.023 × $10^{26}$ per kilo mole and 1 eV = 1.6 × $10^{-19}$ J. The power output of the reactor is close to:

(a) 35 MW (b) 60 MW
(c) 125 MW (d) 54 MW

**14.** A piece of bone of an animal from a ruin is found to have $^{14}C$ activity of 12 disintegrations per minute per gm of its carbon content. The $^{14}C$ activity of a living animal is 16 disintegrations per minute per gm. How long ago nearly did the animal die? (Given half life of $^{14}C$ is $t_{1/2} = 5760$ years)

(a) 1672 years (b) 2391 years
(c) 3291 years (d) 4453 years

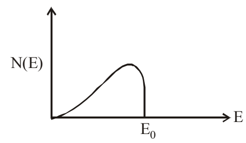
**15.** The binding energy per nucleon of deuteron $\left(^2_1H\right)$ and helium nucleus $\left(^4_2He\right)$ is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is

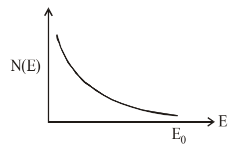
(a) 23.6 MeV (b) 26.9 MeV
(c) 13.9 MeV (d) 19.2 MeV

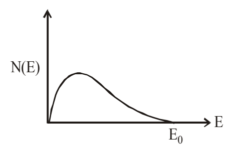
**16.** The activity of a radioactive sample falls from 700 $s^{-1}$ to 500 $s^{-1}$ in 30 minutes. Its half life is close to:

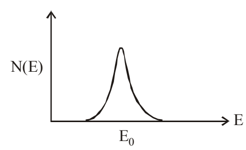
(a) 72 min (b) 62 min
(c) 66 min (d) 52 min

**17.** The energy spectrum of β-particles [number N(E) as a function of β-energy E] emitted from a radioactive source is

(a) 


(b) 


(c) 


(d) 


**18.** The decay constants of a radioactive substance for α and β emission are $\lambda_\alpha$ and $\lambda_\beta$ respectively. If the substance emits α and β simultaneously, then the average half life of the material will be

(a) $\dfrac{2T_\alpha T_\beta}{T_\alpha + T_\beta}$ (b) $T_\alpha + T_\beta$
(c) $\dfrac{T_\alpha T_\beta}{T_\alpha + T_\beta}$ (d) $\dfrac{1}{2}\left(T_\alpha + T_\beta\right)$

**19.** The half-life of a radioactive element A is the same as the mean-life of another radioactive element B. Initially both substances have the same number of atoms, then :

(a) A and B decay at the same rate always.
(b) A and B decay at the same rate initially.
(c) A will decay at a faster rate than B.
(d) B will decay at a faster rate than A.

**20.** In a radioactive decay chain, the initial nucleus is $^{232}_{90}Th$. At the end there are 6 $\alpha$-particles and 4 $\beta$-particles which are emitted. If the end nucleus is $^{A}_{Z}X$, A and Z are given by :

(a) $A = 208 ; Z = 80$ (b) $A = 202 ; Z = 80$
(c) $A = 208 ; Z = 82$ (d) $A = 200; Z = 81$

## Numeric Value Answer

**21.** Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At t = 0 it was 1600 counts per second and t = 8 seconds it was 100 counts per second. The count rate observed, as counts per second, at t = 6 seconds is

**22.** The ratio of the mass densities of nuclei of $^{40}Ca$ and $^{16}O$ is

**23.** The activity of a freshly prepared radioactive sample is $10^{10}$ disintegrations per second, whose mean life is $10^{9}$ s. The mass of an atom of this radioisotope is $10^{-25}$ kg. The mass (in mg) of the radioactive sample is

**24.** The half life of radon is 3.8 days. After how many days will only one twentieth of radon sample be left over?

**25.** The count rate from a radioactive sample falls from $4.0\times 10^{6}$ per second to $1 \times 10^{6}$ per second in 20 hour. What will be the count rate per second 100 hour after the beginning?

**26.** In an ore containing uranium, the ratio of $U^{238}$ to $Pb^{206}$ nuclei is 3. Calculate the age of the ore, (in year) assuming that all the lead present in the ore in the final stable product of $U^{238}$. Take the half life of $U^{238}$ to be $4.5 \times 10^{9}$ year.

**27.** In a nuclear reactor $U^{235}$ undergoes fission liberating 200 MeV of energy. The reactor has 10% efficiency and produces 1000 MW power. If the reactor is to function for 10 year, find the total mass (in kg) of the uranium required.

**28.** The disintegration rate of a certain radioactive sample at any instant is 4750 disintegrations per minute. Five minutes later the rate becomes 2700 disintegrations per minute. Calculate half life of the sample (in minute)

**29.** The mass defect for the nucleus of helium is 0.0303 a.m.u. What is the binding energy per nucleon for helium in MeV?

**30.** If the radius of a nucleus $^{256}X$ is 8 fermi, then the radius (in fermi) of $^{4}He$ nucleus will be

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (d) | **10** | (c) | **13** | (b) | **16** | (b) | **19** | (d) | **22** | (1) | **25** | $(3.91 \times 10^{3})$ | **28** | (6.1) |
| **2** | (c) | **5** | (c) | **8** | (a) | **11** | (d) | **14** | (b) | **17** | (c) | **20** | (c) | **23** | (1) | **26** | $(1.868 \times 10^{9})$ | **29** | (7) |
| **3** | (a) | **6** | (c) | **9** | (b) | **12** | (d) | **15** | (a) | **18** | (c) | **21** | (200) | **24** | (16.45) | **27** | $(3.8 \times 10^{4})$ | **30** | (2) |

# SEMICONDUCTOR ELECTRONICS: MATERIALS, DEVICES AND SIMPLE CIRCUITS

# 28

## MCQs with One Correct Answer

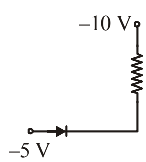
**1.** In a p-type semiconductor the acceptor level is situated 60 meV above the valence band. The maximum wavelength of light required to produce a hole will be

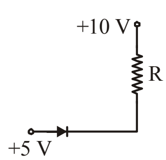
(a) $0.207 \times 10^{-5}$ m (b) $2.07 \times 10^{-5}$ m
(c) $20.7 \times 10^{-5}$ m (d) $2075 \times 10^{-5}$ m

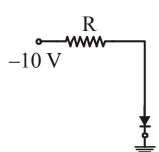
**2.** A half-wave rectifier is being used to rectify an alternating voltage of frequency 50 Hz. The number of pulses of rectified current obtained in one second is

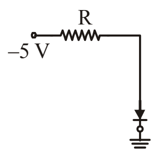
(a) 50 (b) 25
(c) 100 (d) 2000

**3.** Which of the junction diodes shown below are forward biased?

(a)



(b)


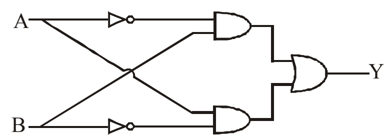

(c)



(d)



**4.** The current gain in the common emitter mode of a transistor is 10. The input impedance is 20kΩ and load of resistance is 100kΩ. The power gain is

(a) 300 (b) 500
(c) 200 (d) 100

**5.** The following circut represents



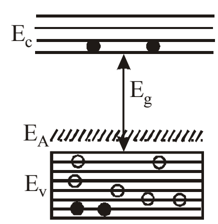

(a) OR gate (b) AND gate
(c) NAND gate (d) None of these

**6.** What is the conductivity (in mho $m^{-1}$)of a semiconductor if electron density = $5 \times 10^{12}$/cm$^3$ and hole density = $8 \times 10^{13}$/cm$^3$ ($\mu_e = 2.3$ m$^2$ V$^{-1}$ s$^{-1}$, $\mu_h = 0.01$ m$^2$V$^{-1}$ s$^{-1}$)

(a) 5.634 (b) 1.968
(c) 3.421 (d) 8.964

**7.** In the energy band diagram of a material shown below, the open circles and filled circles denote holes and electrons respectively.

The material is




(a) an insulator
(b) a metal
(c) an n-type semiconductor
(d) a p-type semiconductor

**8.** A potential barrier of 0.50 V exists across a p-n junction. If the depletion region is $5.0 \times 10^{-7}$ m wide, the intensity of the electric field in this region is

(a) $1.0 \times 10^6$ V/m (b) $1.0 \times 10^5$ V/m
(c) $2.0 \times 10^5$ V/m (d) $2.0 \times 10^6$ V/m

**9.** In case of a common emitter transistor amplifier, the ratio of the collector current to the emitter current $I_c/I_e$ is 0.96. The current gain of the amplifier is
(a) 6 (b) 48
(c) 24 (d) 12

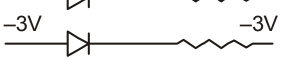
**10.** In common emitter amplifier, the current gain is 62. The collector resistance and input resistance are 5 kΩ an 500 Ω respectively. If the input voltage is 0.01 V, the output voltage is
(a) 0.62 V (b) 6.2 V
(c) 62 V (d) 620 V

**11.** With increasing biasing voltage of a photodiode, the photocurrent magnitude :
(a) remains constant
(b) increases initially and after attaining certain value, it decreases
(c) Increases linearly
(d) increases initially and saturates finally

**12.** If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 nm, then its band gap energy is:
Planck's constant, $h = 6.63 \times 10^{-34}$ J.s.
Speed of light, $c = 3 \times 10^8$ m/s
(a) 1.1 eV (b) 2.0 eV
(c) 1.5 eV (d) 3.1 eV

**13.** A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is :
(a) 5.48 V/m (b) 7.75 V/m
(c) 1.73 V/m (d) 2.45 V/m

**14.** The forward biased diode connection is:
(a) 

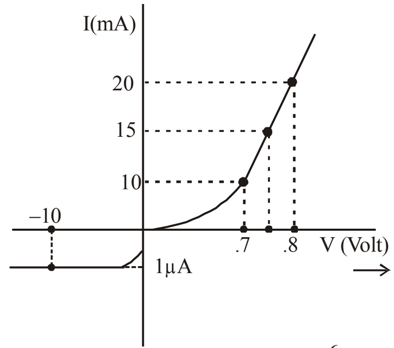
(b) 

(c) 

(d) 


**15.** If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities?
(a) $\frac{5}{8}$ (b) $\frac{4}{5}$
(c) $\frac{5}{4}$ (d) $\frac{4}{7}$

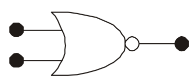
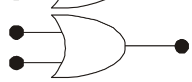
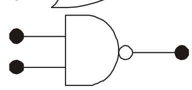
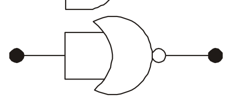
**16.** The V–I characteristic of a diode is shown in the figure. The ratio of forward to reverse bias resistance is :




(a) 10 (b) $10^{-6}$
(c) $10^6$ (d) 100

**17.** In the middle of the depletion layer of a reverse-biased *p-n* junction, the
(a) electric field is zero
(b) potential is maximum
(c) electric field is maximum
(d) potential is zero

**18.** The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm is incident on it. The band gap in (eV) for the semiconductor is
(a) 2.5 eV (b) 1.1 eV
(c) 0.7 eV (d) 0.5 eV

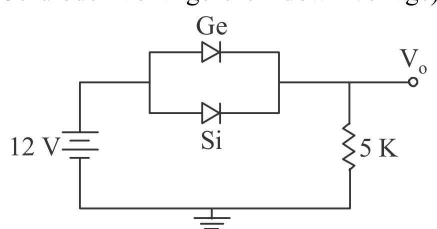
**19.** Which of the following gives a reversible operation?
(a) 
(b) 
(c) 
(d) 

**20.** When a diode is forward biased, it has a voltage drop of 0.5 V. The safe limit of current through the diode is 10 mA. If a battery of emf 1.5 V is used in the circuit, the value of minimum resistance to be connected in series with the diode so that the current does not exceed the safe limit is :
(a) 300 Ω (b) 50 Ω
(c) 100 Ω (d) 200 Ω

## Numeric Value Answer

**21.** Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is $10^{19}$ m $^{-3}$ and their mobility is $1.6 m^2/(V.s)$ then the resistivity (in Ω m) of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is

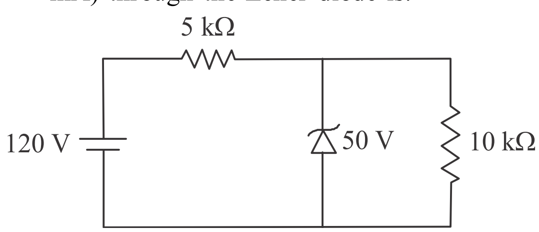
**22.** Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of $V_o$ (in volt) changes by : (assume that the Ge diode has large breakdown voltage)




**23.** Copper, a monovalent, has molar mass 63.54 g/mol and density 8.96 g/cm$^3$. What is the number density n (in m$^{-3}$) of conduction electron in copper?

**24.** An LED is constructed from a p-n junction based on a certain Ga-As -P semiconducting material whose energy gap is 1.9 eV. What is the wavelength (in nm) of the emitted light?

**25.** For the circuit shown below, the current (in mA) through the Zener diode is:

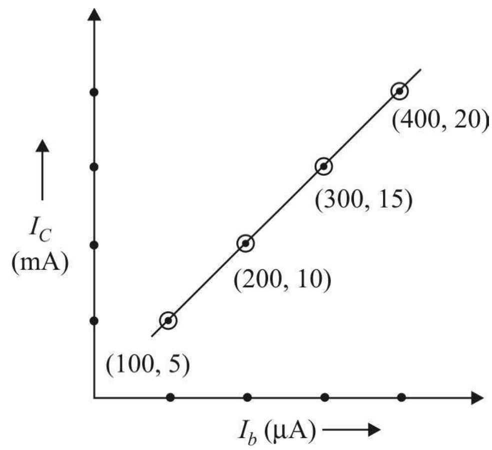



**26.** In half-wave rectification, what is the output frequency (in Hz) if the input frequency is 50 Hz?

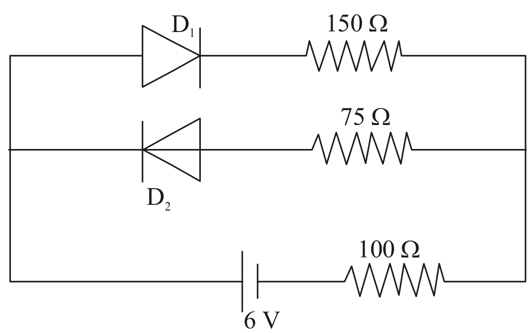
**27.** In a photodiode, the conductivity increases when the material is exposed to light. It is found that the conductivity changes only if the wavelength is less than 620 nm. What is the band gap (in eV)?

**28.** When the base current in a transistor is changed from 30 μA to 80 μA, the collector current is changed from 1.0 mA to 3.5 mA. Find the current gain β.

**29.** The transfer characteristic curve of a transistor, having input and output resistance 100 Ω and 100 k Ω respectively, is shown in the figure. The Voltage gain is




**30.** The circuit shown below contains two ideal diodes, each with a forward resistance of 50 Ω. If the battery voltage is 6V, the current through the 100 Ω resistance (in Ampere) is :




### ANSWER KEY

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (b) | **7** | (d) | **10** | (b) | **13** | (d) | **16** | (b) | **19** | (d) | **22** | (0.04) | **25** | (9) | **28** | (50) |
| **2** | (b) | **5** | (d) | **8** | (a) | **11** | (d) | **14** | (a) | **17** | (a) | **20** | (c) | **23** | $(8.49 \times 10^{26})$ | **26** | (50) | **29** | $(20 \times 10^3)$ |
| **3** | (a) | **6** | (b) | **9** | (c) | **12** | (d) | **15** | (c) | **18** | (d) | **21** | (0.04) | **24** | (650) | **27** | (2.0) | **30** | (0.02) |

# 29 COMMUNICATION SYSTEMS

## MCQs with One Correct Answer

**1.** 100% modulation in FM means
(a) actual frequency deviation > maximum allowed frequency deviation
(b) actual frequency deviation = maximum allowed frequency deviation
(c) actual frequency deviation ≥ maximum allowed frequency deviation
(d) actual frequency deviation < maximum allowed frequency deviation

**2.** If a carrier wave c(t) = A sin $\omega_c$t is amplitude modulated by a modulator signal m(t) = A sin $\omega_m$t then the equation of modulated signal [$C_m$(t)] and its modulation index are respectively
(a) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_c t$ and 2
(b) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_m t$ and 1
(c) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_c t$ and 1
(d) $C_m(t) = A(1 + \sin \omega_c t) \sin \omega_m t$ and 2

**3.** The maximum line-of-sight distance $d_M$ between two antennas having heights $h_T$ and $h_R$ above the earth is
(a) $\sqrt{R(h_T + h_R)}$ (b) $\sqrt{2R(h_T + h_R)}$
(c) $\sqrt{Rh_T} + \sqrt{2Rh_R}$ (d) $\sqrt{2Rh_T} + \sqrt{2Rh_R}$

**4.** Given the electric field of a complete amplitude modulated wave as

$$\vec{E} = \hat{i}E_c\left(1 + \frac{E_m}{E_c}\cos \omega_m t\right)\cos \omega_c t.$$

Where the subscript c stands for the carrier wave and m for the modulating signal. The frequencies present in the modulated wave are
(a) $\omega_c$ and $\sqrt{\omega_c^2 + \omega_m^2}$
(b) $\omega_c, \omega_c + \omega_m$ and $\omega_c - \omega_m$
(c) $\omega_c$ and $\omega_m$
(d) $\omega_c$ and $\sqrt{\omega_c \omega_m}$

**5.** The fundamental radio antenna is a metal rod which has a length equal to
(a) $\lambda$ in free space at the frequency of operation
(b) $\lambda/2$ in free space at the frequency of operation
(c) $\lambda/4$ in free space at the frequency of operation
(d) $3\lambda/4$ in free space at the frequency of operation

**6.** An audio signal represented as 25 sin $2\pi$(2000 t) amplitude modulated by a carrier wave : 60 sin $2\pi$(100, 000)t. The modulation index of the modulated signal is
(a) 25% (b) 41.6%
(c) 50% (d) 75%

**7.** Intensity of electric field obtained at receiver antenna for a space wave propagation is
(a) directly proportional to the perpendicular-distance from transmitter to antenna
(b) inversely proportional to the perpendicular-distance from transmitter to antenna
(c) directly proportional to the square perpendicular-distance from transmitter to antenna
(d) inversely proportional to the square perpendicular-distance from transmitter to antenna

**8.** For sky wave propagation of a 10 MHz signal, what should be the minimum electron density in ionosphere

(a) $\sim 1.2 \times 10^{12}\ m^{-3}$ (b) $\sim 10^{6}\ m^{-3}$

(c) $\sim 10^{14}\ m^{-3}$ (d) $\sim 10^{22}\ m^{-3}$

**9.** An amplitude modulated wave is represented by the expression $v_m = 5(1 + 0.6\cos 6280t)\sin(211 \times 10^4 t)$ volts

The minimum and maximum amplitudes of the amplitude modulated wave are, respectively :

(a) $\frac{3}{2}$ V, 5 V (b) $\frac{5}{2}$ V, 8 V

(c) 5 V, 8 V (d) 3 V, 5 V

**10.** In an amplitude modulator circuit, the carrier wave is given by, $C(t) = 4\sin(20000\ \pi t)$ while modulating signal is given by, $m(t) = 2\sin(2000\ \pi t)$. The values of modulation index and lower side band frequency are :

(a) 0.5 and 10 kHz (b) 0.4 and 10 kHz

(c) 0.3 and 9 kHz (d) 0.5 and 9 kHz

**11.** Television signals on earth cannot be received at distances greater than 100 km from the transmission station. The reason behind this is that

(a) the receiver antenna is unable to detect the signal at a disance greater than 100 km

(b) the TV programme consists of both audio and video signals

(c) the TV signals are less powerful than radio signals

(d) the surface of earth is curved like a sphere

**12.** Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are

(a) 1490, 1510 (b) 1510, 1490

(c) $\frac{1}{1490}, \frac{1}{1510}$ (d) $\frac{1}{1510}, \frac{1}{1490}$

**13.** A radio station has two channels. One is AM at 1020 kHz and the other FM at 89.5 MHz. For good results you will use

(a) longer antenna for the AM channel and shorter for the FM

(b) shorter antenna for the AM channel and longer for the FM

(c) same length antenna will work for both

(d) information given is not enough to say which one to use for which

**14.** An AM- signal is given as

$x_{AM}(t) = 100\,[p(t) + 0.5g(t)]\cos\omega_c t$

in interval $0 \le t \le 1$. One set of possible values of the modulating signal and modulation index would be

(a) t, 0.5 (b) t, 1.0

(c) t, 1.5 (d) $t^2$, 2.0

**15.** A device with input x(t) and outputy(t) is characterized by: $y(t) = x^2$.

An FM signal with frequency deviation of 90 kHz and modulating signal bandwidth of 5 kHz is applid to this device. The bandwidth of the output signal is

(a) 370 kHz (b) 190 kHz

(c) 380 kHz (d) 95 kHz.

**16.** A sinusoidal carrier voltage of frequency 10 MHz and amplitude 200 volts is amplitude modulated by a sinusoidal voltage of frequency 10 kHz producing 40% modulation. Calculate the frequency of upper and lower sidebands.

(a) 10010 kHz, 9990 kHz

(b) 1010 kHz, 990 kHz

(c) 10100 Hz, 9990 Hz

(d) 1010 MHz, 990 MHz

**17.** A modulated signal $C_m(t)$ has the form $C_m(t) = 30\sin 300\pi t + 10(\cos 200\pi t - \cos 400\pi t)$. The carrier frequency $f_c$, the modulating frequency (message frequency) $f_\omega$ and the modulation indix $\mu$ are respectively given by :

(a) $f_c = 200$ Hz; $f_w = 50$ Hz; $\mu = \frac{1}{2}$

(b) $f_c = 150$ Hz; $f_w = 50$ Hz; $\mu = \frac{2}{3}$

(c) $f_c = 150$ Hz; $f_w = 30$ Hz; $\mu = \frac{1}{3}$

(d) $f_c = 200$ Hz; $f_w = 30$ Hz; $\mu = \frac{1}{2}$

**18.** Long range radio transmission is possible when the radio waves are reflected from the ionosphere. For this to happen the frequency of the radio waves must be in the range:

(a) 80 - 150 MHz (b) 8 - 25 MHz

(c) 1 - 3 MHz (d) 150 - 1500 kHz

19. An AM wave is expressed as $e = 10(1 + 0.6 \cos 2000\pi t) \cos 2 \times 10^8 \pi t$ volts, the minimum and maximum value of modulated carrier wave are respectively.
(a) 10 V and 20 V
(b) 4V and 8V
(c) 16 V and 4V
(d) 8 V and 20 V

20. When radio waves passes through ionosphere, phase difference between space current and capacitive displacement current is
(a) 0 rad
(b) $(3\pi/2)$ rad
(c) $(\pi/2)$ rad
(d) $\pi$ rad

## Numeric Value Answer

21. The electron density of a layer of ionosphere at a height 150 km from the earth's surface is $9 \times 10^9$ per $m^3$. For the sky transmission from this layer up to a range of 250 km, the critical frequency (in MHz) of the layer is

22. An amplitude modulated voltage is expressed as $e = 10(1 + 0.8 \cos 2000\pi t) \cos 3 \times 10^6 \pi t$ volt
The peak value (in V) of carrier wave is

23. The rms value of a carrier voltage is 100 volts. Compute its rms value (in V) when it has been amplitude modulated by a sinusoidal audio voltage to a depth of 30%.

24. The electron density of a layer of ionosphere at a height 150 km from the earth's surface is $9 \times 10^9$ per $m^3$. For the sky transmission from this layer up to a range of 250 km, the critical frequency (in MHz) of the layer is

25. Consider the following amplitude modulated (AM) signal , where $f_m < B$
$x_{AM}(t) = 10(1 + 0.5 \sin 2\pi f_m t) \cos 2\pi f_c t$
The average side-band power for the AM signal given above is

26. An audio signal consists of two distinct sounds: one a human speech signal in the frequency band of 200 Hz to 2700 Hz, while the other is a high frequency music signal in the frequency band of 10200 Hz to 15200 Hz. The ratio of the AM signal bandwidth required to send both the signals together to the AM signal bandwidth requried to send just the human speech is :

27. For an AM wave, the maximum voltage was found to be 10 V and minimum voltage was 4 V. The modulation index of the wave is

28. A broadcast radio transmitter radiates 12 kW when percentage of modulation is 50%, then the unmodulated carrier power (in kW) is

29. The area (in $km^2$) of the region covered by the TV broadcast by a TV tower of 100 m height is (Radius of the earth $= 6.4 \times 10^6$ m)

30. 12 signals each band limited to 5 kHz are to be transmitted by frequency-division multiplexer. If AM-SSB modulation guard band of 1 kHz is used then the bandwidth (in kHz) of multiplexed signal is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (b) | **7** | (d) | **10** | (d) | **13** | (b) | **16** | (a) | **19** | (c) | **22** | (10) | **25** | (6.25) | **28** | (9.6) |
| **2** | (c) | **5** | (c) | **8** | (a) | **11** | (d) | **14** | (a) | **17** | (b) | **20** | (a) | **23** | (104.5) | **26** | (6) | **29** | $(1.28\pi \times 10^3)$ |
| **3** | (d) | **6** | (b) | **9** | (b) | **12** | (a) | **15** | (c) | **18** | (b) | **21** | (2.7) | **24** | (2.7) | **27** | (0.43) | **30** | (71) |

# CHAPTER 1

# Physical World, Units and Measurements

1. **(c)** $[x] = [bt^2]$. Hence $[b] = [x/t^2] = km/s^2$.
2. **(b)** According to question

$$E_y \propto J_x B_Z$$

$\therefore$ Constant of proportionality

$$K = \frac{E_y}{B_Z J_x} = \frac{C}{J_x} = \frac{m^3}{As}$$

[As $\frac{E}{B} = C$ (speed of light) and $J = \frac{I}{\text{Area}}$ ]

3. **(a)** The mean value of refractive index,

$$\mu = \frac{1.34 + 1.38 + 1.32 + 1.36}{4} = 1.35$$

and

$$\Delta\mu = \frac{|(1.35-1.34)| + |(1.35-1.38)| + |(1.35-1.32)| + |(1.35-1.36)|}{4}$$

$$= 0.02$$

Thus $\frac{\Delta\mu}{\mu} \times 100 = \frac{0.02}{1.35} \times 100 = 1.48$

4. **(b)** Here, b and $x^2 = L^2$ have same dimensions

Also, $a = \frac{x^2}{E \times t} = \frac{L^2}{(M L^2 T^{-2})T} = M^{-1}T^1$

$a \times b = [M^{-1} L^2 T^1]$

5. **(a)**
6. **(b)** $F = ma = \rho$ volume '$a$'

To write volume in terms of '$a$' and '$f$'

Volume $= L^3 = \left(\frac{L}{T^2}\right)^3 T^6 = a^3 f^{-6}$

$\therefore\ F = \rho\, a^4 f^{-6}$

7. **(b)**
8. **(d)** Reynold's number

= Coefficient of friction $= [M^0 L^0 T^0]$

Curie is the unit of radioactivity (number of atoms decaying per second) and frequency also has the unit per second.

Latent heat $= \frac{Q}{m}$ and Gravitation potential $= \frac{W}{m}$.

9. **(c)** $\frac{A}{B} = \frac{\text{Force}}{\text{Force}} = [M^0 L^0 T^0]$

$Ct = \text{angle} \Rightarrow C = \frac{\text{angle}}{\text{time}} = \frac{1}{T} = T^{-1}$

$Dx = \text{angle} \Rightarrow D = \frac{\text{angle}}{\text{distance}} = \frac{1}{L} = L^{-1}$

$\therefore\ \frac{C}{D} = \frac{T^{-1}}{L^{-1}} = [M^0 L T^{-1}]$

10. **(a)** Number of significant figures in 23.023 = 5

Number of significant figures in 0.0003 = 1

Number of significant figures in $2.1 \times 10^{-3} = 2$

So, the radiation belongs to X-rays part of the spectrum.

11. **(d)** No. of divisions on main scale = $N$

No. of divisions on vernier scale = $N + 1$

size of main scale division = $a$

Let size of vernier scale division be $b$

then we have

$aN = b\,(N+1) \Rightarrow b = \frac{aN}{N+1}$

Least count is $a - b = a - \frac{aN}{N+1}$

$= a\left[\frac{N+1-N}{N+1}\right] = \frac{a}{N+1}$

12. **(a)** Surface tension, $T = \frac{F}{\ell} = \frac{F}{\ell}.\frac{\ell}{\ell}.\frac{T^2}{T^2}$

(As, $F.\ell = K$ (energy); $\frac{T^2}{\ell^2} = V^{-2}$)

Therefore, surface tension $= [KV^{-2}T^{-2}]$

13. **(a)** $X = 5YZ^2$

$$\Rightarrow Y \propto \frac{X}{Z^2} \qquad ...(i)$$

$X$ = Capacitance $= \frac{Q}{V} = \frac{Q^2}{W} = \frac{[A^2T^2]}{[ML^2T^{-2}]}$

$X = [M^{-1}L^{-2}T^4A^2]$

$Z = B = \frac{F}{IL}$ $\quad [\because F = ILB]$

$Z = [MT^{-2}A^{-1}]$

$$Y = \frac{[M^{-1}L^{-2}T^4A^2]}{[MT^{-2}A^{-1}]^2}$$

$Y = [M^{-3}L^{-2}T^8A^4]$ (Using (i))

**14.** **(c)** Relative error in Surface area,

$\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} = \alpha$ and relative error in volume,

$\frac{\Delta v}{v} = 3 \times \frac{\Delta r}{r}$

$\therefore$ Relative error in volume w.r.t. relative error in area,

$\frac{\Delta v}{v} = \frac{3}{2}\alpha$

**15.** **(d)** 30 Divisions of V.S. coincide with 29 divisions of M.S.

$\therefore$ 1 V.S.D $= \frac{29}{30}$ MSD

L.C. = 1 MSD – 1VSD

$= 1 \text{ MSD} - \frac{29}{30} \text{ MSD}$

$= \frac{1}{30}$ MSD

$= \frac{1}{30} \times 0.5° = 1$ minute.

**16.** **(d)** $\left[\sqrt{\frac{\varepsilon_0}{\mu_0}}\right] = \sqrt{\frac{\varepsilon_0^2}{\mu_0\varepsilon_0}} = \left[\frac{\varepsilon_0}{\sqrt{\mu_0\varepsilon_0}}\right]$

$$= \varepsilon_0 C[LT^{-1}] \times [\varepsilon_0] \quad \left[\because \frac{1}{\sqrt{\mu_0\varepsilon_0}} = C\right]$$

$\because F = \frac{q^2}{4\pi\varepsilon_0 r^2}$

$$\Rightarrow [\varepsilon_0] = \frac{[AT]^2}{[MLT^{-2}] \times [L^2]} = [A^2M^{-1}L^{-3}T^4]$$

$$\therefore \left[\sqrt{\frac{\varepsilon_0}{\mu_0}}\right] = [LT^{-1}] \times [A^2M^{-1}L^{-3}T^4]$$

$= [M^{-1}L^{-2}T^3A^2]$

**17.** **(b)** The dimensional formulae of

$e = \left[M^0L^0T^1A^1\right]$

$\varepsilon_0 = \left[M^{-1}L^3T^4A^2\right]$

$G = \left[M^{-1}L^3T^{-2}\right]$ and $m_e = \left[M^1L^0T^0\right]$

Now, $\frac{e^2}{2\pi\varepsilon_0 Gm_e^2}$

$$= \frac{\left[M^0L^0T^1A^1\right]^2}{2\pi\left[M^{-1}L^{-3}T^4A^2\right]\left[M^{-1}L^3T^{-2}\right]\left[M^1L^0T^0\right]^2}$$

$$= \frac{\left[T^2A^2\right]}{2\pi\left[M^{-1-1+2}L^{-3+3}T^{4-2}A^2\right]}$$

$$= \frac{\left[T^2A^2\right]}{2\pi\left[M^0L^0T^2A^2\right]} = \frac{1}{2\pi}$$

$\because \frac{1}{2\pi}$ is dimensionless thus the combination $\frac{e^2}{2\pi\varepsilon_0 Gm_e^2}$ would have the same value in different systems of units.

**18.** **(d)** Average diameter, $d_{av} = 5.5375$ mm

Deviation of data, $\Delta d = 0.07395$ mm

As the measured data are upto two digits after decimal, therefore answer should be in two digits after decimal.

$\therefore d = (5.54 \pm 0.07)$ mm

**19.** **(c)** Dimension of Force $F = M^1L^1T^{-2}$

Dimension of velocity $V = L^1T^{-1}$

Dimension of work $= M^1L^2T^{-2}$

Dimension of length = L

Moment of inertia $= ML^2$

$\therefore x = \frac{IFv^2}{WL^4}$

$$= \frac{(M^1L^2)(M^1L^1T^{-2})(L^1T^{-2})^2}{(M^1L^2T^{-2})(L^4)}$$

$$= \frac{M^1L^{-2}T^{-2}}{L^3} = M^1L^{-1}T^{-2} = \text{Energy density}$$

**20.** **(c)** Take $T \propto r^a M^b G^c$ and solving we get $a = \frac{3}{2}$.

**21.** **(0.2)** The current voltage relation of diode is

$I = (e^{1000\,V/T} - 1)$ mA (given)

When, $I = 5mA, e^{1000\ V/T} = 6mA$

Also, $dI = (e^{1000\ V/T}) \times \frac{1000}{T}$

Error $= \pm\, 0.01$ (By exponential function)

$= (6\,mA) \times \frac{1000}{300} \times (0.01) = 0.2\,\text{mA}$

**22.** **(32)** Given, $P = a^{1/2}\, b^2\, c^2\, d^{-4}$,

Maximum relative error,

$$\frac{\Delta P}{P} = \frac{1}{2}\frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c} + 4\frac{\Delta d}{d}$$

$$= \frac{1}{2} \times 2 + 2 \times 1 + 3 \times 3 + 4 \times 5 = 32\%$$

**23.** **(40)** Density of material in SI unit,

$= \frac{128\text{kg}}{\text{m}^3}$

Density of material in new system

$$= \frac{128(50\,\text{g})(20)}{(25\,\text{cm})^3\,(4)^3} = \frac{128}{64}(20) = 40\,\text{units}$$

**24.** **(0.001)** When screw on a screw-gauge is given six rotations, it moves by 3mm on the main scale

$\therefore$ Pitch $= \frac{3}{6} = 0.5\,\text{mm}$

$\therefore$ Least count L.C. $= \frac{\text{Pitch}}{CSD} = \frac{0.5\,\text{mm}}{50}$

$= \frac{1}{100}\,\text{mm} = 0.01\,mm = 0.001\text{cm}$

**25.** **(6)** According to ohm's law, V = IR

$R = \frac{V}{I}$

$\therefore$ Percentage error $= \frac{\text{Absolute error}}{\text{Measurement}} \times 10^2$

where, $\frac{\Delta V}{V} \times 100 = \frac{\Delta I}{I} \times 100 = 3\%$

then, $\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 10^2 + \frac{\Delta I}{I} \times 10^2$

$= 3\% + 3\% = 6\%$

**26.** **(4.6)** We have

$3.8 \times 10^{-6} + 4.2 \times 10^{-5}$

$= 0.38 \times 10^{-5} + 4.2 \times 10^{-5} = 4.58 \times 10^{-5}$

$= 4.6 \times 10^{-5}$ (upto 2 significant figures)

**27.** **(–1)** $\frac{M}{At} \propto P^x v^y \Rightarrow M\,L^{-2}T^{-1}$

$$= \left[M\,L^{-1}T^{-2}\right]^x \left[L^1 T^{-1}\right]^y$$

$$= M^x L^{-x+y} T^{-2x-y}$$

$x = 1, -x + y = -2$ and $-2x - y = -1$

From here, we get y = –1. Thus, $x \div y = -1$

**28.** **(20)** Required percentage

$$= 2 \times \frac{0.02}{0.24} \times 100 + \frac{1}{30} 100 + \frac{0.01}{4.80} \times 100$$

$$= 16.7 + 3.3 + 0.2 = 20\%$$

**29.** **(0.1)** $Y = \frac{F}{A}.\frac{L}{\Delta L} = \frac{\text{dyne}}{\text{cm}^2} = \frac{10^{-5}\,\text{N}}{10^{-4}\,\text{m}^2}$

$= 0.1\,\text{N}/\text{m}^2$

**30.** **(3)** As, $g = 4\pi^2 \frac{L}{T^2}$

So, $\frac{\Delta g}{g} \times 100 = \frac{\Delta L}{L} \times 100 + 2\frac{\Delta T}{T} \times 100$

$= \frac{0.1}{20} \times 100 + 2 \times \frac{1}{90} \times 100 = 2.72 \simeq 3\%$

CHAPTER 2

# Motion in a Straight Line

1. **(d)** Given $x = ae^{-\alpha t} + be^{\beta t}$

Velocity, $v = \frac{dx}{dt} = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$

$$= -\frac{a\alpha}{e^{\alpha t}} + b\beta e^{\beta t}$$

i.e., go on increasing with time.

2. **(d)** In (a), at the same time particle has two positions which is not possible. In (b), particle has two velocities at the same time. In (c), speed is negative which is not possible.

3. **(c)** We have, $S_n = u + \frac{a}{2}(2n-1)$

or $65 = u + \frac{a}{2}(2\times 5 - 1)$

or $65 = u + \frac{9}{2}a$ ..... (1)

Also, $105 = u + \frac{a}{2}(2\times 9 - 1)$

or $105 = u + \frac{17}{2}a$ ..... (2)

Equation (2) – (1) gives,

$$40 = \frac{17}{2}a - \frac{9}{2}a = 4a \text{ or } a = 10 \text{ m/s}^2.$$

Substitute this value in (1) we get,

$$u = 65 - \frac{9}{2}\times 10 = 65 - 45 = 20 \text{ m/s}$$

$\therefore$ The distance travelled by the body in 20 s is,

$$s = ut + \frac{1}{2}at^2 = 20\times 20 + \frac{1}{2}\times 10\times (20)^2$$
$$= 400 + 2000 = 2400 \text{ m}.$$

4. **(c)** Let $v_A$ and $v_B$ are the velocities of two bodies.

In first case, $v_A + v_B = 6$m/s .....(1)

In second case, $v_A - v_B = 4$m/s .....(2)

From (1) & (2) we get, $v_A = 5$ m/s and $v_B = 1$ m/s.

5. **(b)** Distance in last two second

$$= \frac{1}{2}\times 10\times 2 = 10 \text{ m}.$$

Total distance $= \frac{1}{2}\times 10\times (6+2) = 40$ m.

6. **(b)** Average velocity

$$= \frac{v_1 + v_2 + v_3}{3} = \frac{3+4+5}{3} = 4 \text{ m/s}$$

7. **(b)** Distance along a line i.e., displacement ($s$) $= t^3$ ($\because s \propto t^3$ given)

By double differentiation of displacement, we get acceleration.

$$V = \frac{ds}{dt} = \frac{dt^3}{dt} = 3t^2 \text{ and } a = \frac{dv}{dt} = \frac{d3t^2}{dt} = 6t$$

$a = 6t$ or $a \propto t$

8. **(d)** Let '$S$' be the distance between two ends '$a$' be the constant acceleration

As we know $v^2 - u^2 = 2aS$

or, $aS = \frac{v^2 - u^2}{2}$

Let $v$ be velocity at mid point.

Therefore, $v_c^2 - u^2 = 2a\frac{S}{2}$

$$v_c^2 = u^2 + \frac{v^2 - u^2}{2} \Rightarrow v_c = \sqrt{\frac{u^2 + v^2}{2}}$$

9. **(b)** Time taken by same ball to return to the hands of juggler $= \frac{2u}{g} = \frac{2\times 20}{10} = 4$ s. So he is throwing the balls after each 1 s. Let at some instant he is throwing ball number 4. Before 1 s of it he throws ball. So height of ball 3 :

$$h_3 = 20\times 1 - \frac{1}{2}10(1)^2 = 15 \text{ m}$$

Before 2s, he throws ball 2. So height of ball 2 :

$$h_2 = 20\times 2 - \frac{1}{2}10(2)^2 = 20 \text{ m}$$

Before 3 s, he throws ball 1. So height of ball 1 :

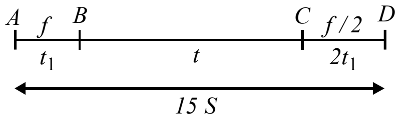
$$h_1 = 20\times 3 - \frac{1}{2}10(3)^2 = 15 \text{ m}$$

10. **(d)** Distance from $A$ to $B = S = \frac{1}{2}ft_1^2$

Distance from $B$ to $C = (ft_1)t$

Distance from $C$ to $D = \frac{u^2}{2a} = \frac{(ft_1)^2}{2(f/2)}$

$$= ft_1^2 = 2S$$




$\Rightarrow \quad S + f\,t_1 t + 2S = 15\,S$

$\Rightarrow \quad f\,t_1 t = 12\,S$ ............. (i)

$\frac{1}{2} f\,t_1^2 = S$ ............ (ii)

Dividing (i) by (ii), we get $t_1 = \frac{t}{6}$

$\Rightarrow \quad S = \frac{1}{2} f\left(\frac{t}{6}\right)^2 = \frac{f\,t^2}{72}$

**11.** **(a)** $v = \alpha\sqrt{x}$,

$\Rightarrow \quad \frac{dx}{dt} = \alpha\sqrt{x} \;\Rightarrow\; \frac{dx}{\sqrt{x}} = \alpha\,dt$

Integrating both sides,

$\int_0^x \frac{dx}{\sqrt{x}} = \alpha\int_0^t dt \;;\; \left[\frac{2\sqrt{x}}{1}\right]_0^x = \alpha[t]_0^t$

$\Rightarrow 2\sqrt{x} = \alpha t \;\Rightarrow x = \frac{\alpha^2}{4}t^2$

**12.** **(c)** Person's speed walking only is

$\frac{1}{60}\frac{\text{"escalator"}}{\text{second}}$

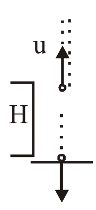
Standing the escalator without walking the speed is $\frac{1}{40}\frac{\text{"escalator"}}{\text{second}}$

Walking with the escalator going, the speed add.

So, the person's speed is $\frac{1}{60} + \frac{1}{40} = \frac{15}{120}$ $\frac{\text{"escalator"}}{\text{second}}$

So, the time to go up the escalator $t = \frac{120}{5} = 24$ second.

**13.** **(c)** Speed on reaching ground




$v = \sqrt{u^2 + 2gh}$

Now, $v = u + at$

$\Rightarrow \quad \sqrt{u^2 + 2gh} = -u + gt$

Time taken to reach highest point is $t = \frac{u}{g}$,

$\Rightarrow t = \frac{u + \sqrt{u^2 + 2gH}}{g} = \frac{nu}{g}$ (from question)

$\Rightarrow \quad 2gH = n(n-2)u^2$

**14.** **(a)** In first case

$u_1 = u \;;\; v_1 = \frac{u}{2}\,,\; s_1 = 3$ cm, $a_1 = ?$

Using, $v_1^2 - u_1^2 = 2a_1 s_1$ ...(i)

$\left(\frac{u}{2}\right)^2 - u^2 = 2 \times a \times 3$

$\Rightarrow \quad a = \frac{-u^2}{8}$

**In second case:** Assuming the same retardation

$u_2 = u/2 \;;\; v_2 = 0 \;;\; s_2 = ?;\; a_2 = \frac{-u^2}{8}$

$v_2^2 - u_2^2 = 2a_2 \times s_2$ ...(ii)

$\therefore \; 0 - \frac{u^2}{4} = 2\left(\frac{-u^2}{8}\right) \times s_2$

$\Rightarrow \; s_2 = 1$ cm

**15.** **(b)** For downward motion $v = -gt$

The velocity of the rubber ball increases in downward direction and we get a straight line between $v$ and $t$ with a negative slope.

Also applying $y - y_0 = ut + \frac{1}{2}at^2$

We get $y - h = -\frac{1}{2}gt^2 \Rightarrow y = h - \frac{1}{2}gt^2$

The graph between $y$ and $t$ is a parabola with $y = h$ at $t = 0$. As time increases $y$ decreases.

**For upward motion.**

The ball suffer elastic collision with the horizontal elastic plate therefore the direction of velocity is reversed and the magnitude remains the same.

Here $v = u - gt$ where $u$ is the velocity just after collision.

As $t$ increases, $v$ decreases. We get a straight line between $v$ and $t$ with negative slope.

Also $y = ut - \frac{1}{2}gt^2$

All these characteristics are represented by graph (b).

**16.** **(b)** $x = at + bt^2 - ct^3$

Velocity, $v = \frac{dx}{dt} = \frac{d}{dt}(at + bt^2 + ct^3)$

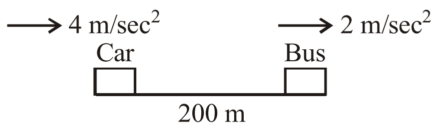
$= a + 2bt - 3ct^2$

Acceleration, $\frac{dv}{dt} = \frac{d}{dt}(a + 2bt - 3ct^2)$

or $0 = 2b - 3c \times 2t \quad \therefore\ t = \left(\frac{b}{3c}\right)$

and $v = a + 2b\left(\frac{b}{3c}\right) - 3c\left(\frac{b}{3c}\right)^2 = \left(a + \frac{b^2}{3c}\right)$

**17.** **(c)**




Given, $u_C = u_B = 0$, $a_C = 4\ m/s^2$, $a_B = 2\ m/s^2$
hence relative acceleration, $a_{CB} = 2\ m/sec^2$

Now, we know, $s = ut + \frac{1}{2}at^2$

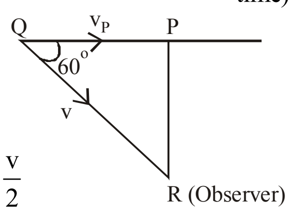
$200 = \frac{1}{2} \times 2t^2 \quad \because u = 0$

Hence, the car will catch up with the bus after time

$t = 10\sqrt{2}$ second

**18.** **(d)** Distance, PQ = $v_p \times t$ (Distance = speed × time)

Distance, QR = V.t



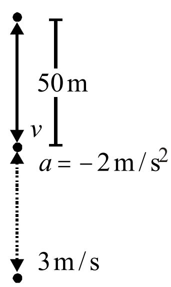

$\cos 60° = \frac{PQ}{QR}$

$\frac{1}{2} = \frac{v_p \times t}{V.t} \Rightarrow v_p = \frac{v}{2}$

**19.** **(a)**

**20.** **(b)** The slope of v-t graph is constant and velocity decreasing for first half. It is positive and constant over next half.

**21.** **(293)** Initial velocity of parachute after bailing out,




$u = \sqrt{2gh}$

$u = \sqrt{2 \times 9.8 \times 50} = 14\sqrt{5}$

The velocity at ground,

$v = 3m/s$

$S = \frac{v^2 - u^2}{2 \times 2} = \frac{3^2 - 980}{4}$

$\approx 243$ m

Initially he has fallen 50 m.

$\therefore$ Total height from where he bailed out = 243 + 50 = 293 m

**22.** **(80)** In first case speed,

$u = 60 \times \frac{5}{18}\ m/s = \frac{50}{3}\ m/s$

$d = 20m$,

Let retardation be a then

$(0)^2 - u^2 = -2ad$

or $\quad u^2 = 2ad \qquad$ ...(i)

In second case speed, $u' = 120 \times \frac{5}{18}$

$= \frac{100}{3}\ m/s$

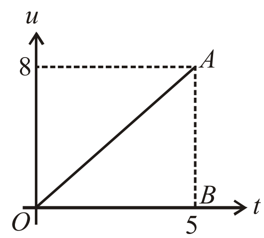
and $(0)^2 - u'^2 = -2ad'$

or $\quad u'^2 = 2ad' \qquad$ ...(ii)

(ii) divided by (i) gives,

$4 = \frac{d'}{d} \Rightarrow d' = 4 \times 20 = 80m$

**23.** **(20)**




Distance travelled = Area of speed-time graph

$= \frac{1}{2} \times 5 \times 8 = 20$ m

**24.** **(3)** Distance X varies with time t as

$x^2 = at^2 + 2bt + c$

$\Rightarrow 2x\frac{dx}{dt} = 2at + 2b$

$\Rightarrow x\frac{dx}{dt} = at + b \Rightarrow \frac{dx}{dt} = \frac{(at+b)}{x}$

$\Rightarrow x\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 = a$

$\Rightarrow \frac{d^2x}{dt^2} = \frac{a - \left(\frac{dx}{dt}\right)^2}{x} = \frac{a - \left(\frac{at+b}{x}\right)^2}{x}$

$= \frac{ax^2 - (at+b)^2}{x^3} = \frac{ac - b^2}{x^3}$

$\Rightarrow \quad a \propto x^{-3} \qquad$ Hence, n = 3

**25.** **(08.00)** Let the ball takes time t to reach the ground

Using, $S = ut + \frac{1}{2}gt^2$

$\Rightarrow S = 0 \times t + \frac{1}{2}gt^2$

$\Rightarrow \quad 200 = gt^2 \qquad [\because 2S = 100m]$

$\Rightarrow t = \sqrt{\frac{200}{g}} \qquad \ldots(i)$

In last $\frac{1}{2}s$, body travels a distance of 19 $m$, so in

$\left(t - \frac{1}{2}\right)$ distance travelled = 81

Now, $\frac{1}{2}g\left(t - \frac{1}{2}\right)^2 = 81$

$\therefore g\left(t - \frac{1}{2}\right)^2 = 81 \times 2$

$\Rightarrow \left(t - \frac{1}{2}\right) = \sqrt{\frac{81 \times 2}{g}}$

$\therefore \frac{1}{2} = \frac{1}{\sqrt{g}}(\sqrt{200} - \sqrt{81 \times 2})$ using (i)

$\Rightarrow \sqrt{g} = 2(10\sqrt{2} - 9\sqrt{2})$

$\Rightarrow \sqrt{g} = 2\sqrt{2}$

$\therefore$ g = 8 $m/s^2$

**26.** **(2)** Given, $\frac{dv}{dt} = -2.5\sqrt{v}$

$\Rightarrow \frac{dv}{\sqrt{v}} = -2.5\, dt$

Integrating,

$\int_{6.25}^{0} v^{-1/2} dv = -2.5\int_0^t dt$

$\Rightarrow \left[\frac{v^{+1/2}}{(1/2)}\right]_{6.25}^{0} = -2.5[t]_0^t$

$\Rightarrow -2(6.25)^{1/2} = -2.5t$

$\Rightarrow -2 \times 2.5 = -2.5t$

$\Rightarrow t = 2$ s

**27.** **(5)** For rat $s = \frac{1}{2}\beta t^2 \qquad \ldots (i)$

for cat $s = d = ut + \frac{1}{2}\alpha t^2 \qquad \ldots (ii)$

Solving (i) and (ii) fot $t$ to be real

$\beta = \alpha + \frac{u^2}{2d}$,

$\Rightarrow \beta = 2.5 + \frac{5^2}{2 \times 5} = 5\ ms^{-2}$

**28.** **(50)** The distance travel in $n^{th}$ second is

$S_n = u + ½\,(2n-1)a \qquad \ldots(1)$

so distance travel in $t^{th}$ & $(t+1)^{th}$ second are

$S_t = u + ½\,(2t-1)a \qquad \ldots(2)$

$S_{t+1} = u + ½\,(2t+1)a \qquad \ldots(3)$

As per question,

$S_t + S_{t+1} = 100 = 2(u + at) \qquad \ldots(4)$

Now from first equation of motion the velocity, of particle after time t, if it moves with an accleration a is

$v = u + a\,t \qquad \ldots(5)$

where u is initial velocity

So from eq(4) and (5), we get v = 50cm./sec.

**29.** **(49)** $S_n$ = Distance covered in $n^{th}$ sec

$\Rightarrow \quad S_n = u + \frac{a}{2}(2n - 1)$

Putting a = – g and n = 5, we get

$\Rightarrow S_5 = u - \frac{g}{2} \times 9 = u - \frac{9g}{2} \qquad \ldots(1)$

Distance covered in two continuous seconds can only be equal when the body reaches the highest point after the fifth second and comes down in the sixth second for which u = 0 and n = 1.

$\Rightarrow S_6 = 0 + \frac{g}{2}(2 \times 1 - 1) = \frac{g}{2} \qquad \ldots(2)$

Equating (1) and (2)

or, $u - \frac{9g}{2} = \frac{g}{2} \Rightarrow u = \frac{10g}{2} = 49\,m/s$

**30.** **(10)** The only force acting on the ball is the force of gravity. The ball will ascend until gravity reduces its velocity to zero and then it will descend. Find the time it takes for the ball to reach its maximum height and then double the time to cover the round trip.

Using $v_{at\ maximum\ height} = v_0 + at = v_0 - gt$, we get:

0 m/s = 50 m/s – (9.8 m/s$^2$) t

Therefore,

t = (50 m/s)/(9.8 m/s$^2$) ~ (50 m/s)/ (10 m/s$^2$) ~ 5s

This is the time it takes the ball to reach its maximum height. The total round trip time is 2t ~ 10s.

# Chapter 3
# Motion in a Plane

1. **(d)** $|B| = \sqrt{7^2 + (24)^2} = \sqrt{625} = 25$

Unit vector in the direction of A will be $\hat{A} = \frac{3\hat{i} + 4\hat{j}}{5}$

So, required vector

$= 25\left(\frac{3\hat{i} + 4\hat{j}}{5}\right) = 15\hat{i} + 20\hat{j}$

2. **(d)** $\frac{H_1}{H_2} = \frac{u^2 \sin^2\theta / 2g}{u^2 \sin^2(90^\circ - \theta)/2g} = \tan^2\theta$

3. **(a)** $H_1 = \frac{u^2 \sin^2\theta}{2g}$

and $H_2 = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2 \cos^2\theta}{2g}$

$$H_1 H_2 = \frac{u^2 \sin^2\theta}{2g} \times \frac{u^2\cos^2\theta}{2g} = \frac{(u^2 \sin 2\theta)^2}{16g^2} = \frac{R^2}{16}$$

$\therefore R = 4\sqrt{H_1 H_2}$

4. **(b)** Comparing the given equation with

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}, \text{ we get}$$

$\tan\theta = \sqrt{3}$ or $\theta = 60^\circ$.

5. **(b)** Circumference of circle is $2\pi r = 40$ m

Total distance travelled in two revolution is 80m.
Initial velocity u = 0, final veloctiy v = 80 m/sec

so from $v^2 = u^2 + 2as$

$\Rightarrow (80)^2 = 0^2 + 2 \times 80 \times a \Rightarrow a = 40 \text{ m/sec}^2$

6. **(d)** $s = t^3 + 5$

$\Rightarrow$ velocity, $v = \frac{ds}{dt} = 3t^2$

Tangential acceleration $a_t = \frac{dv}{dt} = 6t$

Radial acceleration $a_c = \frac{v^2}{R} = \frac{9t^4}{R}$

At $t = 2s$, $a_t = 6 \times 2 = 12 \text{ m/s}^2$

$$a_c = \frac{9\times 16}{20} = 7.2 \text{ m/s}^2$$

$\therefore$ Resultant acceleration

$= \sqrt{a_t^2 + a_c^2} = \sqrt{(12)^2 + (7.2)^2} = \sqrt{144 + 51.84}$

$= \sqrt{195.84} = 14 \text{ m/s}^2$

7. **(d)** $|\vec{A} + \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + 2\vec{A}.\vec{B}$

$= A^2 + B^2 + 2AB\cos\theta$

$|\vec{A} - \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2\vec{A}.\vec{B}$

$= A^2 + B^2 - 2AB\cos\theta$

So, $A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta$

$4AB\cos\theta = 0 \Rightarrow \cos\theta = 0 \quad \therefore \quad \theta = 90^\circ$

So, angle between A & B is 90º.

8. **(c)**

9. **(a)** The angle for which the ranges are same is complementary.

Let one angle be θ, then other is 90° – θ

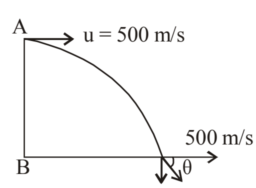
$$T_1 = \frac{2u\sin\theta}{g}, T_2 = \frac{2u\cos\theta}{g}$$

$$T_1 T_2 = \frac{4u^2 \sin\theta\cos\theta}{g} = 2R \quad \left(\because R = \frac{u^2\sin^2\theta}{g}\right)$$

Hence it is proportional to $R$.

10. **(a)** Horizontal component of velocity $v_x = 500$ m/s and vertical component of velocity while striking the ground.

$u_v = 0 + 10 \times 10 = 100$ m/s



$\therefore$ Angle with which it strikes the ground

$$\theta = \tan^{-1}\left(\frac{u_v}{u_x}\right) = \tan^{-1}\left(\frac{100}{500}\right) = \tan^{-1}\left(\frac{1}{5}\right)$$

**11.** **(a)** Given : $\vec{u} = 5\hat{j}$ m/s

Acceleration, $\vec{a} = 10\hat{i} + 4\hat{j}$ and final coordinate $(20, y_0)$ in time $t$.

$$S_x = u_x t + \frac{1}{2}a_x t^2 \qquad [\because u_x = 0]$$

$$\Rightarrow 20 = 0 + \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 2 \text{ s}$$

$$S_y = u_y \times t + \frac{1}{2}a_y t^2$$

$$y_0 = 5 \times 2 + \frac{1}{2} \times 4 \times 2^2 = 18 \text{ m}$$

**12.** **(d)** $\vec{r} = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = 30t\hat{i} - 40t\hat{j}$$

Acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = 30\hat{i} - 40\hat{j}$

$$\therefore a = \sqrt{30^2 + 40^2} = 50 \text{ m/s}^2$$

**13.** **(a)** Let magnitude of two vectors $\vec{A}$ and $\vec{B} = a$

$|\vec{A} + \vec{B}| = \sqrt{a^2 + a^2 + 2a^2\cos\theta}$ and

$$|\vec{A} - \vec{B}| = \sqrt{a^2 + a^2 - 2a^2\left[\cos(180° - \theta)\right]}$$

$$= \sqrt{a^2 + a^2 - 2a^2\cos\theta}$$

and accroding to question,

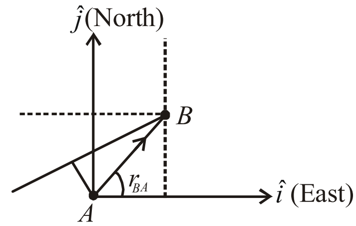
$$|\vec{A} + \vec{B}| = n|\vec{A} - \vec{B}|$$

or, $\dfrac{a^2 + a^2 + 2a^2\cos\theta}{a^2 + a^2 - 2a^2\cos\theta} = n^2$

$$\Rightarrow \frac{\cancel{a^2}(1 + 1 + 2\cos\theta)}{\cancel{a^2}(1 + 1 - 2\cos\theta)} n^2 \Rightarrow \frac{(1 + \cos\theta)}{(1 - \cos\theta)} = n^2$$

using componendo and dividendo theorem, we get $\theta = \cos^{-1}\left(\dfrac{n^2 - 1}{n^2 + 1}\right)$

**14.** **(b)**




$$\vec{v}_A = 30\hat{i} + 50\hat{j} \text{ km/hr}$$

$$\vec{v}_B = (-10\hat{i}) \text{ km/hr}$$

$$r_{BA} = (80\hat{i} + 150\hat{j}) \text{ km}$$

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = -10\hat{i} - 30\hat{i} - 50\hat{i} = 40\hat{i} - 50\hat{j}$$

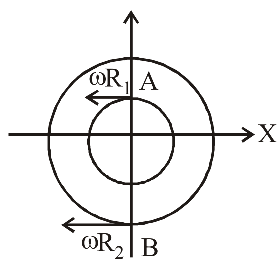
$$t_{\text{minimum}} = \frac{|(\vec{r}_{BA})\cdot(\vec{v}_{BA})|}{|(\vec{v}_{BA})|^2}$$

$$= \frac{|(80\hat{i} + 150\hat{j})(-40\hat{i} - 50\hat{j})|}{(10\sqrt{41})^2}$$

$$\therefore t = \frac{10700}{10\sqrt{41} \times 10\sqrt{41}} = \frac{107}{41} = 2.6 \text{ hrs.}$$

**15.** **(c)** From, $\theta = \omega t = \omega\dfrac{\pi}{2\omega} = \dfrac{\pi}{2}$

So, both have completed quater circle




Relative velocity,

$$v_A - v_B = \omega R_1(-\hat{i}) - \omega R_2(-i) = \omega(R_2 - R_1)i$$

**16.** **(d)** $v = k(yi + xj)$

$v = kyi + kxj$

$$\frac{dx}{dt} = ky, \ \frac{dy}{dt} = kx$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{kx}{ky}$$

$ydy = xdx$ ...(i)

Integrating equation (i)

$\int y dy = \int x \cdot dx$

$y^2 = x^2 + c$

**17.** **(d)** Given, Position vector,

$\vec{r} = \cos\omega t\hat{i} + \sin\omega t\,\hat{j}$

Velocity, $\vec{v} = \frac{d\vec{r}}{dt} = \omega(-\sin\omega t\hat{i} + \cos\omega t\,\hat{j})$

Acceleration,

$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2(\cos\omega t\hat{i} + \sin\omega t\,\hat{j})$

$\vec{a} = -\omega^2\vec{r}$

$\therefore$ $\vec{a}$ is antiparallel to $\vec{r}$

Also $\vec{v}.\vec{r} = 0 \quad \therefore \vec{v} \perp \vec{r}$

Thus, the particle is performing uniform circular motion.

**18.** **(c)** From question,
Horizontal velocity (initial),

$u_x = \frac{40}{2} = 20\,\text{m/s}$

Vertical velocity (initial), $50 = u_y t + \frac{1}{2}gt^2$

$\Rightarrow \quad u_y \times 2 + \frac{1}{2}(-10)\times 4$ or, $\quad 50 = 2u_y - 20$

or, $u_y = \frac{70}{2} = 35\,\text{m/s}$

$\therefore \tan\theta = \frac{u_y}{u_x} = \frac{35}{20} = \frac{7}{4} \Rightarrow \text{Angle } \theta = \tan^{-1}\frac{7}{4}$

**19.** **(c)** $x + u_2\cos\theta_2 t = u_1\cos\theta_1\, t$

$\therefore \quad t = \frac{x}{u_1\cos\theta_1 - u_2\cos\theta_2}$ ...(i)

Also $u_1\sin\theta_1 = u_2\sin\theta_2$ ...(ii)

After solving above equations, we get

$t = \frac{x\sin\theta_2}{u_1\sin(\theta_2 - \theta_1)}$.

**20.** **(c)**

**21.** **(150)** $\frac{B}{2} = \sqrt{A^2 + B^2 + 2AB\cos\theta}$ ... (i)

$\therefore \quad \tan 90° = \frac{B\sin\theta}{A + B\cos\theta} \Rightarrow A + B\cos\theta = 0$

$\therefore \quad \cos\theta = -\frac{A}{B}$

Hence, from (i) $\frac{B^2}{4} = A^2 + B^2 - 2A^2 \Rightarrow A = \sqrt{3}\frac{B}{2}$

$\Rightarrow \quad \cos\theta = -\frac{A}{B} = -\frac{\sqrt{3}}{2} \quad \therefore \theta = 150°$

**22.** **(10)** $S = u \times \sqrt{\frac{2h}{g}} \Rightarrow 10 = u\sqrt{2\times\frac{5}{10}}$

$\Rightarrow u = 10\,\text{m/s}$

**23.** **(8)** Given, $\omega = 2\text{ rad s}^{-1}$, $r = 2$ m, $t = \frac{\pi}{2}$ s

Angular displacement, $\theta = \omega t = 2\times\frac{\pi}{2} = \pi$ rad

Linear velocity, $v = r\times\omega = 2\times 2 = 4\text{ m s}^{-1}$

$\therefore$ change in velocity, $\Delta v = 2v = 2\times 4 = 8\text{ m s}^{-1}$

**24.** **($7\sqrt{2}$)** Given $\vec{u} = 3\hat{i} + 4\hat{j}$, $\vec{a} = 0.4\hat{i} + 0.3\hat{j}$, $t = 10$ s

$\vec{v} = \vec{u} + \vec{a}t = 3\hat{i} + 4\hat{j} + (0.4\hat{i} + 0.3\hat{j})\times 10$

$= 7\hat{i} + 7\hat{j}$

$\therefore \quad |\vec{v}| = \sqrt{7^2 + 7^2} = 7\sqrt{2}$ units

**25.** **(–0.5)** For two vectors to be perpendicular to each other

$\vec{A}\cdot\vec{B} = 0$

$(2\hat{i} + 3\hat{j} + 8\hat{k})\cdot(4\hat{j} - 4\hat{i} + \alpha\hat{k}) = 0$

$-8 + 12 + 8\alpha = 0$

$\alpha = -\frac{4}{8} = -\frac{1}{2}$

**26.** **($20\sqrt{2}$)** As $\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$\vec{S} = (5\hat{i} + 4\hat{j})2 + \frac{1}{2}(4\hat{i} + 4\hat{j})4$

$= 10\hat{i} + 8\hat{j} + 8\hat{i} + 8\hat{j}$

$\vec{r}_f - \vec{r}_i = 18\hat{i} + 16\hat{j}$

[as $\vec{s}$ = change in position $= \vec{r}_f - \vec{r}_i$]

$\vec{r}_r = 20\hat{i} + 20\hat{j}$

$|\vec{r}_r| = 20\sqrt{2}$

**27.** **(60)** Using $S = ut + \frac{1}{2}at^2$

$y = u_y t + \frac{1}{2}a_y t^2$ (along $y$ Axis)

$\Rightarrow \quad 32 = 0 \times t + \frac{1}{2}(4)t^2$

$\Rightarrow \quad \frac{1}{2} \times 4 \times t^2 = 32$

$\Rightarrow \quad t = 4\,s$

$S_x = u_x t + \frac{1}{2}a_x t^2$

(Along $x$ Axis)

$\Rightarrow \quad x = 3 \times 4 + \frac{1}{2} \times 6 \times 4^2 = 60$

**28.** **(580)** For pariticle '$A$' For particle '$B$'

| For particle '$A$' | For particle '$B$' |
|---|---|
| $X_A = -3t^2 + 8t + 10$ | $Y_B = 5 - 8t^3$ |
| $\vec{V}_A = (8-6t)\hat{i}$ | $\vec{V}_B = -24t^2\hat{j}$ |
| $\vec{a}_A = -6\hat{i}$ | $\vec{a}_B = -48t\hat{j}$ |

At $t = 1$ sec

$\vec{V}_A = (8-6t)\hat{i} = 2\hat{i}$ and $\vec{v}_B = -24\hat{j}$

$\therefore \vec{V}_{B/A} = -\vec{v}_A + \vec{v}_B = -2\hat{i} - 24\hat{j}$

$\therefore$ Speed of $B$ w.r.t. $A$, $\sqrt{v} = \sqrt{2^2 + 24^2}$

$= \sqrt{4 + 576} = \sqrt{580}$

$\therefore v = 580$ (m/s)

**29.** **(195)** Given : $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k})$ N

And, $\vec{r} = [(4\hat{i} + 3\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k})]$

$= 3\hat{i} + \hat{j} - 2\hat{k}$

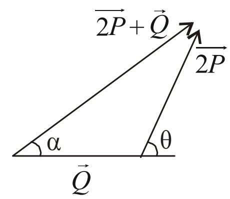
Torque,

$\tau = \vec{r} \times \vec{F} = (3\hat{i} + \hat{j} - 2\hat{k}) \times (\hat{i} + 2\hat{j} + 3\hat{k})$

$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & 2 & 3 \end{vmatrix} = 7\hat{i} - 11\hat{j} + 5\hat{k}$$

Magnitude of torque, $|\vec{\tau}| = \sqrt{195}$.

**30.** **(90)** Given, $|\vec{R}| = |\vec{P}| \Rightarrow |\vec{P} + \vec{Q}| = |\vec{P}|$




$P^2 + Q^2 + 2PQ.\cos\theta = P^2$

$\Rightarrow \quad Q + 2P\cos\theta = 0$

$\Rightarrow \quad \cos\theta = -\frac{Q}{2P}$ ..(i)

$\tan\alpha = \frac{2P\sin\theta}{Q + 2P\cos\theta} = \infty$ ($\because$ 2 P $\cos\theta + Q = 0$)

$\Rightarrow \quad \alpha = 90°$

CHAPTER 4

# Laws of Motion

**1.** **(b)** From Newton's second law

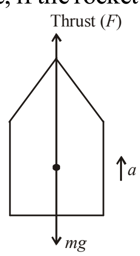
$$\frac{dp}{dt} = F = kt$$

Integrating both sides we get,

$$\int_p^{3p} dp = \int_0^T kt\,dt \Rightarrow [p]_p^{3p} = k\left[\frac{t^2}{2}\right]_0^T$$

$$\Rightarrow 2p = \frac{kT^2}{2} \Rightarrow T = 2\sqrt{\frac{p}{k}}$$

**2.** **(b)** In the absence of air resistance, if the rocket moves up with an acceleration a, then thrust



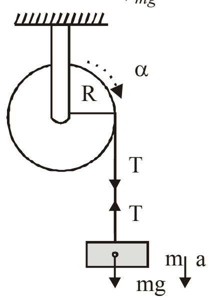

$F = mg + ma$

$\therefore \quad F = m(g + a)$

$= 3.5 \times 10^4\ (10 + 10)$

$= 7 \times 10^5\,N$

**3.** **(b)** From figure,




Acceleration $a = R\alpha$ …(i)

and $mg - T = ma$ …(ii)

From equation (i) and (ii)

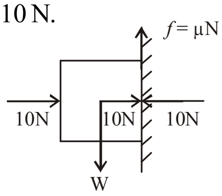
$$T \times R = mR^2\alpha = mR^2\left(\frac{a}{R}\right)$$

or $\quad T = ma$

$\Rightarrow \quad mg - ma = ma$

$\Rightarrow \quad a = \frac{g}{2}$

**4.** **(d)** Horizontal force, $N = 10$ N.




Coefficient of friction $\mu = 0.2$.

The block will be stationary so long as

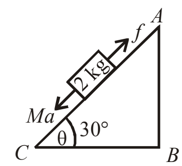
Force of friction = weight of block

$\therefore \quad \mu N = W$

$\Rightarrow 0.2 \times 10 = W$

$\Rightarrow W = 2N$

**5.** **(d)** Equation of motion when the mass slides down




$Mg\sin\theta - f = Ma$

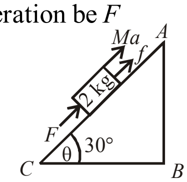
$\Rightarrow \quad 10 - f = 6$

($M = 2$ kg, $a = 3$ m/s$^2$, $\theta = 30°$ given)

$\therefore \quad f = 4$N

Equation of motion when the block is pushed up

Let the external force required to take the block up the plane with same acceleration be $F$




$F - Mg\sin\theta - f = Ma$

$\Rightarrow \quad F - 10 - 4 = 6$

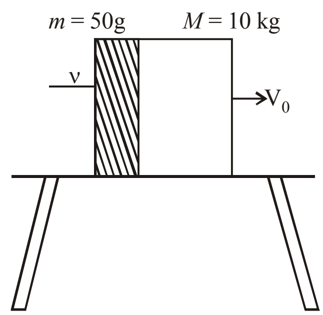
$F = 20$ N

**6.** **(d)** $f = \mu(M + m)\,g$

$$a = \frac{f}{M+m} = \frac{\mu(M+m)g}{(M+m)} = \mu g$$

$= 0.05 \times 10 = 0.5$ ms$^{-2}$

$$V_0 = \frac{\text{Initial momentum}}{(M+m)} = \frac{0.05V}{10.05}$$




$v^2 - u^2 = 2as$

$0 - u^2 = 2as$

$u^2 = 2as$

$$\left(\frac{0.05v}{10.05}\right)^2 = 2 \times 0.5 \times 2$$

Solving we get $v = 201\sqrt{2}$

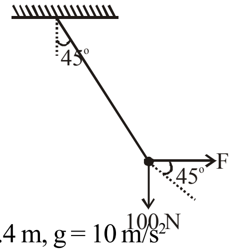
Object falling from height $H$.

$$\frac{V}{10} = \sqrt{2gH}$$

$$\frac{201\sqrt{2}}{10} = \sqrt{2\times10\times H}$$

$H = 40$ m $= 0.04$ km

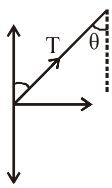
**7.** **(d)** At equilibrium,




$$\tan 45° = \frac{mg}{F} = \frac{100}{F}$$

$\therefore$ F = 100 N

**8.** **(d)** Given, $\theta = 45°$, r = 0.4 m, g = 10 m/s²




$$T\sin\theta = \frac{mv^2}{r} \quad \text{...... (i)}$$

$$T\cos\theta = mg \quad \text{...... (ii)}$$

From equation (i) & (ii) we have,

$$\tan\theta = \frac{v^2}{rg}$$

$v^2 = rg \quad \because \theta = 45°$

Hence, speed of the pendulum in its circular path,

$v = \sqrt{rg} = \sqrt{0.4\times10} = 2$ m/s

**9.** **(c)** Mass $(m) = 0.3$ kg

Force, $F = m.a = -kx$

$\Rightarrow \quad ma = -15x$

$\Rightarrow \quad 0.3a = -15x$

$$\Rightarrow \quad a = -\frac{15}{0.3}x = \frac{-150}{3}x = -50x$$

$a = -50\times0.2 = 10\text{m/s}^2$

**10.** **(a)** When forces $F_1$, $F_2$ and $F_3$ are acting on the particle, it remains in equilibrium. Force $F_2$ and $F_3$ are perpendicular to each other,

$F_1 = F_2 + F_3$

$$\therefore \quad F_1 = \sqrt{F_2^2 + F_3^2}$$

The force $F_1$ is now removed, so, resultant of $F_2$ and $F_3$ will now make the particle move with force equal to $F_1$.

Then, acceleration, $a = \dfrac{F_1}{m}$

**11.** **(c)** • For the man standing in the lift, the acceleration of the ball

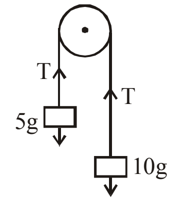
$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - a$$

Where '$a$' is the acceleration of the mass (because the acceleration of the lift is '$a$')

• For the man standing on the ground, the acceleration of the ball

$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - 0 = g$$

**12.** **(a)** Let T be the tension in the string.




$\therefore \quad 10g - T = 10a \quad$ ....(i)

$\quad T - 5g = 5a \quad$ ....(ii)

Adding (i) and (ii),

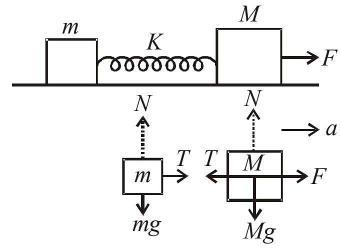
$$5g = 15a \Rightarrow a = \frac{g}{3}\ \text{m/s}^2$$

So, relative acceleration of separation $= \dfrac{2g}{3}$

So, velocity of separation = v

$$= 0 + \left(\frac{2g}{3}\right)\times1 = \frac{2g}{3} = 20/3 \text{ m/s}$$

**13.** **(d)** Making free body-diagrams for $m$ & $M$,




we get $T = ma$ and $F - T = Ma$

where $T$ is force due to spring

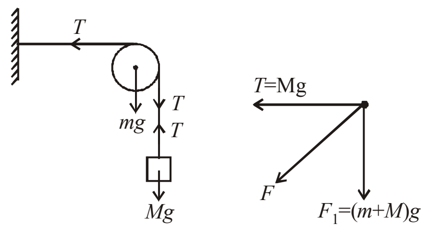
$\Rightarrow F - ma = Ma$ or, $F = Ma + ma$

$$\therefore \quad a = \frac{F}{M+m}.$$

Now, force acting on the block of mass m is

$$ma = m\left(\frac{F}{M+m}\right) = \frac{mF}{m+M}.$$

**14.** **(d)** At equilibrium $T = Mg$




F.B.D. of pulley $\quad F_1 = (m + M)\,g$

The resultant force on pulley is

$$F = \sqrt{F_1^2 + T^2} = [\sqrt{(m+M)^2 + M^2}\,]g$$

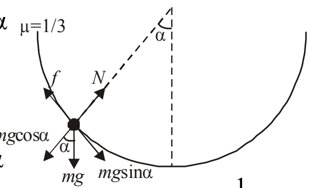
**15.** **(a)** **16.** **(d)**

$$\mu_s = \frac{\text{Length of the chain hanging from the table}}{\text{Length of the chain lying on the table}}$$

$$= \frac{l/3}{l - l/3} = \frac{l/3}{2l/3} = \frac{1}{2}$$

**17.** **(a)** For the maximum possible value of $\alpha$, $mg \sin \alpha$ will also be maximum and equal to the frictional force.
In this case $f$ is the limiting friction. The two forces acting on the insect are $mg$ and $N$. Let us resolve $mg$ into two components.
$mg \cos \alpha$ balances $N$.
$mg \sin \alpha$ is balanced by the frictional force.

$\therefore \quad N = mg \cos \alpha$




$f = mg \sin \alpha$

But $f = \mu N$

$= \mu \, mg \cos \alpha$

$\therefore \quad \mu \, mg \cos \alpha = mg \sin \alpha \Rightarrow \cot \alpha = \dfrac{1}{\mu}$

$\Rightarrow \quad \cot \alpha = 3$

**18.** **(d)** Let the velocity of the ball just when it leaves the hand is u then applying,
$v^2 - u^2 = 2as$ for upward journey

$\Rightarrow -u^2 = 2(-10) \times 2 \Rightarrow u^2 = 40$

Again applying $v^2 - u^2 = 2as$ for the upward journey of the ball, when the ball is in the hands of the thrower,
$v^2 - u^2 = 2as$

$\Rightarrow 40 - 0 = 2\,(a)\,0.2 \Rightarrow a = 100 \text{ m/s}^2$

$\therefore F = ma = 0.2 \times 100 = 20 \; N$

$\Rightarrow N - mg = 20 \Rightarrow N = 20 + 2 = 22N$

**19.** **(d)** The inclination of person from vertical is given by

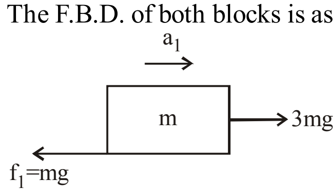
$$\tan\theta = \frac{v^2}{rg} = \frac{(10)^2}{50 \times 10} = \frac{1}{5} \therefore \theta = \tan^{-1}(1/5)$$

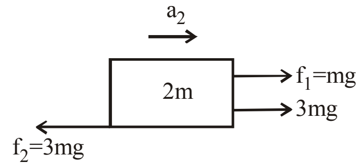
**20.** **(c)** Let T be the tension in the branch of a tree when monkey is descending with acceleration a. Then mg – T = ma; and T = 75% of weight of monkey

$$mg - \frac{75}{100} mg = ma$$

$$\Rightarrow \frac{1}{4} mg = ma \Rightarrow a = \frac{g}{4}$$

**21.** **(b)** The F.B.D. of both blocks is as shown.




$$a_1 = \frac{3mg - mg}{m} = 20 \text{ m/s}^2$$

$$a_2 = \frac{4mg - 3mg}{2m} = 5 \text{ m/s}^2$$

$$\therefore \; a_{pulley} = \frac{a_1 + a_2}{2} = \frac{25}{2} = \frac{X}{2}$$

Hence, $X = 25 \Rightarrow \sqrt{X} = 5$

**22.** **(0.98)** Limiting friction between block and slab
$= \mu_s m_A g = 0.6 \times 10 \times 9.8 = 58.8 \text{ N}$

But applied force on block A is 100 N. So the block will slip over a slab.
Now kinetic friction works between block and slab

$F_k = \mu_k m_A g = 0.4 \times 10 \times 9.8 = 39.2 \text{ N}$

This kinetic friction helps to move the slab

$\therefore$ Acceleration of slab

$$= \frac{39.2}{m_B} = \frac{39.2}{40} = 0.98 \text{ m/s}^2$$

**23.** **(0.5)** Both blocks will move with same acceleration (a) given by

$$a = \frac{F}{m_1 + m_2} = \frac{4}{5+3} = \frac{4}{8} = 0.5 \text{ m/s}^2$$

**24.** **(3.47)** $a = g(\sin\theta - \mu\cos\theta)$

$= 9.8(\sin 45° - 0.5 \cos 45°)$

$= \dfrac{4.9}{\sqrt{2}} \text{ m/sec}^2$

**25.** **(71.8)** Tension in the string when the body is at the top of the circle (T)

$= \dfrac{mv^2}{r} - mg = 71.8 \text{ N}$

**26.** **(346)** Acceleration of block while moving up an inclined plane,

$a_1 = g\sin\theta + \mu g \cos\theta$

$\Rightarrow a_1 = g \sin 30° + \mu g \cos 30°$

$= \dfrac{g}{2} + \dfrac{\mu g \sqrt{3}}{2}$ ...(i) $(\because \theta = 30°)$

Using $v^2 - u^2 = 2a(s)$

$\Rightarrow v_0^2 - 0^2 = 2a_1(s)$ $(\because u = 0)$

$\Rightarrow v_0^2 - 2a_1(s) = 0$

$\Rightarrow s = \dfrac{v_0^2}{a_1}$ ...(ii)

Acceleration while moving down an inclined plane

$a_2 = g\sin\theta - \mu g\cos\theta$

$\Rightarrow a_2 = g\sin 30° - \mu g\cos 30°$

$\Rightarrow a_2 = \dfrac{g}{2} - \dfrac{\mu\sqrt{3}}{2}g$ ...(iii)

Using again $v^2 - u^2 = 2as$ for downward motion

$\Rightarrow \left(\dfrac{v_0}{2}\right)^2 = 2a_2(s) \Rightarrow s = \dfrac{v_0^2}{4a_2}$ ...(iv)

Equating equation (ii) and (iv)

$\dfrac{v_0^2}{a_1} = \dfrac{v_0^2}{4a_2} \Rightarrow a_1 = 4a_2$

$\Rightarrow \dfrac{g}{2} + \dfrac{\mu g\sqrt{3}}{2} = 4\left(\dfrac{g}{2} - \dfrac{\mu\sqrt{3}}{2}\right)$

$\Rightarrow 5 + 5\sqrt{3}\mu = 4(5 - 5\sqrt{3}\mu)$

(Substituting, $g = 10$ m/s$^2$)

$\Rightarrow 5 + 5\sqrt{3}\mu = 20 - 20\sqrt{3}\mu \Rightarrow 25\sqrt{3}\mu = 15$

$\Rightarrow \mu = \dfrac{\sqrt{3}}{5} = 0.346 = \dfrac{346}{1000}$

So, $\dfrac{I}{1000} = \dfrac{346}{1000}$

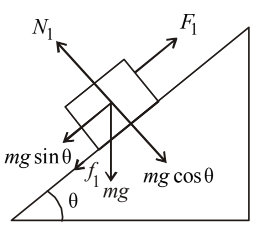
**27. (30)** The maximum velocity of the car is

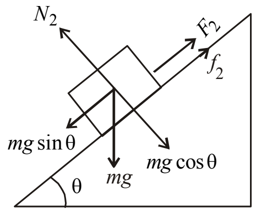
$v_{max} = \sqrt{\mu r g}$

Here $\mu = 0.6$, $r = 150$ m, $g = 9.8$

$v_{max} = \sqrt{0.6 \times 150 \times 9.8} \simeq 30\text{m/s}$

**28. (3)**




When the body slides up the inclined plane, then

mg sin $\theta + f_1 = F_1$

or, $F_1 = $ mg sin $\theta$ + $\mu$mg cos $\theta$

When the body slides down the inclined plane, then

mg sin $\theta - f_2 = F_2$

or $F_2 = $ mg sin $\theta$ – $\mu$mg cos $\theta$

$\therefore \dfrac{F_1}{F_2} = \dfrac{\sin\theta + \mu\cos\theta}{\sin\theta - \mu\cos\theta}$

$\Rightarrow \dfrac{F_1}{F_2} = \dfrac{\tan\theta + \mu}{\tan\theta - \mu} = \dfrac{2\mu + \mu}{2\mu - \mu} = \dfrac{3\mu}{\mu} = 3$

**29. (192)** Acceleration produced in upward direction

$a = \dfrac{F}{M_1 + M_2 + \text{Mass of metal rod}}$

$= \dfrac{480}{20 + 12 + 8} = 12 \text{ ms}^{-2}$

Tension at the mid point

$T = \left(M_2 + \dfrac{\text{Mass of rod}}{2}\right)a$

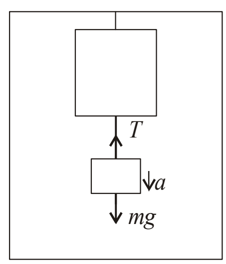
$= (12 + 4) \times 12 = 192\text{ N}$

**30. (24.5)** When lift is stationary, $W_1 = mg$ ...(i)

When the lift descends with acceleration, a

$W_2 = m(g - a)$

$W_2 = \dfrac{49}{10}(10 - 5) = 24.5\ N$



CHAPTER 5

# Work, Energy and Power

**1.** **(c)** Given : retardation $\propto$ displacement

i.e., $a = -kx$ But $a = v\dfrac{dv}{dx}$

$k$ = proportional constant

$\therefore \dfrac{vdv}{dx} = -kx \Rightarrow \int_{v_1}^{v_2} v\,dv = -\int_0^x kxdx$

$\left(v_2^2 - v_1^2\right) = -\dfrac{kx^2}{2}$

$\Rightarrow \dfrac{1}{2}m\left(v_2^2 - v_1^2\right) = \dfrac{1}{2}m\left(\dfrac{-x^2}{2}\right)k$

$\therefore$ Loss in kinetic energy, $\Delta K \propto x^2$

**2.** **(b)** We know that F × v = Power

$\therefore F \times v = c$ where c = constant

$\therefore m\dfrac{dv}{dt} \times v = c \qquad \left(\because F = ma = \dfrac{mdv}{dt}\right)$

$\therefore m\int_0^v vdv = c\int_0^t dt \quad \therefore \dfrac{1}{2}mv^2 = ct$

$\therefore v = \sqrt{\dfrac{2c}{m}} \times t^{1/2}$

$\therefore \dfrac{dx}{dt} = \sqrt{\dfrac{2c}{m}} \times t^{1/2} \qquad$ where $v = \dfrac{dx}{dt}$

$\therefore \int_0^x dx = \sqrt{\dfrac{2c}{m}} \times \int_0^t t^{1/2}dt$

$x = \sqrt{\dfrac{2c}{m}} \times \dfrac{2t^{3/2}}{3} \Rightarrow x \propto t^{3/2}$

**3.** **(a)** In the question, the velocity of the earth before and after the collision may be assumed zero. Hence, coefficient of restitution will be,

$$e^n = \frac{\upsilon_1}{\upsilon_0} \times \frac{\upsilon_2}{\upsilon_1} \times \frac{\upsilon_3}{\upsilon_2} \times ....... \times \frac{\upsilon_n}{\upsilon_n - 1} = \frac{\upsilon_n}{\upsilon_0}$$

where $\upsilon_n$ is the velocity after $n$th rebounding and $\upsilon_0$ is the velocity with which the ball strikes the earth first time.

Hence, $e^n = \dfrac{\upsilon_n}{\upsilon_0} = \dfrac{\sqrt{2gh_n}}{\sqrt{2gh_0}}$

where $h_n$ is the height to which the ball rises after $n$th rebounding.

Hence, $e^n = \dfrac{\upsilon_n}{\upsilon_0} = \sqrt{\dfrac{h_n}{h_0}}$

**4.** **(c)** Acceleration (a) $= \dfrac{\text{v} - \text{u}}{\text{t}}$

$= \dfrac{(0-50)}{(10-0)} = -5 \text{ m/s}^2$

u = 50 m/s

$\therefore$ v = u + at = 50 – 5t

Velocity in first two seconds t = 2

$\text{v}_{(\text{at } t=2)} = 40 \text{ m/s}; \ s = \dfrac{\text{v}^2 - \text{u}^2}{2\text{a}}$

$\text{W} = \text{F.s} = \text{ma}\dfrac{\text{v}^2 - \text{u}^2}{2\text{a}}$

$\text{W} = \dfrac{1}{2}(40^2 - 50^2) \times 10 = -4500 \text{ J}$

**5.** **(b)** Work done in fulling the hanging portion (L/n) on the table W = $\dfrac{mgL^2}{2n^2}$; mass of hanging portion of chain

m = $\dfrac{M}{L}$ Putting the values and solving we get, W = 3.6 J

**6.** **(a)** $U = \dfrac{a}{x^{12}} - \dfrac{B}{x^6}$

$F = -\dfrac{dU}{dx} = +12\dfrac{a}{x^{13}} - \dfrac{6b}{x^7} = 0 \Rightarrow x = \left(\dfrac{2a}{b}\right)^{1/6}$

U (x = ∞) = 0

$\text{U}_{\text{equilibrium}} = \dfrac{a}{\left(\dfrac{2a}{b}\right)^2} - \dfrac{b}{\left(\dfrac{2a}{b}\right)} = -\dfrac{b^2}{4a}$

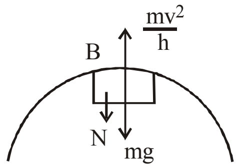
$\therefore \text{U}(x = \infty) - \text{U}_{\text{equilibrium}} = 0 - \left(= \dfrac{b^2}{4a}\right) = \dfrac{b^2}{4a}.$

**7.** **(a)** By conservation of energy

$$mg\,(3h) = mg\,(2h) + \frac{1}{2}mv^2 \quad (v = \text{velocity at B})$$

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh}$$

From free body diagram of block at B




$$N + mg = \frac{mv^2}{h} = 2mg\,;$$

$$N = mg$$

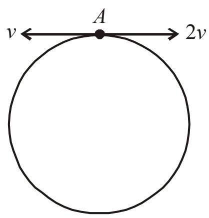
**8.** **(c)** Volume of water to raise $= 22380\, l = 22380 \times 10^{-3} m^3$

$$P = \frac{mgh}{t} = \frac{V\rho gh}{t} \Rightarrow t = \frac{V\rho gh}{P}$$

$$t = \frac{22380 \times 10^{-3} \times 10^3 \times 10 \times 30}{10 \times 746} = 15 \text{ min}$$

**9.** **(c)** Let the radius of the circle be $r$. Then the two distance travelled by the two particles before first collision is $2\pi r$. Therefore

$$2v \times t + v \times t = 2\pi r$$



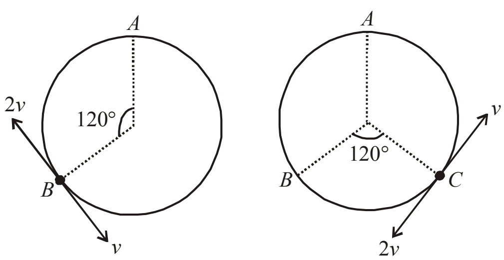

where $t$ is the time taken for first collision to occur.

$$\therefore \quad t = \frac{2\pi r}{3v}$$

$\therefore$ Distance travelled by particle

with velocity $v$ is equal to $v \times \frac{2\pi r}{3v} = \frac{2\pi r}{3}$.

Therefore the collision occurs at $B$.




As the collision is elastic and the particles have equal masses, the velocities will interchange as shown in the figure. According to the same reasoning as above, the 2nd collision will take place at $C$ and the velocities will again interchange.

With the same reasoning the 3rd collision will occur at the point $A$. Thus there will be two elastic collisions before the particles again reach at $A$.

**10.** **(b)** Spring constant, $k = 5 \times 10^3$ N/m

Let $x_1$ and $x_2$ be the initial and final stretched position of the spring, then

$$\text{Work done, } W = \frac{1}{2}k\left(x_2^2 - x_1^2\right)$$

$$= \frac{1}{2} \times 5 \times 10^3 \left[(0.1)^2 - (0.05)^2\right]$$

$$= \frac{5000}{2} \times 0.15 \times 0.05 = 18.75 \text{ Nm}$$

**11.** **(c)** We know area under F-x graph gives the work done by the body

$$\therefore\ W = \frac{1}{2} \times (3+2) \times (3-2) + 2 \times 2$$

$$= 2.5 + 4 = 6.5 \text{ J}$$

Using work energy theorem,

$\Delta$ K.E = work done

$\therefore$ $\Delta$ K.E = 6.5 J

**12.** **(a)** Let u be the initial velocity of the bullet of mass m.

After passing through a plank of width x, its velocity decreases to v.

$$\therefore \quad u - v = \frac{4}{n} \text{ or, } v = u - \frac{4}{n} = \frac{u(n-1)}{n}$$

If F be the retarding force applied by each plank, then using work – energy theorem,

$$Fx = \frac{1}{2}mu^2 - \frac{1}{2}mv^2 = \frac{1}{2}mu^2 - \frac{1}{2}mu^2\,\frac{(n-1)^2}{n^2}$$

$$= \frac{1}{2}mu^2\left[\frac{1-(n-1)^2}{n^2}\right]$$

$$Fx = \frac{1}{2}mu^2\left(\frac{2n-1}{n^2}\right)$$

Let P be the number of planks required to stop the bullet.

Total distance travelled by the bullet before coming to rest = Px

Using work-energy theorem again,

$$F(Px)=\frac{1}{2}mu^2-0$$

or, $P(Fx)=P\left[\frac{1}{2}mu^2\frac{(2n-1)}{n^2}\right]=\frac{1}{2}mu^2$

$\therefore \quad P=\frac{n^2}{2n-1}$

**13.** **(c)** K.E. $\propto t$

K.E. $= ct$ [Here, $c$ = constant]

$\Rightarrow \frac{1}{2}mv^2=ct$

$\Rightarrow \frac{(mv)^2}{2m}=ct$

$\Rightarrow \frac{p^2}{2m}=ct \ (\because p=mv)$

$\Rightarrow p=\sqrt{2ctm}$

$\Rightarrow F=\frac{dp}{dt}=\frac{d(\sqrt{2ctm})}{dt}$

$\Rightarrow F=\sqrt{2\,cm}\times\frac{1}{2\sqrt{t}}$

$\Rightarrow F\propto\frac{1}{\sqrt{t}}$

**14.** **(c)** $mv=(m+M)V'$

or $v=\frac{mv}{m+M}=\frac{mv}{m+4m}=\frac{v}{5}$

Using conservation of ME, we have

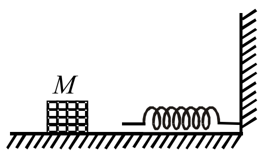
$$\frac{1}{2}mv^2=\frac{1}{2}(m+4m)\left(\frac{v}{5}\right)^2+mgh$$

or $h=\frac{2}{5}\frac{v^2}{g}$

**15.** **(b)** When the spring gets compressed by length L.

K.E. lost by mass M = P.E. stored in the compressed spring.

$\frac{1}{2}Mv^2=\frac{1}{2}kL^2$




$\Rightarrow v=\sqrt{\frac{k}{M}}\cdot L$

Momentum of the block, $= M\times v$

$= M\times\sqrt{\frac{k}{M}}\cdot L=\sqrt{kM}\cdot L$

**16.** **(d)** Considering conservation of momentum along $x$-direction,

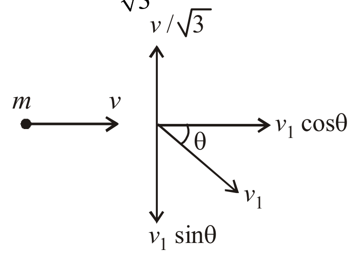
$mv = mv_1\cos\theta$ ...(1)

where $v_1$ is the velocity of second mass

In $y$-direction,

$0=\frac{mv}{\sqrt{3}}-mv_1\sin\theta$

or $m_1v_1\sin\theta=\frac{mv}{\sqrt{3}}$ ...(2)




Squaring and adding eqns. (1) and (2) we get

$$v_1^2=v^2+\frac{v^2}{\sqrt{3}}\Rightarrow v_1=\frac{2}{\sqrt{3}}v$$

**17.** **(b)** As we know, dU = F.dr

$U=\int_0^r \alpha r^2dr=\frac{ar^3}{3}$ ...(i)

As, $\frac{mv^2}{r}=\alpha r^2$

$m^2v^2=m\alpha r^3$

or, 2m(KE) $=\frac{1}{2}\alpha r^3$ ...(ii)

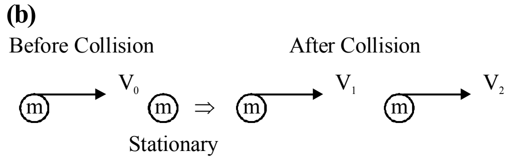
Total energy = Potential energy + kinetic energy

Now, from eqn (i) and (ii)

Total energy = K.E. + P.E.

$=\frac{\alpha r^3}{3}+\frac{\alpha r^3}{2}=\frac{5}{6}\alpha r^3$

**18.** **(b)**




$\frac{1}{2}mv_1^2+\frac{1}{2}mv_2^2=\frac{3}{2}\left(\frac{1}{2}mv_0^2\right)$

$\Rightarrow v_1^2+v_2^2=\frac{3}{2}v_0^2$ ....(i)

From momentum conservation

$mv_0=m(v_1+v_2)$ ....(ii)

Squarring both sides,

$(v_1 + v_2)^2 = v_0^2$

$\Rightarrow \quad v_1^2 + v_2^2 + 2v_1v_2 = v_0^2$

$$2v_1v_2 = -\frac{v_0^2}{2}$$

$$(v_1 - v_2)^2 = v_1^2 + v_2^2 - 2v_1v_2 = \frac{3}{2}v_0^2 + \frac{v_0^2}{2}$$

Solving we get relative velocity between the two particles

$v_1 - v_2 = \sqrt{2}v_0$

**19.** **(b)** Acceleration $= \frac{d^2s}{dt^2} = 2t\ ms^2$

$$w = \int_0^2 ma.ds = \int_0^2 3 \times 2t \times t^2\, dt$$

Solving we get

$w = 24J$

**20.** **(c)** Let m = mass of boy, M = mass of man

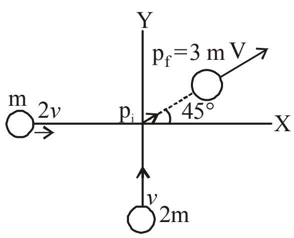
v = velocity of boy, V = velocity of man

$$\frac{1}{2}MV^2 = \frac{1}{2}\left[\frac{1}{2}mv^2\right] \quad ...(i)$$

$$\frac{1}{2}M(V+1)^2 = 1\left[\frac{1}{2}mv^2\right] \quad ...(ii)$$

Dividing eq (i) by (ii) we get, $V = \frac{1}{\sqrt{2}-1}$

**21.** **(56)**




Initial momentum of the system

$p_i = \sqrt{[m(2V)^2 \times 2m(2V)^2]}$

$= \sqrt{2}m \times 2V$

Final momentum of the system = 3mV

By the law of conservation of momentum

$$2\sqrt{2}mv = 3mV \Rightarrow \frac{2\sqrt{2}v}{3} = V_{combined}$$

Loss in energy

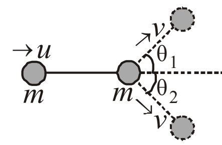
$$\Delta E = \frac{1}{2}m_1V_1^2 + \frac{1}{2}m_2V_2^2 - \frac{1}{2}(m_1 + m_2)V_{combined}^2$$

$$\Delta E = 3mv^2 - \frac{4}{3}mv^2 = \frac{5}{3}mv^2 = 55.55\%$$

Percentage loss in energy during the collision $\simeq$ 56%

**22.** **(0.4)** In an elastic head-on collision, of two equal masses their kinetic energies or velocities are exchanged. Hence when the first ball collides with the second ball at rest, the second ball attains the speed of 0.4 m/s and the first ball comes to rest. This process continues. Thus the velocity of the last ball is 0.4 $ms^{-1}$.

**23.** **(45)**




Applying law of conservation of momentum along horizontal and vertical directions, we get,

$mv \sin\theta_1 - mv \sin\theta_2 = 0$

i.e., $\theta_1 = \theta_2$ ....... (i)

Also, $mu = mv(\cos\theta_1 + \cos\theta_2)$

$= 2\,mv\cos\theta$

$[\theta_1 = \theta_2 = \theta$ (say)$]$

$$\cos\theta = \frac{u}{2v} \quad ...... (ii)$$

According to law of conservation of KE,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}mv^2$$

or $u^2 = 2v^2$ or $u = (\sqrt{2})v$ ..... (iii)

From eqn. (ii) and eqn. (iii), $\cos\theta = \frac{(\sqrt{2})v}{2v}$

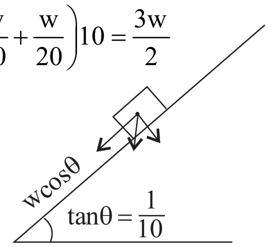
or $\cos\theta = \frac{1}{\sqrt{2}}$ or $\theta = 45°$

Hence, $\theta_1 = \theta_2 = 45°$

**24.** **(15)** While moving uphill power

$$P = \left(w\sin\theta + \frac{w}{20}\right)10 \quad [\because \theta \text{ is very small}]$$

$$P = \left(\frac{w}{10} + \frac{w}{20}\right)10 = \frac{3w}{2}$$



While moving downhill,

$$\frac{P}{2}=\frac{3w}{4}=\left(\frac{w}{10}-\frac{w}{20}\right)V$$

$$\frac{3}{4}=\frac{v}{20}\Rightarrow v=15\ m/s$$

$\therefore$ Speed of car while moving downhill v = 15 m/s.

**25.** **(2)** U = 6x + 8y

$$F_x=-\frac{\partial U}{\partial x}=-6\Rightarrow a_x=-3\ m/s^2$$

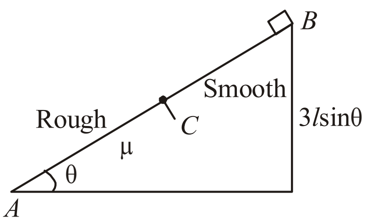
t = 0, x = 6 ; t = $t_0$, x = 0 (i.e. at y-axis)

$$0-6=-\frac{1}{2}a_x t^2\quad [\ s=s_0+ut+\frac{1}{2}at^2\ ]$$

$$\frac{12}{3}=t^2\ \Rightarrow t=2\text{ sec.}$$

**Note :** Although the particle will have acceleration along y-direction also but time will be same.

**26.** **(3)** If $AC = l$ then according to question, $BC = 2l$ and $AB = 3l$.




Here, work done by all the forces is zero.

$$W_{\text{friction}}+W_{mg}=0$$

$$mg(3l)\sin\theta-\mu mg\cos\theta(l)=0$$

$$\Rightarrow \mu mg\cos\theta l=3mgl\sin\theta$$

$$\Rightarrow \mu=3\tan\theta=k\tan\theta$$

$$\therefore k=3$$

**27.** **(150.00)** From work energy theorem,

$$W=F\cdot s=\Delta KE=\frac{1}{2}mv^2$$

Here $V^2=2gh$

$$\therefore F\cdot s=F\times\frac{2}{10}=\frac{1}{2}\times\frac{15}{100}\times 2\times 10\times 20$$

$$\therefore F=150\ \text{N.}$$

**28.** **(10.00)** Kinetic energy = change in potential energy of the particle,

KE = mg$\Delta$h

Given, $m = 1$ kg,

$$\Delta h=h_2-h_1=2-1=1m$$

$$\therefore KE=1\times 10\times 1=10\,J$$

**29.** **(18)** Given, Mass of the body, $m = 2$ kg

Power delivered by engine, $P = 1$ J/s

Time, $t = 9$ seconds

Power, $P = Fv$

$$\Rightarrow P=mav\qquad [\because F=ma]$$

$$\Rightarrow m\frac{dv}{dt}v=P\qquad \left(\because a=\frac{dv}{dt}\right)$$

$$\Rightarrow v\ dv=\frac{P}{m}dt$$

Integrating both sides we get

$$\Rightarrow \int_0^v v\,dv=\frac{P}{m}\int_0^t dt$$

$$\Rightarrow \frac{v^2}{2}=\frac{Pt}{m}\Rightarrow v=\left(\frac{2Pt}{m}\right)^{1/2}$$

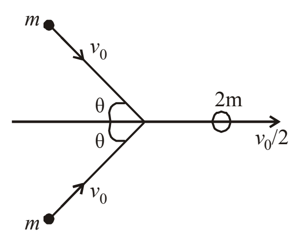
$$\Rightarrow \frac{dx}{dt}=\sqrt{\frac{2P}{m}}t^{1/2}\qquad \left(\because v=\frac{dx}{dt}\right)$$

$$\Rightarrow \int_0^x dx=\sqrt{\frac{2P}{m}}\int_0^t t^{1/2}\,dt$$

$$\therefore \text{ Distance, } x=\sqrt{\frac{2P}{m}}\frac{t^{3/2}}{3/2}=\sqrt{\frac{2P}{m}}\times\frac{2}{3}t^{3/2}$$

$$\Rightarrow x=\sqrt{\frac{2\times 1}{2}}\times\frac{2}{3}\times 9^{3/2}=\frac{2}{3}\times 27=18.$$

**30.** **(120)**




Momentum conservation along $x$ direction,

$$2mv_0\cos\theta=2m\frac{v_0}{2}\Rightarrow\cos\theta=\frac{1}{2}\text{ or }\theta=60^\circ$$

Hence angle between the initial velocities of the two bodies

$$=\theta+\theta=60^\circ+60^\circ=120^\circ.$$

# CHAPTER 6

# System of Particles and Rotational Motion

**1.** **(b)** $x_{cm} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$

$\Rightarrow \quad m_1x_1 + m_2x_2 + m_3x_3 = x_{cm}(m_1 + m_2 + m_3)$

Similarly,

$m_1y_1 + m_2y_2 + m_3y_3 = y_{cm}(m_1 + m_2 + m_3)$

$m_1z_1 + m_2z_2 + m_3z_3 = z_{cm}(m_1 + m_2 + m_3)$

Given, $m_1 = 1\text{kg}, m_2 = 2\text{kg}, m_3 = 3\text{kg}$

$x_{cm} = y_{cm} = z_{cm} = 3\text{m}$

$\therefore \quad x_1 + 2x_2 + 3x_3 = 18$

$y_1 + 2y_2 + 3y_3 = 18$

$z_1 + 2z_2 + 3z_3 = 18$

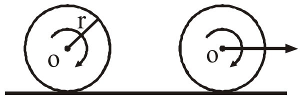
Now, if $m_4 = 4$ kg is introduced in the system,

$$x_{cm} = \frac{x_1 + 2x_2 + 3x_3 + 4x_4}{1+2+3+4} = 1$$

$$\Rightarrow \quad \frac{18 + 4x_4}{10} = 1 \Rightarrow \quad x_4 = -2$$

Similarly, $y_4 = -2$; $\qquad z_4 = -2$

**2.** **(c)**

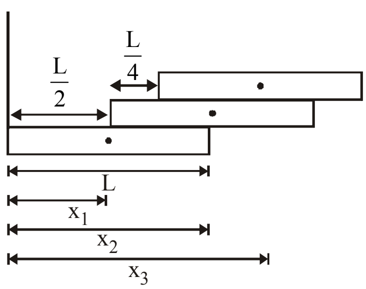



From conservation of angular momentum about any fix point on the surface,

$mr^2\omega_0 = 2mr^2\omega$

$$\Rightarrow \quad \omega = \omega_0/2 \Rightarrow v = \frac{\omega_0 r}{2} \quad [\because v = r\omega]$$

**3.** **(d)**

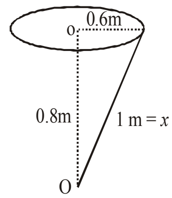



$$x_1 = \frac{L}{2},\ x_2 = L,\ x_3 = \frac{5L}{4}$$

$$\therefore \quad X_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$= \frac{M \times \frac{L}{2} + M \times L + M \times \frac{5L}{4}}{M + M + M} = \frac{11L}{12}$$

**4.** **(a)** Angular momentum,




$L_0 = \text{mvx} \sin 90°$

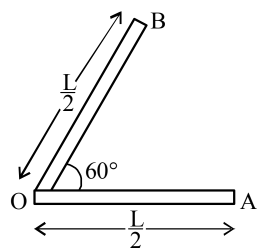
$= 2 \times 0.6 \times 12 \times 1 \times 1$

[As $V = r\omega$, $\sin 90° = 1$,

$r = 0.6$ m, $\omega = 12$ rad/s]

So, $L_0 = 14.4\ \text{kgm}^2/\text{s}$

**5.** **(b)** As the rod is bent at the middle.




Moment of intertia of each part about one end O,

$$= \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2$$

Thus total moment of inertia through the middle point O

$$= \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2 + \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2 = \frac{ML^2}{12}$$

**6.** **(a)** Given, $\theta = 30°$, $v = 5$ m/s

Let $h$ be the height on the plane upto which the cylinder will go up.

$\therefore$ From conservation of energy,

$$\text{Total K.E} = \text{P.E} \Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \text{mgh}$$

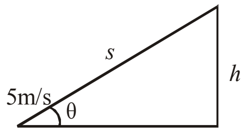
$$\Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 = mgh \quad \left[I_{\text{cyl}} = \frac{1}{2}mr^2\right]$$

$$\Rightarrow \frac{3}{4}mv^2 = mgh \quad [\text{using } v = r\omega]$$

$$\Rightarrow h = \frac{3v^2}{4g} = \frac{3 \times 5^2}{4 \times 9.8} = 1.913 \text{ m.}$$

Let, $s$ = distance moved up by the cylinder on the inclined plane.

$\therefore \sin\theta = \dfrac{h}{s}$



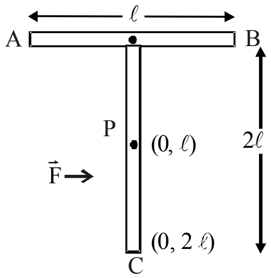

$\Rightarrow \quad s = \dfrac{h}{\sin\theta}$

$= \dfrac{1.913}{\sin 30°} = 3.826$ m

Time taken to return to the bottom = $t$

$$= \sqrt{\frac{2s\left(1+\dfrac{k^2}{r^2}\right)}{g\sin\theta}} = \sqrt{\frac{2\times 3.826\left(1+\dfrac{1}{2}\right)}{9.8\sin 30°}} = 1.53\text{s.}$$

**7.** **(d)**




To have linear motion, the force $\vec{F}$ has to be applied at centre of mass.
i.e. the point 'P' has to be at the centre of mass

$$y_{CM} = \frac{m_1y_1 + m_2y_2}{m_1+m_2} = \frac{m\times 2\ell + 2m\times\ell}{3m} = \frac{4\ell}{3}$$

**8.** **(b)** Moment of Inertia of complete disc about 'O' $I_{total} = \dfrac{MR^2}{2}$

Radius of removed disc = R/4

$\therefore$ Mass of removed disc = M/16 [As $M \propto R^2$]

M.I of removed disc about its own axis (O')

$$= \frac{1}{2}\frac{M}{16}\left(\frac{R}{4}\right)^2 = \frac{MR^2}{512}$$

M.I of removed disc about O

$$I_{\text{removed disc}} = I_{cm} + mx^2 = \frac{MR^2}{512} + \frac{M}{16}\left(\frac{3R}{4}\right)^2 = \frac{19\,MR^2}{512}$$

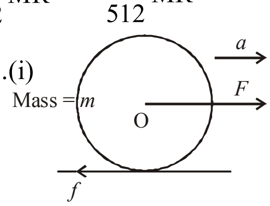
M.I of remaining disc

$$I_{\text{remaining}} = \frac{MR^2}{2} - \frac{19}{512}MR^2 = \frac{237}{512}MR^2$$

**9.** **(c)** From figure,

$ma = F - f$ ....(i)

And, torque $\tau = I\alpha$



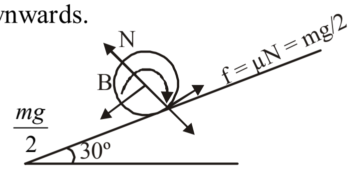

$\dfrac{mR^2}{2}\alpha = fR$

$$\frac{mR^2}{2}\frac{a}{R} = fR \quad \left[\because \alpha = \frac{a}{R}\right] \quad \frac{ma}{2} = f \quad \text{...(ii)}$$

Put this value in equation (i),

$$ma = F - \frac{ma}{2} \text{ or } F = \frac{3ma}{2}$$

**10.** **(a)** Here kinetic friction force will balance the force of gravity. So it will rotate at its initial position and its angular velocity becomes zero (friction also becomes zero), so it will move downwards.




**11.** **(b)** $I = 1.2$ kg m$^2$, $E_r = 1500$ J,
$\alpha = 25$ rad/sec$^2$, $\omega_1 = 0$, $t = ?$

As $E_r = \dfrac{1}{2}I\omega^2$,

$$\omega = \sqrt{\frac{2E_r}{I}} = \sqrt{\frac{2\times 1500}{1.2}} = 50 \text{ rad/sec}$$

From $\omega_2 = \omega_1 + \alpha t$

$50 = 0 + 25t, \quad \therefore \quad t = 2$ seconds

**12.** **(c)** Let v be the velocity of the centre of mass of the sphere and ω be the angular velocity of the body about an axis passing through the centre of mass.

$J = Mv$

$J(h - R) = \dfrac{2}{5}MR^2 \times \omega$

From the above two equations, $v(h - R) = \dfrac{2}{5}r^2\omega$

From the condition of pure rolling, $v = R\omega$

$$h - R = \frac{2R}{5} \Rightarrow h = \frac{7R}{5}$$

**13.** **(d)** By conservation of angular momentum,

$I_1\omega_1 = I_2\omega_2$

$I_1 = \dfrac{2}{5}mR_1^2;\ I_2 = \dfrac{2}{5}mR_2^2$

$$I_2 = \frac{2}{5}m\left(\frac{R_1}{n}\right)^2 \Rightarrow I_2 = \frac{1}{n^2}I_1 \Rightarrow \frac{I_1}{I_2} = n^2$$

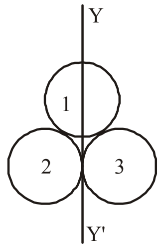
$$\therefore\ \omega_2 = \frac{I_1\omega_1}{I_2} = n^2\omega_1 = n^2\omega \quad [\because \omega_1 = \omega]$$

**14.** **(d)** Moment of inertia of system about YY'

$I = I_1 + I_2 + I_3$

$= \frac{1}{2}MR^2 + \frac{3}{2}MR^2 + \frac{3}{2}MR^2$

$= \frac{7}{2}MR^2$




**15.** **(a)** $\alpha = \frac{\tau}{I} = \frac{30}{2} = 15\,\text{rad}/\text{s}^2$

$\therefore \theta = \omega_0 t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}\times(15)\times(10)^2 = 750\,rad$

**16.** **(b)** For translational motion,

$mg - T = ma$ .....(1)

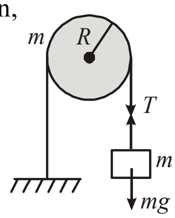
For rotational motion,

$T.R = I\alpha$

$\Rightarrow T.R = \frac{1}{2}mR^2\alpha$

Also, acceleration, $a = R\alpha$

$\therefore T = \frac{1}{2}mR\alpha = \frac{1}{2}ma$




Substituting the value of T is equation (1) we get

$mg - \frac{1}{2}ma = ma \Rightarrow a = \frac{2}{3}g$

**17.** **(c)** For sphere,

$\frac{1}{2}mv^2 + I\omega^2 = \frac{1}{2}mgh$

or $\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\frac{v^2}{R^2} = mgh$

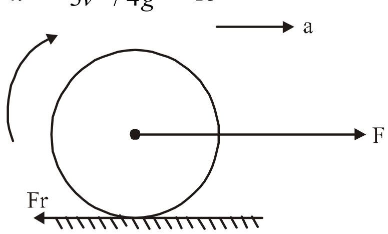
or $h = \frac{7v^2}{10g}$

For cylinder

$\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{mR^2}{2}\right) = mgh'$ or $h' = \frac{3v^2}{4g}$

$\therefore \frac{h}{h'} = \frac{7v^2/10g}{3v^2/4g} = \frac{14}{15}$

**18.** **(d)**




$F - fr = ma$ ...(i)

$f_r R = I\alpha = \frac{mR^2}{2}\alpha$ ...(ii)

for pure rolling

$a = \alpha R$ ...(iii)

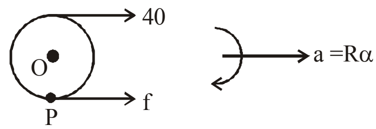
from (1) (2) and (3)

$F - \frac{mR\alpha}{2} = m\alpha R$

$F = \frac{3}{2}mR\alpha$

$\alpha = \frac{2F}{3mR}$

**19.** **(b)**




From newton's second law

$40 + f = m(R\alpha)$ .....(i)

Taking torque about 0 we get

$40 \times R - f \times R = I\alpha$

$40 \times R - f \times R = mR^2\alpha$

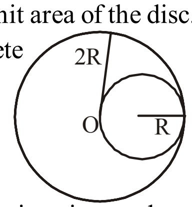
$40 - f = mR\alpha$ ...(ii)

Solving equation (i) and (ii)

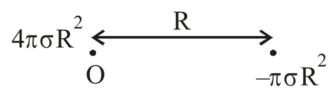
$\alpha = \frac{40}{mR} = 16\,\text{rad}/\text{s}^2$

**20.** **(b)** Let $\sigma$ be the mass per unit area of the disc. Then the mass of the complete disc $= \sigma(\pi(2R)^2)$

The mass of the removed disc $= \sigma(\pi R^2) = \pi\sigma R^2$




Let us consider the above situation to be a complete disc of radius $2R$ on which a disc of radius $R$ of negative mass is superimposed. Let $O$ be the origin. Then the above figure can be redrawn keeping in mind the concept of centre of mass as :




$x_{c.m} = \frac{\left(6\pi(2R)^2\right)\times 0 + \left(-6\left(\pi R^2\right)\right)R}{4\pi\sigma R^2 - \pi\sigma R^2}$

$\therefore x_{c.m} = \frac{-\pi\sigma R^2 \times R}{3\pi\sigma R^2}$

$\therefore x_{c.m} = -\frac{R}{3} = \alpha R \Rightarrow \alpha = \frac{1}{3}$

**21.** **(16)** $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 \Rightarrow \theta = 100\,\text{rad}$

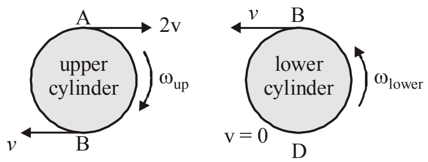
$\therefore$ Number of revolutions $= \frac{100}{2\pi} = 16\,(\text{approx.})$

**22.** **(0.03)** Required M.I. $= \frac{MR^2}{2} + MR^2$

$$= \frac{3}{2}MR^2 = \frac{3}{2} \times 2 \times (0.1)^2 = 0.03 \text{ kg m}^2$$

**23.** **(–3)**

**24.** **(3)**




$$\omega_{up} = \frac{3v}{2R}; \quad \omega_{lower} = \frac{v}{2R} \Rightarrow \frac{\omega_{up}}{\omega_{lower}} = 3$$

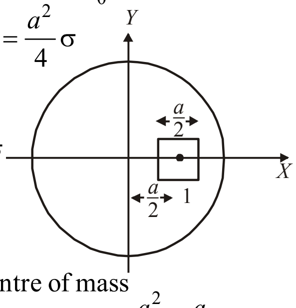
**25.** **(2.08)**

**26.** **(23.00)** Let $\sigma$ be the mass density of circular disc.

Original mass of the disc, $m_0 = \pi a^2 \sigma$

Removed mass, $m = \frac{a^2}{4}\sigma$

Remaining, mass,




$$m' = \left(\pi a^2 - \frac{a^2}{4}\right)\sigma$$

$$= a^2\left(\frac{4\pi - 1}{4}\right)\sigma$$

New position of centre of mass

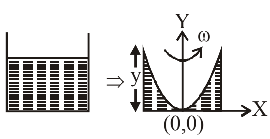
$$X_{CM} = \frac{m_0 x_0 - mx}{m_0 - m} = \frac{\pi a^2 \times 0 - \frac{a^2}{4} \times \frac{a}{2}}{\pi a^2 - \frac{a^2}{4}}$$

$$= \frac{-a^3/8}{\left(\pi - \frac{1}{4}\right)a^2} = \frac{-a}{2(4\pi - 1)} = \frac{-a}{8\pi - 2} = -\frac{a}{23}$$

$\therefore x = 23$

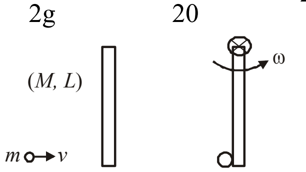
**27.** **(2)** Using $v^2 = u^2 + 2gy$ $\quad [\therefore u = 0 \text{ at } (0,0)]$

$v^2 = 2gy$ $\quad [\therefore v = \omega x]$




$$\Rightarrow y = \frac{\omega^2 x^2}{2g} = \frac{(2 \times 2\pi)^2 \times (0.05)^2}{20} \simeq 2\text{cm}$$

**28.** **(20)**




Before collision After collision

Using principal of conservation of angular momentum we have

$$\vec{L}_i = \vec{L}_f \Rightarrow mvL = I\omega$$

$$\Rightarrow mvL = \left(\frac{ML^2}{3} + mL^2\right)\omega$$

$$\Rightarrow 0.1 \times 80 \times 1 = \left(\frac{0.9 \times 1^2}{3} + 0.1 \times 1^2\right)\omega$$

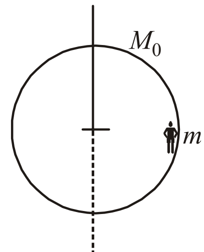
$$\Rightarrow 8 = \left(\frac{3}{10} + \frac{1}{10}\right)\omega \Rightarrow 8 = \frac{4}{10}\omega$$

$$\Rightarrow \omega = 20 \text{ rad/sec.}$$

**29.** **(9.00)** Here $M_0 = 200$ kg, $m = 80$ kg

Using conservation of angular momentum,

$L_i = L_f$




$$I_1\omega_1 = I_2\omega_2$$

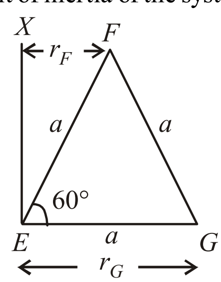
$$I_1 = (I_M + I_m) = \left(\frac{M_0R^2}{2} + mR^2\right)$$

$$I_2 = \frac{1}{2}M_0R^2 \text{ and } \omega_1 = 5 \text{ rpm}$$

$$\therefore \omega_2 = \left(\frac{M_0R^2}{2} + mR^2\right) \times \frac{5}{\frac{M_0R^2}{2}}$$

$$= \frac{5R^2}{R^2} \times \frac{(80 + 100)}{100} = 9 \text{ rpm.}$$

**30.** **(25)** Moment of inertia of the system about axis *XE*.




$$I = I_E + I_F + I_G$$

$$\Rightarrow I = m(r_E)^2 + m(r_F)^2 + m(r_G)^2$$

$$\Rightarrow I = m \times 0^2 + m\left(\frac{a}{2}\right)^2 + ma^2 = \frac{5}{4}ma^2 = \frac{25}{20}ma^2$$

$\therefore N = 25.$

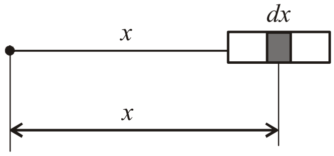
# Chapter 7 Gravitation

**1.** **(c)** $F = KR^{-n} = MR\omega^2 \Rightarrow \omega^2 = KR^{-(n+1)}$

or $\omega = K'R^{\frac{-(n+1)}{2}}$ [where $K' = K^{1/2}$, a constant]

$\frac{2\pi}{T} \alpha R^{\frac{-(n+1)}{2}}$ $\quad \therefore T \alpha R^{\frac{(n+1)}{2}}$

**2.** **(b)** $\because dF = \frac{Gm(\mu dx)}{x^2}$




$$F = \int_a^{L+a} Gm\left(A + Bx^2\right)dx$$

$$F = Gm\left[A\left(\frac{1}{a} - \frac{1}{a+L}\right) + BL\right]$$

**3.** **(b)** Gravitational force will be due to $M_1$ only.

**4.** **(b)**

**5.** **(d)** $\Delta U = U_f - U_i = -\frac{GMm}{nR+h} - \left(-\frac{GMm}{R}\right)$

$= \frac{n}{n+1} \cdot \frac{GMm}{R} = \left(\frac{n}{n+1}\right)mgR$

**6.** **(c)**

**7.** **(d)** $\frac{mv^2}{(R+x)} = \frac{GmM}{(R+x)^2}$ also $g = \frac{GM}{R^2}$

$\therefore \frac{mv^2}{(R+x)} = m\left(\frac{GM}{R^2}\right)\frac{R^2}{(R+x)^2}$

$\therefore \frac{mv^2}{(R+x)} = mg\frac{R^2}{(R+x)^2}$

$\therefore v^2 = \frac{gR^2}{R+x} \Rightarrow v = \left(\frac{gR^2}{R+x}\right)^{1/2}$

**8.** **(c)** $\frac{-Gmm'}{r} \times 3 + \frac{1}{2}m'v_e^2 = 0$

or $\frac{-3Gm}{(\frac{a}{2}/\cos 30°)} + \frac{1}{2}v_e^2 = 0$

$v_e = \sqrt{\frac{6\sqrt{3}\,Gm}{a}}$.

**9.** **(c)** $W = m(V_2 - V_1)$

when, $V_1 = -\left[\frac{GM_1}{a} + \frac{GM_2}{\sqrt{2}a}\right]$,

$V_2 = -\left[\frac{GM_2}{a} + \frac{GM_1}{\sqrt{2}a}\right]$

$\therefore W = \frac{Gm(M_2 - M_1)}{a\sqrt{2}}(\sqrt{2} - 1)$.

**10.** **(b)** $g' = g\left(1 - \frac{d}{R}\right) \Rightarrow \frac{g}{n} = g\left(1 - \frac{d}{R}\right)$

$\Rightarrow d = \left(\frac{n-1}{n}\right)R$

**11.** **(a)** Potential at the given point = Potential at the point due to the shell + Potential due to the particle

$= -\frac{GM}{a} - \frac{4GM}{a} = -\frac{5GM}{a}$

**12.** **(b)**

**13.** **(b)** V is the orbital velocity. If $V_e$ is the escape velocity then $V_e = \sqrt{2}V$. The kinetic energy at the time of ejection

$KE = \frac{1}{2}mV_e^2 = \frac{1}{2}m(\sqrt{2}V)^2 = mV^2$

**14.** **(a)** Volume of removed sphere

$V_{remo} = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{4}{3}\pi R^3\left(\frac{1}{8}\right)$

Volume of the sphere (remaining)

$V_{remain} = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi R^3\left(\frac{1}{8}\right) = \frac{4}{3}\pi R^3\left(\frac{7}{8}\right)$

Therefore mass of sphere carved and remaining sphere are at respectively $\frac{1}{8}$ M and $\frac{7}{8}$ M.

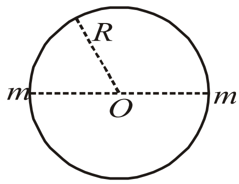
Therefore, gravitational force between these two sphere,

$$F = \frac{GMm}{r^2} = \frac{G\frac{7M}{8}\times\frac{1}{8}M}{(3R)^2} = \frac{7}{64\times 9}\frac{GM^2}{R^2}$$

$$\simeq \frac{41}{3600}\frac{GM^2}{R^2}$$

**15.** **(a)** As two masses revolve about the common centre of mass $O$.

$\therefore$ Mutual gravitational attraction = centripetal force




$$\frac{Gm^2}{(2R)^2} = m\omega^2 R$$

$$\Rightarrow \quad \frac{Gm}{4R^3} = \omega^2$$

$$\Rightarrow \quad \omega = \sqrt{\frac{Gm}{4R^3}}$$

If the velocity of the two particles with respect to the centre of gravity is v then

$v = \omega R$

$$v = \sqrt{\frac{Gm}{4R^3}} \times R = \sqrt{\frac{Gm}{4R}}$$

**16.** **(d)** Value of $g$ with altitude is,

$$g_h = g\left[1 - \frac{2h}{R}\right];$$

Value of $g$ at depth d below earth's surface,

$$g_d = g\left[1 - \frac{d}{R}\right]$$

Equating $g_h$ and $g_d$, we get $d = 2h$

**17.** **(c)** Initial gravitational potential energy, $E_i = -\frac{GMm}{2R}$

Final gravitational potential energy,

$$E_f = -\frac{GMm/2}{2\left(\frac{R}{2}\right)} - \frac{GMm/2}{2\left(\frac{3R}{2}\right)} = -\frac{GMm}{2R} - \frac{GMm}{6R}$$

$$= -\frac{4GMm}{6R} = -\frac{2GMm}{3R}$$

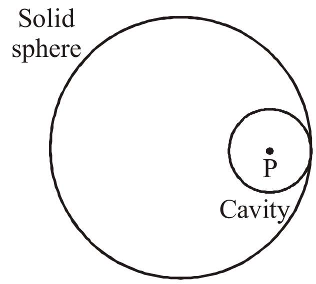
$\therefore$ Difference between initial and final energy,

$$E_f - E_i = \frac{GMm}{R}\left(-\frac{2}{3} + \frac{1}{2}\right) = -\frac{GMm}{6R}$$

**18.** **(d)** Due to complete solid sphere, potential at point $P$

$$V_{sphere} = \frac{-GM}{2R^3}\left[3R^2 - \left(\frac{R}{2}\right)^2\right]$$

$$= \frac{-GM}{2R^3}\left(\frac{11R^2}{4}\right) = -11\frac{GM}{8R}$$




Due to cavity part potential at point $P$

$$V_{cavity} = -\frac{3}{2}\frac{\frac{GM}{8}}{\frac{R}{2}} = -\frac{3GM}{8R}$$

So potential at the centre of cavity

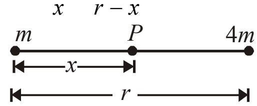
$= V_{sphere} - V_{cavity}$

$$= -\frac{11GM}{8R} - \left(-\frac{3}{8}\frac{GM}{R}\right) = \frac{-GM}{R}$$

**19.** **(c)** Let $P$ be the point where gravitational field is zero.

$$\therefore \quad \frac{Gm}{x^2} = \frac{4Gm}{(r-x)^2}$$

$$\Rightarrow \quad \frac{1}{x} = \frac{2}{r-x} \quad \Rightarrow r - x = 2x \qquad \Rightarrow \quad x = \frac{r}{3}$$




Gravitational potential at $P$,

$$V = -\frac{Gm}{\frac{r}{3}} - \frac{4Gm}{\frac{2r}{3}} = -\frac{9Gm}{r}$$

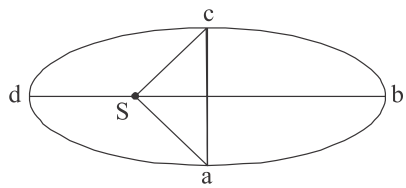
**20.** **(c)** Let area of ellipse abcd = x

Area of SabcS =

$$\frac{x}{2} + \frac{x}{4} \quad \text{(i.e., ar of abca + SacS)}$$

(Area of half ellipse + Area of triangle)

$$= \frac{3x}{4}$$




Area of SadcS $= x - \frac{3x}{4} = \frac{x}{4}$

$$\frac{\text{Area of SabcS}}{\text{Area of SadcS}} = \frac{3x/4}{x/4} = \frac{t_1}{t_2}$$

$\frac{t_1}{t_2} = 3$ or, $t_1 = 3t_2$

**21.** **(129)** From Kepler's law of periods,

$$T_2 = T_1\left(\frac{R_2}{R_1}\right)^{3/2} = 365\left(\frac{\cancel{R}/2}{\cancel{R}}\right)^{3/2}$$

$$= 365 \times \frac{1}{2\sqrt{2}} = 129 \text{ days.}$$

**22.** **(0.5)** $F \propto xM \times (1-x)M = xM^2(1-x)$

For maximum force, $\frac{dF}{dx} = 0$

$$\Rightarrow \frac{dF}{dx} = M^2 - 2xM^2 = 0 \Rightarrow x = 1/2$$

**23.** **(0.15)** $g = \frac{GM}{R^2}$

(Given $M_e = 81\, M_m$, $R_e = 3.5\, R_m$)

Substituting the above values, $\frac{g_m}{g_e} = 0.15$

**24.** **(22.4)**

**25.** **(0.70)** Potential energy at height $R = -\frac{GMm}{2R}$

$v_e$ be the escape velocity, then

$$\frac{1}{2}mv_e^2 = \frac{GM}{2R}.m \Rightarrow v_e = \sqrt{\frac{GM}{R}}$$

or, $v_e = \sqrt{gR}$

Now, $v = \sqrt{gR'} \Rightarrow v = \sqrt{2gR}$ $[\because R' = 2R]$

So, $v = \sqrt{2}v_e$ or, $v_e = \frac{v}{\sqrt{2}}$

Comparing it with given equation, $f = \frac{1}{\sqrt{2}}$.

**26.** **(0.40)**

**27.** **(2)**

**28.** **(0.33)** $F_{min} = \frac{GMm}{r^2} - \frac{GM(2m)}{(2r)^2} = \frac{GMm}{2r^2}$

and $F_{max} = \frac{GMm}{r^2} + \frac{GM(2m)}{(2r)^2} = \frac{3}{2}\frac{GMm}{r^2}$

$\therefore \frac{F_{min}}{F_{max}} = \frac{1}{3}$.

**29.** **(2)** As we know, Gravitational force of attraction,

$$F = \frac{GMm}{R^2}$$

$$F_1 = \frac{GM_e m}{r_1^2} \text{ and } F_2 = \frac{GM_e M_s}{r_2^2}$$

$$\Delta F_1 = \frac{2GM_e m}{r_1^3}\Delta r_1 \text{ and } \Delta F_2 = \frac{GM_e M_s}{r_2^3}\Delta r_2$$

$$\frac{\Delta F_1}{\Delta F_2} = \frac{m\Delta r_1}{r_1^3}\frac{r_2^3}{M_s \Delta r_2} = \left(\frac{m}{M_s}\right)\left(\frac{r_2^3}{r_1^3}\right)\left(\frac{\Delta r_1}{\Delta r_2}\right)$$

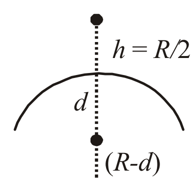
Using $\Delta r_1 = \Delta r_2 = 2\, R_{earth}$; $m = 8 \times 10^{22}$ kg;

$M_s = 2 \times 10^{30}$ kg

$r_1 = 0.4 \times 10^6$ km and $r_2 = 150 \times 10^6$ km

$$\frac{\Delta F_1}{\Delta F_2} = \left(\frac{8 \times 10^{22}}{2 \times 10^{30}}\right)\left(\frac{150 \times 10^6}{0.4 \times 10^6}\right)^3 \times 1 \cong 2$$

**30.** **(0.56)** According to question, $g_h = g_d = g_1$




$$g_h = \frac{GM}{\left(R + \frac{R}{2}\right)^2} \text{ and } g_d = \frac{GM(R-d)}{R^3}$$

$$\frac{GM}{\left(\frac{3R}{2}\right)^2} = \frac{GM(R-d)}{R^3} \Rightarrow \frac{4}{9} = \frac{(R-d)}{R}$$

$$\Rightarrow 4R = 9R - 9d \Rightarrow 5R = 9d$$

$$\therefore \frac{d}{R} = \frac{5}{9}$$

# CHAPTER 8 Mechanical Properties of Solids

1. **(b)** $r\theta = \ell\phi \Rightarrow \phi = \frac{r\theta}{\ell} = \frac{6\text{mm} \times 30°}{1\text{m}} = 0.18°$

2. **(a)** $P = 80$ atm $= 80 \times 1.013 \times 10^5$ Pa

Compressibility $= \frac{1}{B} = 45.8 \times 10^{-11}$/Pa; Density of water at the surface $= \rho = 1.03 \times 10^3$ kg/m$^3$
Let $\rho'$ = density of water at a given depth; $V'$ = Vol. of water of mass $M$ at given depth
$V$ = Vol. of same mass of water at the surface

$\therefore \quad V = \frac{M}{r}$ and $V' = \frac{M}{r'}$

Vol. strain $= \frac{DV}{V} = 1 - \frac{\rho}{\rho'} = 1 - \frac{1.03 \times 10^3}{\rho'}$

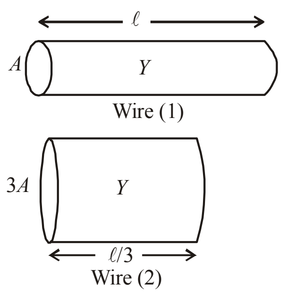
Also $B = \frac{P}{DV/V} \quad \therefore \quad \frac{DV}{V} = \frac{P}{B}$
$= 80 \times 1.013 \times 10^5 \times 45.8 \times 10^{-11} = 3.712 \times 10^{-3}$

$\therefore \quad 1 - \frac{1.03 \times 10^3}{\rho'} = 3.712 \times 10^{-3}$

$\Rightarrow \rho' = \frac{1.03 \times 10^3}{1 - 3.712 \times 10^{-3}} = 1.034 \times 10^3$ kg/m$^3$

3. **(a)** $\tan(90° - \theta) = \frac{\text{stress}}{\text{strain}}$

4. **(c)** As shown in the figure, the wires will have the same Young's modulus (same material) and the length of the wire of area of cross-section $3A$ will be $\ell/3$ (same volume as wire 1).




$$\frac{F}{A} \times \frac{\ell}{\Delta x} = \frac{F'}{3A} \times \frac{\ell}{3\Delta x} \Rightarrow F' = 9F$$

5. **(c)**

6. **(a)** From the graph $l = 10^{-4}$m, $F = 20$ N
$A = 10^{-6}\text{m}^2$, L = 1m

$$\therefore \; Y = \frac{FL}{Al} = \frac{20 \times 1}{10^{-6} \times 10^{-4}}$$

$$= 20 \times 10^{10} = 2 \times 10^{11} \text{ N/m}^2$$

7. **(a)** Force, F = A × Y × strain
$= 1 \times 10^{-4} \times 2 \times 10^{11} \times 0.1$
$= 2 \times 10^6$ N

8. **(b)** Initial length (circumference) of the ring $= 2\pi r$
Final length (circumference) of the ring $= 2\pi R$
Change in length $= 2\pi R - 2\pi r$

$$\text{strain} = \frac{\text{change in length}}{\text{original length}} = \frac{2\pi(R-r)}{2\pi r} = \frac{R-r}{r}$$

Now Young's modulus $E = \frac{F/A}{l/L} = \frac{F/A}{(R-r)/r}$

$$\therefore \; F = AE\left(\frac{R-r}{r}\right)$$

9. **(b)**

10. **(c)** If $\ell$ is the original length of wire, then change in length of wire with tension $T_1$,

$$\Delta\ell_1 = (\ell_1 - \ell)$$

Change in length of wire with tension $T_2$,

$$\Delta\ell_2 = (\ell_2 - \ell)$$

Now, $Y = \frac{T_1}{A} \times \frac{\ell}{\Delta\ell_1} = \frac{T_2}{A} \times \frac{\ell}{\Delta\ell_2}$

$$\Rightarrow \quad \ell = \frac{T_2\ell_1 - T_1\ell_2}{T_2 - T_1}$$

11. **(c)** From formula,

Increase in length $\Delta L = \frac{FL}{AY} = \frac{4FL}{\pi D^2 Y}$

$$\frac{\Delta L_S}{\Delta L_C} = \frac{F_S}{F_C}\left(\frac{D_C}{D_S}\right)^2 \frac{Y_C}{Y_S}\frac{L_S}{L_C} = \frac{7}{5} \times \left(\frac{1}{p}\right)^2 \left(\frac{1}{s}\right) q$$

$$= \frac{7q}{(5sp^2)}$$

**12.** **(c)** Here,

$m = 14.5$ kg, $l = r = 1$m,

$\omega = 2rps = 2 \times 2\pi$ rad/s

$A = 0.065 \times 10^{-4}$ m$^2$

Tension in the wire at the lowest position on the vertical circle $= F = mg + mr\omega^2$

$$= 14.5 \times 9.8 + 14.5 \times 1 \times 4 \times \left(\frac{22}{7}\right)^2 \times 4$$

$$= 142.1 + 2291.6 = 2433.7 \text{ N}$$

$$Y = \frac{Fl}{ADl} \Rightarrow Dl = \frac{Fl}{AY} = \frac{2433.7 \times 1}{0.065 \times 10^{-4} \times 2 \times 10^{11}}$$

$$= 1.87 \times 10^{-3} \text{ m} = 1.87 \text{ mm}$$

**13.** **(a)** For steel wire: Total force $= F_1 = (4 + 6)$ kgwt. $= 10$ kg wt $= 10 \times 9.8$N

$l_1 = 1.5\text{m}$, $r_1 = \frac{0.25}{2}$ cm $= 0.125 \times 10^{-2}$ m;

$Y_1 = 2 \times 10^{11}$ Pa, $Dl_1 = ?$

For brass wire, $F_2 = 6$ kg wt. $= 6 \times 9.8$N, $r_2 = 0.125 \times 10^{-2}$m

$Y_2 = 0.91 \times 10^{11}$ Pa, $l_2 = 1$m

$$\because \quad Y = \frac{Fl}{ADl} \quad \therefore \quad Dl = \frac{Fl}{AY} = \frac{Fl}{pr^2 Y}$$

For steel,

$$Dl_1 = \frac{F_1 l_1}{pr_1^2 Y_1} = \frac{10 \times 9.8 \times 1.5 \times 7}{22 \times (0.125 \times 10^{-2})^2 \times 2 \times 10^{11}}$$

$$= 1.49 \times 10^{-4} \text{ m}$$

For Brass,

$$Dl_2 = \frac{F_2 l_2}{pr_2^2 Y_2} = \frac{6 \times 9.8 \times 1 \times 7}{22 \times (0.125 \times 10^{-2})^2 \times 0.91 \times 10^{11}}$$

$$= 1.3 \times 10^{-4} \text{ m}$$

**14.** **(a)** Given:

$A = 0.1 \times 0.1 = 10^{-2}$ m$^2$ $F = mg = 100 \times 10$N;

$$\text{Shearing strain} = \frac{DL}{L} = \frac{\text{Shearing stress}}{\text{Shear modulus}}$$

$$\therefore \quad \frac{DL}{L} = \frac{F/A}{h} \Rightarrow DL = \frac{FL}{Ah}$$

$$= \frac{100 \times 10 \times 0.1}{10^{-2} \times 25 \times 10^9} \Rightarrow DL = 4 \times 10^{-7} \text{ m}$$

**15.** **(c)** Load on each column $F = \frac{mg}{4}$

$$= \frac{50{,}000 \times 9.8}{4} \text{N}$$

$$A = \pi (r_2^2 - r_1^2) = \frac{22}{7}[(0.60)^2 - (0.30)^2]$$

$$\text{Compressional strain} = \frac{\text{stress}}{Y} = \frac{F/A}{Y} = \frac{F}{AY}$$

$$= \frac{50{,}000 \times 9.8}{4 \times \frac{22}{7} \times [(0.60)^2 - (0.30)^2] \times 2 \times 10^{11}}$$

$$= 7.21 \times 10^{-7}$$

**16.** **(b)** Tension in the wire, $T = \left(\frac{2mM}{m+M}\right)g$

$$\text{Stress} = \frac{\text{Force / Tension}}{\text{Area}} = \frac{2mM}{A(m+M)}g$$

$$= \frac{2(m \times 2m)g}{A(m+2m)} = \frac{4mg}{3A} \quad (M = 2\text{ m given})$$

**17.** **(a)** Here, $r = \frac{6}{2} = 3\text{mm} = 3 \times 10^{-3}$m;

Max. stress $= 6.9 \times 10^7$ Pa

Max. load on a rivet

$=$ Max. stress $\times$ area of cross section

$$= 6.9 \times 10^7 \times \frac{22}{7} \times (3 \times 10^{-3})^2$$

$\therefore$ Max. tension $= 4 \times$ max. load

$$= 4 \times 6.9 \times 10^7 \times \frac{22}{7} \times 9 \times 10^{-6}$$

$$= 7.8 \times 10^3 \text{ N}$$

**18.** **(a)**

**19.** **(d)** $\text{Stress} = \frac{\text{Normal force}}{\text{Area}} = \frac{N}{A} = \frac{N}{(2\pi a)b}$

Stress $=$ B$\times$strain

$$\frac{N}{(2\pi a)b} = B\frac{2\pi a \Delta a \times b}{\pi a^2 b}$$

$$\Rightarrow N = B\frac{(2\pi a)^2 \Delta a b^2}{\pi a^2 b}$$

Force needed to push the cork.

$$f = \mu N = \mu 4\pi b \Delta a B = (4\pi\mu Bb)\Delta a$$

**20.** **(d)** $Y_c \times (\Delta L_c / L_c) = Y_s \times (\Delta L_s / L_s)$

$$\Rightarrow 1 \times 10^{11} \times \left(\frac{1 \times 10^{-3}}{1}\right) = 2 \times 10^{11} \times \left(\frac{\Delta L_s}{0.5}\right)$$

$$\therefore\ \Delta L_s = \frac{0.5\times10^{-3}}{2} = 0.25 \text{ mm}$$

Therefore, total extension of the composite wire

$= \Delta L_c + \Delta L_s = 1\text{ mm} + 0.25\text{ mm} = 1.25\text{ mm}$

**21.** **(0.04)** $r\theta = L\phi \Rightarrow 10^{-2}\times0.8 = 2\times\phi$

$\Rightarrow \phi = 0.004$

**22.** **($2\times10^9$)** $K = \dfrac{P}{\Delta V/V} = \dfrac{h\rho g}{\Delta V/V}$

$= \dfrac{200\times10^3\times10}{0.1/100} = 2\times10^9$

**23.** **(0.03)** If side of the cube is L then

$V = L^3 \Rightarrow \dfrac{dV}{V} = 3\dfrac{dL}{L}$

$\therefore$ % change in volume = 3 × (% change in length)

= 3 × 1% = 3%

$\therefore$ Bulk strain $\dfrac{\Delta V}{V} = 0.03$

**24.** **($2.026\times10^9$)** Here,

$\Delta V = 100.5 - 100 = 0.5$ litre $= 0.5\times10^{-3}$ m$^3$;

$P = 100$ atm $= 100\times1.013\times10^5$ Pa

$V = 100$ litre $= 100\times10^{-3}$m$^3$

Bulk modulus $= B = \dfrac{P}{DV/V} = \dfrac{PV}{\Delta V}$

$= \dfrac{100\times1.013\times10^5\times100\times10^{-3}}{0.5\times10^{-3}}$

$\Rightarrow\ B = 2.026\times10^9$ Pa

**25.** **(200)** Breaking stress $= \dfrac{\text{Force}}{\text{area}}$

The breaking force will be its own weight.

$F = mg = V\rho g = \text{area}\times\ell\rho g$

Breaking stress $= 6\times10^6 = \dfrac{\text{area}\times\ell\times\rho g}{\text{area}}$

or $\quad \ell = \dfrac{6\times10^6}{3\times10^3\times10} = 200\text{ m}.$

**26.** **(4)** Given : Wire length, $l = 0.3$ m

Mass of the body, m = 10 kg

Breaking stress, $\sigma = 4.8\times10^7$ Nm$^{-2}$

Area of cross-section, a = $10^{-2}$ cm$^2$

Maximum angular speed $\omega$ = ?

$T = Ml\omega^2$

$$\sigma = \frac{T}{A} = \frac{ml\omega^2}{A}$$

$$\frac{ml\omega^2}{A} \le 48\times10^7 \Rightarrow \omega^2 \le \frac{\left(48\times10^7\right)A}{ml}$$

$$\Rightarrow \omega^2 \le \frac{\left(48\times10^7\right)\left(10^{-6}\right)}{10\times3} = 16$$

$\Rightarrow \omega_{max} = 4$ rad/s

**27.** **(100)** Breaking force $\alpha$ area of cross section of wire

Load hold by wire is independent of length of the wire.

**28.** **(1.41)** If force $F$ acts along the length L of the wire of cross-section $A$, then energy stored in unit volume of wire is given by

Energy density $= \dfrac{1}{2}$ stress × strain

$= \dfrac{1}{2}\times\dfrac{F}{A}\times\dfrac{F}{AY}$

$\left(\because \text{ stress} = \dfrac{F}{A} \text{ and strain} = \dfrac{X}{AY}\right)$

$= \dfrac{1}{2}\dfrac{F^2}{A^2Y} = \dfrac{1}{2}\dfrac{F^2\times16}{(\pi d^2)^2Y} = \dfrac{1}{2}\dfrac{F^2\times16}{\pi d^4Y}$

If $u_1$ and $u_2$ are the densities of two wires, then

$$\frac{u_1}{u_2} = \left(\frac{d_2}{d_1}\right)^4 \Rightarrow \frac{d_1}{d_2} = (4)^{1/4} \Rightarrow \frac{d_1}{d_2} = \sqrt{2}:1$$

**29.** **(1.75)** $\Delta_1 = \Delta_2$

or $\dfrac{Fl_1}{\pi r_1^2 y_1} = \dfrac{Fl_2}{\pi r_2^2 y_2}$ or $\dfrac{2}{R^2\times7} = \dfrac{1.5}{2^2\times4}$

$\therefore R = 1.75$ mm

**30.** **(1.15)** Stress $= \dfrac{F}{A} = \dfrac{400\times4}{\pi d^2} = 379\times10^6$ N/m$^2$

$$\Rightarrow d^2 = \frac{400\times4}{379\times10^6\pi}$$

$d = 1.15$ mm

CHAPTER 9

# Mechanical Properties of Fluids

**1.** **(d)** Weight of submerged part of the block

$W = \frac{1}{3}v$ (Density of water) g ...(i)

Excess weight, = weight of water having $\frac{2}{3}$ volume of the block.

$W' = \frac{2}{3}v$ (Density of water) g ...(ii)

Dividing (ii) by (i),

$\frac{W'}{W} = \frac{2/3}{1/3}$

$\therefore \; W' = 2W \Rightarrow W' = 2 \times 6 = 12$ kg

**2.** **(b)** The maximum force, which the bigger piston can bear, m = 3000 kg, F = 3000 × 9.8 N

Area of piston, $A = 425\ cm^2 = 425 \times 10^{-4}\ m^2$

$\therefore$ Maximum pressure on the bigger piston,

$P = \frac{F}{A} = \frac{3000 \times 9.8}{425 \times 10^{-4}} = 6.92 \times 10^5$ Pa

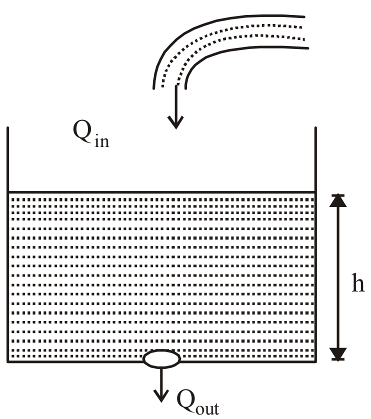
$\therefore$ The maximum pressure the smaller piston can bear is $6.92 \times 10^5$ Pa.

**3.** **(a)**

**4.** **(c)** $P_2 - P_1 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) = 7500\ Nm^{-2}$

**5.** **(d)** Reynold's no. $N_R = \frac{\rho v d}{\eta}$ and $V = \frac{4Q}{d^2\pi}$

**6.** **(a)**




Since height of water column is constant therefore, water inflow rate ($Q_{in}$)
= water outflow rate

$Q_{in} = 10^{-4}\ m^3s^{-1}$

$Q_{out} = Au = 10^{-4} \times \sqrt{2gh}$

$\therefore\ 10^{-4} = 10^{-4} \times \sqrt{20 \times h}$

$\therefore\ h = \frac{1}{20}\ m = 5$ cm

**7.** **(c)** Volume of 8 small droplets = Volume of 1 big drop

$\Rightarrow \frac{4}{3}\pi r^3 \times 8 = \frac{4}{3}\pi R^3 \quad \Rightarrow r = \frac{R}{(8)^{1/3}}$

Work done = (Change in area) × surface tension

$= (4\pi r^2 n - 4\pi R^2)T$

$= \left\{4\pi\left(\frac{R^2}{(8)^{2/3}}\right) \times 8 - 4\pi R^2\right\}T = 4\pi R^2 T$

**8.** **(a)** Since, soap film has two free surfaces, so total length of the film $= 2\ell = 2 \times 30 = 60$ cm = 0.6 m.

Total force on the slider due to surface tension,

$F = S \times 2\ell = S \times 0.6$ N

In equilibrium, F = mg

$\therefore\ S \times 0.6 = 1.5 \times 10^{-2} \Rightarrow S = \frac{1.5 \times 10^{-2}}{0.6} = 2.5 \times 10^{-2}\ Nm^{-1}$

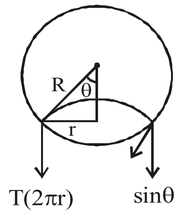
**9.** **(a)** Weight of the liquid column $= T\cos\theta \times 2\pi r$.

For water θ= 0°. Here weight of liquid column $W = 7.5 \times 10^{-4}$N and $T = 6 \times 10^{-2}$N/m. Then circumference, $2\pi r = W/T = 1.25 \times 10^{-2}$m

**10.** **(b)** When the bubble gets detached,

Bouyant force = force due to surface tension

$2\pi r T \sin\theta = \frac{4}{3}\pi R^3 \rho_\omega g$




$$\Rightarrow \ T \times \frac{r}{R} \times 2\pi r = \frac{4}{3}\pi R^3 \rho_\omega g$$

or, $$\frac{2T}{R}(\pi r^2) = \frac{4\pi R^3}{3T}\rho_w g$$

$$\Rightarrow r = R^2\sqrt{\frac{2\rho_w g}{3T}}$$

**11.** **(a)**

**12.** **(a)** As $A_1 v_1 = A_2 v_2$ (Principle of continuity)

or, $\ell^2\sqrt{2gh} = \pi r^2\sqrt{2g \times 4h}$

(Efflux velocity $= \sqrt{2gh}$ )

$$\therefore \quad r^2 = \frac{\ell^2}{2\pi} \quad \text{or} \quad r = \sqrt{\frac{\ell^2}{2\pi}} = \frac{\ell}{\sqrt{2\pi}}$$

**13.** **(c)** Given,
Density of gold, $\rho_G = 19.5\ \text{kg/m}^3$
Density of silver, $\rho_5 = 10.5\text{kg/m}^3$
Density of liquid, $\sigma = 1.5\text{kg/m}^3$

Terminal velocity, $$v_T = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

$$\therefore \frac{v_{T_2}}{0.2} = \frac{(10.5-1.5)}{(19.5-1.5)} \Rightarrow v_{T_2} = 0.2 \times \frac{9}{18}$$

$$\therefore v_{T_2} = 0.1\ \text{m/s}$$

**14.** **(d)** The volume of liquid flowing through both the tubes i.e., rate of flow of liquid is same.
Therefore, $V = V_1 = V_2$

i.e., $$\frac{\pi P_1 r_1^4}{8\eta l_1} = \frac{\pi P_2 r_2^4}{8\eta l_2}$$

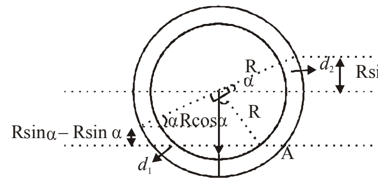
or $$\frac{P_1 r_1^4}{l_1} = \frac{P_2 r_2^4}{l_2}$$

$\because \quad P_2 = 4\ P_1$ and $l_2 = l_1/4$

$$\frac{P_1 r_1^4}{l_1} = \frac{4P_1 r_2^4}{l_1/4} \Rightarrow r_2^4 = \frac{r_1^4}{16}$$

$$r_2 = r_1/2$$

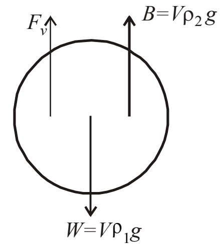
**15.** **(c)** Pressure at interface A must be same from both the sides to be in equilibrium.




$$\therefore \ (R\cos\alpha + R\sin\alpha)d_2 g$$
$$= (R\cos\alpha - R\sin\alpha)d_1 g$$

$$\Rightarrow \frac{d_1}{d_2} = \frac{\cos\alpha + \sin\alpha}{\cos\alpha - \sin\alpha} = \frac{1+\tan\alpha}{1-\tan\alpha}$$

**16.** **(a)** When the ball attains terminal velocity
Weight of the ball = Buoyant force + Viscous force

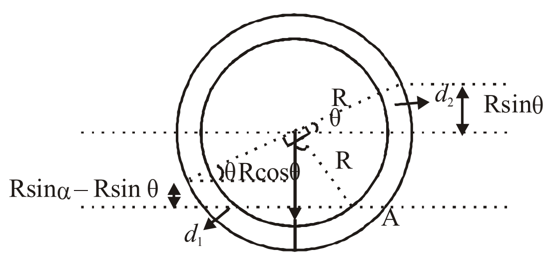



$$\therefore \ V\rho_1 g = V\rho_2 g + kv_t^2 \Rightarrow Vg(\rho_1 - \rho_2)g = kv_t^2$$

$$\Rightarrow v_t = \sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$$

**17.** **(d)** As liquid 1 floats over liquid 2. The lighter liquid floats over heavier liquid. So, $\rho_1 < \rho_2$
Also $\rho_3 < \rho_2$ because the ball of density $\rho_3$ does not sink to the bottom of the jar.
Also $\rho_3 > \rho_1$ otherwise the ball would have floated in liquid 1. we conclude that
$\rho_1 < \rho_3 < \rho_2$.

**18.** **(d)** Pressure at interface A must be same from both the sides to be in equilibrium.




$$\therefore \quad (R\cos\theta + R\sin\theta)\rho_2 g$$

$$= (R\cos\theta - R\sin\theta)\rho_1 g$$

$$\Rightarrow \frac{d_1}{d_2} = \frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} = \frac{1+\tan\theta}{1-\tan\theta}$$

$$\Rightarrow r_1 - \rho_1 \tan\theta = \rho_2 + \rho_2 \tan\theta$$

$$\Rightarrow (\rho_1 + \rho_2)\tan\theta = \rho_1 - \rho_2$$

$$\therefore \quad \theta = \tan^{-1}\left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_2}\right)$$

**19.** **(c)**

**20.** **(a)** Weight of cylinder = upthrust due to both liquids

$$V \times D \times g = \left(\frac{A}{5} \times \frac{3}{4}L\right) \times d \times g + \left(\frac{A}{5} \times \frac{L}{4}\right) \times 2d \times g$$

$$\Rightarrow \left(\frac{A}{5} \times L\right) \times D \times g = \frac{A \times L \times d \times g}{4} \Rightarrow \frac{D}{5} = \frac{d}{4}$$

$$\therefore D = \frac{5}{4}d$$

**21.** **(6)** Initially, the pressure of air column above water is $P_1 = 10^5\ \text{Nm}^{-2}$ and volume $V_1 = (500 - H)A$, where $A$ is the area of cross-section of the vessel.

Finally, the volume of air column above water is 300 A. If $P_2$ is the pressure of air then

$$P_2 + \rho g h = 10^5 \quad \therefore \quad P_2 + 10^3 \times 10 \times \frac{200}{1000} = 10^5$$

$$\therefore \quad P_2 = 9.8 \times 10^4\ N/m^2$$

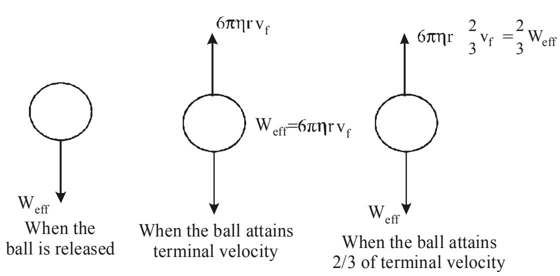
As the temperature remains constant

$$10^5 \times (500 - H)A = (9.8 \times 10^4) \times 300A$$

$\Rightarrow$ H = 206 mm

$\therefore$ The fall of height of water level due to the opening of orifice = 206 – 200 = 6 mm

**22.** **(3)**




When the ball is just released, the net force on ball is $W_{eff}$ (= mg – buoyant force)

The terminal velocity $v_f$ of the ball is attained when net force on the ball is zero.

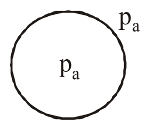
$\therefore$ Viscous force $6\pi\eta r\, v_f = W_{eff}$

When the ball acquires $\frac{2}{3}$rd of its maximum velocity $v_f$ the viscous force is $= \frac{2}{3}W_{eff}$

Hence net force is $W_{eff} - \frac{2}{3}W_{eff} = \frac{1}{3}W_{eff}$

$\therefore$ Required acceleration is a/3

**23.** **(8)** Inside pressure must be $\frac{4T}{r}$ greater than outside pressure in bubble.




This excess pressure is provided by charge on bubble.

$$\frac{4T}{r} = \frac{\sigma^2}{2\varepsilon_0}$$

$$\Rightarrow \frac{4T}{r} = \frac{Q^2}{16\pi^2 r^4 \times 2\varepsilon_0} \quad \left[\sigma = \frac{Q}{4\pi r^2}\right]$$

$$Q = 8\pi r\sqrt{2rT\varepsilon_0}$$

**24.** **(20)** Water fills the tube entirely in gravityless condition i.e., 20 cm.

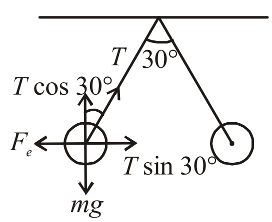
**25.** **(20)** Given, Height of cylinder, h=20 cm
Acceleration due to gravity, g=10 $ms^{-2}$
Velocity of efflux

$$v = \sqrt{2gh}$$

Where $h$ is the height of the free surface of liquid from th e hole

$$\Rightarrow v = \sqrt{2 \times 10 \times 20} = 20\,\text{m/s}$$

**26.** **(2)**



$F_e = T\sin 30°$

$mg = T\cos 30°$

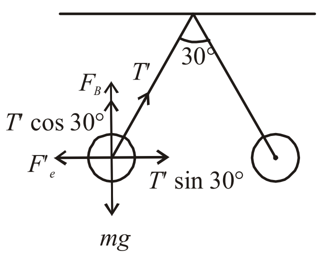
$\Rightarrow \tan 30° = \frac{F_e}{mg}$ ...(1)

In liquid,

$F_e' = T'\sin 30°$ ...(A)

$mg = F_B + T'\cos 30°$

But $F_B$ = Buoyant force




$= V(d-\rho)g = V(1.6-0.8)g = 0.8\,Vg$

$= 0.8\frac{m}{d}g = \frac{0.8mg}{1.6} = \frac{mg}{2}$

$\therefore mg = \frac{mg}{2} + T'\cos 30°$

$\Rightarrow \frac{mg}{2} = T'\cos 30°$ ...(B)

From (A) and (B), $\tan 30° = \frac{2F_e'}{mg}$

From (1) and (2)

$\frac{F_e}{mg} = \frac{2F_e'}{mg}$ (2)

$\Rightarrow F_e = 2F_e'$

If $K$ be the dielectric constant, then

$F_e' = \frac{F_e}{K}$

$\therefore F_e = \frac{2F_e}{K} \Rightarrow K = 2$

**27.** **(101)**

Given : Radius of capillary tube,

$r = 0.015$ cm $= 15 \times 10^{-5}$ mm

$h = 15$ cm $= 15 \times 10^{-2}$ mm

Using, $h = \frac{2T\cos\theta}{\rho g r}$ $[\cos\theta = \cos 0° = 1]$

Surface tension,

$T = \frac{rh\rho g}{2} = \frac{15\times10^{-5}\times15\times10^{-2}\times900\times10}{2}$

$= 101$ milli newton $m^{-1}$

**28.** **(0.1)** **Given:** Radius of air bubble,

$r = 0.1$ cm $= 10^{-3}$ m

Surface tension of liquid,

$S = 0.06$ N/m $= 6 \times 10^{-2}$ N/m

Density of liquid, $\rho = 10^3$ kg/m$^3$

Excess pressure inside the bubble,

$\rho_{exe} = 1100$ Nm$^{-2}$

Depth of bubble below the liquid surface,

$h = ?$

As we know,

$\rho_{Excess} = h\rho g + \frac{2s}{r}$

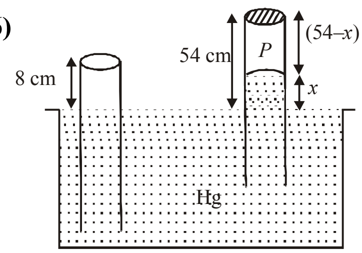
$\Rightarrow 1100 = h \times 10^3 \times 9.8 + \frac{2\times6\times10^{-2}}{10^{-3}}$

$\Rightarrow 1100 = 9800\,h + 120$

$\Rightarrow 9800h = 1100 - 120$

$\Rightarrow h = \frac{980}{9800} = 0.1$ m

**29.** **(16)**




Length of the air column above mercury in the tube is,

$P + x = P_0$

$\Rightarrow P = (76 - x)$

$\Rightarrow 8 \times A \times 76 = (76 - x) \times A \times (54 - x)$

$\therefore x = 38$

Thus, length of air column

$= 54 - 38 = 16$ cm.

**30.** **($10^{-2}$)** $\eta = 10^{-2}$ poise; $v = 18$ km/h $= \frac{18000}{3600} = 5$ m/s, $l = 5$ m

Strain rate $= \frac{v}{l}$

$\therefore$ Shearing stress $= \eta \times$ strain rate

$= 10^{-2} \times \frac{5}{5} = 10^{-2}$ Nm$^{-2}$

# CHAPTER 10

# Thermal Properties of Matter

**1.** **(c)** The lengths of each rod increases by the same amount

$\therefore \quad \Delta\ell_a = \Delta\ell_s \Rightarrow \ell_1\alpha_a t = \ell_2\alpha_s t$

$\Rightarrow \dfrac{\ell_1}{\ell_1+\ell_2} = \dfrac{\alpha_s}{\alpha_a+\alpha_s}$

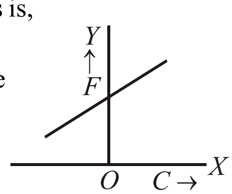
**2.** **(a)** We know that,

$\dfrac{C}{100} = \dfrac{F-32}{180}$ or $F = \dfrac{9}{5}C + 32$

Equation of straight lines is,

$y = mx + c$

Hence, $m = (9/5)$, positive and $c = 32$ positive.
The graph is shown in figure.




**3.** **(b)** Young's modulus

$= \dfrac{\text{Thermal stress}}{\text{Strain}} = \dfrac{F/A}{\Delta L/L}$

$Y = \dfrac{F}{A.\alpha.\Delta\theta} \qquad \left(Q\,\dfrac{\Delta L}{L} = \alpha\,\Delta\theta\right)$

Force developed in the rail $F = YA\alpha\Delta\theta$

$= 2 \times 10^{11} \times 40 \times 10^{-4} \times 1.2 \times 10^{-5} \times 10$

$= 9.6 \times 10^4 \simeq 1 \times 10^5$ N

**4.** **(b)** Due to volume expansion of both liquid and vessel, the change in volume of liquid relative to container is given by

$\Delta V = V_0[\gamma_L - \gamma_g]\Delta\theta$

Given $V_0 = 1000cc, \alpha_g = 0.1\times10^{-4}/°C$

$\therefore \gamma_g = 3\alpha_g = 3\times0.1\times10^{-4}/°C$

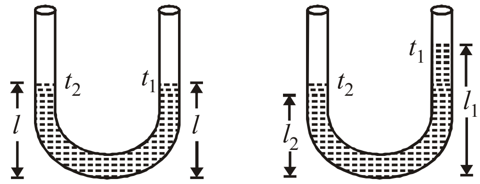
$= 0.3\times10^{-4}/°C$

$\therefore \Delta V = 1000[1.82\times10^{-4} - 0.3\times10^{-4}]\times100$

$= 15.2cc$

**5.** **(b)**

**6.** **(a)** Suppose, height of liquid in each arm before rising the temperature is $l$.




With temperature rises height of liquid in each arm increases *i.e*, $l_1 > l$ and $l_2 > l$

Also $l = \dfrac{l_1}{1+\gamma t_1} = \dfrac{l_2}{1+\gamma t_2}$

$\Rightarrow l_1 + \gamma l_1 t_2 = l_2 + \gamma l_2 t_1 \Rightarrow \gamma = \dfrac{l_1 - l_2}{l_2 t_1 - l_1 t_2}.$

**7.** **(d)** According to principle of calorimetry,

$Q_{given} = Q_{used}$

$0.2 \times S \times (150-40) = 150 \times 1 \times (40-27) + 25 \times (40-27)$

$S = \dfrac{13\times25\times7}{0.2\times110} = 434$ J/kg-°C

**8.** **(c)** Applying Wein's displacement law,

$\lambda_m T =$ constant

$5000$ Å $\times (1227+273) = (2227+273) \times \lambda_m$

$\lambda_m = \dfrac{5000\times1500}{2500} = 3000$Å

**9.** **(d)** $t \propto \dfrac{\ell}{A}$ and $t' \propto \dfrac{2\ell}{A/2} \qquad \Rightarrow \dfrac{t'}{t} = 4\dfrac{\ell/A}{\ell/A}$

$t' = 4 \times t \Rightarrow t' = 48s$

**10.** **(c)** Heat lost by He = Heat gained by $N_2$

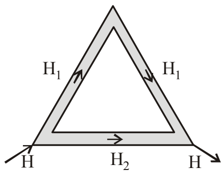
$n_1 C_{v_1}\Delta T_1 = n_2 C_{v_2}\Delta T_2$

$\dfrac{3}{2}R\left[\dfrac{7}{3}T_0 - T_f\right] = \dfrac{5}{2}R\left[T_f - T_0\right]$

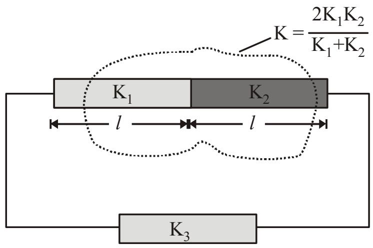
$7T_0 - 3T_f = 5T_f - 5T_0 \Rightarrow T_f = \dfrac{3}{2}T_0.$

**11.** **(c)**

**12.** **(c)**




The given arrangement of rods can be redrawn as follows




It is given that $H_1 = H_2$

$$\Rightarrow \frac{KA(\theta_1 - \theta_2)}{2l} = \frac{K_3 A(\theta_1 - \theta_2)}{l}$$

$$\Rightarrow K_3 = \frac{K}{2} = \frac{K_1 K_2}{K_1 + K_2}$$

**13.** **(b)** By Newton's law of cooling,

$$\frac{\theta_1 - \theta_2}{t} = -k\left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right] \quad ....(1)$$

A sphere cools from 62°C to 50°C in 10 min.

$$\frac{62 - 50}{10} = -k\left[\frac{62 + 50}{2} - \theta_0\right] \quad ....(2)$$

Now, sphere cools from 50°C to 42°C in next 10 min.

$$\frac{50 - 42}{10} = -k\left[\frac{50 + 42}{2} - \theta_0\right] \quad ....(3)$$

Dividing eq$^n$. (2) by (3) we get,

$$\frac{56 - \theta_0}{46 - \theta_0} = \frac{1.2}{0.8} \text{ or } 0.4\theta_0 = 10.4 \text{ hence } \theta_0 = 26°C$$

**14.** **(d)** Consider a concentric spherical shell of thickness ($dr$) and of radiius ($r$) and let the temperature of inner and outer surfaces of this shell be $T$ and $(T - dT)$ respectively.

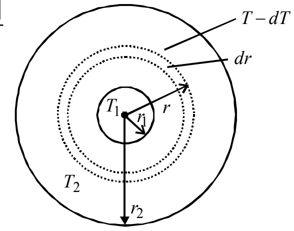
$\frac{dQ}{dt}$ = rate of flow of heat through it

$$= \frac{KA[(T - dT) - T]}{dr}$$

$$= \frac{-KAdT}{dr}$$

$$= -4\pi K r^2 \frac{dT}{dr}$$

$(\because A = 4\pi r^2)$




To measure the radial rate of heat flow, integration technique is used, since the area of the surface through which heat will flow is not constant.

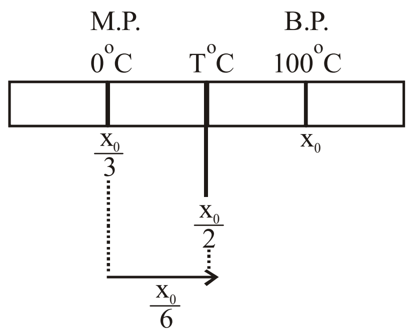
$$\text{Then, } \left(\frac{dQ}{dt}\right)\int_{r_1}^{r_2} \frac{1}{r^2} dr = -4\pi K \int_{T_1}^{T_2} dT$$

$$\frac{dQ}{dt}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = -4\pi K\left[T_2 - T_1\right]$$

$$\text{or} \quad \frac{dQ}{dt} = \frac{-4\pi K r_1 r_2 (T_2 - T_1)}{(r_2 - r_1)}$$

$$\therefore \quad \frac{dQ}{dt} \propto \frac{r_1 r_2}{(r_2 - r_1)}$$

**15.** **(a)** Let required temperature = T°C




$$\Rightarrow T°C = \frac{x_0}{2} - \frac{x_0}{3} = \frac{x_0}{6}$$

$$\& \left(x_0 - \frac{x_0}{3}\right) = (100 - 0°C)$$

$$\Rightarrow \frac{2x_0}{3} = 100 \Rightarrow x_0 = \frac{300}{2}$$

$$\Rightarrow T°C = \frac{x_0}{6} = \frac{150}{6} = 25°C$$

**16.** **(a)** Let $Q$ be the temperature at a distance $x$ from hot end of bar. Let $Q$ is the temperature of hot end.

The heat flow rate is given by

$$\frac{dQ}{dt} = \frac{kA(\theta_1 - \theta)}{x}$$

$$\Rightarrow \theta_1 - \theta = \frac{x}{kA}\frac{dQ}{dt} \Rightarrow \theta = \theta_1 - \frac{x}{kA}\frac{dQ}{dt}$$

Thus, the graph of $Q$ versus $x$ is a straight line with a positive intercept and a negative slope.
The above equation can be graphically represented by option (a).

**17.** **(b)** As 1g of steam at 100°C melts 8g of ice at 0°C.
10 g of steam will melt 8× 10 g of ice at 0°C
Water in calorimeter = 500 + 80 + 10g = 590g

**18.** **(a)** Change in length in both rods are same i.e.

$$\Delta\ell_1 = \Delta\ell_2$$

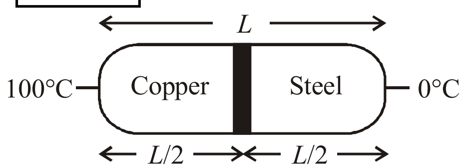
$$\ell\alpha_1\Delta\theta_1 = \ell\alpha_2\Delta\theta_2$$

$$\frac{\alpha_1}{\alpha_2} = \frac{\Delta\theta_2}{\Delta\theta_1} \qquad \left[\because \frac{\alpha_1}{\alpha_2} = \frac{4}{3}\right]$$

$$\frac{4}{3} = \frac{\theta - 30}{180 - 30}$$

$$\boxed{\theta = 230°C}$$

**19.** **(a)**




Let conductivity of steel $K_{steel} = k$ then from question
Conductivity of copper $K_{copper} = 9k$
$\theta_{copper} = 100°C$
$\theta_{steel} = 0°C$
$l_{steel} = l_{copper} = \frac{L}{2}$
From formula temperature of junction;

$$\theta = \frac{K_{copper}\theta_{copper}\, l_{steel} + K_{steel}\,\theta_{steel}\, l_{copper}}{K_{copper}\, l_{steel} + K_{steel}\, l_{copper}}$$

$$= \frac{9k \times 100 \times \frac{L}{2} + k \times 0 \times \frac{L}{2}}{9k \times \frac{L}{2} + k \times \frac{L}{2}}$$

$$= \frac{\frac{900}{2}kL}{\frac{10kL}{2}} = 90°C$$

**20.** **(a)** According to newton's law of cooling

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

$$\Rightarrow \frac{d\theta}{(\theta - \theta_0)} = -kdt$$

$$\Rightarrow \int_{\theta_0}^{\theta} \frac{d\theta}{(\theta - \theta_0)} = -k\int_{\theta}^{t} dt$$

$$\Rightarrow \log(\theta - \theta_0) = -kt + c$$

Which represents an equation of straight line.
Thus the option (a) is correct.

**21.** **($9 \times 10^{-6}$)** $\gamma_{real} = \gamma_{app.} + \gamma_{vessel}$

So $(\gamma_{app.} + \gamma_{vessel})_{glass} = (\gamma_{app} + \gamma_{vessel})_{steel}$

$$\Rightarrow 153 \times 10^{-6} + (\gamma_{vessel})_{glass}$$

$$= 144 \times 10^{-6} + (\gamma_{vessel})_{steel}$$

Further,

$$(\gamma_{vessel})_{steel} = 3\alpha = 3 \times (12 \times 10^{-6})$$

$$= 36 \times 10^{-6} / °C$$

$$\Rightarrow 153 \times 10^{-6} + (\gamma_{vessel})_{glass}$$

$$= 144 \times 10^{-6} + 36 \times 10^{-6}$$

$$\Rightarrow (\gamma_{vessel})_{glass} = 3\alpha = 27 \times 10^{-6} / °C$$

$$\Rightarrow \alpha = 9 \times 10^{-6} / °C$$

**22.** **(25)** Time lost/gained per day $= \frac{1}{2}\alpha\,\Delta\theta \times 86400$ second

$$12 = \frac{1}{2}\alpha(40 - \theta) \times 86400 \qquad \text{.... (i)}$$

$$4 = \frac{1}{2}\alpha(\theta - 20) \times 86400 \qquad \text{....(ii)}$$

On dividing we get, $3 = \frac{40 - \theta}{\theta - 20}$

$$3\theta - 60 = 40 - \theta$$

$$4\theta = 100 \Rightarrow \theta = 25°C$$

**23.** **(0.33)** For slab in series, we have

$$R_{eq} = R_1 + R_2 = \frac{x}{KA} + \frac{4x}{2KA} = \frac{3x}{KA}$$

Now, in a steady state rate of heat transfer through the slab is given by

$$\frac{dQ}{dt} = \frac{T_2 - T_1}{R_{eq}} = \frac{(T_2 - T_1)}{3x}KA \qquad \text{...(i)}$$

Given $\frac{dQ}{dt} = \left(\frac{A(T_2 - T_1)K}{x}\right) f$ ...(ii)

Comparing (i) and (ii), we get f = 1/3

**24.** **(10)** Rate of cooling $\propto$ temperature difference between system and surrounding.

As the temperature difference is halved, so the rate of cooling will also be halved.

So time taken will be doubled

**25.** **(20.00)**

Volume capacity of beaker, $V_0 = 500$ cc

$$V_b = V_0 + V_0 \gamma_{\text{beaker}} \Delta T$$

When beaker is partially filled with $V_m$ volume of mercury,

$$V_b^1 = V_m + V_m \gamma_m \Delta T$$

Unfilled volume $(V_0 - V_m) = (V_b - V_m^1)$

$$\Rightarrow V_0 \gamma_{\text{beaker}} = V_m \gamma_M$$

$$\therefore V_m = \frac{V_0 \gamma_{\text{beaker}}}{\gamma_M}$$

or, $V_m = \frac{500 \times 6 \times 10^{-6}}{1.5 \times 10^{-4}} = 20$ cc.

**26.** **(40)** Using the principal of calorimetry

$M_{\text{ice}} L_f + m_{\text{ice}} (40 - 0)\, C_w$

$= m_{\text{stream}} L_v + m_{\text{stream}} (100 - 40)\, C_w$

$\Rightarrow M(540) + M \times 1 \times (100 - 40)$

$= 200 \times 80 + 200 \times 1 \times 40$

$\Rightarrow 600\,M = 24000$

$\Rightarrow M = 40g$

**27.** **(1)** From Newton's law of cooling

$$-\frac{dQ}{dt} \propto (\Delta\theta)$$

**28.** **(64)** From stefan's law, energy radiated by sun per second

$E = \sigma A T^4$ ;

$\therefore A \propto R^2$

$\therefore E \propto R^2 T^4$

$$\therefore \frac{E_2}{E_1} = \frac{R_2^2\, T_2^4}{R_1^2\, T_1^4}$$

put $R_2 = 2R, R_1 = R$ ; $T_2 = 2T, T_1 = T$

$$\Rightarrow \frac{E_2}{E_1} = \frac{(2R)^2 (2T)^4}{R^2 T^4} = 64$$

**29.** **(1)** From stefan's law, the energy radiated per second is given by $E = e\sigma T^4 A$

Here, $T$ = temperature of the body

$A$ = surface area of the body

For same material e is same. $\sigma$ is stefan's constant

Let $T_1$ and $T_2$ be the temperature of two spheres. $A_1$ and $A_2$ be the area of two spheres.

$$\therefore \frac{E_1}{E_2} = \frac{T_1^4 A_1}{T_2^4 A_2} = \frac{T_1^4\, 4\pi r_1^2}{T_2^4\, 4\pi r_2^2}$$

$$= \frac{(4000)^4 \times 1^2}{(2000)^4 \times 4^2} = \frac{1}{1}$$

**30.** **(6.28)** $\Delta_{\text{temp}} = \Delta_{\text{load}}$

and $A = \pi r^2 = \pi(10^{-3})^2 = \pi \times 10^{-6}$

$$L\, \alpha\, \Delta T = \frac{FL}{AY}$$

or $\quad 0.2 \times 10^{-5} \times 20 = \frac{F \times 0.2}{(\pi \times 10^{-6}) \times 10^{11}}$

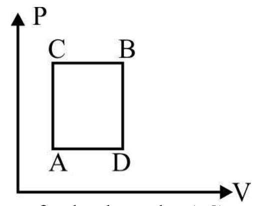
$\therefore \quad F = 20\pi\, N \;\therefore\; m = \frac{f}{g} = 2\pi = 6.28$ kg

# Chapter 11

# Thermodynamics

1. **(b)** Work done is not a thermodynamical function.

2. **(a)**




$\Delta U$ remains same for both paths ACB and ADB

$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$

$\Rightarrow \quad 60\text{ J} = 30\text{ J} + \Delta U_{ACB}$

$\Rightarrow \quad U_{ACB} = 30\text{ J}$

$\therefore \quad \Delta U_{ADB} = \Delta U_{ACB} = 30\text{ J}$

$\Delta Q_{ADB} = \Delta U_{ADB} + \Delta W_{ADB}$

$= 10\text{ J} + 30\text{ J} = 40\text{ J}$

3. **(c)** $Q = mL = 1 \times L = L$; $W = P(V_2 - V_1)$

Now $Q = \Delta U + W$

or $\quad L = \Delta U + P(V_2 - V_1)$

$\therefore \quad \Delta U = L - P(V_2 - V_1)$

4. **(a)** $dQ = dU + dW$

or $nCdT = nC_v dT + PdV$

$$\therefore \quad C = C_v + \frac{P}{n}\left(\frac{dV}{dT}\right)$$

Differentiating $TV^2 =$ constant, w.r.t. $T$, we get

$$\frac{dV}{dT} = -\frac{V}{2T}$$

Also, $PV = nRT \ \Rightarrow \ \dfrac{P}{n} = \dfrac{RT}{V}$

Now, $C = Cv + \dfrac{RT}{V} \times \left(-\dfrac{V}{2T}\right) = \dfrac{3R}{2} - \dfrac{R}{2} = R.$

5. **(c)** Curve A, B shows expansion. For expansion of a gas,

$W_{isothermal} > W_{adiabatic}$

$P_{isothermal} > P_{adiabatic}$

$T_{isothermal} > T_{adiabatic}$

6. **(a)** $T_1 = T$, $W = 6R$ joules, $\gamma = \dfrac{5}{3}$

$$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nRT_1 - nRT_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

$$n = 1,\ T_1 = T \Rightarrow \frac{R(T - T_2)}{5/3 - 1} = 6R$$

$\Rightarrow \quad T_2 = (T - 4)\text{K}$

7. **(b)** Suppose amount of water evaporated be M gram.

Then (150 – M) gram water converted into ice.

So, heat consumed in evoporation = Heat released in fusion

$M \times L_v = (150 - M) \times L_s$

$M \times 2.1 \times 10^6 = (150 - M) \times 3.36 \times 10^5$

$\Rightarrow \quad M \simeq 20\text{ g}$

8. **(d)** Efficiency of engine $A$, $\eta_1 = 1 - \dfrac{T}{T_1}$,

Efficiency of engine $B$, $\eta_2 = 1 - \dfrac{T_2}{T}$

Here, $\eta_1 = \eta_2 \quad \therefore \ \dfrac{T}{T_1} = \dfrac{T_2}{T} \Rightarrow T = \sqrt{T_1T_2}$

9. **(a)** $\dfrac{Q_2}{Q_1} = \dfrac{T_2}{T_1}$

or $\quad Q_2 = \dfrac{T_2Q_1}{T_1} = \dfrac{375 \times 600}{500} = 450\text{ J}$

10. **(a)**

11. **(a)** Initially the efficiency of the engine was $\dfrac{1}{6}$ which increases to $\dfrac{1}{3}$ when the sink temperature reduces by 62ºC.

$\eta = \dfrac{1}{6} = 1 - \dfrac{T_2}{T_1}$, when $T_2 =$ sink temperature

$T_1 =$ source temperature

$\Rightarrow \quad T_2 = \dfrac{5}{6}T_1$

Secondly,

$$\frac{1}{3} = 1 - \frac{T_2 - 62}{T_1} = 1 - \frac{T_2}{T_1} + \frac{62}{T_1} = 1 - \frac{5}{6} + \frac{62}{T_1}$$

or, $\quad T_1 = 62 \times 6 = 372\text{K} = 372 - 273 = 99^{\circ}\text{C}$

& $\quad T_2 = \dfrac{5}{6} \times 372 = 310\text{ K} = 310 - 273 = 37^{\circ}\text{C}$

**12.** **(b)** Adiabatic modulus of elasticity of gas,

$K_a = -\gamma P$

Isothermal, $K_i = -P$

$$\therefore \quad K_i = \frac{K_a}{\gamma} = \frac{K_a}{C_p/C_v}$$

$$= \frac{2.1\times10^5}{1.4} = 1.5\times10^5\ N/m^2 . \left[\because \gamma = \frac{C_p}{C_v}\right]$$

**13.** **(a)** $PV^{3/2} = K$, $\log P + \frac{3}{2}\log V = \log K$

$$\frac{\Delta P}{P} + \frac{3}{2}\frac{\Delta V}{V} = 0$$

$$\frac{\Delta V}{V} = -\frac{2}{3}\frac{\Delta P}{P} \text{ or } \frac{\Delta V}{V} = \left(-\frac{2}{3}\right)\left(\frac{2}{3}\right) = -\frac{4}{9}$$

**14.** **(a)** For a perfect gas, $PV = \mu RT$

$$P_1 = \frac{\mu RT}{V} = \frac{2\times8.31\times(273+27)}{20\times10^{-3}}$$

$P_1 = 2.5\times10^5\ N/m^2$

At constant pressure, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$$\therefore\ T_2 = \left(\frac{V_2}{V_1}\right)T_1 = 2\times300 = 600\ K$$

The gas now undergoes an adiabatic change.

$T_1 = 600\,K$, $T_2 = 300\,K$, $V_1 = 40$ lit., $V_2 = ?$

$\gamma - 1 = 5/3 - 1 = 2/3$

$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$600\,(40)^{2/3} = 300\,(V_2)^{2/3}$

$(2)^{3/2}\times 40 = V_2$ or $V_2 = 112.4$ lit.

**15.** **(a)** $U = a + bPV$

In adiabatic change,

$$dU = -dW = \frac{nR}{\gamma-1}(T_2 - T_1) = \frac{nR}{\gamma-1}(dT)$$

$$\therefore \quad b = \frac{1}{\gamma-1} \Rightarrow \gamma = \frac{b+1}{b}.$$

**16.** **(d)** Work done by the system in the cycle

= Area under P-V curve and V-axis

$$= \frac{1}{2}(2P_0 - P_0)(2V_0 - V_0) + \left[-\left(\frac{1}{2}\right)(3P_0 - 2P_0)(2V_0 - V_0)\right]$$

$$= \frac{P_0V_0}{2} - \frac{P_0V_0}{2} = 0$$

**17.** **(b)** In *VT* graph

ab-process : Isobaric, temperature increases.

bc process : Adiabatic, pressure decreases.

cd process : Isobaric, volume decreases.

da process : Adiabatic, pressure increases.

The above processes correctly represented in *P-V* diagram (b).

**18.** **(d)** For isothermal process :

$PV = P_i\,2V$

$P = 2P_i$ ...(i)

For adiabatic process $PV^\gamma = P_a(2V)^\gamma$

($\because$ for monatomic gas $\gamma = 5/3$)

or, $2P_iV^{\frac{5}{3}} = P_a(2V)^{\frac{5}{3}}$ [From (i)]

$$\Rightarrow \frac{P_a}{P_i} = \frac{2}{2^{\frac{5}{3}}} \Rightarrow \frac{P_a}{P_i} = 2^{\frac{-2}{3}}$$

**19.** **(b)** Let the initial pressure of the three samples be $P_A$, $P_B$ and $P_C$ then

$$P_A(V)^{3/2} = (2V)^{3/2}P \qquad (\because P_B = P)$$

or $P_A = P(2)^{3/2}$

$P_C(V) = P(2V)$ or $P_C = 2P$ $\therefore$ $P_A : P : P_C$

$$= (2)^{3/2} : 1 : 2 = 2\sqrt{2} : 1 : 2$$

**20.** **(d)** $T_2 = 7°C = (7+273) = 280\,K$

$$\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = 1 - \eta$$

$$= 1 - \frac{50}{100} = \frac{50}{100} = \frac{1}{2}$$

$\therefore T_1 = 2\times T_2 = 2\times280 = 560\,K$

New efficiency, $\eta' = 70\%$

$$\therefore \frac{T_2}{T_1} = 1 - \eta' = 1 - \frac{70}{100} = \frac{30}{100} = \frac{3}{10}$$

$$\therefore T_1' = \frac{10}{3}\times280 = \frac{2800}{3} = 933.3\,K$$

$\therefore$ Increase in the temperature of high temp. reservoir $= 933.3 - 560 = 373.3\,K$

**21.** **(402)** $T = 27°C = 300\,K$

$$\gamma = \frac{5}{3};\quad V_2 = \frac{8}{27}V_1;\quad \frac{V_1}{V_2} = \frac{27}{8}$$

From adiabatic process we know that

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{27}{8}\right)^{\frac{5}{3}-1}$$

$$\frac{T_2}{T_1} = \frac{9}{4} \Rightarrow T_2 = \frac{9}{4}\times T_1 = \frac{9}{4}\times 300 = 675\,K$$

$$T_2 = 675 - 273°C = 402°C$$

**22.** **(1.5)** As $P \propto \frac{1}{V^{1.5}}$, So $PV^{1.5}$ = constant

$\therefore \gamma = 1.5$ ($\because$ Process is adiabatic)

As we know, $\frac{C_p}{C_v} = \gamma$

$$\therefore \frac{C_p}{C_v} = 1.5$$

**23.** **(1.5)** $P \propto T^3 \Rightarrow PT^{-3}$ = constant ....(i)

But for an adiabatic process, the pressure temperature relationship is given by

$P^{1-\gamma}\ T^{\gamma}$ = constant $\Rightarrow PT^{\frac{\gamma}{1-\gamma}}$ = constt. ....(ii)

From (i) and (ii)

$$\frac{\gamma}{1-\gamma} = -3 \Rightarrow \gamma = -3+3\gamma \Rightarrow \gamma = \frac{3}{2}$$

**24.** **($1.67\times 10^5$)** $\Delta H = mL = 5\times 336\times 10^3 = Q_{sink}$

$$\frac{Q_{sink}}{Q_{source}} = \frac{T_{sink}}{T_{source}}$$

$$\therefore Q_{source} = \frac{T_{source}}{T_{sink}}\times Q_{sink}$$

Energy consumed by freezer

$$\therefore w_{output} = Q_{source} - Q_{sink}$$

$$= Q_{sink}\left(\frac{T_{source}}{T_{sink}} - 1\right)$$

Given: $T_{source} = 27°C + 273 = 300K$,

$T_{sink} = 0°C + 273 = 273\,k$

$$W_{output} = 5\times 336\times 10^3\left(\frac{300}{273} - 1\right) = 1.67\times 10^5\,J$$

**25.** **(400)** $\eta = 1 - \frac{T_2}{T_1}$ or $\frac{50}{100} = 1 - \frac{500}{T_1}$

$$\Rightarrow T_1 = 1000\ K$$

Also, $\frac{60}{100} = 1 - \frac{T_2}{1000} \Rightarrow T_2 = 400\ K$

**26.** **(46)** For adiabatic process, $TV^{\gamma-1}$ = constant

or, $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$T_1 = 20°C + 273 = 293$ K, $V_2 = \frac{V_1}{10}$ and $\gamma = \frac{7}{5}$

$$T_1(V_1)^{\gamma-1} = T_2\left(\frac{V_1}{10}\right)^{\gamma-1}$$

$$\Rightarrow 293 = T_2\left(\frac{1}{10}\right)^{2/5} \Rightarrow T_2 = 293(10)^{2/5} \simeq 736\ K$$

$$\Delta T = 736 - 293 = 443\ K$$

During the process, change in internal energy

$$\Delta U = NC_V\Delta T = 5\times\frac{5}{2}\times 8.3\times 443$$

$$\simeq 46\times 10^3\ J = X\ kJ$$

$$\therefore X = 46.$$

**27.** **(1818)** For an adiabatic process,

$TV^{\gamma-1}$ = constant

$$\therefore T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\Rightarrow T_2 = (300)\times\left(\frac{V_1}{\frac{V_1}{16}}\right)^{1.4-1}$$

$$\Rightarrow T_2 = 300\times(16)^{0.4}$$

Ideal gas equation, $PV = nRT$

$$\therefore V = \frac{nRT}{P}$$

$\Rightarrow V = kT$ (since pressure is constant for isobaric process)

So, during isobaric process

$V_2 = kT_2$ ...(i)

$2V_2 = kT_f$ ...(ii)

Dividing (i) by (ii)

$$\frac{1}{2} = \frac{T_2}{T_f}$$

$$T_f = 2T_2 = 300\times 2\times(16)^{0.4} = 1818\,K$$

**28. (600.00)**

Given; $T_1 = 900\,K$, $T_2 = 300K$, $W = 1200\,J$

Using, $1 - \frac{T_2}{T_1} = \frac{W}{Q_1}$

$\Rightarrow 1 - \frac{300}{900} = \frac{1200}{Q_1}$

$\Rightarrow \frac{2}{3} = \frac{1200}{Q_1} \Rightarrow Q_1 = 1800$

Therefore heat energy delivered by the engine to the low temperature reservoir, $Q_2 = Q_1 - W$ $= 1800 - 1200 = 600.00\,J$

**29. (500)** $\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1}$

and, $\eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$

According to question,

$W_A = W_B$

$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2}$

$\therefore T_2 = \frac{T_1 + T_3}{2}$

$= \frac{600 + 400}{2}$

$= 500K$

**30. (980)** Efficiency, $\eta = \frac{\text{Work done}}{\text{Heat absorbed}} = \frac{W}{\Sigma Q}$

$= \frac{Q_1 + Q_2 + Q_3 + Q_4}{Q_1 + Q_3} = 0.5$

Here, $Q_1 = 1915$ J, $Q_2 = -40$ J and $Q_3 = 125$ J

$\therefore \frac{1915 - 40 + 125 + Q_4}{1915 + 125} = 0.5$

$\Rightarrow 1915 - 40 + 125 + Q_4 = 1020$

$\Rightarrow Q_4 = 1020 - 2000$

$\Rightarrow Q_4 = -Q = -980$ J

$\Rightarrow Q = 980$ J

# CHAPTER 12

# Kinetic Theory

1. **(d)** $PV = NkT \Rightarrow \dfrac{N_A}{N_B} = \dfrac{P_A V_A}{P_B V_B} \times \dfrac{T_B}{T_A}$

$$\Rightarrow \frac{N_A}{N_B} = \frac{P \times V \times (2T)}{2P \times \frac{V}{4} \times T} = \frac{4}{1}$$

2. **(b)** For an ideal gas PV = constant i.e. PV doesn't vary with V.
3. **(b)**
4. **(a)** According to given Vander Waal's equation

$$P = \frac{nRT}{V - n\beta} - \frac{\alpha n^2}{V^2}$$

Work done,

$$W = \int_{V_1}^{V_2} PdV = nRT \int_{V_1}^{V_2} \frac{dV}{V - n\beta} - \alpha n^2 \int_{V_1}^{V_2} \frac{dV}{V^2}$$

$$= nRT\left[\log_e(V - n\beta)\right]_{V_1}^{V_2} + \alpha n^2 \left[\frac{1}{V}\right]_{V_1}^{V_2}$$

$$= nRT \log_e\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) + \alpha n^2 \left[\frac{V_1 - V_2}{V_1 V_2}\right]$$

5. **(d)** Length of air column on both side is $\dfrac{100-10}{2} = 45$ cm when one side is at 0°C and the other is at 273°C

The pressure must be same on both sides. Hence

$$\frac{l_1}{T_1} = \frac{l_2}{T_2} \Rightarrow \frac{l_1}{273} = \frac{l_2}{(273+273)} \Rightarrow l_1 = \frac{l_2}{2}$$

Also $l_1 + l_2 = 90$ cm $\Rightarrow l_1 = 30$ cm

and $l_2 = 60$ cm

Applying gas equation to the side at 0°C, we get

$$\frac{P_1 l_1}{T_1} = \frac{Pl}{T} \Rightarrow \frac{P_1 \times 30}{273} = \frac{76 \times 45}{273 + 31}$$

$\Rightarrow P_1 = 102.4$ cm of Hg.

6. **(a)** For mixture of gas, $C_v = \dfrac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$

$$= \frac{4 \times \frac{3}{2}R + \frac{1}{2} \times \frac{5}{2}R}{\left(4 + \frac{1}{2}\right)} = \frac{6R + \frac{5}{4}R}{\frac{9}{2}} = \frac{29R \times 2}{9 \times 4} = \frac{29R}{18}$$

and $C_p = \dfrac{n_1 C_{p_1} + n_2 C_{p_2}}{(n_1 + n_2)} = \dfrac{4 \times \frac{5R}{2} + \frac{1}{2} \times \frac{7R}{2}}{\left(4 + \frac{1}{2}\right)}$

$$= \frac{10R + \frac{7}{4}R}{\frac{9}{2}} = \frac{47R}{18}$$

$$\therefore \frac{C_p}{C_v} = \frac{47R}{18} \times \frac{18}{29R} = 1.62$$

7. **(c)** Mean free path of a gas molecule is given by

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$$

Here, $n$ = number of collisions per unit volume

$d$ = diameter of the molecule

If average speed of molecule is v then

Mean free time, $\tau = \dfrac{\lambda}{v}$

$$\Rightarrow \tau = \frac{1}{\sqrt{2}\pi n d^2 v} = \frac{1}{\sqrt{2}\pi n d^2}\sqrt{\frac{M}{3RT}}$$

$$\left(\because v = \sqrt{\frac{3RT}{M}}\right)$$

$$\therefore \tau \propto \frac{\sqrt{M}}{d^2} \quad \therefore \frac{\tau_1}{\tau_2} = \frac{\sqrt{M_1}}{d_1^2} \times \frac{d_2^2}{\sqrt{M_2}}$$

$$= \sqrt{\frac{40}{140}} \times \left(\frac{0.1}{0.07}\right)^2 = 1.09$$

8. **(c)** For a given pressure volume will be more if temperature is more (charles's law) From graph $V_2 > V_1 \therefore T_2 > T_1$
9. **(d)** K.E. of molecules is dependends on temperature. Since,

$$\frac{E_{k_1}}{E_{k_2}} = \frac{T_1}{T_2} = \frac{T}{T} = \frac{1}{1} = 1:1$$

(Since temperature of jar is same)

Hence K.E. of $H_2$ and $O_2$ will be found in ratio 1 : 1.

10. **(c)** As temperature is constant during the process, $\Rightarrow V_{rms}$ remains constant so $\Delta V_{rms} = 0$

**11.** **(a)** When temperature is same according to kinetic theory of gases, kinetic energy of molecules will be same.

$$K.E. = \frac{1}{2} \times 32 \times \left(\frac{1}{2}\right)^2 = \frac{1}{2} \times 2 \times v^2$$

RMS velocity of hydrogen molecules = 2 km/sec.

**12.** **(b)**

**13.** **(d)** Kinetic energy of each molecule,

$$K.E. = \frac{3}{2} K_B T$$

In the given problem,
Temperature, T = 0°C = 273 K
Height attained by the gas molecule, h = ?

$$K.E. = \frac{3}{2} K_B (273) = \frac{819 K_B}{2}$$

K.E. = P.E.

$$\Rightarrow \frac{819 K_B}{2} = Mgh \quad \text{or} \quad h = \frac{819 K_B}{2Mg}$$

**14.** **(d)** $C_v$ for hydrogen = $\frac{5}{2}R$ ;

$C_v$ for helium = $\frac{3R}{2}$

$C_v$ for water vapour = $\frac{6R}{2} = 3R$

$$\therefore (C_v)_{mix} = \frac{4 \times \frac{5}{2}R + 2 \times \frac{3}{2}R + 1 \times 3R}{4 + 2 + 1} = \frac{16}{7}R$$

$\therefore C_p = C_v + R$

$$C_p = \frac{16}{7}R + R \quad \text{or} \quad C_p = \frac{23}{7}R$$

**15.** **(a)**

**16.** **(c)** $v_{rms} = v_e$

$$\sqrt{\frac{3RT}{M}} = 11.2 \times 10^3 \quad \text{or} \quad \sqrt{\frac{3kT}{m}} = 11.2 \times 10^3$$

$$\text{or} \quad \sqrt{\frac{3 \times 1.38 \times 10^{-23} T}{2 \times 10^{-3}}} = 11.2 \times 10^3 \quad \therefore v = 10^4 K$$

**17.** **(d)** Kinetic energy of each molecule,

$$K.E. = \frac{3}{2} K_B T$$

In the given problem,
Temperature, T = 0°C = 273 K
Height attained by the gas molecule, h = ?

$$K.E. = \frac{3}{2} K_B (273) = \frac{819 K_B}{2}$$

K.E. = P.E.

$$\Rightarrow \frac{819 K_B}{2} = Mgh \quad \text{or} \quad h = \frac{819 K_B}{2Mg}$$

**18.** **(c)** From P-V graph,

$$P \propto \frac{1}{V}, \quad T = \text{constant}$$

and Pressure is increasing from 2 to 1

**19.** **(a)** $\because \quad C = \sqrt{\frac{3RT}{M}}$

$$(1930)^2 = \frac{3 \times 8.314 \times 300}{M}$$

$$M = \frac{3 \times 8.314 \times 300}{1930 \times 1930} \approx 2 \times 10^{-3} kg$$

The gas is $H_2$.

**20.** **(a)** Number of moles of first gas = $\frac{n_1}{N_A}$

Number of moles of second gas = $\frac{n_2}{N_A}$

Number of moles of third gas = $\frac{n_3}{N_A}$

If there is no loss of energy then

$P_1V_1 + P_2V_2 + P_3V_3 = PV$

$$\frac{n_1}{N_A} RT_1 + \frac{n_2}{N_A} RT_2 + \frac{n_3}{N_A} RT_3$$

$$= \frac{n_1 + n_2 + n_3}{N_A} RT_{mix}$$

$$T_{mix} = \frac{n_1T_1 + n_2T_2 + n_3T_3}{n_1 + n_2 + n_3}$$

**21.** **(2)** Degree of freedom is the number of independent variables required to define body's motion completely. Here f = 2 (1 translational + 1 rotational)

**22.** **(1.2)** Internal energy of n moles of an ideal gas at temperature T is given by

$$U = \frac{f}{2} nRT \text{ (f = degrees of freedom)}$$

$U_1 = U_2 \quad \Rightarrow \quad f_1n_1T_1 = f_2n_2T_2$

$$\therefore \quad \frac{n_1}{n_2} = \frac{f_2T_2}{f_1T_1} = \frac{3 \times 2}{5 \times 1} = \frac{6}{5}$$

Here $f_2$ = degrees of freedom of He = 3
and $f_1$ = degrees of freedom of $H_2$ = 5

**23.** **(1.5)** $\gamma_1 = \frac{5}{3}$ $\gamma_2 = \frac{7}{5}$

$n_1 = 1, n_2 = 1$

$$\frac{n_1+n_2}{\gamma-1} = \frac{n_1}{\gamma_1-1} + \frac{n_2}{\gamma_2-1}$$

$$\Rightarrow \frac{1+1}{\gamma-1} = \frac{1}{\frac{5}{3}-1} + \frac{1}{\frac{7}{5}-1} = \frac{3}{2}+\frac{5}{2} = 4$$

$$\therefore \frac{2}{\gamma-1} = 4 \Rightarrow \gamma = \frac{3}{2}$$

**24.** **(925)** Using equipartition of energy, we have

$$\frac{6}{2}KT = mCT$$

$$C = \frac{3\times1.38\times10^{-23}\times6.02\times10^{23}}{27\times10^{-3}}$$

$\therefore$ C = 925 J/kgK

**25.** **($-2.5\times10^{25}$)** Given: Temperature $T_i = 17+273 = 290\,K$

Temperature $T_f = 27+273 = 300\,K$

Atmospheric pressure, $P_0 = 1\times10^5\,Pa$

Volume of room, $V_0 = 30\,m^3$

Difference in number of molecules, $n_f - n_i = ?$

Using ideal gas equation, $PV = nRT(N_0)$,

$N_0$ = Avogadro's number

$$\Rightarrow n = \frac{PV}{RT}(N_0)$$

$$\therefore n_f - n_i = \frac{P_0V_0}{R}\left(\frac{1}{T_f}-\frac{1}{T_i}\right)N_0$$

$$= \frac{1\times10^5\times30}{8.314}\times6.023\times10^{23}\left(\frac{1}{300}-\frac{1}{290}\right)$$

$= -2.5\times10^{25}$

**26.** **(5)** Using ideal gas equation, $PV = nRT$

$$\Rightarrow P_1V_1 = nR\times250 \;[\because T_1 = 250\text{ K}] \quad ...(i)$$

$$P_2(2V_1 = \frac{5n}{4}R\times2000$$

$[\because T_2 = 2000\text{ K}]$ ...(ii)

Dividing eq. (i) by (ii),

$$\frac{P_1}{2P_2} = \frac{4\times250}{5\times2000} \Rightarrow \frac{P_1}{P_2} = \frac{1}{5}$$

$$\therefore \frac{P_2}{P_1} = 5.$$

**27.** **(150)** In first case,

From ideal gas equation

$PV = nRT$

$P\Delta V + V\Delta P = 0$ (As temperature is constant)

$$\Delta V = -\frac{\Delta P}{P}V \quad ...(i)$$

In second case, using ideal gas equation again

$P\Delta V = -nR\Delta T$

$$\Delta V = -\frac{nR\Delta T}{P} \quad ...(ii)$$

Equating (i) and (ii), we get

$$\frac{nR\Delta T}{P} = -\frac{\Delta P}{P}V \Rightarrow \Delta T = \Delta P\frac{V}{nR}$$

Comparing the above equation with $|\Delta T| = C|\Delta P|$, we have

$$C = \frac{V}{nR} = \frac{\Delta T}{\Delta P} = \frac{300\text{ K}}{2\text{ atm}} = 150\text{ K/atm}$$

**28.** **(41)** Room mean square speed is given by

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Here, $M$ = Molar mass of gas molecule

$T$ = temperature of the gas molecule

We have given $v_{N_2} = v_{H_2}$

$$\therefore \sqrt{\frac{3RT_{N_2}}{M_{N_2}}} = \sqrt{\frac{3RT_{H_2}}{M_{H_2}}}$$

$$\Rightarrow \frac{T_{H_2}}{2} = \frac{573}{28} \Rightarrow T_{H_2} = 41\text{ K}$$

**29.** **(3.16)** Using $\frac{V_{1rms}}{V_{2rms}} = \sqrt{\frac{M_2}{M_1}}$

$$\frac{V_{rms}(He)}{V_{rms}(Ar)} = \sqrt{\frac{M_{Ar}}{M_{He}}} = \sqrt{\frac{40}{4}} = 3.16$$

**30.** **(266.67)** Here work done on gas and heat supplied to the gas are zero.

Let $T$ be the final equilibrium temperature of the gas in the vessel.

Total internal energy of gases remain same.

i.e., $u_1 + u_2 = u'_1 + u'_2$

or, $n_1C_v\Delta T_1 + n_2C_v\Delta T_2 = (n_1+n_2)C_vT$

$$\Rightarrow (0.1)C_v(200) + (0.05)C_v(400) = (0.15)C_vT$$

$$\therefore T = \frac{800}{3} = 266.67\text{ K}$$

# Chapter 13 Oscillations

1. **(d)** The motion of particle is S.H.M. with
$x = A \sin \omega t + B \cos \omega t$
$= a \sin (\omega t + \theta)$

Where $a = \sqrt{A^2 + B^2}$, $A = a \cos\theta$, $B = a \sin\theta$, $\theta = \tan^{-1} B/A$.

2. **(b)** The particles will meet at the mean position when P completes one oscillation and Q completes half an oscillation

So $\frac{v_P}{v_Q} = \frac{a\omega_P}{a\omega_Q} = \frac{T_Q}{T_P} = \frac{6}{3} = \frac{2}{1}$

3. **(b)**

$$E_1 = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{\frac{2E_1}{k}},\ E_2 = \frac{1}{2}ky^2 \Rightarrow y = \sqrt{\frac{2E_2}{k}}$$

and $E = \frac{1}{2}k(x+y)^2 \Rightarrow x + y = \sqrt{\frac{2E}{k}}$

$$\Rightarrow \sqrt{\frac{2E_1}{k}} + \sqrt{\frac{2E_2}{k}} = \sqrt{\frac{2E}{k}} \Rightarrow \sqrt{E_1} + \sqrt{E_2} = \sqrt{E}$$

4. **(a)** When the mass $m_1$ is removed, only mass $m_2$ remains. Therefore, its angular frequency is

$\omega = \sqrt{\frac{k}{m_2}}$

5. **(d)** In damped oscillation, amplitude goes on decaying exponentially.
$a = a_0 e^{-bt}$ where $b$ = damping coefficient.

Initially, $\frac{a_0}{3} = a_0 e^{-b \times 100T}$, T = time of one oscillation or $\frac{1}{3} = e^{-100bT}$

Finally, $a = a_0 e^{-b \times 200T}$ or $a = a_0[e^{-100bT}]^2$

or $a = a_0 \times \left[\frac{1}{3}\right]^2$ [from (*i*)] or $a = a_0/9$.

6. **(a)** Displacement $y(t) = A \sin (\omega t + \phi)$ [Given]

For $\phi = \frac{2\pi}{3}$

at $t = 0$; $y = A \sin \phi = A \sin \frac{2\pi}{3}$
$= A \sin 120° = 0.87 A$ [$\because \sin 120° \simeq 0.866$]

Graph (a) depicts $y = 0.87A$ at $t = 0$

7. **(b)** At the topmost position normal reaction $= 0$

$\therefore a_{max} = g$

$\Rightarrow \omega^2 A = g$

$\Rightarrow \omega = \sqrt{\frac{g}{A}}$

$= \sqrt{\frac{g}{0.05}}$

now, $v_{max} = \omega A = \sqrt{\frac{g}{0.05}} \times 0.05 = \frac{1}{\sqrt{2}}$ m/s

8. **(d)**

9. **(b)** $\frac{d\theta}{dt} = 2 \ \therefore\ \theta = 2t$

Let BP = a,

$\therefore\ x = OM = a \sin \theta = a \sin (2t)$

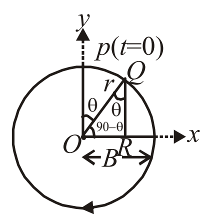
$\Rightarrow \frac{dx}{dt} = a \times 2 \cos (2t)$

$\Rightarrow \frac{d^2x}{dt^2} = -4a \sin 2t$

Hence M executes SHM within the given time period and its acceleration is opposite to x that means towards left.

10. **(a)** As the particle ($P$) is executing circular motion with radius $B$.

Consider angular velocity of the particle executing circular motion is $\omega$ and when it is at $Q$ makes an angle $\theta$ as shown in the diagram.




Clearly, $\theta = \omega t$

Now, we can write

$OR = OQ \cos(90 - \theta)$ ($\because OR = X$)

$x = OQ\sin\theta = OQ\sin\omega t$

$= r\sin\omega t \qquad [\because OQ = r]$

$\therefore x = B\sin\frac{2\pi}{T}t = B\sin\left(\frac{2\pi}{30}t\right)$

$\Rightarrow x = B\sin\left(\frac{2\pi}{30}t\right)$

Hence, this equation represents SHM.

**11.** **(a)** The displacement of a particle in S.H.M. is given by :

$y = a\sin(\omega t + \phi)$

velocity $= \frac{dy}{dt} = \omega a\cos(\omega t + \phi)$

given $\phi = \frac{\pi}{4}$

The velocity is maximum when the particle passes through the mean position i.e., $\left(\frac{dy}{dt}\right)_{max} = \omega a$

The kinetic energy at this instant is given by

$\frac{1}{2}m\left(\frac{dy}{dt}\right)^2_{max} = \frac{1}{2}m\omega^2 a^2 = 8 \times 10^{-3}$ joule

or $\frac{1}{2} \times (0.1)\,\omega^2 \times (0.1)^2 = 8 \times 10^{-3} \Rightarrow \omega = \pm 4$

Substituting the values of a, $\omega$ and $\phi$ in the equation of S.H.M., we get $y = 0.1\sin(\pm 4t + \pi/4)$ metre.

**12.** **(a)** KE and PE completes two vibration in a time during which SHM completes one vibration. Thus frequency of PE or KE is double than that of SHM.

**13.** **(a)** Two perpendicular S.H.Ms are

$x = a_1\cos\omega t$ ....(1)

and $y = a_2\cos 2\omega t$ ....(2)

From eqn (1) $\frac{x}{a_1} = \cos\omega t$

and from eqn (2) $\frac{y}{a_2} = \cos 2\omega t = [2\cos^2\omega t - 1]$

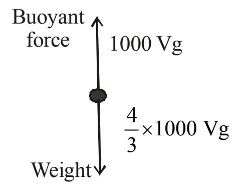
$y = 2a_2\left(\frac{x}{a_1}\right)^2 - a_2$

**14.** **(b)** Minimum normal reaction is, $N = mg - mA\omega^2$

For N = 0, $\omega = \sqrt{\frac{g}{A}}$, $T = 2\pi\sqrt{\frac{A}{g}}$

For N to be positive, T must be greater than $2\pi\sqrt{\frac{A}{g}}$

**15.** **(a)** $t = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$ ; $t_0 = 2\pi\sqrt{\frac{\ell}{g}}$




Net force $= \left(\frac{4}{3} - 1\right) \times 1000\ Vg = \frac{1000}{3}Vg$

$g_{eff} = \frac{1000\ Vg}{3 \times \frac{4}{3} \times 1000\ V} = \frac{g}{4}$

$\therefore t = 2\pi\sqrt{\frac{\ell}{g/4}} \Rightarrow t = 2t_0$

**16.** **(b)** Equation of displacement is given by $x = A\sin(\omega t + \phi)$

where $A = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega^2)^2}} = \frac{F_0}{m(\omega_0^2 - \omega^2)}$

here damping effect is considered to be zero

$\therefore x \propto \frac{1}{m(\omega_0^2 - \omega^2)}$

**17.** **(d)** As we know, $E = E_0 e^{-\frac{bt}{m}}$

$\Rightarrow 15 = 45e^{-\frac{b15}{m}}$

[As no. of oscillations = 15 so $t$ = 15sec]

$\frac{1}{3} = e^{-\frac{b15}{m}}$

Taking log on both sides $\frac{b}{m} = \frac{1}{15}\ell n\,3$

**18.** **(a)** (i) Restoring force $= -k_1x - k_2x$;

$k_{eq} = k_1 + k_2 \therefore f = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{m}}$

(ii) Here, extensions are different. Total extension = $x$

$$= x_1 + x_2$$

$$\Rightarrow \frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\therefore f = \frac{1}{2\pi}\sqrt{\frac{k_1 k_2}{m(k_1 + k_2)}}$$

So, ratio $= \frac{k_1 + k_2}{\sqrt{k_1 k_2}}$

**19.** **(b)** $U = U_0 + \alpha x^2 \Rightarrow F = -\frac{dU}{dx} = -2\alpha x$

$$a = \frac{F}{m} = -\frac{2\alpha}{m}x$$

$$\Rightarrow \omega^2 = \frac{2\alpha}{m} \Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{2\alpha}}$$

**20.** **(c)** As we know, time period, $T = 2p\sqrt{\frac{\ell}{g}}$

When additional mass M is added then

$$T_M = 2p\sqrt{\frac{\ell + D\ell}{g}}$$

$$\frac{T_M}{T} = \sqrt{\frac{\ell + D\ell}{\ell}} \text{ or } \left(\frac{T_M}{T}\right)^2 = \frac{\ell + \Delta\ell}{\ell}$$

$$\text{or, } \left(\frac{T_M}{T}\right)^2 = 1 + \frac{Mg}{Ay} \quad \left[\because \Delta\ell = \frac{Mg\ell}{Ay}\right]$$

$$\therefore \frac{1}{y} = \left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{A}{Mg}$$

**21.** **(0.0314)** Slope of F - x curve = – k = $-\frac{80}{0.2} \Rightarrow$ k = 400 N/m,

Time period, $T = 2\pi\sqrt{\frac{m}{k}} = 0.0314$ sec.

**22.** **(1.9)** Washer contact with piston $\Rightarrow N = 0$

Given Amplitude A = 7 cm = 0.07 m.

$a_{max} = g = \omega^2 A$

The frequency of piston

$$f = \frac{\omega}{2\pi} = \sqrt{\frac{g}{A}}\frac{1}{2\pi} = \sqrt{\frac{1000}{7}}\frac{1}{2\pi} = 1.9 \text{ Hz.}$$

**23.** **(9)** At resonance, amplitude of oscillation is maximum.

$\Rightarrow$ $2\omega^2 - 36\omega + 9$ is minimum $\Rightarrow$ derivative must be zero

$\Rightarrow$ $4\omega - 36 = 0$ ($\because$ derivative is zero)

$\Rightarrow$ $\omega = 9$

**24.** **(0.33)** Kinetic energy, $k = \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t$

Potential energy, $U = \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t$

$$\frac{k}{U} = \cot^2 \omega t = \cot^2 \frac{\pi}{90}(210) = \frac{1}{3}$$

**25.** **(3.5)** Since system dissipates its energy gradually, and hence amplitude will also decreases with time according to

$a = a_0 e^{-bt/m}$ ....... (i)

$\because$ Energy of vibration drop to half of its initial value ($E_0$), as $E \propto a^2 \Rightarrow a \propto \sqrt{E}$

$$a = \frac{a_0}{\sqrt{2}} \Rightarrow \frac{bt}{m} = \frac{10^{-2}t}{0.1} = \frac{t}{10}$$

From eq$^n$ (i),

$$\frac{a_0}{\sqrt{2}} = a_0 e^{-t/10}$$

$$\frac{1}{\sqrt{2}} = e^{-t/10} \text{ or } \sqrt{2} = e^{\frac{t}{10}}$$

$\ln\sqrt{2} = \frac{t}{10}$ $\therefore$ t = 3.5 seconds

**26.** **(7)** Amplitude of vibration at time t = 0 is given by

$A = A_0 e^{-0.1\times 0} = 1 \times A_0 = A_0$

also at t = t, if $A = \frac{A_0}{2}$

$$\Rightarrow \frac{1}{2} = e^{-0.1t}$$

$t = 10 \ln 2 \simeq 7$s

**27.** **(0.729)** $\because A = A_0 e^{-\frac{bt}{2m}}$

(where, $A_0$ = maximum amplitude)

According to the questions, after 5 second,

$$0.9A_0 = A_0 e^{-\frac{b(5)}{2m}} \ldots\text{(i)}$$

After 10 more second,

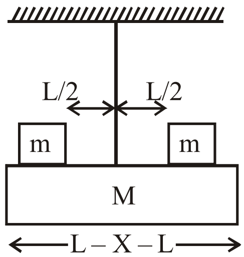
$$A = A_0 e^{-\frac{b(15)}{2m}} \ldots\text{(ii)}$$

From eq$^n$s (i) and (ii)

$A = 0.729\,A_0$

$\therefore \alpha = 0.729$

**28.** **(0.37)**




$$f_1 = \frac{1}{2\pi}\sqrt{\frac{C}{1}} \qquad \text{...(i)}$$

$$= \frac{1}{2}\sqrt{\frac{3C}{ML^2}}$$

$$f_2 = \frac{1}{2\pi}\sqrt{\frac{C}{L^2\left(\frac{M}{3}+\frac{M}{2}\right)}} \qquad \text{...(ii)}$$

As frequency reduces by 80%

$$\therefore\ f_2 = 0.8\ f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \qquad \text{...(iii)}$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio}\frac{\text{m}}{\text{M}} = 0.37$$

**29.** **(0.1)** As we know, Time-period of simple pendulum, $T \propto \sqrt{l}$

differentiating both side, $\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta l}{l}$

$\because$ change in length $\Delta l = r_1 - r_2$

$$5 \times 10^{-4} = \frac{1}{2}\frac{r_1 - r_2}{1}$$

$r_1 - r_2 = 10 \times 10^{-4}$

$10^{-3}\,\text{m} = 10^{-1}\,\text{cm} = 0.1\,\text{cm}$

**30.** **($4\sqrt{2}$)** Displacement, $x = 4(\cos \pi t + \sin \pi t)$

$$= \sqrt{2} \times 4\left(\frac{\sin \pi t}{\sqrt{2}} + \frac{\cos \pi t}{\sqrt{2}}\right)$$

$$= 4\sqrt{2}(\sin \pi\, t \cos 45° + \cos \pi\, t \sin 45°)$$

$$x = 4\sqrt{2}\sin(\pi\, t + 45°)$$

On comparing it with standard equation

$x = A\sin(\omega t + \phi)$

we get $A = 4\sqrt{2}$

# Chapter 14

# Waves

1. **(a)** $y(x,t) = 0.005 \cos(\alpha x - \beta t)$ (Given)
Comparing it with the standard equation of wave
$y(x,t) = a \cos(kx - \omega t)$ we get
$k = \alpha$ and $\omega = \beta$

But $k = \frac{2\pi}{\lambda}$ and $\omega = \frac{2\pi}{T}$ $\Rightarrow$ $\frac{2\pi}{\lambda} = \alpha$ and $\frac{2\pi}{T} = \beta$

Given that $\lambda = 0.08$ m and $T = 2.0$s

$\therefore$ $\alpha = \frac{2\pi}{0.08} = 25\pi$ and $\beta = \frac{2\pi}{2} = \pi$

2. **(a)**

3. **(c)** The frequency that the observer receives directly from the source has frequency $n_1$ = 500 Hz. As the observer and source both move towards the fixed wall with velocity u, the apparent frequency of the reflected wave coming from the wall to the observer will have frequency

$$n_2 = \left(\frac{V}{V-u}\right) 500 \text{ Hz}$$

where V is the velocity of sound wave in air. The apparent frequency of this reflected wave as heard by the observer will then be

$$n_3 = \left(\frac{V+u}{V}\right) n_2 = \left(\frac{V+u}{V}\right)\left(\frac{V}{V-u}\right) 500 = \left(\frac{V+u}{V-u}\right) 500$$

It is given, that the number of beat per second is $n_3 - n_1 = 10$

$$\therefore \quad (n_3 - n_1) = 10 = \left(\frac{V+u}{V-u}\right) 500 - 500$$

$$\Rightarrow 10 = 500\left[\frac{V+u}{V-u} - 1\right]$$

$$\Rightarrow 10 = \frac{2 \times u \times 500}{V-u}$$

Hence, $10V = 1000u + 10u = 1010u$
Putting $u = 4$ m/s,

we have $V = \frac{1}{10}[4040] = 404\, m/s \simeq 400\, m/s$

4. **(c)** Using $n_{Last} = n_{First} + (N-1)\,x$
$\Rightarrow$ $3n = n + (26-1) \times 4$
$\Rightarrow$ $n = 50$ Hz

5. **(d)** $\nu' = \nu\left(\frac{v + v_o}{v - v_S}\right)$

Here, $\nu = 600$ Hz, $\nu_o = 15$ m/s
$v_s = 20$ m/s, $v = 340$ m/s

$\therefore$ $\nu' = 600\left(\frac{355}{320}\right) \approx 666$ Hz

6. **(d)** $T = \frac{1}{f} = \frac{1}{500}$ s;
Since compression alternates with rarefaction so again compression appears after
$t = \frac{T}{2} = \frac{1}{1000}$ s

7. **(a)** Equation of the harmonic progressive wave given by :
$y = a \sin 2\pi(bt - cx)$.
Here $\upsilon = b$

$k = \frac{2\pi}{\lambda} = 2\pi c$ take, $\frac{1}{\lambda} = c$

$\therefore$ Velocity of the wave $= \upsilon\lambda = b\frac{1}{c} = \frac{b}{c}$

$$\frac{dy}{dt} = a\, 2\pi b \cos 2\pi(bt - cx) = a\omega \cos(\omega t - kx)$$

Maximum particle velocity $= a\omega = a 2\pi b = 2\pi ab$

given this is $2 \times \frac{b}{c}$ i.e. $2\pi a = \frac{2}{c}$ or $c = \frac{1}{\pi a}$

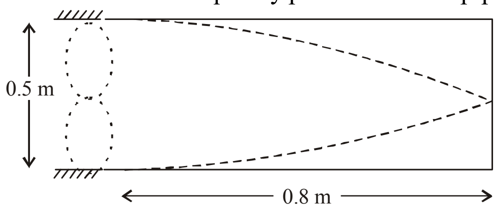
8. **(d)** In case of closed organ pipe frequency,

$f_n = (2n+1)\,\frac{v}{4l}$

for $n = 0, f_0 = 100$ Hz; $n = 1, f_1 = 300$ Hz
$n = 2, f_2 = 500$ Hz; $n = 3, f_3 = 700$ Hz
$n = 4, f_4 = 900$ Hz; $n = 5, f_5 = 1100$ Hz
$n = 6, f_6 = 1300$ Hz

Hence possible natural oscillation whose frequencies < 1250 Hz = 6(n = 0, 1, 2, 3, 4, 5)

**9.** **(b)** Frequency of 2nd harmonic of string = Fundamental frequency produced in the pipe




$$\therefore\ 2\times\left[\frac{1}{2l_1}\sqrt{\frac{T}{\mu}}\right]=\frac{v}{4l_2}$$

$$\therefore\ \frac{1}{0.5}\sqrt{\frac{50}{\mu}}=\frac{320}{4\times 0.8}$$

$\therefore \mu = 0.02$ kg m$^{-1}$

The mass of the string $= \mu l_1 = 0.02 \times 0.5$ kg $= 10$g

**10.** **(d)**

**11.** **(b)** $n_1 = n_2$, $T \rightarrow$ Same, $r \rightarrow$ Same, $l \rightarrow$ Same

Frequency of vibration

$$n=\frac{p}{2l}\sqrt{\frac{T}{\pi r^2 \rho}}$$

As T, r, and $l$ are same for both the wires; $n_1 = n_2$

$$\frac{p_1}{\sqrt{\rho_1}}=\frac{p_2}{\sqrt{\rho_2}} \Rightarrow \quad \frac{p_1}{p_2}=\frac{1}{2} \quad \because\ \rho_2 = 4\rho_1$$

**12.** **(d)** We know that the apperent frequency

$$f'=\left(\frac{v-v_0}{v-v_s}\right)f \qquad \text{from Doppler's effect}$$

where $v_0 = v_s = 30$ m/s, velocity of observer and source

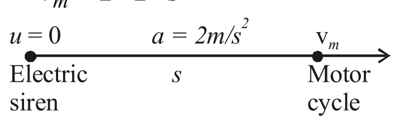
Speed of sound v = 330 m/s

$$\therefore\ f'=\frac{330+30}{330-30}\times 540=648\text{ Hz.}$$

($\because$ Frequency of whistle (f) = 540 Hz.)

**13.** **(a)** $v_m^2 - u^2 = 2as$

$\therefore\ v_m^2 = 2\times 2\times s$




$\therefore\ v_m = 2\sqrt{s}$

According to Doppler's effect $v' = v\left[\dfrac{v-v_m}{v}\right]$

$$0.94v = v\left[\frac{330-2\sqrt{s}}{330}\right] \Rightarrow s = 98.01\ m$$

**14.** **(a)** Using, $\beta = 10$

or $120 = 10\ \log 10\left(\dfrac{I}{10^{-12}}\right)$ ...(i)

Also $I=\dfrac{P}{4\pi r^2}=\dfrac{2}{4\pi r^2}$ ...(ii)

On solving above equations, we get $r = 40$ cm.

**15.** **(d)** $y = A\sin(\omega t - kx) + A\sin(\omega t + kx)$

$y = 2A\sin\omega t\cos kx$

This is an equation of standing wave. For position of nodes

$\cos kx = 0$

$$\Rightarrow\ \frac{2\pi}{\lambda}.x=(2n+1)\frac{\pi}{2}$$

$$\Rightarrow x=\frac{(2n+1)\lambda}{4},\ n=0,1,2,3,..........$$

**16.** **(a)** Fundamental frequency, $f = 70$ Hz.

The fundamental frequency of wire vibrating under tension $T$ is given by

$$f=\frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

Here, $\mu$ = mass per unit length of the wire

$L$ = length of wire

$$70=\frac{1}{2L}\sqrt{\frac{540}{6\times 10^{-3}}}$$

$\Rightarrow L \approx 2.14$ m

**17.** **(b)** Given : Frequency of tuning fork, $n = 264$ Hz

Length of column $L$ = ?

For closed organ pipe

$$n=\frac{v}{4l}$$

$$\Rightarrow l=\frac{v}{4n}=\frac{330}{4\times 264}=0.3125$$

or, $l = 0.3125 \times 100 = 31.25$ cm

In case of closed organ pipe only odd harmonics are possible.

Therefore value of $l$ will be $(2n - 1)\, l$
Hence option (b) i.e. $3 \times 31.25 = 93.75$ cm is correct.

**18.** **(b)** Standard equation

$$y(x,t) = A\cos\left(\frac{\omega}{V}x - \omega t\right)$$

From any of the displacement equation
Say $y_1$

$$\frac{\omega}{V} = 0.50\pi \text{ and } \omega = 100\pi$$

$$\therefore \quad \frac{100\pi}{V} = 0.5\pi$$

$$\therefore \; V = \frac{100\pi}{0.5\pi} = 200\,\text{m/s}$$

**19.** **(c)** According to Doppler's effect,

Apparent, frequency $f = \left(\frac{V + V_0}{V - V_S}\right)f_0$

Now, $f = \left(\frac{f_0}{V - V_S}\right)V_0 + \frac{V f_0}{V - V_s}$

So, slope $= \frac{f_0}{V - V_S}$

Hence, option (c) is the correct answer.

**20.** **(a)** It is given that 315 Hz and 420 Hz are two resonant frequencies, let these be $n^{th}$ and $(n + 1)^{th}$ harmonies, then we have $\frac{nv}{2\ell} = 315$

and $(n+1)\frac{v}{2\ell} = 420$

$$\Rightarrow \frac{n+1}{n} = \frac{420}{315}$$

$$\Rightarrow n = 3$$

Hence $3 \times \frac{v}{2\ell} = 315 \Rightarrow \frac{v}{2\ell} = 105\ Hz$

The lowest resonant frequency is when
$n = 1$
Therefore lowest resonant frequency
$= 105$ Hz.

**21.** **(0.85)** $x_1$ and $x_2$ are in successive loops of stationary waves.

So $\quad \phi_1 = \pi$ and $\phi_2 = K(\Delta x)$

$$= k\left(\frac{3\pi}{2k} - \frac{\pi}{3k}\right) = \frac{7\pi}{6} \quad = \frac{\phi_1}{\phi_2} = \frac{6}{7}$$

**22.** **(0.5)** The equation of wave at any time is obtained by putting $X = x - vt$

$$y = \frac{1}{1+X^2} = \frac{1}{1+(x-vt)^2} \qquad \text{...(i)}$$

We know at $t = 2$ sec,

$$y = \frac{1}{1+(x-1)^2} \qquad \text{...(ii)}$$

On comparing (i) and (ii) we get $vt = 1$;

$$v = \frac{1}{t}$$

As $t = 2$ sec $\qquad \therefore\ v = \frac{1}{2} = 0.5$ m/s.

**23.** **(0.375)** $3 \times \frac{v}{4l_c} = 4 \times \frac{v}{2l_0}$ or $\frac{l_c}{l_0} = \frac{3v}{4} \times \frac{2}{4v} = \frac{3}{8}$

**24.** **(0.1)** Velocity of wave on string $= \sqrt{T/\mu}$
$= 8$ m/s
The pulse gets inverted after reflection from the fixed end, so for constructive interference to take place between successive pulses, the first pulse has to undergo two reflections from the fixed end.

$$\Delta t = \frac{2L}{v} = \frac{2 \times 0.4}{8} = 0.1\text{s}$$

**25.** **(7.7)** When the source S is between the wall (W) and the observer (O) For direct sound the source is moving away from the observer, therefore the apparent frequency

$$n'' = \frac{v}{v + v_s}n = \frac{330}{330+5} \times 256 = 252.2$$

and frequency of reflected sound

$$n' = \frac{v}{v - v_s}n = \frac{330}{330-5} \times 256 = 259.9$$

Number of beats/sec $= n' - n'' = 259 - 252.2 = 7.7$

**26.** **(8516)** Reflected frequency of sound reaching bat

$$= \left[\frac{V - (-V_0)}{V - V_s}\right]f = \left[\frac{V + V_0}{V - V_s}\right]f = \frac{V+10}{V-10}f$$

$$= \left(\frac{320+10}{320-10}\right) \times 8000 \simeq 8516 \text{ Hz}$$

**27.** **(4)** For observer, tone of B will not change due to zero relative motion.
Observed frequency of sound produced by A

$$= 660\frac{(330-30)}{330} = 600\,Hz$$

$\therefore \quad$ No. of beats $= 600 - 596 = 4$

**28.** **(35.00)** Given,

Denisty of wire, $\sigma = 9\times10^{-3}$ kg cm$^{-3}$

Young's modulus of wire, $Y = 9\times10^{10}$ Nm$^{-2}$

Strain $= 4.9\times10^{-4}$

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\text{Strain}}$$

$$\therefore \frac{T}{A} = Y\times\text{Strain} = 9\times10^{9}\times4.9\times10^{-4}$$

Also, mass of wire, $m = Al\sigma$

Mass per unit length, $\mu = \frac{m}{J} = A\sigma$

Fundamental frequency in the string

$$f = \frac{1}{2l}\sqrt{\frac{T}{\mu}} = \frac{1}{2l}\sqrt{\frac{T}{\sigma A}}$$

$$= \frac{1}{2\times1}\sqrt{\frac{9\times10^{9}\times4.9\times10^{-4}}{9\times10^{3}}}$$

$$= \frac{1}{2}\sqrt{49\times10^{9-4-3}} = \frac{1}{2}\times70 = 35\text{ Hz}$$

**29.** **(106)** Given : $V_{\text{air}} = 300$ m/s, $\rho_{\text{gas}} = 2\,\rho$ air

Using, $V = \sqrt{\frac{B}{\rho}}$

$$\frac{V_{\text{gas}}}{V_{\text{air}}} = \frac{\sqrt{\frac{B}{2\rho_{\text{air}}}}}{\sqrt{\frac{B}{\rho_{\text{air}}}}} \Rightarrow V_{\text{gas}} = \frac{V_{\text{air}}}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 150\sqrt{2}\text{m/s}$$

And $f_{\text{nth}}$ harmonic $= \frac{nv}{2L}$ (in open organ pipe)

(L = 1 metre given)

$$\therefore f_{\text{2nd}}\text{ harmonic} - f_{\text{fundamental}} = \frac{2v}{2\times1} - \frac{v}{2\times1} = \frac{v}{2}$$

$$\therefore f_{\text{2n}}\text{ harmonic} - f_{\text{fundamental}}$$

$$= \frac{150\sqrt{2}}{2} = \frac{150}{\sqrt{2}} \approx 106\,Hz$$

**30.** **(2.25)** Using $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$,

where, $T$ = tension and $\mu = \frac{\text{mass}}{\text{length}}$

$$f_x = \frac{1}{2\ell}\sqrt{\frac{T_x}{\mu}} \text{ and } f_z = \frac{1}{2\ell}\sqrt{\frac{T_z}{\mu}}$$

$$\frac{f_x}{f_z} = \frac{450}{300} = \sqrt{\frac{T_x}{T_z}}$$

$$\therefore \frac{T_x}{T_z} = \frac{9}{4} = 2.25.$$

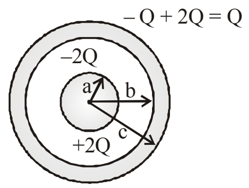
# CHAPTER 15

# Electric Charges and Fields

**1.** **(a)** Surface charge density $(\sigma) = \dfrac{\text{Charge}}{\text{Surface area}}$

So $\sigma_{\text{inner}} = \dfrac{-2Q}{4\pi b^2}$

and $\sigma_{\text{Outer}} = \dfrac{Q}{4\pi c^2}$



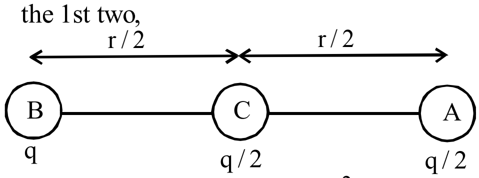

**2.** **(d)**

**3.** **(a)** Initial force between the two spheres carrying charge (say q) is,

$F = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q^2}{r^2}$ (r is the distance between them)

Further when an uncharged sphere is kept in touch with the sphere of charge q, the net charge on both become $\dfrac{q+0}{2} = \dfrac{q}{2}$.

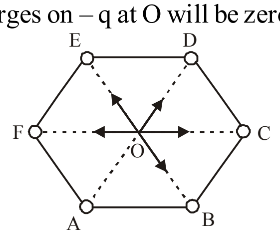
Force on the 3rd charge, when placed in center of the 1st two,




$$F_3 = \frac{1}{4\pi\varepsilon_0}\frac{q\left(\frac{q}{2}\right)}{\left(\frac{r}{2}\right)^2} - \frac{1}{4\pi\varepsilon_0}\frac{\left(\frac{q}{2}\right)^2}{\left(\frac{r}{2}\right)^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2}[2-1] = F$$

**4.** **(d)** If there had been a sixth charge +q at the remaining vertex of hexagon, force due to all the six charges on – q at O will be zero.



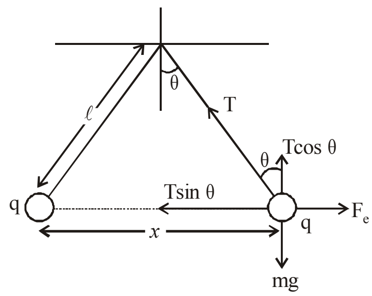

Now if $f$ is the force due to the sixth charge and F due to the remaining five charge then,

$\vec{F} + \vec{f} = 0$ i.e., $\vec{F} = -\vec{f}$

$$|\vec{F}| = |\vec{f}| = \frac{1}{4\pi\varepsilon_0}\frac{q \times q}{L^2} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{L}\right)^2$$

**5.** **(d)** In equilibrium, $F_e = T\sin\theta$

$mg = T\cos\theta$




$$\tan\theta = \frac{F_e}{mg}$$

$$= \frac{q^2}{4\pi \in_0 x^2 \times mg}$$

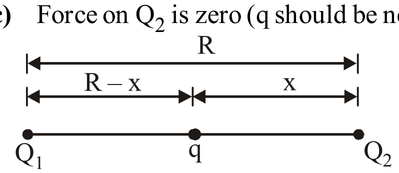
also $\tan\theta \approx \sin\theta = \dfrac{x/2}{\ell}$

Hence, $\dfrac{x}{2\ell} = \dfrac{q^2}{4\pi \in_0 x^2 \times mg}$

$$\Rightarrow x^3 = \frac{2q^2\ell}{4\pi \in_0 mg} \qquad \therefore\ x = \left(\frac{q^2\ell}{2\pi \in_0 mg}\right)^{1/3}$$

Therefore, $x \propto \ell^{1/3}$

**6.** **(c)** Force on $Q_2$ is zero (q should be negative)




$$\frac{kQ_1Q_2}{R^2} = \frac{kqQ_2}{x^2} \text{ or } \frac{x}{R} = \sqrt{\frac{q}{Q_1}} = \frac{\sqrt{q}}{\sqrt{Q_1}}$$

Force on q is zero,

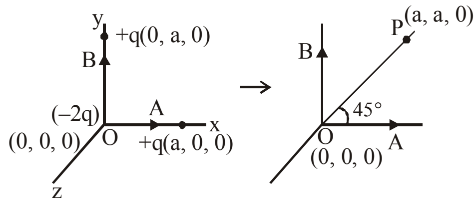
$$\frac{kQ_1q}{(R-x)^2}=\frac{kqQ_2}{x^2} \quad \text{or} \quad \frac{R-x}{x}=\frac{\sqrt{Q_1}}{\sqrt{Q_2}}$$

or $$\frac{R}{x}=\frac{\sqrt{Q_1}+\sqrt{Q_2}}{\sqrt{Q_2}} \text{ or } \frac{\sqrt{Q_1}}{\sqrt{q}}=\frac{\sqrt{Q_1}+\sqrt{Q_2}}{\sqrt{Q_2}}$$

or $$q=\frac{Q_1Q_2}{\left(\sqrt{Q_1}+\sqrt{Q_2}\right)^2}$$

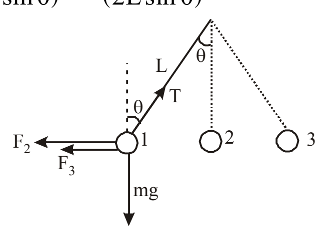
**7.** **(a)** Three point charges +q, –2q and +q are placed at points B (x = 0, y = a, z = 0), O (x = 0, y = 0, z = 0)
and A(x = a, y = 0, z = 0)
The system consists of two dipole moment vectors due to (+q and –q) and again due to (+q and –q) charges having equal magnitudes qa units i one along $\overrightarrow{OA}$ and other along $\overrightarrow{OB}$.
Hence, net dipole moment,

$p_{net}=\sqrt{(qa)^2+(qa)^2}=\sqrt{2}qa$ along $\overrightarrow{OP}$ at an angle 45° with positive X-axis.




**8.** **(a)** $F_e = F_2 + F_3$

$$=\frac{kq^2}{(L\sin\theta)^2}+\frac{kq^2}{(2L\sin\theta)^2}$$




$F_e=\frac{5}{4}\frac{kq^2}{L^2\sin^2\theta}$ ...(i)
$T\sin\theta = Fe$ ... (ii)
$T\cos\theta = mg$ ... (iii)
From (i), (ii), and (iii)

$$q=\sqrt{\frac{16}{5}\pi\varepsilon_0 mgL^2\sin^2\theta\tan\theta}$$

**9.** **(b)** Electric field at a point inside a charged conducting spherical shell is zero.

**10.** **(b)** Due to induction net charge on outer surfaces of shpere

$$\sigma=\frac{Q_1}{4\pi R^2}=\frac{Q_1+Q_2}{4\pi(2R)^2}=\frac{Q_1+Q_2+Q_3}{4\pi(3R)^2}$$

$$\Rightarrow Q_1=\frac{Q_1+Q_2}{4}=\frac{Q_1+Q_2+Q_3}{9}$$

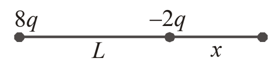
$\Rightarrow Q_2 = 3Q_1$ and $Q_3 = 5Q_1$
$\Rightarrow Q_1 : Q_2 : Q_3 = 1 : 3 : 5$

**11.** **(c)**

**12.** **(c)** $E_1 = E_2$ or

$$\frac{1}{4\pi\varepsilon_0}\frac{8q}{(L+x)^2}=\frac{1}{4\pi\varepsilon_0}\frac{2q}{x^2}$$

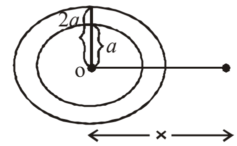
$\therefore \quad x = L$
Thus distance from origin is $= L + L = 2L$




**13.** **(a)** Electric field due to complete disc ($R = 2a$) at a distance $x$ and on its axis

$$E_1=\frac{\sigma}{2\varepsilon_0}\left[1-\frac{x}{\sqrt{R^2+x^2}}\right]$$

$$E_1=\frac{\sigma}{2\varepsilon_0}\left[1-\frac{h}{\sqrt{4a^2+h^2}}\right]$$




$$=\frac{\sigma}{2\varepsilon_0}\left[1-\frac{h}{2a}\right] \quad \left[\begin{array}{l}\text{here } x=h\\ \text{and, } R=2a\end{array}\right]$$

Similarly, electric field due to disc ($R = a$)

$$E_2=\frac{\sigma}{2\varepsilon_0}\left(1-\frac{h}{a}\right)$$

Electric field due to given disc
$E = E_1 - E_2$

$$\frac{\sigma}{2\varepsilon_0}\left[1-\frac{h}{2a}\right]-\frac{\sigma}{2\varepsilon_0}\left[1-\frac{h}{a}\right]=\frac{\sigma h}{4\varepsilon_0 a}$$

Hence, $c=\frac{\sigma}{4a\varepsilon_0}$

**14.** **(b)** Flux $= \vec{E}\,.\,\vec{A}$.

$\vec{E}$ is electric field vector & $\vec{A}$ is area vector.

Here, angle between $\vec{E}$ & $\vec{A}$ is 90º.

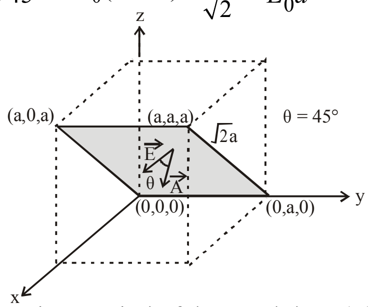
So, $\vec{E}\,.\,\vec{A} = 0$; Flux $= 0$

**15.** **(c)** Given $\vec{E} = E_o\,\hat{x}$

This shows that the electric field acts along + $x$ direction and is a constant. The area vector makes an angle of 45° with the electric field. Therefore the electric flux through the shaded portion whose area is

$a \times \sqrt{2}\,a = \sqrt{2}\,a^2$ is $\phi = \vec{E}.\vec{A} = EA\cos\theta = E_0(\sqrt{2}a^2)$

$\cos 45° = E_0(\sqrt{2}\,a^2) \times \dfrac{1}{\sqrt{2}} = E_0 a^2$




**16.** **(d)** Time period of the pendulum (T) is given by

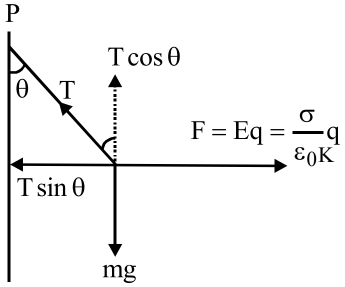
$$T = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}}$$

$$g_{\text{eff}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$\Rightarrow g_{\text{eff}} = \sqrt{g^2 + \left(\frac{gE}{m}\right)^2}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

**17.** **(c)**




$T\sin\theta = qE$ .... (i)

$T\cos\theta = mg$ .... (ii)

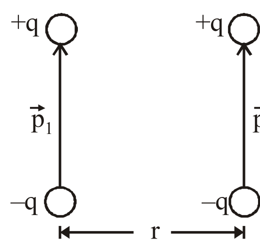
Dividing (i) by (ii),

$$\tan\theta = \frac{qE}{mg} = \frac{q}{mg}\left(\frac{\sigma}{\varepsilon_0 K}\right)\frac{\sigma q}{\varepsilon_0 K.mg}$$

$\therefore\ \sigma \propto \tan\theta$

**18.** **(c)** Force of interaction

$$F = \frac{1}{4\pi\in_0}.\frac{6p_1p_2}{r^4}$$

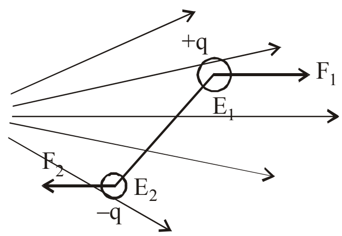



**19.** **(a)** The electric flux $\phi_1$ entering an enclosed surface is taken as negative and the electric flux $\phi_2$ leaving the surface is taken as positive, by convention. Therefore the net flux leaving the enclosed surface, $\phi = \phi_2 - \phi_1$

According to Gauss theorem

$$\phi = \frac{q}{\in_0} \Rightarrow q = \in_0\phi = \in_0(\phi_2 - \phi_1)$$

**20.** **(c)**



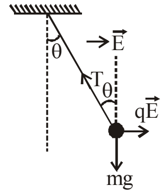

As the dipole is placed in non-uniform field, so the force acting on the dipole will not cancel each other. This will result in a force as well as torque.

**21.** **($12 \times 10^{-3}$)** Charges (q) = $2 \times 10^{-6}$ C, Distance (d) = 3 cm = $3 \times 10^{-2}$ m and electric field (E) = $2 \times 10^5$ N/C. Torque ($\tau$) = q.d.E

$\tau = (2 \times 10^{-6}) \times (3 \times 10^{-2}) \times (2 \times 10^5)$

$= 12 \times 10^{-3}$ N–m.

**22.** **($5 \times 10^{-4}$)** At equilibrium



$T\cos\theta = mg$ and $T\sin\theta = qE$

$mg = 30.7 \times 10^{-6} \times 9.8 = 3 \times 10^{-4}\,N$

$qE = 2 \times 10^{-8} \times 20{,}000 = 4 \times 10^{-4}\,N$

$\therefore\ T = \sqrt{(3\times10^{-4})^2 + (4\times10^{-4})^2} = 5 \times 10^{-4}\,N$

**23.** ($7.8 \times 10^{-7}$) $F = qE = mg\ (q = 6e = 6 \times 1.6 \times 10^{-19})$

Density (d) $= \dfrac{\text{mass}}{\text{volume}} = \dfrac{m}{\frac{4}{3}\pi r^3}$ or $r^3 = \dfrac{m}{\frac{4}{3}\pi d}$

Putting the value of d and m $\left(= \dfrac{qE}{g}\right)$ and solving we get $r = 7.8 \times 10^{-7}$m

**24.** **(0.125)** $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$

**Given,** $E_0 = 100\,N/c$

So, $\vec{E} = 100\hat{i} + 200\hat{j}$

Radius of circular surface = 0.02 m

Area $= \pi r^2 = \dfrac{22}{7} \times 0.02 \times 0.02$

$= 1.25 \times 10^{-3}\,\hat{i}\ m^2$ [Loop is parallel to Y-Z plane]

Now, flux ($\phi$) = EA cos$\theta$

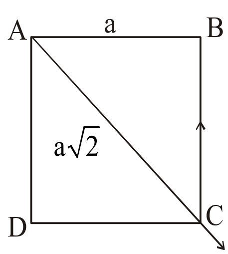
$= \left(100\hat{i} + 200\hat{j}\right).1.25 \times 10^{-3}\,\hat{i}\cos\theta°\ [\theta = 0°]$

$= 125 \times 10^{-3}\ Nm^2/C = 0.125\ Nm^2/C$

**25.** **($-2\sqrt{2}$)** It is given that $\vec{F}_A + \vec{F}_B + \vec{F}_D = 0$

Where $\vec{F}_A$, $\vec{F}_B$ and $\vec{F}_D$ are the forces applied by charges placed at A, B and D on the charge placed at C.

$\Rightarrow \vec{F}_B + \vec{F}_D = -\vec{F}_A$




$|\vec{F}_B + \vec{F}_D| = \sqrt{2}\dfrac{KqQ}{a^2}$

$\Rightarrow\ |\vec{F}_A| = \dfrac{KQ^2}{2a^2}$

$\therefore\ \sqrt{2}\dfrac{KqQ}{a^2} = -\dfrac{KQ^2}{2a^2} \Rightarrow \dfrac{Q}{q} = -2\sqrt{2}$

**26.** **(2)** $\rho = Cr^2$

$q = \int_0^r 4\pi r^2 dr\rho = \int_0^r 4\pi C r^4 dr = \dfrac{4}{5}\pi C r^5$

$E_{r=2R} = \dfrac{kq_{(r=R)}}{(2R)^2} = \dfrac{k(4/5)\pi CR^5}{4R^2}$

$E_r = \dfrac{R}{2} = \dfrac{kq_{(r=R)}}{(R/2)^2} = \dfrac{k(4/5)\pi C(R/2)^5}{(R/2)^2}$

Now solve to get $\dfrac{E_{r=2R}}{E_{r=R/2}} = 2$

**27.** **(70.7%)** Electric field intensity at the centre of the disc.

$E = \dfrac{\sigma}{2\epsilon_0}$ (given)

Electric field along the axis at any distance x from the centre of the disc

$E' = \dfrac{\sigma}{2\epsilon_0}\left(1 - \dfrac{x}{\sqrt{x^2 + R^2}}\right)$

From question, x = R (radius of disc)

$\therefore\ E' = \dfrac{\sigma}{2\epsilon_0}\left(1 - \dfrac{R}{\sqrt{R^2 + R^2}}\right)$

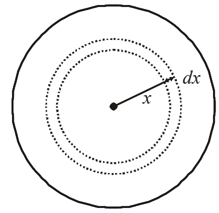
$= \dfrac{\sigma}{2\epsilon_0}\left(\dfrac{\sqrt{2}R - R}{\sqrt{2}R}\right)$

$\simeq \dfrac{4}{14}E$

$\therefore$ % reduction in the value of electric field

$= \dfrac{\left(E - \frac{4}{14}E\right) \times 100}{E} = \dfrac{1000}{14}\% \simeq 70.7\%$

**28.** **(2)** Let us consider a spherical shell of radius *x* and thickness *dx*. The volume of this shell is $4\pi x^2(dx)$. The charge enclosed in this spherical shell is



$dq = (4\pi x^2)dx \times kx^a$

$\therefore \quad dq = 4\pi k x^{2+a}\, dx.$

**For r = R :**

The total charge enclosed in the sphere of radius $R$ is

$$Q = \int_0^R 4\pi k\, x^{2+a}\, dx = 4\pi k \frac{R^{3+a}}{(3+a)}.$$

$\therefore$ The electric field at $r = R$ is

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{4\pi k R^{3+a}}{(3+a)R^2} = \frac{1}{4\pi\varepsilon_0}\frac{4\pi k}{(3+a)}R^{1+a}$$

**For r = R/2 :**

The total charge enclosed in the sphere of radius $R/2$ is

$$Q' = \int_0^{R/2} 4\pi k\, x^{2+a} dx = \frac{4\pi k (R/2)^{3+a}}{(3+a)}$$

$\therefore$ The electric field at $r = R/2$ is

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{4\pi k}{3+a}\frac{(R/2)^{3+a}}{(R/2)^2} = \frac{1}{4\pi\varepsilon_0}\frac{4\pi k}{(3+a)}\left(\frac{R}{2}\right)^{1+a}$$

Given, $E_2 = \frac{1}{8}E_1$

$\therefore$

$$\frac{1}{4\pi\varepsilon_0}\frac{4\pi k}{(3+a)}\left(\frac{R}{2}\right)^{1+a} = \frac{1}{2^3}\times\frac{1}{4\pi\varepsilon_0}\frac{4\pi k}{(3+a)}R^{1+a}$$

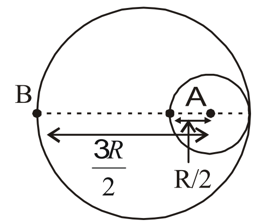
$\Rightarrow \quad 1 + a = 3 \Rightarrow a = 2$

**29. (–670)**

**30. (0.53)** Electric field at A $\left(R' = \frac{R}{2}\right)$

$$E_A . ds = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \vec{E}_A = \frac{\rho \times \frac{4}{3}\pi\left(\frac{R}{2}\right)^3}{\varepsilon_0 \cdot 4\pi\left(\frac{R}{2}\right)^2}$$




$$\Rightarrow \vec{E}_A = \frac{\sigma(R/2)}{3\varepsilon_0} = \left(\frac{\sigma R}{6\varepsilon_0}\right)$$

Electric fields at '$B$'

$$\vec{E}_B = \frac{k\times\rho\times\frac{4}{3}\pi R^3}{R^2} - \frac{k\times\rho\times\frac{4}{3}\pi\left(\frac{R}{2}\right)^3}{\left(\frac{3R}{2}\right)^2}$$

$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\varepsilon_0} - \left(\frac{1}{4\pi\varepsilon_0}\right)\frac{(\sigma)}{\left(\frac{3R}{2}\right)^2}\frac{4\pi}{3}\left(\frac{R}{2}\right)^3$$

$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\varepsilon_0} - \frac{\sigma R}{54\varepsilon_0}$$

$$\Rightarrow E_B = \frac{17}{54}\left(\frac{\sigma R}{\varepsilon_0}\right)$$

$$\left|\frac{E_A}{E_B}\right| = \frac{1\times 54}{6\times 17} = \left(\frac{9}{17}\right) = \frac{9}{17}\times\frac{2}{2} = \frac{18}{34}$$

Chapter 16

# Electrostatic Potential and Capacitance

**1.** **(b)** Potential at the centre of the triangle,

$$V = \frac{\Sigma q}{4\pi\varepsilon_0 r} = \frac{2q - q - q}{4\pi\varepsilon_0 r} = 0$$

Obviously, $E \neq 0$

**2.** **(c)** Volume of big drop = 1000 × volume of each small drop

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3 \Rightarrow R = 10r$$

$\because \quad V = \dfrac{kq}{r}$ and $V' = \dfrac{kq}{R} \times 1000$

Total charge on one small droplet is q and on the big drop is 1000q.

$$\Rightarrow \quad \frac{V'}{V} = \frac{1000r}{R} = \frac{1000}{10} = 100$$

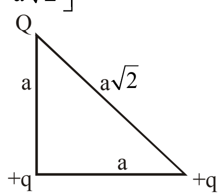
$\therefore \quad V' = 100V$

**3.** **(b)** Net electrostatic energy for the system

$$U = K\left[\frac{q^2}{a} + \frac{Qq}{a} + \frac{Qq}{a\sqrt{2}}\right] = 0$$

$$\Rightarrow q = -Q\left[1 + \frac{1}{\sqrt{2}}\right]$$

$$\Rightarrow Q = \frac{-q\sqrt{2}}{\sqrt{2}+1}$$




**4.** **(d)** $V = 5 + 4x^2 \quad \therefore \dfrac{dV}{dx} = 8x \quad \ldots(1)$

Force on a charge is

$$F = qE = q\left(-\frac{dV}{dx}\right) = q(-8x); \text{ from (1)}$$

$$= -2 \times 10^{-6} \times (-8 \times 0.5) = 8 \times 10^{-6} \text{ N}$$

**5.** **(a)** The electric potential $V(x, y, z) = 4x^2$ volt

Now $\vec{E} = \left(\hat{i}\dfrac{\partial V}{\partial x} + \hat{j}\dfrac{\partial V}{\partial y} + \hat{k}\dfrac{\partial V}{\partial z}\right)$

Now $\dfrac{\partial V}{\partial x} = 8x, \dfrac{\partial V}{\partial y} = 0$ and $\dfrac{\partial V}{\partial z} = 0$

Hence $\vec{E} = -8x\hat{i}$, so at point (1m, 0, 2m)

$\vec{E} = -8\hat{i}$ volt/metre or 8 along negative X-axis.

**6.** **(d)** Current will flow in connecting wire so that energy decreases in the form of heat through the connecting wire.

**7.** **(c)**

**8.** **(d)** For a parallel plate capacitor $C = \dfrac{\varepsilon_0 A}{d}$

$$\therefore \quad A = \frac{Cd}{\varepsilon_0} = \frac{1 \times 10^{-3}}{8.85 \times 10^{-12}} = 1.13 \times 10^8 \text{ m}^2$$

This corresponds to area of square of side 10.6 km which shows that one farad is very large unit of capacitance.

**9.** **(d)** As system is isolated so charge remains constant,

$$C = \frac{\varepsilon_0 A}{d}$$

so, if d increases then C decreases.

Now, for keeping the charge constant V increases.

**10.** **(c)** $C_2$ and $C_3$ are parallel so $V_2 = V_3$

$C_1$ and combination of $C_2$ & $C_3$ is in series.

So, $V = V_2 + V_1$ or $V = V_3 + V_1$

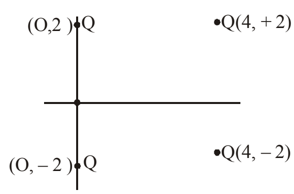
and also $Q_1 = Q_2 + Q_3$

**11.** **(b)** $\omega = \omega_f - v_i = \dfrac{q}{2}\left(\dfrac{1}{C_f} - \dfrac{1}{C_i}\right)$

$$= \frac{(5 \times 10)^2}{2}\left(\frac{1}{2} - \frac{1}{5}\right) \times 10^6$$

$= 3.75 \times 10^{-6}$J

**39.** **(b)**




Potential at origin

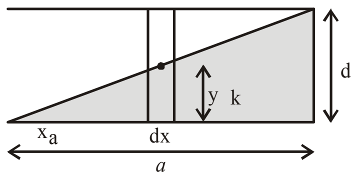
$$v = \frac{KQ}{2} + \frac{KQ}{2} + \frac{KQ}{\sqrt{20}} + \frac{KQ}{\sqrt{20}}$$

and potential at $\infty = 0 = KQ\left(1 + \dfrac{1}{\sqrt{5}}\right)$

$\therefore$ Work required to put a fifth charge Q at origin

$$W = VQ = \frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{5}}\right)$$

**13. (b)**




From figure, $\frac{y}{x} = \frac{d}{a} \Rightarrow y = \frac{d}{a}x$

$$dy = \frac{d}{a}(dx) \Rightarrow \frac{1}{dc} = \frac{y}{K\varepsilon_0 adx} + \frac{(d-y)}{\varepsilon_0 adx}$$

$$\frac{1}{dc} = \frac{y}{\varepsilon_0 abx}\left(\frac{y}{k}+d-y\right)$$

$$\int dc = \int \frac{\varepsilon_0 adx}{\frac{y}{k}+d-y}$$

or, $$c = \varepsilon_0 a.\frac{a}{d}\int_0^d \frac{dy}{d+y\left(\frac{1}{k}-1\right)}$$

$$= \frac{\varepsilon_0 a^2}{\left(\frac{1}{k}-1\right)d}\left[\ell n\left(d+y\left(\frac{1}{k}-1\right)\right)\right]_0^d$$

$$= \frac{k\in_0 a^2}{(1-k)d}\ell n\left(\frac{d+d\left(\frac{1}{k}-1\right)}{d}\right)$$

$$= \frac{k\in_0 a^2}{(1-k)d}\ell n\left(\frac{1}{k}\right) = \frac{k\in_0 a^2 \ell nk}{(k-1)d}$$

**14. (b)** $W = -\Delta u$

$$= (-1)\left|\frac{(c\varepsilon)^2}{2kc} - \frac{(c\varepsilon)^2}{2c}\right|$$

$$= \frac{\varepsilon^2 c}{2}\frac{k-1}{k}$$

$= 508$ J

**15. (d)** When two capacitors with capacitance $C_1$ and $C_2$ at potential $V_1$ and $V_2$ connected to each other by wire, charge begins to flow from higher to lower potential till they acquire common potential. Here, some loss of energy takes place which is given by.

Heat loss, $H = \frac{C_1C_2}{2(C_1+C_2)}(V_1-V_2)^2$

In the equation, put $V_2 = 0$, $V_1 = V_0$

$$C_1 = C,\ C_2 = \frac{C}{2}$$

$$\text{Loss of heat} = \frac{C\times\frac{C}{2}}{2\left(C+\frac{C}{2}\right)}(V_0-0)^2 = \frac{C}{6}V_0^2$$

$$H = \frac{1}{6}CV_0^2$$

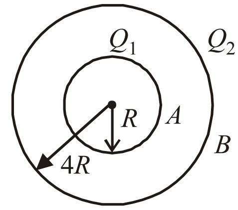
**16. (a)** We have given two metallic hollow spheres of radii $R$ and $4R$ having charges $Q_1$ and $Q_2$ respectively.

Potential on the surface of inner sphere (at $A$)

$$V_A = \frac{kQ_1}{R} + \frac{kQ_2}{4R}$$

Potential on the surface of outer sphere (at $B$)

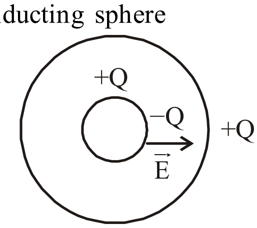
$$V_B = \frac{kQ_1}{4R} + \frac{kQ_2}{4R} \quad \left(\text{Here, } k = \frac{1}{4\pi\varepsilon_0}\right)$$




Potential difference,

$$\Delta V = V_A - V_B = \frac{3}{4}\cdot\frac{kQ_1}{R} = \frac{3}{16\pi\in_0}\cdot\frac{Q_1}{R}$$

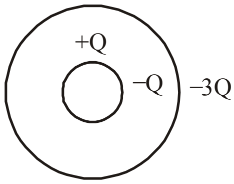
**17. (d)** When charge Q is on inner solid conducting sphere




Electric field between spherical surface

$$\vec{E} = \frac{KQ}{r^2}\ So\int \vec{E}.d\vec{r} = V\ given$$

Now when a charge – 4Q is given to hollow shell




Electric field between surface remain unchanged.

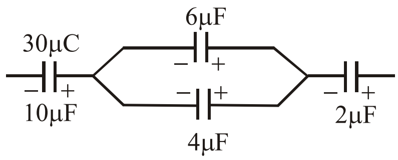
$$\vec{E} = \frac{KQ}{r^2}$$

as, field inside the hollow spherical shell = 0

$\therefore$ Potential difference between them remain unchanged

i.e. $\int \vec{E}.d\vec{r} = V$

**18.** **(18)**




As given in the figure, 6μF and 4μF are in parallel. Now using charge conservation

Charge on 6μF capacitor $= \frac{6}{6+4} \times 30 = 18\mu C$

Since charge is asked on right plate therefore is +18μC

**19.** **(8.5)** Capacitance of a capacitor with a dielectric of dielectric constant k is given by

$$C = \frac{k \epsilon_0 A}{d}$$

$$\because\ E = \frac{V}{d} \qquad \therefore\ C = \frac{k \epsilon_0 AE}{V}$$

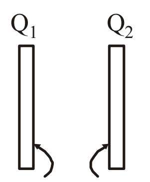
$$15 \times 10^{-12} = \frac{k \times 8.86 \times 10^{-12} \times 10^{-4} \times 10^{6}}{500}$$

$k = 8.5$

**20.** **(1)** $V = \frac{Q}{C}$

$$= \left(\frac{Q_1 - Q_2}{2C}\right)$$

$$= \left(\frac{4-2}{2 \times 1}\right) = 1\ \text{V}$$




$$\frac{Q_1 - Q_2}{2} - \frac{(Q_1 - Q_2)}{2}$$

**21.** **(180)** Given, $\vec{E} = (Ax + B)\hat{i}$

or $E = 20x + 10$

Using $V = \int E dx$, we have

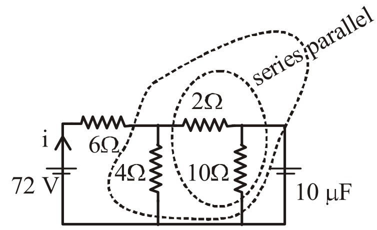
$$V_2 - V_1 = \int_{-5}^{1} (20x + 10) dx = -180\ \text{V}$$

or $V_1 - V_2 = 180$ V

**22.** **(200)** At steady state, there is no current in capacitor.

2Ω and 10Ω are in series. There equivalent resistance is 12Ω. This 12Ω is in parallel with 4Ω and there combined resistance is 12 × 4/(12 + 4). This resistance is in series with 6Ω. Therefore, current drawn from battery

$$i = \frac{V}{R} = \left(\frac{72}{6 + \frac{12 \times 4}{12 + 4}}\right) = 8A$$




Current in 10Ω resistor

$$i' = \left(\frac{4}{4+12}\right)8 = 2A$$

Pd across capacitor, V = $i'$R = 2 × 10 = 20V

$\therefore$ Charge on the capacitor, q = CV = 10 × 20 = 200 μC.

**23.** **(100)** Volume of big drop = 1000 × volume of each small drop

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3 \Rightarrow R = 10r$$

$$\because \quad V = \frac{kq}{r} \text{ and } V' = \frac{kq}{R} \times 1000$$

Total charge on one small droplet is q and on the big drop is 1000q.

$$\Rightarrow \frac{V'}{V} = \frac{1000r}{R} = \frac{1000}{10} = 100$$

$$\therefore \quad V' = 100V$$

**24.** **(10)** Potential at any point inside the sphere = potential at the surface of the sphere = 10V.

**25.** **(4)** The inner sphere is grounded, hence its potential is zero. The net charge on isolated outer sphere is zero.

Let the charge on inner sphere be q'.

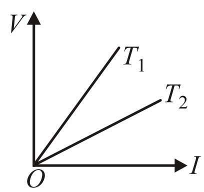
$\therefore$ Potential at centre of inner sphere is

$$= \frac{1}{4\pi\varepsilon_0}\frac{q'}{a} + 0 + \frac{1}{4\pi\varepsilon_0}\frac{q}{4a} = 0 \quad \therefore\ q' = -\frac{q}{4}$$

# Chapter 17 Current Electricity

1. **(b)** $R_t = R_0(1 + \alpha t)$ at t°C $R_t = 3R_0$
$\alpha = 4 \times 10^{-3}$ /°C
$3R_0 = R_0(1 + 4 \times 10^{-3} \times t)$
$\therefore \quad 3 - 1 = 4 \times 10^{-3} t \quad \therefore t = \dfrac{2}{4 \times 10^{-3}} = 500°C$

2. **(a)** The slope of $V - I$ graph gives the resistance of a conductor at a given temperature.
From the graph, it follows that resistance of a conductor at temperature $T_1$ is greater than at temperature $T_2$. As the resistance of a conductor is more at higher temperature and less at lower temperature, hence $T_1 > T_2$




3. **(b)** $I = \dfrac{E}{R + r} \quad \Rightarrow \quad V = \dfrac{E}{R + r} R.$
$[\because V = IR]$
$\Rightarrow \quad r = \dfrac{(E - V)}{V} R.$

4. **(b)** Since, the voltage is same for the two combinations, therefore $H \propto \dfrac{1}{R}$. Hence, the combination of 39 bulbs will glow more.

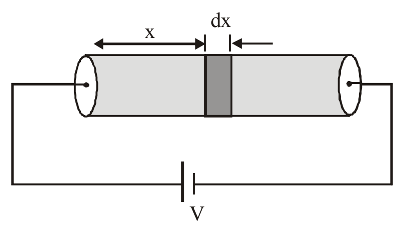
5. **(b)** $\dfrac{P}{Q} = \dfrac{R}{S}$ where $S = \dfrac{S_1 S_2}{S_1 + S_2}$

6. **(c)** The resistance of the square Y is given by
$R_Y = \rho \dfrac{2L}{(2L)t} = \dfrac{\rho}{t}$
same as before. Hence, $R_x/R_Y = 1$

7. **(d)** $R_1 = R_{10}(1 + \alpha_1 t_1)$
$R_2 = R_{20}(1 + \alpha_2 t_2)$
As $R_1 = R_2$ and $R_{10} = R_{20}$, $\Rightarrow \quad \dfrac{\alpha_1}{\alpha_2} = \dfrac{t_2}{t_1}$

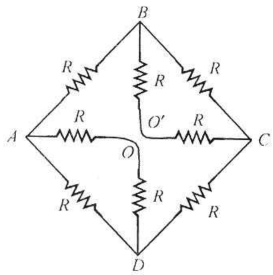
8. **(a)** Let the resistance of single copper wire be $R_1$. If $\rho$ is the specific resistance of copper wire, then
$$R_1 = \frac{\rho \times \ell}{A_1} = \frac{\rho \times \ell}{\pi r_1^2} \qquad \text{...(1)}$$
When the wire is replaced by six wires, let the resistance of each wire be $R_2$. Then
$$R_2 = \frac{\rho \times \ell}{A_2} = \frac{\rho \times \ell}{\pi r_2^2} \qquad \text{...(2)}$$
From eqs. (1) and (2), we get
$$\frac{R_1}{R_2} = \frac{r_2^2}{r_1^2} \text{ or } \frac{5}{R_2} = \frac{(3 \times 10^{-3})^2}{(9 \times 10^{-3})^2}; \quad R_2 = 45\,\Omega$$
These six wires are in parallel. Hence the required resistance would be $R_2 = 7.5\ \Omega$

9. **(a)** Consider an element part of solid at a distance x from left end of width dx.




Resistance of this elemental part is,
$$dR = \frac{\rho dx}{\pi a^2} = \frac{\rho_0 x dx}{\pi a^2}$$
$$R = \int dR = \int_0^L \frac{\rho_0 x dx}{\pi a^2} = \frac{\rho_0 L^2}{2\pi a^2}$$
Current through cylinder is, $I = \dfrac{V}{R} = \dfrac{V \times 2\pi a^2}{\rho_0 L^2}$
Potential drop across element is, $dV = I\,dR = \dfrac{2V}{L^2} x\,dx$
$$E(x) = \frac{dV}{dx} = \frac{2V}{L^2} x$$

**10.** **(d)** The equivalent circuit is as shown in figure.




The resistance of arm AOD (= R + R) is in parallel to the resistance R of arm AD.

Their effective resistance

$$R_1 = \frac{2R \times R}{2R + R} = \frac{2}{3}R$$

The resistance of arms AB, BC and CD is

$$R_2 = R + \frac{2}{3}R + R = \frac{8}{3}R$$

The resistance $R_1$ and $R_2$ are in parallel. The effective resistance between A and D is

$$R_3 = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{\frac{2}{3}R \times \frac{8}{3}R}{\frac{2}{3}R + \frac{8}{3}R} = \frac{8}{15}R.$$

**11.** **(d)** Let internal resistance of source = R

Current in coil of resistance $R_1 = I_1 = \frac{V}{R + R_1}$

Current in coil of resistance $R_2 = I_2 = \frac{V}{R + R_2}$

Further, as heat generated is same, so

$$I_1^2 R_1 t = I_2^2 R_2 t$$

or $\left(\frac{V}{R + R_1}\right)^2 R_1 = \left(\frac{V}{R + R_2}\right)^2 R_2$

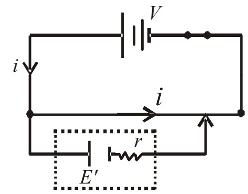
$$\Rightarrow \quad R_1(R + R_2)^2 = R_2(R + R_1)^2$$

$$\Rightarrow R^2(R_1 - R_2) = R_1 R_2 (R_1 - R_2)$$

$$\Rightarrow R = \sqrt{R_1 R_2}$$

**12.** **(d)** From the principle of potentiometer, $V \propto l$

$\Rightarrow \frac{V}{E} = \frac{l}{L}$; where




V = emf of battery,

E = emf of standard cell.

L = length of potentiometer wire

$$V = \frac{El}{L} = \frac{30E}{100}$$

**13.** **(a)**

**14.** **(c)** Balancing length $l$ will give emf of cell

$\therefore\ E = Kl$

Here K is potential gradient.

If the cell is short circuited by resistance '$R$'

Let balancing length obtained be $l'$ then

$V = kl'$

$$r = \left(\frac{E - V}{V}\right)R$$

$\Rightarrow \quad V = E - V \quad \Rightarrow 2V = E \quad [\because\ r = R \text{ given}]$

or, $2Kl' = Kl \quad \therefore \quad l' = \frac{l}{2}$

**15.** **(c)** $\rho = \frac{m}{ne^2\tau}$

$$= \frac{9.1 \times 10^{-31}}{8.5 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 25 \times 10^{-15}}$$

$= 10^{-8}\ \Omega\text{-m}$

**16.** **(a)** Using, $I = neAv_d$

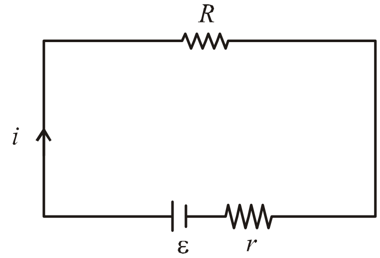
$\therefore$ Drift speed $v_d = \frac{1}{neA}$

$$\frac{1.5}{9 \times 10^{28} \times 1.6 \times 10^{-19} \times 5 \times 10^{-6}} = 0.02\ \text{mms}^{-1}$$

**17.** **(d)** $i = \left(\frac{\varepsilon}{R + r}\right)$

Power delivered to R.

$$P = i^2 R = \left(\frac{\varepsilon}{R + r}\right)^2 R$$




P to be maximum, $\frac{dP}{dR} = 0$

or $\frac{d}{dR}\left[\left(\frac{\varepsilon}{R+r}\right)^2 R\right] = 0$

or $R = r$

**18.** **(c)** Current passing through resistance $R_1$,

$i_1 = \frac{v}{R_1} = \frac{10}{20} = 0.5A$

and, $i_2 = 0$

**19.** **(d)** Net Power, $P$

$= 15 \times 45 + 15 \times 100 + 15 \times 10 + 2 \times 1000$

$= 15 \times 155 + 2000$ W

Power, $P = VI$

$\Rightarrow \quad I = \frac{P}{V}$

$\therefore \quad I_{\text{main}} = \frac{15 \times 155 + 2000}{220} = 19.66\ A \approx 20A$

**20.** **(c)** We have given

$\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}} \Rightarrow \frac{dR}{d\ell} = k \times \frac{1}{\sqrt{\ell}}$ (where k is constant)

$dR = k\frac{d\ell}{\sqrt{\ell}}$

Let $R_1$ and $R_2$ be the resistance of AP and PB respectively.

Using wheatstone bridge principle

$\therefore \quad \frac{R'}{R'} = \frac{R_1}{R_2}$ or $R_1 = R_2$

Now, $\int dR = k\int \frac{d\ell}{\sqrt{\ell}}$

$\therefore \quad R_1 = k\int_0^{\ell} \ell^{-1/2} d\ell = k.2.\sqrt{\ell}$

$R_2 = k\int_{\ell}^{1} \ell^{-1/2} d\ell = k.(2 - 2\sqrt{\ell})$

Putting $R_1 = R_2$

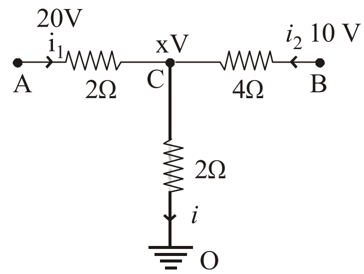
$k2\sqrt{\ell} = k(2 - 2\sqrt{\ell})$

$\therefore \quad 2\sqrt{\ell} = 1$

$\sqrt{\ell} = \frac{1}{2}$

i.e., $\ell = \frac{1}{4} m \Rightarrow 0.25$ m

**21.** **(5)**




Let voltage at C = xV

From kirchhoff's current law,

KCL : $i_1 + i_2 = i$

$\frac{20 - x}{2} + \frac{10 - x}{4} = \frac{x - 0}{2} \Rightarrow x = 10$

$\therefore \quad i = \frac{V}{R} = \frac{X}{R} = \frac{10}{2} = 5A$

**22.** **(1)** $P = V^2/R$

$\Rightarrow \quad R = V^2/P \times 10^4 / 100 = 400\ \Omega$

$S = 10^4/200 = 0.5 \times 10^2 = 50\ \Omega$

$R / S = 8$

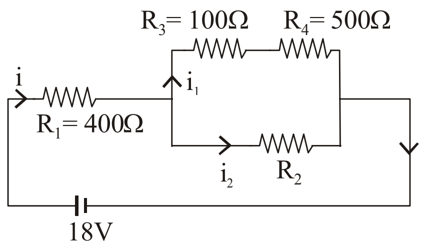
**23.** **(1)** Resistance, $R = \frac{\rho\ell}{A}$

$R = \rho\frac{\ell}{A} \times \frac{\ell}{\ell} = \frac{\rho\ell^2}{V}$ [$\because$ Volume (V) = $A\ell$.]

Since resistivity and volume remains constant therefore % change in resistance

$\frac{\Delta R}{R} = \frac{2\Delta\ell}{\ell} = 2 \times (0.5) = 1\%$

**24. (300)**




Across $R_4$ reading of voltmeter, $V_4 = 5V$

Current, $i_4 = \frac{V_4}{R_4} = 0.01\,A$

$V_3 = i_1 R_3 = 1V$

$V_3 + V_4 = 6V = V_2$

$V_1 + V_3 + V_4 = 18V$

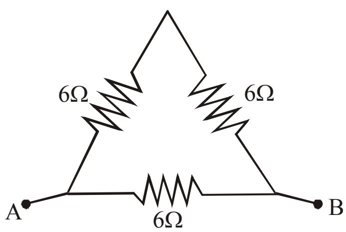
$\Rightarrow V_1 = 12\,V$

$i = \frac{V_1}{R_1} = 0.03A$

$i = i_1 + i_2 \Rightarrow i_2 = i - i_1 = 0.03 - 0.01A = 0.02\,A$

$\therefore R_2 = \frac{V_2}{i_2} = \frac{6}{0.02} = 300\Omega$

**25. (4)**




Resistance, $R \propto l$ so resistance of each side of the equilateral triangle = 6 Ω

Resistance $R_{eq}$ between any two vertices

$\frac{1}{R_{eq}} = \frac{1}{12} + \frac{1}{6} \Rightarrow R_{eq.} = 4\Omega$

**26. (20)** Colour code for carbon resistor

| Bl, | Br, | R, | O, | Y, | G, | Blue, | V, | Gr, | W |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

Resistance, $R = AB \times C \pm D$

$\therefore$ Resistance, $R = 50 \times 10^2\,\Omega$

Now using formula, Power, $P = i^2R$

$\therefore \quad i = \sqrt{\frac{P}{R}} = \sqrt{\frac{2}{50 \times 10^2}} = 20mA$

**27. ($11 \times 10^{-5}$)** Power, $P = I^2R$

$4.4 = 4 \times 10^{-6} \times R$

$\Rightarrow R = 1.1 \times 10^6\Omega$

When supply of 11 v is connected

Power, $P' = \frac{v^2}{R} = \frac{11^2}{1.1} \times \frac{11^2}{1.1} \times 10^{-6} = 11 \times 10^{-5}\,W$

**28. (53)** $I = \frac{dQ}{dt} = 10t + 3 = 10 \times 5 + 3 = 53\,A$

**29. ($6.25 \times 10^{15}$)** Current, $I = \frac{\text{Charge}}{\text{Time}}$;

as charge $q = n \times 1.6 \times 10^{-19}$

$10^{-3}\,\text{amp} = \frac{n \times 1.6 \times 10^{-19}}{1\,\text{sec}} \Rightarrow n = 6.25 \times 10^{15}$

**30. (3.6)** The potential difference across 4Ω resistance is given by $V = 4 \times i_1 = 4 \times 1.2 = 4.8$ volt

So, the potential across 8Ω resistance is also 4.8 volt.

Current $i_2 = \frac{V}{8} = \frac{4.8}{8} = 0.6\,\text{amp}$

Current in 2Ω resistance $i = i_1 + i_2$

$\therefore i = 1.2 + 0.6 = 1.8\,\text{amp}$

Potential difference across 2Ω resistance

$V_{BC} = 1.8 \times 2 = 3.6$ volts

# CHAPTER 18 Moving Charges and Magnetism

**1.** **(c)** As electron move with constant velocity without deflection. Hence, force due to magnetic field is equal and opposite to force due to electric field.

$$qvB = qE \Rightarrow v = \frac{E}{B} = \frac{20}{0.5} = 40 \text{ m/s}$$

**2.** **(b)** $M = iA = i \times \pi R^2$

also $i = \frac{Q\omega}{2\pi} \Rightarrow M = \frac{1}{2}Q\omega R^2 \left[\because i = \frac{Q}{t}\right]$

**3.** **(d)** $F = qvB\sin\theta \Rightarrow B = \frac{F}{qv\sin\theta}$

$$B_{\min} = \frac{F}{qv} \quad (\text{when } \theta = 90°)$$

$$= \frac{10^{-10}}{10^{-12} \times 10^5} = 10^{-3} \text{ Tesla}$$

**4.** **(a)**

**5.** **(d)** $B = \frac{\mu_0}{4\pi}.\frac{2\pi n i_1}{r_1} - \frac{\mu_0}{4\pi}.\frac{2\pi n i_2}{r_2}$

$$= \frac{\mu_0}{2}\left[\frac{ni_1}{r_1} - \frac{ni_2}{r_2}\right]$$

**6.** **(a)** $\frac{\mu_0 I_c}{2R} = \frac{\mu_0 I_e}{2\pi H} \Rightarrow H = \frac{I_e R}{\pi I_c}$

**7.** **(d)** $B = \mu_0 n\mathrm{I}$

**8.** **(a)** Force between two long conductor carrying current,

$$F = \frac{\mu_0}{4\pi}\frac{2I_1I_2}{d} \times \ell \quad ; \quad F' = -\frac{\mu_0}{4\pi}\frac{2(2I_1)I_2}{3d}\ell$$

$$\therefore \frac{F'}{F} = \frac{-2}{3}$$

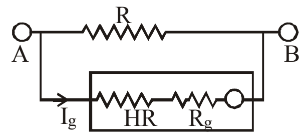
**9.** **(a)** $\tau_{max} = MB = niAB = ni\,(l \times b)\,B$

$\tau_{max} = 600 \times 10^{-5} \times 5 \times 10^{-2} \times 12 \times 10^{-2} \times 0.10 = 3.6 \times 10^{-6}$ N-m.

**10.** **(a)** $R_g = 50\Omega$, $I_g = 25 \times 4 \times 10^{-4}\Omega = 10^{-2}$ A

Range of V = 25 volts

$V = I_g(HR + R_g)$




$$\therefore HR = \frac{V}{I_g} - R_g = 2450\Omega$$

**11.** **(a)** In mass spectrometer, when ions are accelerated through potential V

$$\frac{1}{2}mv^2 = qV \quad \text{..........(i)}$$

As the magnetic field curves the path of the ions in a semicircular orbit

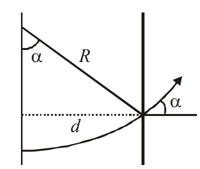
$$Bqv = \frac{mv^2}{R} \Rightarrow v = \frac{BqR}{m} \quad \text{.......... (ii)}$$

Substituting (ii) in (i)

$$\frac{1}{2}m\left[\frac{BqR}{m}\right]^2 = qV \quad \text{or} \quad \frac{q}{m} = \frac{2V}{B^2R^2}$$

Since V and B are constants, $\therefore \frac{q}{m} \propto \frac{1}{R^2}$

**12.** **(d)** From figure,
$\sin\alpha = d/R$




we know, $\frac{mv^2}{R} = qvB$

$$\Rightarrow R = \frac{mv}{qB} \quad \therefore \sin\alpha = \frac{dqB}{mv}$$

$$\sin\alpha = Bd\sqrt{\frac{q}{2mV}} \left[\because qV = \frac{1}{2}mv^2\right]$$

**13.** **(b)** $\frac{mv^2}{r} = qvB \Rightarrow r = \frac{mv}{qB}$

$$\Rightarrow r_p = \frac{m_p v_p}{q_p B}; \; r_d = \frac{m_d v_d}{q_d B}; \; r_\alpha = \frac{m_\alpha v_\alpha}{q_\alpha B}$$

$$m_\alpha = 4m_p, \; m_d = 2m_p$$

$q_\alpha = 2q_p\,,\ q_d = q_p$

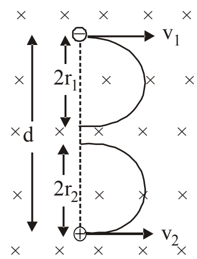
From the problem

$$E_p = E_d = E_\alpha = \frac{1}{2}m_p v_p{}^2 = \frac{1}{2}m_d v_d{}^2 = \frac{1}{2}m_\alpha v_\alpha^2$$

$$\Rightarrow v_p^2 = 2v_d^2 = 4v_\alpha^2$$

Thus we have, $r_\alpha = r_p < r_d$

**14. (c)** The particles will not collide if




$d > (2r_1 + 2r_2)$

$d > 2\,(r_1 + r_2)$

$$d > 2\left(\frac{mv_1}{qB} + \frac{mv_2}{qB}\right)$$

or $$d > \frac{2m}{qB}(v_1 + v_2)$$

**15. (d)** $$B_1 = -\frac{\mu_0 i_1}{2\pi r/2} + \frac{\mu_0 i_2}{2\pi r/2} = \frac{\mu_0}{\pi r}(i_2 - i_1)$$

$$= 6\times10^{-6}\,\mathrm{T} \quad \text{...(i)}$$

When the current is reversed in $I_2$,

$$B_2 = -\frac{\mu_0(i_1+i_2)}{2\pi r/2} = -\frac{\mu_0(i_1+i_2)}{\pi r} = 3\times10^{-5}\,\mathrm{T} \quad \text{...(ii)}$$

Dividing (ii) by (i) we get

$$\frac{-(i_1+i_2)}{i_2-i_1} = \frac{30}{6} = 5$$

$$-(i_1+i_2) = 5i_2 - 5i_1 \Rightarrow 6i_2 = 4i_1$$

$$\frac{i_1}{i_2} = \frac{3}{2}$$

**16. (a)**

**17. (a)** $$B_{solenoid} = \mu_0\, n_s\, i_s = \frac{\mu_0 N_S i_S}{L_S},$$

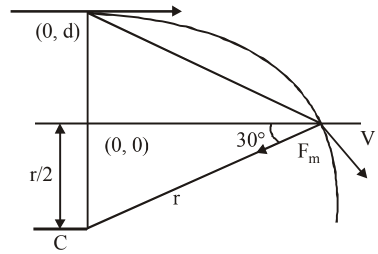
$$\tau = B_S\, iNA = \frac{\mu_0 N_S i_S\, iN\pi r^2}{L_S}$$

$$\tau = \frac{4\pi\times10^{-7}\times500\times3\times0.4\times10\times\pi\times(0.01)^2}{0.4}$$

$$= 6\pi^2\times10^{-7}\ \text{N-m.}$$

**18. (Bonus)** Assuming particle enters from (0, d)

$$r = \frac{mv}{qB}, \qquad d = \frac{r}{2}$$



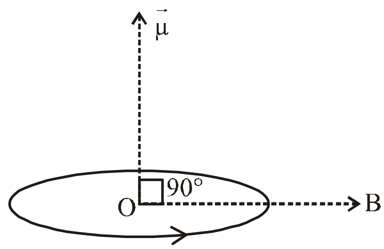

$$a = \frac{qVB}{m}\left[\frac{-\sqrt{3}i - j}{2}\right]$$

this option is not given in the all above four choices.

**19. (b)** $|\vec{\tau}| = |\vec{\mu}\times\vec{B}| \quad [\mu = NIA]$

$= NIA \times B\sin 90^\circ \qquad [A = \pi r^2]$

$\Rightarrow \tau = NI\pi r^2 B$




**20.** (a) $I_1 I_2$ = Positive

(attract) $F$ = Negative

$I_1 I_2$= Negative

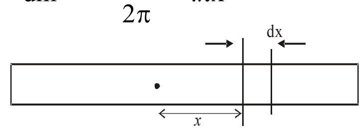
(repell) $F$ = Positive

Hence, option (a) is the correct answer.

**21. (6)** At a distance x consider small element of width dx.

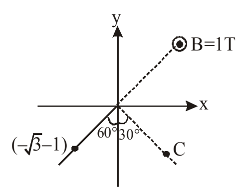
Magnetic moment of the small element is

$$dm = \frac{\left(\frac{q}{\ell}dx\right)\omega}{2\pi}.\pi x^2$$




$$M = \int_{-\ell/2}^{\ell/2}\frac{q\omega}{2\ell}x^2 dx\ ;\ M = \frac{q\omega\ell^2}{24} = \frac{q\pi f\ell^2}{12}$$

**22. (5)** The centre will be at C as shown :



Coordinates of the centre are (r cos 60°, – r sin 60°)
where r = radius of circle

$$= \frac{mv}{Bq} = \frac{1\times1}{1\times1} = 1 \text{ i.e.,} \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

**23.** **(27)** Given : Radius = R; Distance $x = 2\sqrt{2}R$

$$\frac{B_{centre}}{B_{axis}} = \left(1+\frac{x^2}{R^2}\right)^{3/2} = \left(1+\frac{(2\sqrt{2}R)^2}{R^2}\right)^{3/2}$$

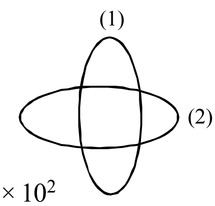
$$= (9)^{3/2} = 27$$

**24.** **($5 \times 10^{-5}$)** The magnetic field due to circular coil,

$$B_1 = \frac{\mu_0 i_1}{2r} = \frac{\mu_0 i_1}{2(2\pi\times10^{-2})} = \frac{\mu_0\times3\times10^2}{4\pi}$$

$$B_2 = \frac{\mu_0 i_2}{2(2\pi\times10^{-2})}$$

$$= \frac{\mu_0\times4\times10^2}{4\pi}$$




$$B = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0}{4\pi}.5\times10^2$$

$$\Rightarrow B = 10^{-7}\times5\times10^2 \Rightarrow B = 5\times10^{-5}\ \text{Wb/m}^2$$

**25.** **(7)** $B = \frac{\mu_0 I}{4\pi\frac{R}{\sqrt{2}}}(\sin 90° + \sin 135°)$

$$= \frac{\mu_0 I}{4\pi R}(\sqrt{2}-1)$$

**26.** **(7)** Magnetic field strength at P due to $I_1$

$$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi(AP)}\hat{k} = \frac{4\pi\times10^{-7}\times2}{2\pi\times1\times10^{-2}}\hat{k}$$

$$= (4\times10^{-5}\,T)\,\hat{k}$$

Magnetic field strength at P due to $I_2$

$$\vec{B}_2 = \frac{\mu_0 I_2}{2\pi(BP)}\hat{j} = \frac{4\pi\times10^{-7}\times3}{2\pi\times2\times10^{-2}}\hat{j}$$

$$= (3\times10^{-5}\,T)\,\hat{j}$$

Hence, $\vec{B} = (3\times10^{-5}\,T)\,\hat{j} + (4\times10^{-5}\,T)\,\hat{k}$

**27.** **($2\times10^{-24}$)** As particle is moving along a circular path

$$\therefore R = \frac{mv}{qB} \qquad ...(i)$$

Path is straight line, then
qE = qvB

$$E = vB \Rightarrow v = \frac{E}{B} \qquad ....(ii)$$

From equation (i) and (ii)

$$m = \frac{qB^2R}{E} = \frac{1.6\times10^{-19}\times(0.5)^2\times0.5\times10^{-2}}{100}$$

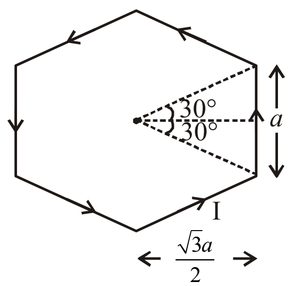
$$\therefore m = 2.0\times10^{-24}\ \text{kg}$$

**28.** **(4)** Pitch $= (v\cos\theta)T$ and $T = \frac{2\pi m}{qB}$

$$\therefore \text{Pitch} = (V\cos\theta)\frac{2\pi m}{qB}$$

$$= (4\times10^5\cos60°)\frac{2\pi}{0.3}\left(\frac{1.67\times10^{-27}}{1.69\times10^{-19}}\right) = 4\ \text{cm}$$

**29.** **($500\sqrt{3}$)**




Magnetic field due to one side of hexagon

$$B = \frac{\mu_0 I}{4\pi\frac{\sqrt{3}a}{2}}(\sin30° + \sin30°)$$

$$\Rightarrow B = \frac{\mu_0 I}{2\sqrt{3}a}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}a\pi}$$

Now, magnetic field due to one hexagon coil

$$B = 6\times\frac{\mu_0 I}{2\sqrt{3}a\pi}$$

Again magnetic field at the centre of hexagonal shape coil of 50 turns,

$$B = 50\times6\times\frac{\mu_0 I}{2\sqrt{3}a\pi} \qquad \left[\because a = \frac{10}{100} = 0.1\ \text{m}\right]$$

$$\text{or, } B = \frac{150\mu_0 I}{\sqrt{3}\times0.1\times\pi} = 500\sqrt{3}\frac{\mu_0 I}{\pi}$$

**30.** **(20)** Given,
Area of galvanometer coil, $A = 3\times10^{-4}\ \text{m}^2$
Number of turns in the coil, $N = 500$
Current in the coil, $I = 0.5$ A

Torque $\tau = |\vec{M}\times\vec{B}| = NiAB\sin(90°) = NiAB$

$$\Rightarrow B = \frac{\tau}{NiA} = \frac{1.5}{500\times0.5\times3\times10^{-4}} = 20T$$

CHAPTER 19

# Magnetism and Matter

**1.** **(c)** Here, $\theta = 30°$, $\tau = 0.018$ N-m, $B = 0.06$ T
Torque on a bar magnet :
$\tau = MB\sin\theta$
$0.018 = M \times 0.06 \times \sin 30°$
$\Rightarrow 0.018 = M \times 0.06 \times \frac{1}{2} \Rightarrow M = 0.6 \text{ A-m}^2$
Position of stable equilibrium ($\theta = 0°$)
Position of unstable equilibrium ($\theta = 180°$)
Minimum work required to rotate bar magnet from stable to unstable equilibrium
$\Delta U = U_f - U_i = -MB\cos 180° - (-MB\cos 0°)$
$W = 2MB = 2 \times 0.6 \times 0.06$
$\therefore\ W = 7.2 \times 10^{-2}$ J

**2.** **(a)** According to Curie law for paramagnetic substance,

$$\chi \propto \frac{1}{T_C} \Rightarrow \frac{\chi_1}{\chi_2} = \frac{T_{C_2}}{T_{C_1}}$$

$$\frac{2.8 \times 10^{-4}}{\chi_2} = \frac{300}{350}$$

$$\chi_2 = \frac{2.8 \times 350 \times 10^{-4}}{300} = 3.266 \times 10^{-4}$$

**3.** **(d)** Magnetic susceptibility, $\chi = \frac{I}{H}$

where, $I = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{20 \times 10^{-6}}{10^{-6}}$

$= 20 \text{ N/m}^2$

Now, $\chi = \frac{20}{60 \times 10^3} = \frac{1}{3} \times 10^{-3} = 3.3 \times 10^{-4}$

**4.** **(b)** Corecivity, $H = \frac{B}{\mu_0}$

and $B = \mu_0 ni \left( n = \frac{N}{\ell} \right)$

or, $H = \frac{N}{\ell} i = \frac{100}{0.2} \times 5.2 = 2600$ A/m

**5.** **(Bonus)** We have, $T = 2\pi\sqrt{\frac{I}{MB_x}}$

$$\therefore\ \frac{T_1^2}{T_2^2} = \frac{Bx_2}{Bx_1}$$

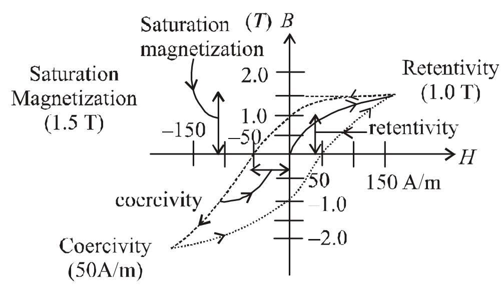
or $\left(\frac{2}{1.5}\right)^2 = \frac{B_2 \cos 45°}{B_1 \cos 30°} = \frac{B_2 \times 2}{\sqrt{2} \times B_1 \times \sqrt{3}}$

$$\left(\frac{4}{3}\right)^2 = \frac{B_2}{B_1} \times \frac{2}{\sqrt{6}}$$

$$\therefore\ \frac{B_1}{B_2} = \frac{9}{8\sqrt{6}} = 0.46$$

**6.** **(b)** Electromagnet should be amenable to magnetisation & demagnetization.
$\therefore$ Materials suitable for making electromagnets should have low retentivity and low coercivity should be low.

**7.** **(d)**




**8.** **(b)** Given,
Volume of iron rod, $V = 10^{-3}\ \text{m}^3$
Relative permeability, $\mu_r = 1000$
Number of turns per unit length, $n = 10$
Magnetic moment of an iron core solenoid,
$M = (\mu_r - 1) \times NiA$

$$\Rightarrow M = (\mu_r - 1) \times Ni\frac{V}{l} \Rightarrow M = (\mu_r - 1) \times \frac{N}{l} iV$$

$$\Rightarrow M = 999 \times \frac{10}{10^{-2}} \times 0.5 \times 10^{-3} = 499.5 \approx 500.$$

9. **(d)** For paramagnetic material. According to curies law

$$\chi \propto \frac{1}{T}$$

For two temperatures $T_1$ and $T_2$

$$\chi_1 T_1 = \chi_2 T_2$$

But $\chi = \frac{I}{B}$

$$\therefore \frac{I_1}{B_1} T_1 = \frac{I_2}{B_2} T_2$$

$$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I_2}{0.3} \times 24 \Rightarrow I_2 = \frac{0.3}{0.4} = 0.75 \text{ A/m}$$

10. **(b)** When magnetic field is applied to a diamagnetic substance, it produces magnetic field in opposite direction so net magnetic field inside the cavity of sphere will be zero. So, field inside the paramagnetic substance kept inside the cavity is zero.

11. **(b)** Work done
$= MB (\cos\theta_1 - \cos\theta_2)$
$= MB (\cos 0° - \cos 60°)$

$$= MB\left(1 - \frac{1}{2}\right) = \frac{2 \times 10^4 \times 6 \times 10^{-4}}{2} = 6 \text{ J}$$

12. **(b)**

13. **(a)**

14. **(d)** $\tan\theta = \frac{V}{H} = \frac{3}{4}$ $\left[\because \tan 37° = \frac{3}{4}\right]$

$$\therefore V = \frac{3}{4} H$$

$V = 6 \times 10^{-5}$ T

$$H = \frac{4}{3} \times 6 \times 10^{-5} \text{ T} = 8 \times 10^{-5} \text{ T}$$

$$\therefore B_{total} = \sqrt{V^2 + H^2} = \sqrt{(36 + 64)} \times 10^{-5}$$
$$= 10 \times 10^{-5} = 10^{-4} \text{T}.$$

15. **(d)** Given, $I = 9 \times 10^{-5}$ kg m$^2$, $B = 16\pi^2 \times 10^{-5}$ T

$$T = \frac{15}{20} = \frac{3}{4} \text{s}$$

In a vibration magnetometer

Time period, $T = 2p\sqrt{\frac{I}{MB}}$ or $M = \frac{4p^2 I}{BT^2}$

$$M = \frac{4\pi^2 \times 9 \times 10^{-5}}{16\pi^2 \times 10^{-5} \times \left(\frac{3}{4}\right)^2} = 4 \text{ A m}^2$$

16. **(a)** $W = MB[1 - \cos\theta]$

$W_{90°} = MB$

$$W_{60°} = MB\left(\frac{1}{2}\right) \Rightarrow \frac{MB}{2}$$

$\Rightarrow MB = nMB/2 \Rightarrow n = 2$

17. **(c)** The potential energy of a magnetic dipole m placed in an external magnetic dipole is $U = -\vec{m}.\vec{B}$.
Therefore, work done in rotating the dipole is-
$W = \Delta U = 2mB = 2 \times 5.4 \times 10^{-6} \times 0.8$
$= 8.6 \times 10^{-6}$ Joule.

18. **(c)** Along the equatorial line, magnetic field strength

$$B = \frac{\mu_0}{4\pi} \frac{M}{\left(r^2 + \ell^2\right)^{3/2}}$$

Given: $M = 4JT^{-1}$ $r = 200$ cm $= 2$ m

$$\ell = \frac{6\text{cm}}{2} = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$$

$$\therefore B = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{4}{\left[2^2 + \left(3 \times 10^{-2}\right)^2\right]^{3/2}}$$

Solving we get, $B = 5 \times 10^{-8}$ tesla

19. **(a)** Magnetic field due to a bar magnet in the broad-side on position is given by

$$B = \frac{\mu_0}{4\pi} \frac{M}{\left[r^2 + \frac{\ell^2}{4}\right]^{3/2}}; \; M = m\ell.$$

20. **(a)** Couple acting on a bar magnet of dipole moment M when placed in a magnetic field, is given by $\tau = MB\sin\theta$
where $\theta$ is the angle made by the axis of magnet with the direction of field.
Given the $m = 5$Am, $2\ell = 0.2$m, $\theta = 30°$
and $B = 15$ Wbm$^{-2}$
$\therefore \tau = MB\sin\theta = (m \times 2\ell) B \sin\theta$

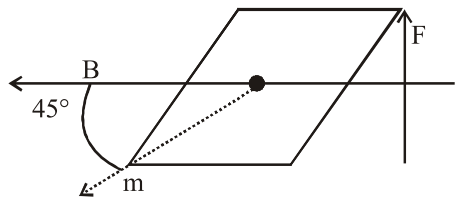
$$= 5 \times 0.2 \times 15 \times \frac{1}{2} = 7.5 \text{ Nm}.$$

21. **(2600)** Corecivity, $H = \frac{B}{\mu_0}$

and $B = \mu_0 ni \left(n = \frac{N}{\ell}\right)$

or, $H = \frac{N}{\ell} i = \frac{100}{0.2} \times 5.2 = 2600$ A/m

**22.** **($6.5 \times 10^{-5}$)** Using, $MB \sin\theta = F \ell \sin\theta \, (\tau)$




$$MB \sin 45° = F\frac{\ell}{2}\sin 45°$$

$F = 2MB = 2 \times 1.8 \times 18 \times 10^{-6} = 6.5 \times 10^{-5}N$

**23.** **($3.3 \times 10^{-4}$)** Magnetic susceptibility, $\chi = \frac{I}{H}$

where, $I = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{20\times10^{-6}}{10^{-6}}$

$= 20\, N/m^2$

Now, $\chi = \frac{20}{60\times10^3} = \frac{1}{3}\times10^{-3} = 3.3\times10^{-4}$

**24.** **($3.266 \times 10^{-4}$)** According to Curie law for paramagnetic substance,

$$\chi \propto \frac{1}{T_C}$$

$$\frac{\chi_1}{\chi_2} = \frac{T_{C_2}}{T_{C_1}}$$

$$\frac{2.8\times10^{-4}}{\chi_2} = \frac{300}{350}$$

$$\chi_2 = \frac{2.8\times350\times10^{-4}}{300} = 3.266 \times 10^{-4}$$

**25.** **(25)**

**26.** **(0.1)**

**27.** **($10^{23}$)** Given, $B = 4 \times 10^{-5}$ T

$R_E = 6.4 \times 10^6$ m

Dipole moment of the earth M = ?

$$B = \frac{\mu_0}{4\pi}\frac{M}{d^3}$$

$$4\times10^{-5} = \frac{4\pi\times10^{-7}\times M}{4\pi\times\left(6.4\times10^6\right)^3}$$

$\therefore$ $M \cong 10^{23}\, Am^2$

**28.** **($\sqrt{3}$)** In the first galvanometer

$i_1 = K_1 \tan\theta_1 = K_1 \tan 60^o = K_1\sqrt{3}$

In the second galvanometes,

$i_2 = K_2 \tan\theta_2 = K_2 \tan 45^o = K_2$

In series $i_1 = i_2 \Rightarrow K_1\sqrt{3} = K_2$

$$\Rightarrow \frac{K_1}{K_2} = \frac{1}{\sqrt{3}}$$

But, $K \alpha \frac{1}{n} \Rightarrow \frac{K_1}{K_2} = \frac{n_2}{n_1}$ $\quad \therefore \frac{n_1}{n_2} = \frac{\sqrt{3}}{1}$

**29.** **(0.51)** For null deflection,

$$\frac{M_1}{M_2} = \left(\frac{d_1}{d_2}\right)^3 = \left(\frac{40}{50}\right)^3 = \frac{64}{125}$$

**30.** **(15)** In tangent galvanometer, $I \propto \tan\theta$

$$\frac{I_1}{I_2} = \frac{\tan\theta_1}{\tan\theta_2} \qquad \Rightarrow \frac{I_1}{I_2/\sqrt{3}} = \frac{\tan 45}{\tan\theta_2}$$

$\theta_2 = 30^o$

so, deflection will decrease by $45^o - 30^o = 15^o$

# Chapter 20 Electromagnetic Induction

1. **(c)** Total number of turns in the solenoid, N = 500; I = 2A.
Magnetic flux linked with each turn
$= 4 \times 10^{-3}$ Wb
As, $N\phi = LI \Rightarrow L = \frac{Nf}{1} = 1$ H.

2. **(b)** $\phi_{total} = B_{\text{large}}A_{\text{small}}$
$$= \left[\frac{\mu_0}{4\pi}\frac{i}{L/2}(2\sin 45°)\times \ell^2\right]\times 4 = \frac{\mu_0 \times 8\sqrt{2}i\ell^2}{4\pi\times(L)}$$
On comparing with $\phi_{total} = Mi$, we get $M = \frac{\mu_0}{4\pi}.\frac{8\sqrt{2}\ell^2}{L}$

3. **(a)** Emf induced $|e| = B\ell v$
$$I = \frac{B\ell v}{r}$$
Force act on the rod due to magnetic field in opposite direction of velocity
$$F = BI\ell = \frac{B^2\ell^2 v}{r}$$
Therefore, an equal force must be provided to move the rod with velocity v.

4. **(a)** Take a very small element dx at a distance x from one end then
$$\varepsilon = -B\omega\int_0^{\ell} x\,dx = -\frac{1}{2}B\omega\ell^2$$

5. **(b)** Rate of work $= \frac{W}{t} = P = Fv$;
also $F = Bil = B\left(\frac{Bvl}{R}\right)l$
$$\Rightarrow P = \frac{B^2v^2l^2}{R} = \frac{(0.5)^2 \times (2)^2 \times (1)^2}{6} = \frac{1}{6}W$$
[Here W = Watts]

6. **(d)** According to Lenz's law, when switch is closed, the flux in the loop increases out of plane of paper, so induced current will be clockwise and simillarly anticlock wise when switch will be open.

7. **(a)** The flux linked with the coil when the plane of the coil is perpendicular to the magnetic field is
$\phi = nAB\cos\theta = nAB.$
the change in flux on rotating the coil by 180° is $d\phi = nAB - (-nAB) = 2nAB$
$\therefore$ induced charge $= \frac{d\phi}{R}$
$$= \frac{2nAB}{R} = \frac{2 \times 100 \times 0.001 \times 1}{10}$$
$\therefore$ Induced charge = 0.02 C.

8. **(b)** $Q = \frac{\Delta\phi}{R} = \frac{\phi_2 - \phi_1}{R} = \frac{BA - 0}{R} = \frac{BA}{R}$

9. **(a)** The direction of current in the loop will be such as to oppose the increase in the magnetic field.

10. **(d)** $|e| = BlV\sin\theta = 0.1 \times 2 \times 20 \sin 30^{o} = 2$ volt

11. **(b)** According to faraday's law of electromagnetic induction, $e = \frac{-d\phi}{dt}$
$$L\times\frac{di}{dt} = 25 \Rightarrow L\times\frac{15}{1} = 25 \text{ or } L = \frac{5}{3}H$$
Change in the energy of the inductance,
$$\Delta E = \frac{1}{2}L\left(i_1^2 - i_2^2\right) = \frac{1}{2}\times\frac{5}{3}\times(25^2 - 10^2)$$
$$= \frac{5}{6}\times 525 = 437.5\,J$$

12. **(a)** According to Faraday's law of electromagnetic induction, $\varepsilon = \frac{d\phi}{dt}$
Also, $\varepsilon = iR$
$\therefore$ $iR = \frac{d\phi}{dt} \Rightarrow \int d\phi = R\int i\,dt$

Magnitude of change in flux ($d\phi$) = $R$ × area under current vs time graph

or, $d\phi = 100 \times \frac{1}{2} \times \frac{1}{2} \times 10 = 250$ Wb

**13.** **(d)** According to Faraday's law of electromagnetic induction,

Induced emf, $e = \frac{Ldi}{dt}$

$$50 = L\left(\frac{5-2}{0.1\text{sec}}\right)$$

$$\Rightarrow \quad L = \frac{50 \times 0.1}{3} = \frac{5}{3} = 1.67 \text{ H}$$

**14.** **(a)** Potential difference between two faces perpendicular to x-axis

$= lV.B = 2 \times 10^{-2}(6 \times 0.1) = 12\text{mV}$

**15.** **(a)** Induced, emF, $\varepsilon = Bv\ell$
$= 0.3 \times 10^{-4} \times 5 \times 20$
$= 3 \times 10^{-3}\text{V} = 3 \text{ mV}.$

**16.** **(d)** The rate of mutual inductance is given by

$M = \mu_0 n_1 n_2 \pi r_1^2$ ...(i)

The rate of self inductance is given by

$L = \mu_0 n_1^2 \pi r_1^2$ ...(ii)

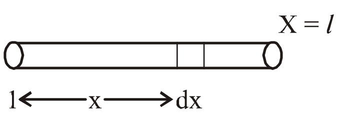
Dividing (i) by (ii) $\Rightarrow \frac{M}{L} = \frac{n_2}{n_1}$

**17.** **(a)** $Q_{\text{coil}} = (NQ) \propto i$

So, $\frac{Q_1}{Q_2} = \frac{i_1}{i_2} = \frac{3}{2}$

or $Q_2 = \frac{2}{3}Q_1 = \frac{2}{3} \times 10^{-3} = 6.67 \times 10^{-4}$ Wb

**18.** **(c)**




Magnetic moment, M = NIA
$dQ = \rho\, dx$

$$dI = \frac{dQ}{2\pi}.\omega$$

$dM = dI \times A$

$$= \frac{\omega}{2\pi}.\frac{\rho_0}{\ell}.x\pi x^2\, dx \Rightarrow M = \frac{\rho_0}{\ell} n\pi \int_0^\ell x^3\, dx$$

$$= \frac{\pi}{4}.n\rho\ell^3$$

**19.** **(b)** In the given question,

Current flowing through the wire, I = 1A

Speed of the frame, v = 10 ms$^{-1}$

Side of square loop, $l$ = 10 cm

Distance of square frame from current carrying wires x = 10 cm.

We have to find, e.m.f induced e = ?

According to Biot-Savart's law

$$B = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{x^2}$$

$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{1 \times 10^{-1}}{\left(10^{-1}\right)^2}$$

$= 10^{-6}$

Induced e.m.f. e = Blv
$= 10^{-6} \times 10^{-1} \times 10 = 1\ \mu\text{v}$

**20.** **(a)** Induced emf in a coil,

$$e = -\frac{d\phi}{dt} = \text{NBA}\sin\omega t$$

Also, $e = e_0 \sin\omega t$

∴ Maximum emf induced, $e_0 = nBA\omega$

**21.** **(1.25)** The resistance of the loops,

$R_1 = 2\pi r_1 \times 10 = 2\pi \times 0.1 \times 10$
$= 6.28\,\Omega.$

and $R_2 = 2\pi r_2 = 2\pi \times 1 \times 10$
$= 62.8\,\Omega.$

Flux in the smaller loop, $\phi = B_2 A_1$

$$= \frac{\mu_0 i_2}{2r_2}\pi r_1^2$$

$$= \frac{\mu_0\left[\frac{V}{R_2}\right]\pi r_1^2}{2r_2}$$

$$= \frac{\mu_0 \frac{[4+2.5t]}{R_2}\pi r_1^2}{2r_2}$$

The induced current, $i_1 = \frac{e}{R_1} = \frac{[d\phi/dt]}{R_1}$

After substituting the value and simplifying we get

$i = 1.25$ A.

**22.** **(4)** For $0 = 2\sin t^2 \Rightarrow t = 0$

and $2 = 2\sin^2 t \Rightarrow t^2 = \frac{\pi}{2}$,

$\therefore \quad t = \sqrt{\frac{\pi}{2}}$

The energy spent, $E = \frac{1}{2}Li^2$

$= \frac{1}{2}L(2\sin t^2)^2$

$= 2L\sin^2 t$

$= 2 \times 2\sin^2 \pi/2 = 4J.$

**23.** **(437.5)** According to faraday's law of electromagnetic induction, $e = \frac{-d\phi}{dt}$

$L \times \frac{di}{dt} = 25 \Rightarrow L \times \frac{15}{1} = 25$

or, $L = \frac{5}{3}H$

Change in the energy of the inductance,

$\Delta E = \frac{1}{2}L\left(i_1^2 - i_2^2\right) = \frac{1}{2} \times \frac{5}{3} \times (25^2 - 10^2)$

$= \frac{5}{6} \times 525 = 437.5\,J$

**24.** **($1.5 \times 10^{-3}$)** Induced emf, $\varepsilon = Bv\ell$

$= 0.3 \times 10^{-4} \times 5 \times 10$

$= 1.5 \times 10^{-3}\,V$

**25.** **($7.85 \times 10^{-2}$)** The magnetic flux, $\phi_B = BA = B \times \pi r^2$

The induced emf, $|e| = \frac{\Delta\phi_B}{\Delta t} = \frac{B \times \pi r^2}{\Delta t}$

$= \frac{0.50 \times \pi(0.05)^2}{0.50}$

$= 7.85 \times 10^{-2}\,V$

**26.** **($9.3 \times 10^{-2}$)** The resistance of the wire

$R = \frac{\rho\ell}{\pi r^2} = \frac{(1.7 \times 10^{-8}) \times (0.40)}{\pi(10^{-3})^2}$

$= 2.16 \times 10^{-3}\,\Omega$

The area of the loop $= 0.10 \times 0.10 = 10^{-2}\,m^2$.

The induced emf, $|e| = A\left(\frac{dB}{dt}\right)$

$= 10^{-2} \times (0.02) = 2 \times 10^{-4}\,V$

The induced current, $i = \frac{e}{R} = \frac{2 \times 10^{-4}}{2.16 \times 10^{-3}}$

$= 9.3 \times 10^{-2}\,A.$

**27.** **(0.36)** The required ratio, $\frac{E_{\text{stored}}}{E_{\text{Supplied}}} = \frac{V^2/R}{V_0^2/R}$

$= \frac{V^2}{V_0^2} = \frac{V_0^2(1 - e^{-t/\tau})}{V_0^2}$

$= (1 - e^{-t/\tau})^2$

$= \left[1 - e^{-\left(\frac{0.1 \times 10}{1}\right)}\right]^2$

$= 0.36$

**28.** **(0.048)** $B = \mu_0 n i$

$= (4\pi \times 10^{-7})(200 \times 10^{-2}) \times 1.5$

$= 3.8 \times 10^{-2}\,Wb/m^2$

Magnetic flux through each turn of the coil

$\phi = BA = (3.8 \times 10^{-2})(3.14 \times 10^{-4})$

$= 1.2 \times 10^{-5}$ weber

When the current in the solenoid is reversed, the change in magnetic flux

$= 2 \times (1.2 \times 10^{-5}) = 2.4 \times 10^{-5}$ weber

Induced e.m.f.

$= N\frac{d\phi}{dt} = 100 \times \frac{2.4 \times 10^{-5}}{0.05} = 0.048\,V.$

**29.** **(0.173)** We have, $\tan\theta = \frac{B_V}{B_H}$

$\therefore \quad B_V = B_H \tan\theta$

$= 2 \times 10^{-5} \times \tan 60°$

$= 2\sqrt{3} \times 10^{-5}\,T$

The induced emf, $e = B_v v\ell$

$= (2\sqrt{3} \times 10^{-5}) \times 250 \times 20$

$= 0.173\,V$

**30.** **($9 \times 10^{-3}$)** Power $= \frac{B^2 v^2 l^2}{R}$

$= \frac{0.5 \times 0.5 \times 12 \times 12 \times 15 \times 15 \times 10^{-8}}{9 \times 10^{-3}}$

$= 9 \times 10^{-3}$ watt

# CHAPTER 21

# Alternating Current

**1.** **(d)** $I = 2\sin\omega t$

$$V = 5\cos\omega t = 5\sin\left(\frac{\pi}{2} - \omega t\right)$$

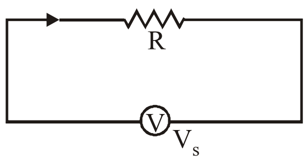
Since, there is a phase difference of $\frac{\pi}{2}$ between the current and voltage

$\therefore$ Average power over a complete cycle is zero.

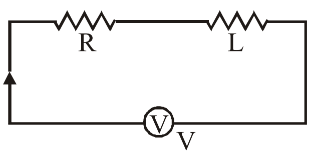
**2.** **(d)**

**3.** **(d)**

Pure resistor



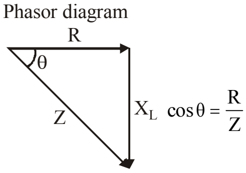

L-R series circuit




For pure resistor circuit, power

$$P = \frac{V^2}{R} \Rightarrow V^2 = PR$$

Phasor diagram




Z = impedance

For L-R series circuit, power

$$P^1 = \frac{V^2}{Z}\cos\theta = \frac{V^2}{Z}\cdot\frac{R}{Z} = \frac{PR}{Z^2}\cdot R = P\left(\frac{R}{Z}\right)^2$$

**4.** **(c)** Impedance at resonant frequency is minimum in series LCR circuit.

$$\text{So, } Z = \sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}$$

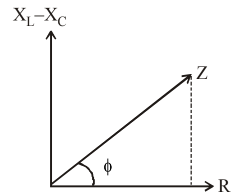
**5.** **(b)** $R = X_L = 2X_C$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{(2X_C)^2 + (2X_C - X_C)^2}$$

$$= \sqrt{4X_C^2 + X_C^2}$$

$$= \sqrt{5}X_C = \frac{\sqrt{5}R}{2}$$




$$\tan\phi = \frac{X_L - X_C}{R} = \frac{2X_C - X_C}{2X_C}$$

$$\Rightarrow \phi = \tan^{-1}\left(\frac{1}{2}\right)$$

**6.** **(c)** $P = V_{r.m.s} \times I_{r.m.s} \times \cos\phi = \frac{1}{2}V_0 I_0 \cos\phi$

$$= \frac{1}{2} \times 100 \times (100 \times 10^{-3}) \cos\pi/3$$

$$= 2.5\text{ W}$$

**7.** **(c)** Frequency $f = \frac{1}{2\pi\sqrt{LC}}$

$$= \frac{1}{2 \times 3.14\sqrt{24 \times 2 \times 10^{-6}}} \simeq 23\text{Hz}$$

**8.** **(d)** The transformer converts A.C. high voltage into A.C. low voltage, but it does not cause any change in frequency. The formula for voltage is

$$E_s = \frac{N_s}{N_p} \times E_p = \frac{5000}{500} \times 20 = 200\text{V}$$

Thus, output has voltage 200 V and frequency 50 Hz.

**9.** **(d)** In case of series RLC circuit,
Equation of voltage is given by

$$V^2 = V_R^2 + (V_L - V_C)^2$$

Here, $V = 220\ V$; $V_L = V_C = 300$ V

$\therefore V_R = \sqrt{V^2} = 220\text{V}$

Current $i = \dfrac{V}{R} = \dfrac{220}{100} = 2.2\text{A}$

**10.** **(a)** Since $\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p}$

Where

$N_p = 50 \qquad N_s = 1500$

and $V_p = \dfrac{d\phi}{dt} = \dfrac{d}{dt}(\phi_0 + 4t) = 4$

$\Rightarrow V_s = \dfrac{1500}{50} \times 4 = 120 \text{ V}$

**11.** **(a)** Quality factor,

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{100}\sqrt{\frac{80 \times 10^{-3}}{2 \times 10^{-6}}}$$

$$= \frac{1}{100}\sqrt{40 \times 10^3} = \frac{200}{100} = 2$$

**12.** **(a)** Given, Inductance, $L = 40$ mH

Capacitance, $C = 100\ \mu F$

Impedance, $Z = X_C - X_L$

$$\Rightarrow Z = \frac{1}{\omega C} - \omega L \left(\because X_c = \frac{1}{\omega C} \text{ and } X_L = \omega L\right)$$

$$= \frac{1}{314 \times 100 \times 10^{-6}} - 314 \times 40 \times 10^{-3}$$

$= 19.28\Omega$

Current, $i = \dfrac{V_0}{Z}\sin(\omega t + \pi/2)$

$\Rightarrow i = \dfrac{10}{19.28}\cos \omega t = 0.52 \cos (314\ t)$

**13.** **(b)** Efficiency, $\eta = \dfrac{P_{out}}{P_{in}} = \dfrac{V_s I_s}{V_p I_p}$

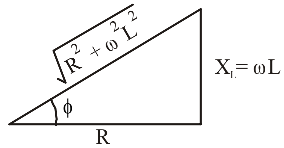
$\Rightarrow 0.9 = \dfrac{230 \times I_s}{2300 \times 5}$

$\Rightarrow I_s = 0.9 \times 50 = 45\text{A}$

Output current $= 45$A

**14.** **(b)** Resistance of the inductor, $X_L = \omega L$

The impedance triangle for resistance ($R$) and inductor ($L$) connected in series is shown in the figure.




Net impedance of circuit $Z = \sqrt{X_L^2 + R^2}$

Power factor, $\cos\phi = \dfrac{R}{Z}$

$\Rightarrow \cos\phi = \dfrac{R}{\sqrt{R^2 + \omega^2 L^2}}$

**15.** **(d)** As $V(t) = 220 \sin 100\ \pi t$

so, $I(t) = \dfrac{220}{50} \sin 100\ \pi t$

i.e., $I = I_m = \sin (100\ \pi t)$

For $I = I_m$

$t_1 = \dfrac{\pi}{2} \times \dfrac{1}{100\pi} = \dfrac{1}{200}$ sec.

and for $I = \dfrac{I_m}{2}$

$\Rightarrow \dfrac{I_m}{2} = I_m \sin(100\ \pi t_2) \Rightarrow \dfrac{\pi}{6} = 100\ \pi t_2$

$\Rightarrow t_2 = \dfrac{1}{600} s$

$\therefore t_{req} = \dfrac{1}{200} - \dfrac{1}{600} = \dfrac{2}{600} = \dfrac{1}{300} s = 3.3$ ms

**16.** **(a)** The current (I) in LR series circuit is given by

$$I = \frac{V}{R}\left(1 - e^{-\frac{tR}{L}}\right)$$

At $t = \infty$,

$$I_\infty = \frac{20}{5}\left(I - e^{\frac{-\infty}{L/R}}\right) = 4 \qquad \text{...(i)}$$

At $t = 40$s,

$$\left(1 - e^{\frac{-40 \times 5}{10 \times 10^{-3}}}\right) = 4(1 - e^{-20,000}) \qquad \text{...(ii)}$$

Dividing (i) by (ii) we get

$$\Rightarrow \frac{I_\infty}{I_{40}} = \frac{1}{1 - e^{-20,000}},$$

**17.** **(a)** Quality factor $Q = \dfrac{\omega_0}{2\Delta\omega} = \dfrac{\omega_0 L}{R}$

**18.** **(a)** Current in inductor circuit is given by,

$$i = i_0\left(1 - e^{-\frac{Rt}{L}}\right)$$

$$\frac{i_0}{2} = i_0\left(1 - e^{-\frac{Rt}{L}}\right) \Rightarrow e^{-\frac{Rt}{L}} = \frac{1}{2}$$

Taking log on both the sides,

$$-\frac{Rt}{L} = \log 1 - \log 2$$

$$\Rightarrow t = \frac{L}{R}\log 2 = \frac{300 \times 10^{-3}}{2} \times 0.69$$

$\Rightarrow t = 0.1$ sec.

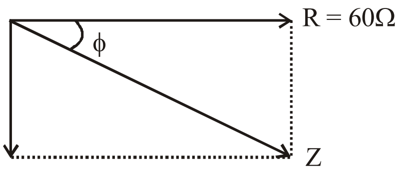
**19.** **(c)** Given: R = 60Ω, f = 50 Hz, ω = 2 πf = 100 π and v = 24v

C = 120 μf = 120 × $10^{-6}$f

$$x_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 120 \times 10^{-6}} = 26.52\Omega$$

$x_L = \omega L = 100\,\pi \times 20 \times 10^{-3} = 2\,\pi\Omega$

$x_C - x_L = 20.24 \approx 20$




$$z = \sqrt{R^2 + (x_C - x_L)^2}$$

$$z = 20\sqrt{10}\Omega$$

$$\cos\phi = \frac{R}{z} = \frac{60}{20\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$P_{avg} = VI\cos\phi, I = \frac{v}{z} = \frac{v^2}{z}\cos\phi = 8.64\text{ watt}$$

Energy dissipated (Q) in time t = 60s is

Q = P.t = 8.64 × 60 = 5.17 × $10^2$J

**20.** **(c)** Across resistor, $I = \dfrac{V}{R} = \dfrac{100}{1000} = 0.1\ A$

At resonance,

$$X_L = X_C = \frac{1}{\omega C} = \frac{1}{200 \times 2 \times 10^{-6}} = 2500$$

Voltage across $L$ is

$I X_L = 0.1 \times 2500 = 250\text{ V}$

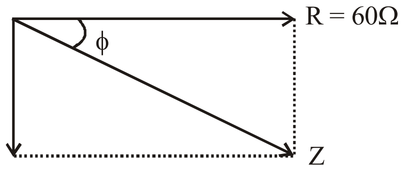
**21.** **($5.17 \times 10^2$)** Given: R = 60Ω, f = 50 Hz, ω = 2 πf = 100 π and v = 24v

C = 120 μf = 120 × $10^{-6}$f

$$x_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 120 \times 10^{-6}} = 26.52\Omega$$

$x_L = \omega L = 100\,\pi \times 20 \times 10^{-3} = 2\,\pi\Omega$

$x_C - x_L = 20.24 \approx 20$




$$z = \sqrt{R^2 + (x_C - x_L)^2}$$

$$z = 20\sqrt{10}\Omega$$

$$\cos\phi = \frac{R}{z} = \frac{60}{20\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$P_{avg} = VI\cos\phi, I = \frac{v}{z} = \frac{v^2}{z}\cos\phi = 8.64\text{ watt}$$

Energy dissipated (Q) in time t = 60s is

Q = P.t = 8.64 × 60 = 5.17 × $10^2$J

**22.** **(45)** Efficiency, $\eta = \dfrac{P_{out}}{P_{in}} = \dfrac{V_s I_s}{V_p I_p}$

$$\Rightarrow 0.9 = \frac{230 \times I_s}{2300 \times 5}$$

$$\Rightarrow I_s = 0.9 \times 50 = 45\ A$$

Output current = 45A

**23.** **(3.3)** As $V(t)$ = 220 sin 100 πt

so, $I(t) = \dfrac{220}{50}$ sin 100 πt

i.e., $I = I_m$ = sin (100 πt)

For $I = I_m$

$$t_1 = \frac{\pi}{2} \times \frac{1}{100\pi} = \frac{1}{200}\text{ sec.}$$

and for $I = \dfrac{I_m}{2}$

$$\Rightarrow \frac{I_m}{2} = I_m \sin(100\,\pi t_2)$$

$$\Rightarrow \frac{\pi}{6} = 100\,\pi t_2 \Rightarrow t_2 = \frac{1}{600}s$$

$$\therefore t_{req} = \frac{1}{200} - \frac{1}{600} = \frac{2}{600} = \frac{1}{300}s = 3.3\text{ ms}$$

**24.** **($63 \times 10^{-9}$)** For the maximum current,

$X_C = X_L$

or $\frac{1}{\omega C} = \omega L$

$\therefore \quad C = \frac{1}{\omega^2 L} = \frac{1}{(2\pi f)^2 L}$

$= \frac{1}{(2\pi \times 2 \times 10^3)^2 \times 100 \times 10^{-3}}$

$= 63 \times 10^{-9}\, F$

**25.** **(125)** The impedance, $Z = \frac{V_{rms}}{i} = \frac{160}{2} = 80\,\Omega$

We know that,Power,

$P = \frac{V_{\text{rms}}^2 R}{Z^2}$

or $200 = \frac{160^2 \times R}{80^2}$

$\therefore \quad R = 50\,\Omega$

We know that, $Z = \sqrt{R^2 + X_L^2}$

or $80 = \sqrt{50^2 + X_L{}^2}$

$\therefore \quad X_L = 62.5\,\Omega$

The back emf, $V_L = iX_L = 2 \times 62.5 = 125$ V.

**26.** **(100)** $X_L = \omega L = (2\pi \times 500) \times 8.1 = 25.4\,\Omega$

and $X_C = \frac{1}{\omega C} = \frac{1}{(2\pi \times 500) \times (12.5 \times 10^{-6})}$

$= 25.4\,\Omega$

As $X_L = X_C$, so resonance will occur and $V_R = 100V$.

**27.** **(0.625)** Average power consumed is given by

$P = \frac{V_{\text{rms}}^2 R}{Z^2}$,

where $Z = \sqrt{R^2 + (\omega L)^2}$

$= \sqrt{10^2 + (2\pi \times \frac{30}{\pi} \times 0.4)^2} = 26\,\Omega$

$\therefore \quad P = \frac{6.5^2 \times 10}{26^2} = 0.625\,\text{W}$

**28.** **(5)** From Kirchoff's current law,

$i_3 = i_1 + i_2 = 3\sin\omega t + 4\sin(\omega t + 90°)$

$\Rightarrow i_3 = i_0 \sin(\omega t + \phi)$

where $i_0 = \sqrt{3^2 + 4^2 + 2(3)(4)\cos 90°}$

and $\tan\phi = \frac{4\sin 90°}{3 + 4\cos 90°} = \frac{4}{3}$

$\therefore \quad i_3 = 5\sin(\omega t + 53°)$

**29.** **(25)** $f = \frac{1}{2\pi\sqrt{LC}} = \frac{2}{2\pi\sqrt{5 \times 80 \times 10^{-6}}} = \frac{25}{\pi}\, Hz$

**30.** **(2)** If first case,$\tan 60° = \frac{X_L}{R} = \frac{\omega L}{R}$

or $\sqrt{3} = \frac{300L}{100}$

$\therefore \quad L = 0.58$ H

In second case, $\tan 60° = \frac{X_C}{R} = \frac{1}{\omega CR}$

or $\sqrt{3} = \frac{1}{300C \times 100}$

$\therefore \quad C = 19.2\,\mu\text{F}$

The impedance of the circuit is

$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

$= \sqrt{100^2 + \left(300 \times 0.58 - \frac{1}{300 \times 19.2 \times 10^{-6}}\right)^2}$

$= 100\,\Omega$

Current, $i = \frac{V}{Z} = \frac{200}{100} = 2\text{A}$

# Chapter 22

# Electromagnetic Waves

1. **(d)** Direction of propagation of electromagnetic waves is perpendicular to Electric field and Magnetic field. Hence, direction is given by Poynting vector $\vec{S} = \vec{E} \times \vec{H} = \dfrac{\vec{E} \times \vec{B}}{\mu_0}$.

2. **(c)** $\dfrac{E}{B} = c \Rightarrow B = \dfrac{E}{c} = \dfrac{24}{3 \times 10^8} = 8 \times 10^{-8}$ T

3. **(c)**

4. **(d)** Ultravoilet radiations are used in the detection of invisible writing, forged documents, finger prints in forensic lab. While microwaves are used in microwave oven.

5. **(d)** Intensity of EM wave is given by

$$I = \frac{P}{4\pi R^2} = U_{av}.c = \frac{1}{2}\varepsilon_0 E_0^2 \times c \Rightarrow E_0 = \sqrt{\frac{P}{2\pi R^2 \varepsilon_0 c}}$$

$$= \sqrt{\frac{800}{2 \times 3.14 \times (4)^2 \times 8.85 \times 10^{-12} \times 3 \times 10^8}}$$

$$= 54.77 \frac{V}{m}$$

6. **(a)** Both magnetic and electric fields have zero average value in a plane e.m. wave.

7. **(a)**

8. **(d)** The power per unit area carried by an E.M. wave i.e., energy transported per unit time across a unit crossection area is perpendicular to the direction of both electric and magnetic field i.e. in the direction of which the wave is travelling.

9. **(a)**

10. **(d)** G > X > U > V > I > M > R [frequency order] reverse is true for wavelength

11. **(a)** Relation between electric field $E_0$ and magnetic field $B_0$ of an electromagnetic wave is given by

$$c = \frac{E_0}{B_0} \qquad \text{(Here, } c = \text{Speed of light)}$$

$$\Rightarrow E_0 = B_0 \times c = 1.2 \times 10^{-7} \times 3 \times 10^8 = 36$$

As the wave is propagating along $x$-direction, magnetic field is along $z$-direction

and $(\hat{E} \times \hat{B}) \parallel \hat{C}$

$\therefore \vec{E}$ should be along $y$-direction.

So, electric field $\vec{E} = E_0 \sin \vec{E} \cdot (x, t)$

$$= [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{j}] \frac{V}{m}$$

12. **(d)** Average energy density of magnetic field,

$$u_B = \frac{B_0^2}{4\mu_0}.$$

Average energy density of electric field,

$$u_E = \frac{\varepsilon_0 E_0^2}{4}$$

Now, $E_0 = CB_0$ and $C^2 = \dfrac{1}{\mu_0 \in_0}$

$$u_E = \frac{\varepsilon_0}{4} \times C^2 B_0^2 = \frac{\varepsilon_0}{4} \times \frac{1}{\mu_0 \varepsilon_0} \times B_0^2 = \frac{B_0^2}{4\mu_0} = u_B$$

$\therefore u_E = u_B$

Since energy density of electric and magnetic field is same, so energy associated with equal volume will be equal i.e. $u_E = u_B$

13. **(c)** **Given:** Amplitude of electric field, $E_0 = 4$ v/m

Absolute permitivity, $\varepsilon_0 = 8.8 \times 10^{-12}$ c$^2$/N-m$^2$

Average energy density $u_E$ = ?

Applying formula,

Average energy density $u_E = \dfrac{1}{4}\varepsilon_0 E^2$

$$\Rightarrow u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2 = 35.2 \times 10^{-12} \text{ J/m}^3$$

14. **(b)**

15. **(b)** $\because$ The E.M. wave are transverse in nature *i.e.*,

$$= \frac{\vec{k} \times \vec{E}}{\mu} = \vec{H} \qquad \text{...(i)}$$

where $\vec{H} = \dfrac{\vec{B}}{\mu}$

and $\dfrac{\vec{k} \times \vec{H}}{\omega\varepsilon} = -\vec{E}$ ...(ii)

$\vec{k}$ is $\perp$ $\vec{H}$ and $\vec{k}$ is also $\perp$ to $\vec{E}$
The direction of wave propagation is parallel to $\vec{E} \times \vec{B}$.
The direction of polarization is parallel to electric field.

**16.** **(b)** The orderly arrangement of different parts of EM wave in decreasing order of wavelength is as follows:

$\lambda_{\text{radiowaves}} > \lambda_{\text{microwaves}} > \lambda_{\text{visible}} > \lambda_{\text{X-rays}}$

**17.** **(a)** $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} \Rightarrow \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{11 \times 1000 \times 1.6 \times 10^{-19}}$
$= 12.4$ Å

Increasing order of frequency →
x-rays | u-v rays | visible | Infrared

wavelength range of visible region is 4000Å to 7800Å.

**18.** **(d)** $I = \frac{B_0^2}{2\mu_0} \cdot C \Rightarrow \frac{B_0^2}{2} = \frac{I\mu_0}{C}$

$\Rightarrow B_{rms} = \sqrt{\frac{I\mu_0}{C}} = \sqrt{\frac{10^8 \times 4\pi \times 10^{-7}}{3 \times 10^8}}$

$\simeq 6 \times 10^{-4}$ T
Which is closest to $10^{-4}$.

**19.** **(c)** $\xrightarrow{\text{E, Decreases}}$
γ-rays X-rays uv-rays Visible rays VIBGYOR IR rays Microwaves Radio waves

Radio wave < yellow light < blue light < X-rays
(Increasing order of energy)

**20.** **(a)** Frequency range of γ-ray,
$b = 10^{18} - 10^{23}$ Hz
Frequency range of X-ray, $a = 10^{16} - 10^{20}$ Hz
Frequency range of ultraviolet ray,
$c = 10^{15} - 10^{17}$ Hz $\therefore$ $a < b$; $b > c$

**21.** **(2.1)** As we know,

$|\vec{B}| = \frac{|\vec{E}|}{C} = \frac{6.3}{3 \times 10^8} = 2.1 \times 10^{-8}$T and $\hat{E} \times \hat{B} = \hat{C}$

$\hat{j} \times \hat{B} = \hat{i}$ [$\because$ EM wave travels along +(ve) $x$-direction.]

$\therefore$ $\hat{B} = \hat{k}$ or $\vec{B} = 2.1 \times 10^{-8}\hat{k}$T

**22.** **($3 \times 10^4$ N/C)** Using, formula $E_0 = B_0 \times C$
$= 100 \times 10^{-6} \times 3 \times 10^8 = 3 \times 10^4$ N/C
Here we assumed that
$B_0 = 100 \times 10^{-6}$ is in tesla (T) units

**23.** **(1.4)** EM wave intensity

$\Rightarrow I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2}\varepsilon_0 E_0^2 c$
[where $E_0$ = maximum electric field]

$\Rightarrow \frac{27 \times 10^{-3}}{10 \times 10^{-6}} = \frac{1}{2} \times 9 \times 10^{-12} \times E_0^2 \times 3 \times 10^8$

$\Rightarrow E_0 = \sqrt{2} \times 10^3$ kV/m $= 1.4$ kV/m

**24.** **($6 \times 10^{-4}$)** $I = \frac{B_0^2}{2\mu_0} \cdot C \Rightarrow \frac{B_0^2}{2} = \frac{I\mu_0}{C}$

$\Rightarrow B_{rms} = \sqrt{\frac{I\mu_0}{C}} = \sqrt{\frac{10^8 \times 4\pi \times 10^{-7}}{3 \times 10^8}}$

$\simeq 6 \times 10^{-4}$ T

**25.** **(0.64)** $B_0 = \sqrt{B_0^2 + B_1^2} = \sqrt{30^2 + 2^2} \times 10^{-6}$

$\approx 30 \times 10^{-6}$T $\therefore E_0 = cB = 3 \times 10^8 \times 30 \times 10^{-6}$
$= 9 \times 10^3$ V/m

$E_{rms} = \frac{E_0}{\sqrt{2}} = \frac{9}{\sqrt{2}} \times 10^3 V/m$

Force on the charge,

$F = E_{rms}Q = \frac{9}{\sqrt{2}} \times 10^3 \times 10^{-4} \simeq 0.64N$

**26.** **($20 \times 10^{-8}$)** $F = (1 + r)\frac{IA}{C}$

$= \frac{(1 + 0.25) \times 50 \times 1}{3 \times 10^8} \simeq 20 \times 10^{-8}$N

**27.** **($4.8 \times 10^{-7}$)** $E_0 = 300$ V/m,

$\therefore B_0 = \frac{E_0}{C} = \frac{300}{3 \times 10^8} = 1 \times 10^{-6}$ N/A-m

The maximum electric force,
$F_0 = E_0 q = 300 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-7}$ N.

**28.** **($1.33 \times 10^{-10}$)**

**29.** **(6)** $E_0 = B_0 C = 20 \times 10^{-9} \times 3 \times 10^8 = 6$ v/m

**30.** **($5 \times 10^{-3}$)** Pressure, $P = \frac{I}{C}$

$\Rightarrow \frac{F}{A} = \frac{I}{C} \Rightarrow F = \frac{IA}{C} = \frac{\Delta p}{\Delta t} \Rightarrow \Delta p = \frac{I}{C} A\Delta t$

$= \frac{(25 \times 25) \times 10^4 \times 10^{-4} \times 40 \times 60}{3 \times 10^8}$ N-s

$= 5 \times 10^{-3}$ N-s

# CHAPTER 23

# Ray Optics and Optical Instruments

**1.** **(d)** Let d be the depth of two liquids.
Then apparant depth

$$\frac{(d/2)}{\mu}+\frac{(d/2)}{1.5\mu}=\frac{d}{2} \text{ or } \frac{1}{\mu}+\frac{2}{3\mu}=1$$

Solving we get $\mu = 1.671$

**2.** **(a)** $\frac{|P_1|}{|P_2|}=\frac{2}{3} \Rightarrow \frac{f_2}{f_1}=\frac{2}{3}$ ...(i)

Focal length of their combination

$$\frac{1}{f}=\frac{1}{f_1}-\frac{1}{f_2} \Rightarrow \frac{1}{30}=\frac{1}{f_1}-\frac{1\times 3}{2f_1}$$

from (i)

$$\Rightarrow \frac{1}{30}=\frac{1}{f_1}\left[1-\frac{3}{2}\right]=\frac{1}{f_1}\times\left(-\frac{1}{2}\right)$$

$\therefore \quad f_1 = -15$ cm and

$$\therefore\ f_2=\frac{2}{3}\times f_1=\frac{2}{3}\times 15=10\,\text{cm}$$

**3.** **(c)** The focal length($F$) of the final mirror is

$$\frac{1}{F}=\frac{2}{f_\ell}+\frac{1}{f_m}$$

Here $\frac{1}{f_\ell}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

$$=(1.5-1)\left[\frac{1}{\infty}-\frac{1}{-30}\right]=\frac{1}{60}$$

$$\therefore\ \frac{1}{F}=2\times\frac{1}{60}+\frac{1}{30/2}=\frac{1}{10}\ \therefore\ F=10\,\text{cm}$$

The combination acts as a converging mirror. For the object to be of the same size of mirror,
$u = 2F = 20$ cm

**4.** **(b)** Incident angle > critical angle, $i > i_c$

$\therefore \sin i > \sin i_c$ *or* $\sin 45 > \sin i_c$

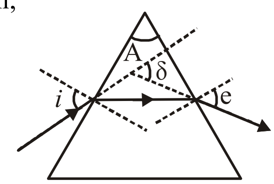
$$\sin i_c = \frac{1}{n}$$

$$\therefore\ \sin 45^\circ > \frac{1}{n}\ \text{or}\ \frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2}$$

**5.** **(b)** Magnifying power of telescope is

$$M=\frac{f_0}{f_e}=\frac{225}{5} \Rightarrow M = 45\ \text{cm.}$$

**6.** **(b)** From the fig.
Angle of deviation,




$$\delta = i + e - A$$

Here, e = i

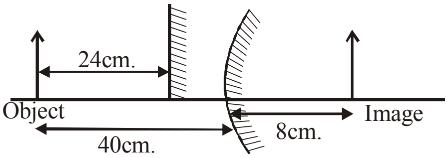
and $e=\frac{3}{4}A$

$$\therefore\ \delta=\frac{3}{4}A+\frac{3}{4}A-A=\frac{A}{2}$$

For equilateral prism, A = 60°

$$\therefore\ \delta=\frac{60^\circ}{2}=30^\circ$$

**7.** **(b)**




$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v} \Rightarrow \frac{1}{10}=\frac{1}{-40}+\frac{1}{v} \Rightarrow \frac{1}{v}=\frac{4+1}{40}$$

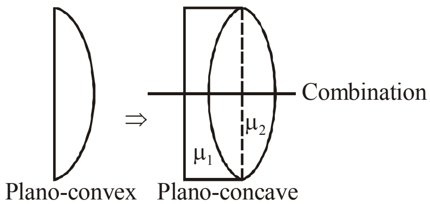
$\Rightarrow$ v = 8 cm.

Hence, plane mirror must be at 24cm.

**8.** **(b)** $\text{Sin C}=\frac{V_P}{V_Q}=\frac{2.2\times 10^8\,\text{m/sec}}{2.4\times 10^8\,\text{m/sec}}=\frac{11}{12}$

$$\Rightarrow C = \sin^{-1}\left(\frac{11}{12}\right)$$

**9. (b)**




$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$= (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right) + (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$$

Solving we get, $f = \dfrac{R}{\mu_1 - \mu_2}$

**10. (d)** The deviation produced as light passes through a thin prism of angle A and refractive index $\mu$ is $\delta = A(\mu - 1)$. We want deviation produced by both prism to be zero.

$$\delta = \delta'$$

$$\Rightarrow \quad A(\mu - 1) = A'(\mu' - 1)$$

$$\Rightarrow \quad A' = \frac{4 \times (1.54 - 1)}{(1.72 - 1)} = 4 \times 0.75 = 3^\circ$$

**11. (b)** Using lens maker's formula

$$\frac{1}{f} = \left(\frac{\mu_g}{\mu_a} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

Here, $\mu_g$ and $\mu_a$ are the refractive index of glass and air respectively

$$\Rightarrow \quad \frac{1}{f} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \text{...(i)}$$

When immersed in liquid

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

[Here, $\mu_l$ = refractive index of liquid]

$$\Rightarrow \frac{1}{f_l} = \left(\frac{1.5}{1.42} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \text{...(ii)}$$

Dividing (i) by (ii)

$$\Rightarrow \quad \frac{f_l}{f} = \frac{(1.5 - 1)1.42}{0.08} = \frac{1.42}{0.16} = \frac{142}{16} \approx 9$$

**12. (b)** Using, $M = \dfrac{v}{u}$

or $\quad -2 = \dfrac{v_1}{x_1} \Rightarrow v_1 = -2x_1$

We have $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

or $\quad \dfrac{1}{-2x_1} - \dfrac{1}{x_1} = \dfrac{1}{20}$

$x_1 = 30$ cm

And $\dfrac{1}{2x_2} - \dfrac{1}{x_2} = \dfrac{1}{20}$

or $x_2 = -10$ cm

So, $\dfrac{x_1}{x_2} = \dfrac{30}{10} = 3$

**13. (c)** When two thin lenses are in contact coaxially, power of combination is given by

$P = P_1 + P_2$

$= (-15 + 5)D$

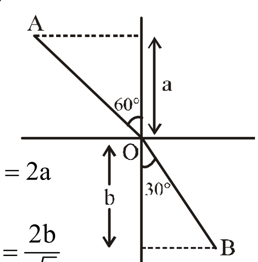
$= -10D.$

Also, $P = \dfrac{1}{f}$

$\Rightarrow f = \dfrac{1}{P} = \dfrac{1}{-10}$ metre

$\therefore \quad f = -\left(\dfrac{1}{10} \times 100\right)$ cm $= -10$ cm.

**14. (d)** From the given figure




As $\sin 60^\circ = \mu \sin 30^\circ$

$$\Rightarrow \mu = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

$$\frac{a}{AO} = \cos 60^\circ \Rightarrow AO = 2a$$

$$\frac{b}{BO} = \cos 30^\circ \Rightarrow BO = \frac{2b}{\sqrt{3}}$$

Optical path length $= AO + \mu BO$

$$= 2a + (\sqrt{3})\frac{2b}{\sqrt{3}} = 2a + 2b$$

**15.** **(a)** According question, $M = 375$

$L = 150$ mm, $f_0 = 5$ mm and $f_e = ?$

Using, magnification, $M \simeq \frac{L}{f_0}\left(1+\frac{D}{f_e}\right)$

$$\Rightarrow 375 = \frac{150}{5}\left(1+\frac{250}{f_e}\right) \quad (\because D = 25\text{ cm} = 250\text{ mm})$$

$$\Rightarrow 12.5 = 1 + \frac{250}{f_e}$$

$$\Rightarrow f_e = \frac{250}{11.5} = 21.7 \approx 22\,mm$$

**16.** **(d)** Given, using lens maker's formula

$$\frac{1}{f} = (k-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Here, $R_1 = R_2 = R$ (For double convex lens)

$$\therefore \frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$$

$$\Rightarrow P = \frac{1}{f} = (\mu-1)\frac{2}{R} \quad \text{...(i)}$$

For plano convex lens,

$R_1 = R'$, $R_2 = \infty$

Using lens maker's formula again, we have

$$1.5P = (\mu-1)\left(\frac{1}{R'} - \frac{1}{\infty}\right) \quad \text{...(ii)}$$

$$\Rightarrow \frac{3}{2}P = \frac{\mu-1}{R'}$$

From (i) and (ii),

$$\frac{3}{2} = \frac{R'}{2R} \Rightarrow R' = \frac{R}{3}$$

**17.** **(a)** **Given :** $f_0 = 1.2$ cm; $f_e = 3.0$ cm

$u_0 = 1.25$ cm; $M_\infty = ?$

From $\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$

$$\Rightarrow \frac{1}{1.2} = \frac{1}{v_0} - \frac{1}{(-1.25)}$$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25}$$

$\Rightarrow v_0 = 30$ cm

Magnification at infinity,

$$M_\infty = -\frac{v_0}{u_0} \times \frac{D}{f_e}$$

$$= \frac{30}{1.25} \times \frac{25}{3}$$

($\because$ $D = 25$ cm least distance of distinct vision)

$= 200$

Hence the magnifying power of the compound microscope is 200

**18.** **(b)** Velocity of light in medium

$$V_{med} = \frac{3\text{ cm}}{0.2\text{ns}} = \frac{3\times10^{-2}\text{ m}}{0.2\times10^{-9}\text{ s}} = 1.5\text{ m/s}$$

Refractive index of the medium

$$\mu = \frac{V_{air}}{V_{med}} = \frac{3\times10^8}{1.5} = 2\text{ m/s}$$

As $\mu = \frac{1}{\sin C}$

$$\therefore \sin C = \frac{1}{\mu} = \frac{1}{2} = 30°$$

Condition of TIR is angle of incidence i must be greater than critical angle. Hence ray will suffer TIR in case of (B) ($i = 40° > 30°$) only.

**19.** **(b)** The number of images formed is given by

$$n = \frac{360}{\theta} - 1$$

$$\Rightarrow \frac{360}{\theta} - 1 = 3$$

$$\Rightarrow \theta = \frac{360°}{4} = 90°$$

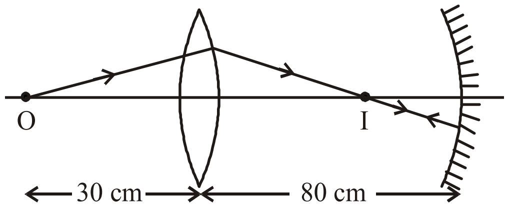
**20.** **(a)** When angle of prism is small,

Angle of deviation, $D = (\mu - 1)A$

Since $\lambda_b < \lambda_r$

$\Rightarrow \mu_r < \mu_b \Rightarrow D_1 < D_2$

**21. (10)** For lens




$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

or $$\frac{1}{v}-\frac{1}{-30}=\frac{1}{20}$$

$\therefore \quad v = +60$ cm

According to the condition, image formed by lens should be the centre of curvature of the mirror, and so

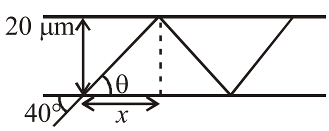
$2f' = 20$ or $f' = 10$ cm

**22. (57000)** Using Snell's law of refraction,

$1 \times \sin 40° = 1.31 \sin\theta$

$$\Rightarrow \quad \sin\theta = \frac{0.64}{1.31} = 0.49 \approx 0.5$$

$\Rightarrow \quad \theta = 30°$




$x = 20\ \mu m \times \cot\theta$

$$\therefore \text{ Number of reflections } = \frac{2}{20\times 10^{-6}\times \cot\theta}$$

$$= \frac{2\times 10^{6}}{20\times\sqrt{3}} = 57735 \approx 57000$$

**23. (0.32)** $+5 = -\dfrac{v}{u} \Rightarrow v = -5u$

Using $\dfrac{1}{v}+\dfrac{1}{u}=\dfrac{1}{f} \Rightarrow \dfrac{1}{-5u}+\dfrac{1}{u}=\dfrac{-1}{0.4}$

$\therefore$ u = – 0.32 m.

**24. (8.8)** If $v$ is the distance of image formed by mirror, then

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

or $$\frac{1}{v}+\frac{1}{-5}=\frac{1}{-20}$$

$$\therefore \quad v = \frac{20}{3}\text{cm}$$

Distance of this image from water surface

$$= \frac{20}{3}+5 = \frac{35}{3}\text{cm}$$

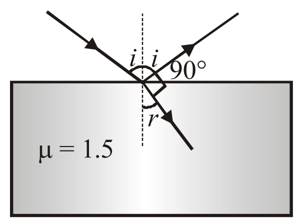
Using, $\dfrac{RD}{AD} = \mu$

$$\therefore \quad AD = d = \frac{RD}{\mu} = \frac{(35/3)}{1.33} = 8.8 \text{ cm}$$

**25. (57)** If $i$ and $r$ are the angle of incident and angle of refraction respectively, then

$i + r = 90°$

$\therefore \quad r = 90° - i$




By Snell's law, $\dfrac{\sin i}{\sin r} = \mu$

or $\dfrac{\sin i}{\sin(90°-i)} = \mu$

or $\dfrac{\sin i}{\cos i} = \mu$

or $\tan i = \mu$

$\therefore \quad i = \tan^{-1}(\mu) = \tan^{-1}(1.5)$

$\simeq 57°$

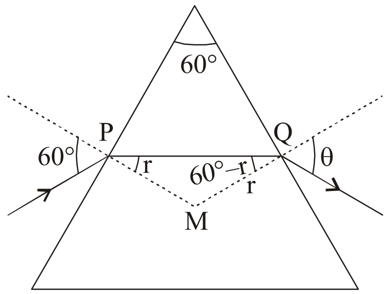
**26. (2)** Here $\angle MPQ + \angle MQP = 60°$. If $\angle MPQ = r$ then $\angle MQP = 60 - r$

Applying Snell's law at $P$

$\sin 60° = n \sin r$ ...(i)

Differentiating w.r.t '$n$' we get

$$O = \sin r + n \cos r \times \frac{dr}{dn} \quad ...(ii)$$




Applying Snell's law at $Q$

$\sin \theta = n \sin (60° - r)$ ...(iii)

Differentiating the above equation w.r.t '$n$' we get

$$\cos \theta \frac{d\theta}{dn} = \sin (60° - r) + n \cos (60° - r)\left[-\frac{dr}{dn}\right]$$

$$\therefore \cos \theta \frac{d\theta}{dn} = \sin (60° - r) - n \cos (60° - r)$$

$$\left[-\frac{\tan r}{n}\right] \quad \text{[from (ii)]}$$

$$\therefore \quad \frac{d\theta}{dn} = \frac{1}{\cos\theta}[\sin (60° - r) + \cos (60° - r) \tan r]$$

...(iv)

From eq. (i), substituting $n = \sqrt{3}$ we get $r = 30°$

From eq (iii), substituting $n = \sqrt{3}$, $r = 30°$

we get $\theta = 60°$

On substituting the values of $r$ and $\theta$ in eq (iv) we get

$$\frac{d\theta}{dn} = \frac{1}{\cos 60°}[\sin 30° + \cos 30° \tan 30°] = 2$$

**27.** **(2.5)** $M = -\frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$

or $\quad -40 = -\frac{20}{5}\left(1 + \frac{25}{f_e}\right)$

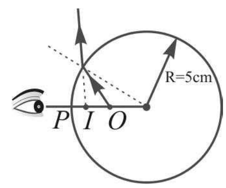
or $\quad f_e = 2.5$ cm

**28.** **(2.5)** For the object $O$,

$u = -(PO)$

$= -3$ cm

By refraction formula,




$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \text{ we have}$$

$\mu_1 = 1.5$, $\mu_2 = 1$, and $R = -5$cm

$$\therefore \quad \frac{1}{v} - \frac{1.5}{-3} = \frac{1 - 1.5}{-5}$$

$\Rightarrow \quad v = -2.5$ cm

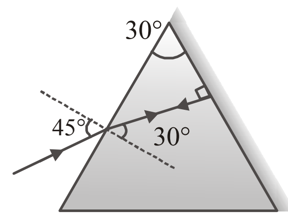
**29.** **(30)** For concave lens, $u = +10$ cm (virtual object)

and $v = +15$ cm

We have $\frac{1}{+15} - \frac{1}{+10} = \frac{1}{f}$

$\therefore \quad f = -30$ cm.

**30.** **($\sqrt{2}$)**




$$\mu = \frac{\sin 45°}{\sin 30°} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2}.$$

CHAPTER 24

# Wave Optics

1. **(b)** $\phi = \frac{\pi}{3}$, $a_1 = 4$, $a_2 = 3$

So, $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi} \approx 6$

2. **(b)** Here, $D = 1.25\text{m}$, $\mu_w = 4/3$, $\theta_w = 0.2^\circ$

$$\mu_w = \frac{\lambda_a}{\lambda_w} = \frac{4}{3} \qquad \text{...(i)}$$

Angular width $\theta_a = \frac{\beta}{D} = \frac{(\lambda_a D/d)}{D} = \frac{\lambda_a}{d}$

As d remains the same

$$\therefore \frac{\theta_a}{\theta_w} = \frac{\lambda_a}{\lambda_w} \text{ or } \theta_a = \theta_w \times \frac{\lambda_a}{\lambda_w} = 0.2^\circ \times \frac{4}{3} = 0.27^\circ$$

3. **(c)** Let $\mu_1$ be refractive index of denser medium and that of rarer medium be $\mu_2$ for total internal reflection,

$$\mu_1 \sin\theta_{iC} = \mu_2 \Rightarrow \sin\theta_{iC} = \frac{\mu_2}{\mu_1}$$

Brewsters angle

$$\theta_{iB} = \tan^{-1}\left(\frac{\mu_2}{\mu_1}\right) \Rightarrow \tan\theta_{iB} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \sin\theta_{iC} = \tan\theta_{iB}$$

$$\frac{\sin\theta_{iC}}{\sin\theta_{iB}} = 1.28 \Rightarrow \frac{\tan\theta_{iB}}{\sin\theta_{iB}} = 1.28$$

$$\Rightarrow \cos\theta_{iB} = \frac{1}{1.28}$$

Relative refractive index

$$= \frac{\mu_2}{\mu_1} = \tan\theta_{iB} = \frac{\sin\theta_{iB}}{\cos\theta_{iB}} = \frac{\sqrt{1 - \frac{1}{1.28^2}}}{\frac{1}{1.28}}$$

$= 0.624 \times 1.28 = 0.8$

4. **(c)** $\Delta = x\frac{d}{D}$, where $\Delta$ is path difference between two waves.

$\therefore$ phase difference $= \phi = \frac{2\pi}{\lambda}\Delta.$

Let a = amplitude at the screen due to each slit.

$\therefore\ I_0 = k(2a)^2 = 4ka^2$, where k is a constant.

For phase difference $\phi$,

amplitude $= A = 2a\cos(\phi/2)$.

[Since, $a^2 = a_1^2 + a_2^2 + 2a_1a_2\cos\phi$, here $a_1 = a_2$]

Intensity I,

$$I = kA^2 = k(4a^2)\cos^2(\phi/2) = I_0\cos^2\left(\frac{\pi x}{\beta}\Delta\right)$$

$$= I_0\cos^2\left(\frac{\pi}{\lambda}.\frac{xd}{D}\right) = I_0\cos^2\left(\frac{\pi x}{\beta}\right)$$

5. **(c)** $\mu = \tan i$

$\Rightarrow i = \tan^{-1}(\mu) = \tan^{-1}(\sqrt{3}) = 60^\circ.$

6. **(c)** $\because i_p = \phi$, therefore, angle between reflected and refracted rays is $90^\circ$.

7. **(c)** Given, refractive index, $\mu = \frac{4}{3}$

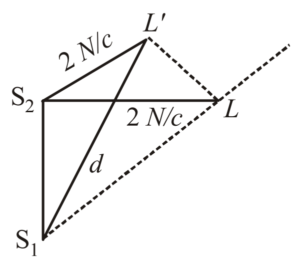
According to Brewster's law when unpolarised light strikes at polarising angle $i_p$ on an interface then reflected and refracted rays are normal to each other and is given by :

$$\tan i_p = \mu \ \therefore\ i_p = \tan^{-1}\left(\frac{4}{3}\right)$$

8. **(d)** Initially, $S_2L = 2$ m

$$S_1L = \sqrt{2^2 + \left(\frac{3}{2}\right)^2} = \frac{5}{2} = 2.5 \text{ m}$$

Path difference, $\Delta x = S_1L - S_2L = 0.5 \text{ m} = \frac{\lambda}{2}$




When the listner move from $L$, first maxima will appear if path difference is integral multiple of wavelength.

For example

$\Delta x = n\lambda = 1\lambda$ ($n = 1$ for first maxima)

$\therefore \ \Delta x = \lambda = S_1L' - S_2L$

$\Rightarrow 1 = d - 2 \Rightarrow d = 3$ m

**9.** **(d)** The resolving power of an optical instrument is inversely proportional to the wavelength of light used.

$$\frac{(R.P)_1}{(R.P)_2} = \frac{\lambda_2}{\lambda_1} = \frac{5}{4}$$

**10.** **(b)** According to Brewster's law, refractive index of material ($\mu$) is equal to tangent of polarising angle

$$\because \tan i_b = \mu = \frac{1.5}{\mu}$$

$$\frac{1}{\mu} < \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}} \ (\because \sin i_c < \sin i_b)$$

$$\therefore \sin i_b = \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}}$$

or, $\sqrt{\mu^2 + (1.5)^2} < 1.5 \times \mu$

$\Rightarrow \mu^2 + (1.5)^2 < (\mu \times 1.5)^2$

$\Rightarrow \mu < \frac{3}{\sqrt{5}}$ i.e. minimum value of $\mu$ should be $\frac{3}{\sqrt{5}}$

**11.** **(a)** $kx\Delta x + \pi = (2n+1)\pi \Rightarrow k\Delta x = 2n\pi$

$$\Rightarrow \Delta x = \frac{2n\pi}{k} = \frac{2n\pi}{2\pi} \times \lambda = n\lambda$$

$\Delta x = n\lambda$ but $\Delta x = BC + CD = \frac{x}{\sqrt{3}} = \sqrt{3}x$

So, $\sqrt{3}x = n\lambda \Rightarrow \lambda = \frac{\sqrt{3}x}{n}$

for $n = 1$

$\lambda = \sqrt{3}x$

**12.** **(d)** Let $I_0$ be the intensity of incident light. Then the intensity of light from the 1st polaroid is

$$I_1 = \frac{I_0}{2}$$

Intensity of light from the 2nd polaroid is

$$I_2 = I_1 \cos^2 60° = \frac{I_0}{2}\left(\frac{1}{2}\right)^2 = \frac{I_0}{8}$$

Intensity of light from the third polaroid is :

$$I_3 = I_2 \cos^2 60° = \frac{I_0}{8}\left(\frac{1}{2}\right)^2 = \frac{I_0}{32}$$

Intensity of light from the 4th polaroid is

$$I_4 = I_3 \cos^2 60° = \frac{I_0}{32}\left(\frac{1}{2}\right)^2 = \frac{I_0}{128}$$

Intensity of light from 5th polaroid is

$$I_5 = I_4 \cos^2 60° = \frac{I_0}{128}\left(\frac{1}{2}\right)^2 = \frac{I_0}{512}$$

**13.** **(d)** If unpolarised light is passed through a polaroid $P_1$, its intensity will become half.

So $I_1 = \frac{I_0}{2}$

Now this light will pass through the second polaroid $P_2$ whose axis is inclined at an angle of 30° to the axis of $P_1$ and hence, vibrations of $I_1$. So in accordance with Malus law, the intensity of light emerging from $P_2$ will be

$$I_2 = I_1 \cos^2 30° = \left(\frac{1}{2}I_0\right)\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{8}I_0$$

So the fractional transmitted light

$$\frac{I_2}{I_0} \times 100 = \frac{3}{8} \times 100 = 37.5\%$$

**14.** **(c)** $\Delta x = (SS_1 + S_{10}) - (SS_2 + S_{20})$

or $\frac{\lambda}{2} = 2\sqrt{D^2 + d^2} - 2D$

$\therefore \ d = \sqrt{\frac{\lambda D}{2}}$

**15.** **(d)** Conditions for diffraction minima are
Path diff. $\Delta x = n\lambda$ and Phase diff. $\delta\phi = 2n\pi$
Path diff. $= n\lambda = 2\lambda$
Phase diff. $= 2n\pi = 4\pi$ ($\because n = 2$)

**16.** **(c)**

**17.** **(b)** $a = 0.1$ mm $= 10^{-4}$ cm,
$\lambda = 6000 \times 10^{-10}$ cm $= 6 \times 10^{-7}$ cm, $D = 0.5$ m
for 3rd dark band, $a \sin\theta = 3\lambda$

or $\sin\theta = \frac{3\lambda}{a} = \frac{x}{D}$

The distance of the third dark band from the central bright band

$$x = \frac{3\lambda D}{a} = \frac{3 \times 6 \times 10^{-7} \times 0.5}{10^{-4}} = 9 \text{ mm}$$

**18.** **(a)**

**19.** **(b)**

**20.** **(c)** For first minimum, $a \sin\theta = n\lambda = 1\lambda$

$$\sin\theta = \frac{\lambda}{a} = \frac{5000\times10^{-10}}{0.001\times10^{-3}} = 0.5$$

$\theta = 30°$

**21. (641)** For 'n' number of maximas

$d\sin\theta = n\lambda$

$0.32\times10^{-3}\sin 30° = n\times500\times10^{-9}$

$$\therefore\ n = \frac{0.32\times10^{-3}}{500\times10^{-9}}\times\frac{1}{2} = 320$$

Hence total no. of maximas observed in angular range $-30° \le \theta \le 30°$

$= 320 + 1 + 320 = 641$

**22. (0.85)** Given, path difference, $\Delta x = \dfrac{\lambda}{8}$

Phase difference ($\Delta\phi$) is given by

$$\Delta\phi = \frac{2\pi}{\lambda}(\Delta x)$$

$$\Delta\phi = \frac{(2\pi)}{\lambda}\frac{\lambda}{8} = \frac{\pi}{4}$$

For two sources in different phases,

$$I = I_0\cos^2\left(\frac{\pi}{8}\right)$$

$$\frac{I}{I_0} = \cos^2\left(\frac{\pi}{8}\right)$$

$$= \frac{1+\cos\frac{\pi}{4}}{2} = \frac{1+\frac{1}{\sqrt{2}}}{2} = 0.85$$

**23. (5)**

**24. (305 × 10⁻⁹)** $\theta = \dfrac{1.22\lambda}{d} = \dfrac{1.22\times500\times10^{-9}}{2}$

$= 305\times10^{-9}$ rad.

**25. (0.24)** $x = \dfrac{1.22\lambda}{2\mu\sin\theta}$

$= 0.24\ \mu m$

**26. (3)** $\Delta x_{max} = 0$ and $\Delta x_{max} = 2\lambda$

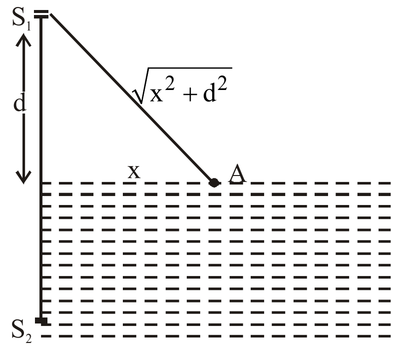
Theortical maximas are $= 2n + 1 = 2\times2 + 1 = 5$

But on the screen there will be three maximas.

**27. (3)** For maxima

Path defference $= m\lambda$

$\therefore\quad S_2A - S_1A = m\lambda$




$$\therefore\ \left[(n-1)\sqrt{d^2+x^2} + \sqrt{d^2+x^2}\right] - \sqrt{d^2-x^2} = m\lambda$$

$$\therefore\quad (n-1)\sqrt{(d^2+x^2)} = m\lambda$$

$$\therefore\quad \left(\frac{4}{3}-1\right)\sqrt{d^2+x^2} = m\lambda$$

$$\therefore\quad \sqrt{d^2+x^2} = 3m\lambda$$

$\therefore\quad d^2 + x^2 = 9m^2\lambda^2$

$\therefore\quad x^2 = 9m^2\lambda^2 - d^2$

$\therefore\quad p^2 = 9 \Rightarrow p = 3$

**28. ($\sqrt{28}$)** The resultant amplitude is given by

$$R = (a_1^2 + a_2^2 + 2a_1a_2\cos\phi)^{1/2}$$

$$= (2^2 + 4^2 + 2\times2\times4\cos60°)^{1/2}$$

$$= \sqrt{28}\ .$$

**29. (3 × 10⁻⁷)** $\theta = \dfrac{1.22\lambda}{d} = \dfrac{1.22\times600\times10^{-9}}{250\times10^{-2}}$

$= 3.0\times10^{-7}$ rad

**30. (20)** The distance of $n^{th}$ maxima from central maxima is given by

$$y_n = n\frac{D\lambda}{d},$$

For $y_n$ to be constant, $n\lambda$ = constant. Thus

$n_1\lambda_1 = n_2\lambda_2$

$$\therefore\quad n_2 = \frac{n_1\lambda_1}{\lambda_2} = \frac{16\times6000}{4800} = 20$$

# CHAPTER 25 Dual Nature of Radiation and Matter

1. **(a)** According to De-broglie $p = \frac{h}{\lambda}$ or $P \propto \frac{1}{\lambda}$

$P \propto \frac{1}{\lambda}$ represents rectangular hyperbola.

2. **(a)** $E_k = E - \phi_0 = 6.2 - 4.2 = 2.0$ eV,
$E_k = 2 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19}$ J

3. **(d)** $I \propto \frac{1}{d^2}$

4. **(b)** Photoelectrons are emitted in A alone. Energy of electron needed if emitted from

$A = \frac{h\upsilon}{e}$ eV

$$\therefore \quad E_A = \frac{\left(6.6\times10^{-34}\right)\times\left(1.8\times10^{14}\right)}{1.6\times10^{-19}} = 0.74\,\text{eV}$$

$$E_B = \frac{\left(6.6\times10^{-34}\right)\times\left(2.2\times10^{14}\right)}{1.6\times10^{-19}} = 0.91\text{ eV}$$

Incident energy 0.825 eV is greater than $E_A$ (0.74 eV) but less than $E_B$ (0.91 eV).

5. **(b)** The momentum of the photon is energy/ speed of light. In black holes the gravity pull is so high that even photon cannot escapes.

6. **(c)** Given, Initial velocity, $u = v_0\hat{i} + v_0\hat{j}$

Acceleration, $a = \frac{qE_0}{m} = \frac{eE_0}{m}$

Using $v = u + at$

$$v = v_0\hat{i} + v_0\hat{j} + \frac{eE_0}{m}t\hat{k}$$

$$\therefore |\vec{v}| = \sqrt{2v_0^2 + \left(\frac{eE_0t}{m}\right)^2}$$

de-Broglie wavelength, $\lambda = \frac{h}{p}$

$$\Rightarrow \lambda = \frac{h}{mv} \quad (\because p = mv)$$

Initial wavelength, $\lambda_0 = \frac{h}{mv_0\sqrt{2}}$

Final wavelength,

$$\lambda = \frac{h}{m\sqrt{2v_0^2 + \left(\frac{eE_0t}{m}\right)^2}}$$

$$\frac{\lambda}{\lambda_0} = \frac{1}{\sqrt{1+\left(\frac{eE_0t}{\sqrt{2}mv_0}\right)^2}}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{2m^2v_0^2}}}$$

7. **(b)** $P_1 - P_2 = (P_1 + P_2) = P$

As $P \propto \frac{1}{\lambda}$

or $\frac{1}{\lambda_x} - \frac{1}{\lambda_y} = \frac{1}{\lambda}$

or $\frac{\lambda_y - \lambda_x}{\lambda_x\lambda_y} = \frac{1}{\lambda}$

8. **(b)** $f_0 = 4 \times 10^{14}$ Hz
$W_0 = hf_0 = 6.63 \times 10^{-34} \times (4 \times 10^{14})$ J

$$= \frac{(6.63\times10^{-34})\times(4\times10^{14})}{1.6\times10^{-19}}$$

$= 1.66$ eV

9. **(c)** using, intensity $I = \frac{nE}{At}$

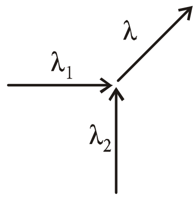
n = no. of photoelectrons

$$\Rightarrow 16\times10^{-3} = \left(\frac{n}{t}\right)\times\frac{10\times1.6\times10^{-19}}{10^{-4}} \text{ or, } \frac{n}{t} = 10^{12}$$

So, effective number of photoelectrons ejected per unit time $= 10^{12} \times 10/100 = 10^{11}$

**10.** **(a)** From the de-Broglie relation,

$$p_1 = \frac{h}{\lambda_1}$$




$$p_2 = \frac{h}{\lambda_2}$$

Momentum of the final particle ($p_f$) is given by

$$\therefore \quad p_f = \sqrt{p_1^2 + p_2^2}$$

$$\Rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_1^2} + \frac{h^2}{\lambda_2^2}}$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

**11.** **(d)** According to question, there are two EM waves with different frequency,

$B_1 = B_0 \sin(\pi \times 10^7 c)t$

and $B_2 = B_0 \sin(2\pi \times 10^7 c)t$

To get maximum kinetic energy we take the photon with higher frequency

using, $B = B_0 \sin \omega t$ and $\omega = 2\pi\nu \Rightarrow \nu = \dfrac{\omega}{2\pi}$

$B_1 = B_0 \sin(\pi \times 10^7 c)t \Rightarrow \nu_1 = \dfrac{10^7}{2} \times c$

$B_2 = B_0 \sin(2\pi \times 10^7 c)t \Rightarrow \nu_2 = 10^7 c$

where c is speed of light $c = 3 \times 10^8$ m/s

Clearly, $\nu_2 > \nu_1$

so KE of photoelectron will be maximum for photon of higher energy.

$\nu_2 = 10^7 c$ Hz

$h\nu = \phi + KE_{max}$

energy of photon

$E_{ph} = h\nu = 6.6 \times 10^{-34} \times 10^7 \times 3 \times 10^9$

$E_{ph} = 6.6 \times 3 \times 10^{-19}$ J

$$= \frac{6.6 \times 3 \times 10^{-19}}{1.6 \times 10^{-19}} \text{eV} = 12.375 \text{eV}$$

$KE_{max} = E_{ph} - \phi$

$= 12.375 - 4.7 = 7.675 \text{ eV} \approx 7.7 \text{ eV}$

**12.** **(d)** $\dfrac{hc}{\lambda} - \phi = eV_0$

$$v_0 = \frac{hc}{e\lambda} - \frac{\phi}{e}$$

For metal A $\qquad$ For metal B

$\dfrac{\phi A}{hc} = \dfrac{1}{\lambda} \qquad \dfrac{\phi B}{hc} = \dfrac{1}{\lambda}$

As the value of $\dfrac{1}{\lambda}$ (increasing and decreasing) is not specified hence we cannot say that which metal has comparatively greater or lesser work function ($\phi$).

**13.** **(a)** Work function, $\phi = 6.2 \text{ eV} = 6.2 \times 1.6 \times 10^{-19}$ J

Stopping potential, $V = 5$ volt

From the Einstein's photoelectric equation

$$\frac{hc}{\lambda} - \phi = eV_0$$

$$\Rightarrow \lambda = \frac{hc}{\phi + eV_0}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}(6.2 + 5)} \approx 10^{-7} \text{ m}$$

This range lies in ultra violet range.

**14.** **(d)** From the Einstein photoelectric equation $K.E. = h\nu - \phi$

Here, $\phi$ = work function of metal

$h$ = Plank's constant

slope of graph of $K.E.$ & $\nu$ is $h$ (Plank's constant) which is same for all metals.

**15.** **(c)** According to photo-electric equation :

$K.E_{max} = h\nu - h\nu_0$ (Work function)

Some sort of energy is used in ejecting the photoelectrons.

**16.** **(c)** Applying Einstein's formula for photo-electricity

$$h\nu = \phi + \frac{1}{2}mv^2 ; \; h\nu = \phi + K ; \; \phi = h\nu - K$$

If we use $2\nu$ frequency then let the kinetic energy becomes $K'$

So, $h \cdot 2\nu = \phi + K' \Rightarrow 2h\nu = h\nu - K + K'$

$\Rightarrow \quad K' = h\nu + K$

**17.** **(a)** Given, $\lambda = 660$ nm, Power $= 0.5$ kW, $t = 60$ ms

Power $P = \frac{nhc}{\lambda t} \Rightarrow n = \frac{p\lambda t}{hc} = 10^{20}$

**18.** **(a)** As we know, $h\upsilon = h\upsilon_0 + K.E._{max}$

or $\frac{hc}{\lambda_{incident}} = \frac{hc}{\lambda_0} + eV_s$

when $\lambda_{incident} = \lambda$, $V_s = 3V$ and for $\lambda_{incident} = 2\lambda$, $V_5 = 1V$.

$\therefore \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + 3eV$ ...(i) and $\frac{hc}{2\lambda} = \frac{hc}{\lambda_0} + 1eV$ ...(ii)

On simplifying (i) and (ii)

$\frac{2hc}{\lambda_0} = \frac{1}{2}\frac{hc}{\lambda} \Rightarrow \lambda_0 = 4\lambda.$

**19.** **(b)** By using $hv - hv_0 = K_{max}$

$\Rightarrow h(v_1 - v_0) = K_1$ ..... (i)

And $h(v_2 - v_0) = K_2$ ..... (ii)

$\Rightarrow \frac{v_1 - v_0}{v_2 - v_0} = \frac{K_1}{K_2} = \frac{1}{K}$, Hence $v_0 = \frac{kv_1 - v_2}{K-1}.$

**20.** **(b)** $E = W_0 + K_{max} \Rightarrow hf = W_A + K_A$ ...(i)

and $2hf = W_B + K_B = 2W_A + K_B$ ...(ii)

$\left(\because \frac{W_A}{W_B} = \frac{1}{2}\right)$

Dividing equation (i) by (ii) $\frac{K_A}{K_B} = \frac{1}{2}$

**21.** **(1.8)** From Einstein's photoelectric equation,

$\frac{hc}{\lambda_1} = \phi + \frac{1}{2}m(2v)^2$ ....(i)

and $\frac{hc}{\lambda_2} = \phi + \frac{1}{2}mv^2$ ....(ii)

As per question, maximum speed of photoelectrons in two cases differ by a factor 2

From eqn. (i) & (ii)

$$\Rightarrow \frac{\frac{hc}{\lambda_1} - \phi}{\frac{hc}{\lambda_2} - \phi} = 4 \Rightarrow \frac{hc}{\lambda_1} - \phi = \frac{4hc}{\lambda_2} - 4\phi$$

$$\Rightarrow \frac{4hc}{\lambda_2} - \frac{hc}{\lambda_1} = 3\phi \Rightarrow \phi = \frac{1}{3}hc\left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

$$= \frac{1}{3} \times 1240\left(\frac{4 \times 350 - 540}{350 \times 540}\right) = 1.8\text{ eV}$$

**22.** **(7.7)** According to question, there are two EM waves with different frequency,

$B_1 = B_0 \sin(\pi \times 10^7 c)t$

and $B_2 = B_0 \sin(2\pi \times 10^7 c)t$

To get maximum kinetic energy we take the photon with higher frequency

using, $B = B_0 \sin \omega t$ and $\omega = 2\pi v \Rightarrow v = \frac{\omega}{2\pi}$

$B_1 = B_0 \sin(\pi \times 10^7 c)t \Rightarrow v_1 = \frac{10^7}{2} \times c$

$B_2 = B_0 \sin(2\pi \times 10^7 c)t \Rightarrow v_2 = 10^7 c$

where c is speed of light $c = 3 \times 10^8$ m/s

Clearly, $v_2 > v_1$

so KE of photoelectron will be maximum for photon of higher energy.

$v_2 = 10^7 c$ Hz

$hv = \phi + KE_{max}$

energy of photon

$E_{ph} = hv = 6.6 \times 10^{-34} \times 10^7 \times 3 \times 10^9$

$E_{ph} = 6.6 \times 3 \times 10^{-19}$ J

$= \frac{6.6 \times 3 \times 10^{-19}}{1.6 \times 10^{-19}}$ eV $= 12.375$ eV

$KE_{max} = E_{ph} - \phi$

$= 12.375 - 4.7 = 7.675$ eV $\approx 7.7$ eV

**23.** **($10^{11}$)** Using, intensity $I = \frac{nE}{At}$

n = no. of photoelectrons

$\Rightarrow 16 \times 10^{-3} = \left(\frac{n}{t}\right) \times \frac{10 \times 1.6 \times 10^{-19}}{10^{-4}}$ or, $\frac{n}{t} = 10^{12}$

So, effective number of photoelectrons ejected per unit time $= 10^{12} \times 10/100 = 10^{11}$

**24.** **($1.45 \times 10^6$)** de-Broglie wavelength,

$\lambda = \frac{h}{mv} = 10^{-3}\left(\frac{3 \times 10^8}{6 \times 10^{14}}\right)$ $\left[\because \lambda = \frac{c}{v}\right]$

$v = \frac{6.63 \times 10^{-34} \times 6 \times 10^{14}}{9.1 \times 10^{-31} \times 3 \times 10^5}$

$v = 1.45 \times 10^6$ m/s

**25.** **(1)** Let $\phi$ = work function of the metal,

$$\frac{hc}{\lambda_1} = \phi + eV_1 \qquad ......(i)$$

$$\frac{hc}{\lambda_2} = \phi + eV_2 \qquad ......(ii)$$

Sutracting (ii) from (i) we get

$$hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = e(V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{hc}{e}\left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \cdot \lambda_2}\right) \quad \left[\begin{array}{l} \lambda_1 = 300\,\text{nm} \\ \lambda_2 = 400\,\text{nm} \\ \frac{hc}{e} = 1240\,\text{nm} - \text{V} \end{array}\right]$$

$$= (1240\,\text{nm} - \text{v})\left(\frac{100\text{nm}}{300\ \text{nm} \times 400\,\text{nm}}\right)$$

$$= 1.03\ \text{V} \approx 1\text{V}$$

**26.** **(14.14)** de-Broglie wavelength ($\lambda$) is given by

$K = qV$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \left(\because p = \sqrt{2mK}\right)$$

Substituting the values we get

$$\therefore \ \frac{\lambda_A}{\lambda_B} = \frac{\sqrt{2m_B q_B V_B}}{\sqrt{2m_A q_A V_A}} = \sqrt{\frac{4\text{m.q.}2500}{\text{m.q.}50}}$$

$$= 2\sqrt{50} = 2 \times 7.07 = 14.14$$

**27.** **(400)** If $E$ is the energy of each photon, then

$$nE = P$$

$$\therefore \quad E = \frac{P}{n} = \frac{200}{4 \times 10^{20}} = 50 \times 10^{-20} J$$

If $\lambda$ is the wavelength of light, then

$$E = \frac{hc}{\lambda}$$

$$\therefore \quad \lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{500 \times 10^{-20}}$$

$$\simeq 400\ \text{nm}$$

**28.** **($5 \times 10^{15}$)** Energy of photon (E) is given by $E = \dfrac{hc}{\lambda}$

Number of photons of wavelength $\lambda$ emitted in $t$ second from laser of power P is given by

$$n = \frac{Pt\lambda}{hc}$$

$$\Rightarrow \quad n = \frac{2 \times \lambda}{hc} = \frac{2 \times 10^{-3} \times 5 \times 10^{-7}}{2 \times 10^{-25}} \quad (\because t = 1\text{S})$$

$$\Rightarrow \quad n = 5 \times 10^{15}$$

**29.** **(0.149)** If $v_{max}$ is the speed of the fastest electron emitted from the metal surface, then

$$\frac{hc}{\lambda} = W_0 + \frac{1}{2} m v_{max}^2$$

$$\frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{(180 \times 10^{-9})}$$

$$= 2 \times (1.6 \times 10^{-19}) + \frac{1}{2}(9.1 \times 10^{-31}) v_{max}^2$$

$$\therefore \quad v = 1.31 \times 10^6\ \text{m/s}$$

The radius of the electron is given by

$$r = \frac{mv}{qB} = \frac{(9.1 \times 10^{-31}) \times (1.31 \times 10^6)}{(1.6 \times 10^{-19}) \times (5 \times 10^{-9})}$$

$$= 0.149\ \text{m}$$

**30.** **(1.5)** $KE_{max} = E - \phi_0$

(where E = energy of incident light $\phi_0$ = work function) $= \dfrac{hc}{\lambda} - \dfrac{hc}{\lambda_0}$

$$= 1237\left[\frac{1}{260} - \frac{1}{380}\right]$$

$$= \frac{1237 \times 120}{380 \times 260} = 1.5 eV$$

# CHAPTER 26 Atoms

1. **(a)** The significant result deduced from the Rutherford's scatter ing is that whole of the positive charge is concentrated at the centre of atom i.e. nucleus.

2. **(a)** Distance of closest approach

$$r_0 = \frac{Ze(2e)}{4\pi\varepsilon_0\left(\frac{1}{2}mv^2\right)}$$

Energy, $E = 5\times10^6\times1.6\times10^{-19}$ J

$$\therefore r_0 = \frac{9\times10^9\times(92\times1.6\times10^{-19})(2\times1.6\times10^{-19})}{5\times10^6\times1.6\times10^{-19}}$$

$\Rightarrow r = 5.2\times10^{-14} m = 5.3\times10^{-12}$ cm.

3. **(a)** $\lambda_{min} = \frac{n^2}{R}[\because n = 1 \text{ for Lymen series}]$

$\lambda_{min} = \frac{1}{R} \simeq 910\,\text{Å}$

4. **(d)** Circumference, $2\pi rn = n\lambda$

5. **(c)** $N \propto \frac{1}{\sin^4\theta/2}$ ; $\frac{N_2}{N_1} = \frac{\sin^4(\theta_1/2)}{\sin^4(\theta_2/2)}$

or $\frac{N_2}{5\times10^6} = \frac{\sin^4(60°/2)}{\sin^4(120°/2)}$

or $\frac{N_2}{5\times10^6} = \frac{\sin^4 30°}{\sin^4 60°}$

or $N_2 = 5\times10^6\times\left(\frac{1}{2}\right)^4\left(\frac{2}{\sqrt{3}}\right)^4 = \frac{5}{9}\times10^6$

6. **(c)** As $\alpha$-particles are doubly ionised helium $He^{++}$ i.e. Positively charged and nucleus is also positively charged and we know that like charges repel each other.

7. **(a)** Number of emission spectral lines

$N = \frac{n(n-1)}{2}$ $\quad\therefore 3 = \frac{n_1(n_1-1)}{2}$, in first case.

Or $n_1^2 - n_1 - 6 = 0$ or $(n_1-3)(n_1+2) = 0$

Take positive root. $\quad\therefore n_1 = 3$

Again, $6 = \frac{n_2(n_2-1)}{2}$, in second case.

Or $n_2^2 - n_2 - 12 = 0$ or $(n_2-4)(n_2+3) = 0$.

Take positive root, or $n_2 = 4$

Now velocity of electron, $v = \frac{2\pi KZe^2}{nh}$

$\therefore \frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{4}{3}$.

8. **(a)** Energy of ground state 13.6 eV

Energy of first excited state $= -\frac{13.6}{4} = -3.4$ eV

Energy of second excited state $= -\frac{13.6}{9} = -1.5$ eV

Difference between ground state and 2nd excited state = 13.6 – 1.5 = 12.1 eV

So, electron can be excited upto 3rd orbit

No. of possible transition

$3\to2, 3\to1, 2\to1$

So, three lines are possible.

9. **(d)** $[E = -13.6/n^2]$

$\Delta E = 13.6 - \frac{13.6}{9} = 13.6\times\frac{8}{9}$ eV

= 12.08 eV

$= 12.08\times1.6\times10^{-19}$

$= 19.34\times10^{-19}$ J

10. **(c)** Transition A ($n = \infty$ to 1) : Series lime of Lyman series

Transition B (n = 5 to n = 2) : Third spectral line of Balmer series

Transition C (n = 5 to n = 3) : Second spectral line of Paschen series.

11. **(b)** Work done to stop the $\alpha$ particle is equal to K.E.

$\therefore qV = \frac{1}{2}mv^2 \Rightarrow q\times\frac{K(Ze)}{r} = \frac{1}{2}mv^2$

$$\Rightarrow r = \frac{2(2e)K(Ze)}{mv^2} = \frac{4KZe^2}{mv^2}$$

$$\Rightarrow r \propto \frac{1}{v^2} \text{ and } r \propto \frac{1}{m}.$$

**12.** **(b)** According to Bohr's Theory the wavelength of the radiation emitted from hydrogen atom is given by

$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$\because \quad Z = 3$

$$\therefore \quad \frac{1}{\lambda} = 9R\left(1 - \frac{1}{9}\right)$$

$$\Rightarrow \quad \lambda = \frac{1}{8R} = \frac{1}{8 \times 10973731.6}$$

(R = 10973731.6 m–1)

$\Rightarrow \ \lambda = 11.39$ nm

**13.** **(b)** Total energy of electron in $n^{th}$ orbit of hydrogen atom

$$E_n = -\frac{Rhc}{n^2}$$

Total energy of electron in $(n+1)^{th}$ level of hydrogen atom

$$E_{n+1} = -\frac{Rhc}{(n+1)^2}$$

When electron makes a transition from $(n+1)^{th}$ level to $n^{th}$ level

Change in energy,

$$\Delta E = E_{n+1} - E_n$$

$$h\nu = Rhc \cdot \left[\frac{1}{n^2} - \frac{1}{(n+1)^2}\right] \qquad (\because E = h\nu)$$

$$\nu = R \cdot c\left[\frac{(n+1)^2 - n^2}{n^2(n+1)^2}\right]$$

$$\nu = R \cdot c\left[\frac{1+2n}{n^2(n+1)^2}\right]$$

For $n >> 1$

$$\Rightarrow \nu = R \cdot c\left[\frac{2n}{n^2 \times n^2}\right] = \frac{2RC}{n^3}$$

$$\Rightarrow \nu \propto \frac{1}{n^3}$$

**14.** **(b)** Energy required to remove $e^-$ from singly ionized helium atom $= \frac{(13.6)Z^2}{1^2} = 54.4$ eV

$(\because Z = 2)$

Energy required to remove $e^-$ from helium atom = x eV

According to question, 54.4 eV = 2.2x

$\Rightarrow$ x = 24.73 eV

Therefore, energy required to ionize helium atom = (54.4 + 24.73) eV = 79.12 eV

**15.** **(b)** For first excited state, n = 2 and for $Li^{++}$ Z = 3

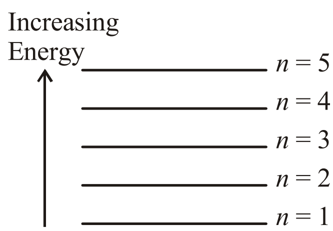
$$E_n = \frac{13.6}{n^2} \times Z^2 = \frac{13.6}{4} \times 9 = 30.6 \text{ eV}$$

**16.** **(b)** If $n_1 = n$ and $n_2 = n + 1$

$$\text{Maximum wavelength } \lambda_{max} = \frac{n^2(n+1)^2}{(2n+1)R}$$

Therefore, for large n, $\lambda_{max} \propto n^3$

**17.** **(c)** It is given that transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom result in ultraviolet radiation. For infrared radiation $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ should be less. The only option is $5 \rightarrow 4$.




**18.** **(d)** We have to find the frequency of emitted photons. For emission of photons electron should makes a transition from higher energy level to lower energy level. so, option (a) and (b) are incorrect.

Frequency of emitted photon is given by

$$h\nu = -13.6\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$$

For transition from $n = 6$ to $n = 2$,

$$\nu_1 = \frac{-13.6}{h}\left(\frac{1}{6^2}-\frac{1}{2^2}\right) = \frac{2}{9}\times\left(\frac{13.6}{h}\right)$$

For transition from $n = 2$ to $n = 1$,

$$\nu_2 = \frac{-13.6}{h}\left(\frac{1}{2^2}-\frac{1}{1^2}\right) = \frac{3}{4}\times\left(\frac{13.6}{h}\right).$$

$\therefore \nu_1 < \nu_2$

**19.** **(d)** When electron jumps from $M \to L$ shell

$$\frac{1}{\lambda} = K\left(\frac{1}{2^2}-\frac{1}{3^2}\right) = \frac{K\times 5}{36} \quad \text{..... (i)}$$

When eletron jumps from $N \to L$ shell

$$\frac{1}{\lambda'} = K\left(\frac{1}{2^2}-\frac{1}{4^2}\right) = \frac{K\times 3}{16} \quad \text{....(ii)}$$

solving equation (i) and (ii) we get

$$\lambda' = \frac{20}{27}\lambda$$

**20.** **(d)** $v = \frac{c}{137n} = \frac{c}{137\times 3}$

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\left(\frac{m\times c}{3\times 137}\right)} = \frac{h}{mc}\times(3\times 137)$$

$= 9.7$ Å

**21.** **(488.9)** $\frac{1}{\lambda_1} = R\left(\frac{1}{2^2}-\frac{1}{3^2}\right) = \frac{5R}{36}$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{2^2}-\frac{1}{4^2}\right) = \frac{3R}{16}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{80}{108}$$

$$\lambda_2 = \frac{80}{108}\lambda_1 = \frac{80}{108}\times 660 = 488.9\text{nm}.$$

**22.** **(5)** $E = E_1 + E_2$

$$13.6\frac{z^2}{n^2} = \frac{1240}{\lambda_1} + \frac{1240}{\lambda_2}$$

or $\frac{13.6(2)^2}{n^2} = 1240\left(\frac{1}{108.5}+\frac{1}{30.4}\right)\times\frac{1}{10^{-9}}$

On solving, $n = 5$

**23.** **(823.5)** The smallest frequency and largest wavelength in ultraviolet region will be for transition of electron from orbit 2 to orbit 1.

$$\therefore \frac{1}{\lambda} = R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

$$\Rightarrow \frac{1}{122\times 10^{-9}m}$$

$$= R\left[\frac{1}{1^2}-\frac{1}{2^2}\right] = R\left[1-\frac{1}{4}\right] = \frac{3R}{4}$$

$$\Rightarrow R = \frac{4}{3\times 122\times 10^{-9}}m^{-1}$$

The highest frequency and smallest wavelength for infrared region will be for transition of electron from $\infty$ to 3rd orbit.

$$\therefore \frac{1}{\lambda} = R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{4}{3\times 122\times 10^{-9}}\left(\frac{1}{3^2}-\frac{1}{\infty}\right)$$

$$\therefore \lambda = \frac{3\times 122\times 9\times 10^{-9}}{4} = 823.5\text{nm}$$

**24.** **(25)** If $\lambda$ is the de-Broglie wavelength, then for $n^{\text{th}}$ orbit

$$2\pi r_n = n\lambda$$

where $r_n = \frac{\in_0 h^2 n^2}{\pi m e^2 Z}$

$$\therefore \frac{1}{\lambda} = \frac{me^2 Z}{2\in_0 h^2 n} \quad \ldots\text{(i)}$$

For Lyman series

$$\frac{1}{\lambda} = Z^2 R\left(\frac{1}{1^2}-\frac{1}{n^2}\right) \quad \ldots\text{(ii)}$$

From equations (i) and (ii), we have

$$Z^2 R\left(1-\frac{1}{n^2}\right) = \frac{me^2 Z}{2\in_0 h^2 n} \quad \ldots\text{(iii)}$$

where $R = \frac{me^4}{8\in_0^2 ch^3}$ ...(iv)

After substituting the values in (iii) & (iv), we get

$$n = 25$$

**25.** **(0.34)** From energy level diagram, using $\Delta E = \frac{hc}{\lambda}$

For wavelength $\lambda_1$ $\Delta E = -E-(-2E) = \frac{hc}{\lambda_1}$

$\therefore \quad \lambda_1 = \frac{hc}{E}$

For wavelength $\lambda_2$ $\Delta E = -E - \left(-\frac{4E}{3}\right) = \frac{hc}{\lambda_2}$

$\therefore \quad \lambda_2 = \frac{hc}{\left(\frac{E}{3}\right)} \qquad \therefore \quad r = \frac{\lambda_1}{\lambda_2} = \frac{1}{3}$

**26.** **(0.18)** Wavelength of first line of Lyman series

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$$

or $\quad \frac{1}{\lambda_1} = \frac{3R}{4}$

or $\quad \lambda_1 = \frac{4}{3R}$.

and $\quad \frac{1}{\lambda_2} = R\left(\frac{1}{2^2}-\frac{1}{3^2}\right) = \frac{5R}{36}$

or $\quad \lambda_2 = \frac{36}{5R}$

$\therefore \quad \frac{\lambda_1}{\lambda_2} = \frac{5}{27}$

**27.** **(122.4)** $E = 13.6\, Z^2$ eV

$= 13.6 \times (3)^2$

$= 122.4$ eV

**28.** **(0.66)** $E_3 = \frac{-13.6}{3^2} = -1.51 eV$

and $\quad E_4 = \frac{-13.6}{4^2} = -0.85 eV$

$\therefore \quad E_4 - E_3 = 0.66$ eV

**29.** **(30.6)** For lithium, $E_2 = -13.6\frac{z^2}{n^2}$

$= -\frac{13.6 \times 3^2}{2^2}$

$= -30.6$ eV

So energy needed to remove the electron

$= 30.6$ eV.

**30.** **($6.8 \times 10^{-27}$)** $\Delta E = 13.6\left(1-\frac{1}{16}\right)$

$= \frac{13.6 \times 15}{16} = \frac{51}{4}$

$= 12.75 \text{ eV} = \frac{12.75 \times 1.6 \times 10^{-19}}{3 \times 10^8}$

Photon will take away almost all of the energy

$\frac{hc}{\lambda} = 12.75$ eV

$\Rightarrow \frac{h}{\lambda} = \left(\frac{12.75}{c}\right) = P_{\text{photon}} = P_{\text{revoled atom}}$

$= 6.8 \times 10^{-27}$ kg m/s

# CHAPTER 27 Nuclei

**1.** **(a)** As we know, $R = R_0\ (A)^{1/3}$
where A = mass number
$R_{A\ell} = R_0\,(27)^{1/3} = 3R_0$
$R_{Te} = R_0\,(125)^{1/3} = 5R_0 = \frac{5}{3}R_{A\ell}$

**2.** **(c)** Here, conservation of linear momentum can be applied
$238 \times 0 = 4u + 234v$
$\therefore\ v = -\frac{4}{234}u$
$\therefore$ speed $= |\vec{v}| = \frac{4}{234}u$

**3.** **(a)** For substance A :
$$2N_0 \to N_A = 2N_0\left(\frac{1}{2}\right)^{48/12} = \frac{N_0}{2^3} = \frac{N_0}{8}$$
For substance B :
$$N_0 \to N_B = N_0\left(\frac{1}{2}\right)^{48/16} = \frac{N_0}{2^3} = \frac{N_0}{8}$$
$N_A : N_B = 1 : 1$

**4.** **(d)** $p_1 = p_2 \Rightarrow m_1v_1 = m_2v_2$
$2m_1 = m_2$
$$2\rho.\frac{4}{3}\pi R_1^3 = \rho.\frac{4}{3}\pi R_2^3\ ;\ \frac{R_1^3}{R_2^3} = 1:2\ ;$$
$R_1 : R_2 = 1 : 2^{1/3}$

**5.** **(c)** Radius R of a nucleus changes with the nucleon number A of the nucleus as
$R = 1.3 \times 10^{-15} \times A^{1/3}$ m
Hence,
$$\frac{R_2}{R_1} = \left(\frac{A_2}{A_1}\right)^{1/3} = \left(\frac{128}{16}\right)^{1/3} = (8)^{1/3} = 2$$
$\therefore\ R_2 = 2R_1 = 2\,(3 \times 10^{-15})\text{m} = 6 \times 10^{-15}$ m

**6.** **(c)** Binding energy
$= [ZM_P + (A - Z)M_N - M]c^2$
$= [8M_P + (17 - 8)M_N - M]c^2$
$= [8M_P + 9M_N - M]c^2$
$= [8M_P + 9M_N - M_o]c^2$

**7.** **(d)** It has been known that a nucleus of mass number $A$ has radius
$R = R_0A^{1/3}$,
where $R_0 = 1.2 \times 10^{-15}$m and $A$ = mass number
In case of $^{27}_{13}A\ell$, let nuclear radius be $R_1$
and for $^{125}_{32}Te$, nuclear radius be $R_2$
For $^{27}_{13}Al$, $R_1 = R_0(27)^{1/3} = 3R_0$
For $^{125}_{32}Te$, $R_2 = R_0(125)^{1/3} = 5R_0$
$$\frac{R_2}{R_1} = \frac{5R_0}{3R_0} \Rightarrow R_2 = \frac{5}{3}R_1 = \frac{5}{3} \times 3.6 = 6\,\text{fm}.$$

**8.** **(a)** The binding energy per nucleon is lowest for very light nuclei such as $^4_2$He and is greatest around A = 56, and then decreases with increasing A.

**9.** **(b)** Effective half life is calculated as
$$\frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$$
$$\frac{1}{T} = \frac{1}{16} + \frac{1}{48} \Rightarrow T = 12 \text{ years}$$
Time in which $\frac{3}{4}$ will decay is 2 half lives = 24 years

**10.** **(c)** $A = A_0e^{-\lambda t}$;
$2100 = 16000e^{-12\lambda} \to e^{12\lambda} = 7.6$
$\Rightarrow\ 12\lambda = \log_e 7.6 = 2 \Rightarrow \lambda = \frac{2}{12} = \frac{1}{6}$
$\therefore\ T = \frac{0.6931 \times 6}{1} = 4$

**11.** **(d)** Density of nucleus, $\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{mA}{\frac{4}{3}\pi R^3}$
$$\Rightarrow \rho = \frac{mA}{\frac{4}{3}\pi(R_0A^{1/3})^3} \qquad (\because R = R_0A^{1/3})$$
Here m = mass of a nucleon
$$\therefore \rho = \frac{3 \times 1.67 \times 10^{-27}}{4 \times 3.14 \times (1.3 \times 10^{-15})^3}$$
(Given, $R_0 = 1.3 \times 10^{-15}$)
$\Rightarrow \rho = 2.38 \times 10^{17}$ kg/m$^3$

**12.** **(d)** Mass defect,

$\Delta m = (50m_p + 70m_n) - (m_{sn})$

$= (50 \times 1.00783 + 70 \times 1.008) - (119.902199)$

$= 1.096$

Binding energy

$= (\Delta m)C^2 = (\Delta m) \times 931 = 1020.56$

$\frac{\text{Binding energy}}{\text{Nucleon}} = \frac{1020.5631}{120} = 8.5 \text{ MeV}$

**13.** **(b)** Power output of the reactor,

$P = \frac{\text{energy}}{\text{time}}$

$= \frac{2}{235} \times \frac{6.023 \times 10^{26} \times 200 \times 1.6 \times 10^{-19}}{30 \times 24 \times 60 \times 60}$

$\simeq 60 \text{MW}$

**14.** **(b)** Given, for $^{14}C$

$A_0 = 16$ dis $\min^{-1}$ $g^{-1}$

$A = 12$ dis $\min^{-1}$ $g^{-1}$

$t_{1/2} = 5760$ years

Now, $\lambda = \frac{0.693}{t_{1/2}}$

$\lambda = \frac{0.693}{5760}$ per year

Then, from, $t = \frac{2.303}{\lambda} \log_{10} \frac{A_0}{A}$

$= \frac{2.303 \times 5760}{0.693} \log_{10} \frac{16}{12}$

$= \frac{2.303 \times 5760}{0.693} \log_{10} 1.333$

$= \frac{2.303 \times 5760 \times 0.1249}{0.693}$

$= 2390.81 \approx 2391$ years.

**15.** **(a)** The chemical reaction of process is

$2\,_1^2H \rightarrow \,_2^4He$

Binding energy of two deuterons,

$4 \times 1.1 = 4.4$ MeV

Binding energy of helium nucleus $= 4 \times 7$

$= 28$ MeV

Energy released $= 28 - 4.4 = 23.6$ MeV

**16.** **(b)** We know that

Activity, $A = A_0 e^{-\lambda t}$

$A = A_0 e^{-t In 2/T_{1/2}}$ $\left(\because \lambda = \frac{In_2}{T_{1/2}}\right)$

$\Rightarrow 500 = 700\, e^{-t In 2/T_{1/2}}$

$\Rightarrow In \frac{7}{5} = \frac{30 In 2}{T_{1/2}}$ $(\because t = 30 \text{ minute})$

$\Rightarrow T_{1/2} = 30 \frac{In\, 2}{In\, 1.4} = 61.8$ minute

$(\because \ln 2 = 0.693 \text{ and } \ln.1.4 = 0.336)$

$\Rightarrow T_{1/2} \approx 62$ minute

**17.** **(c)** The range of energy of $\beta$-particles is from zero to some maximum value.

**18.** **(c)** $T_{av} = \frac{T_\alpha T_\beta}{T_\alpha + T_\beta}$

If $\alpha$ and $B$ are emitted simultaneously.

**19.** **(d)** $(T_{1/2})_A = (t_{mean})_B$

$\Rightarrow \frac{0.693}{\lambda_A} = \frac{1}{\lambda_B} \Rightarrow \lambda_A = 0.693 \lambda_B$

or $\lambda_A < \lambda_B$

Also rate of decay $= \lambda N$

Initially number of atoms (N) of both are equal but since $\lambda_B > \lambda_A$, therefore B will decay at a faster rate than A

**20.** **(c)** When one $\alpha$- particle emitted then danghter nuclei has 4 unit less mass number (A) and 2 unit less atomic (z) number (z).

$_{90}^{232}Th \rightarrow \,_{78}^{208}Y + 6\,_2^4He$

$_{78}^{208}Y \rightarrow \,_{82}^{208}X + 4\beta$ praticle

**21.** **(200)** According to question, at t = 0, $A_0 = \frac{dN}{dt}$

$= 1600$ C/s

and at t = 8s, A = 100 C/s

$\therefore \frac{A}{A_0} = \left(\frac{1}{16}\right)$

Therefore half life period, $t_{1/2} = 2$s

$\therefore$ Activity at t = 6s = $1600 \left(\frac{1}{2}\right)^3 = 200$ C/s

**22.** **(1)** Nuclear density is independent of atomic number.

**23.** **(1)** We know that, $\left|\frac{dN}{dt}\right| = \lambda N = \frac{1}{T_{\text{mean}}} N$

$\therefore\ 10^{10} = \frac{1}{10^9} \times N$

$\therefore\ N = 10^{19}$

*i.e.* $10^{19}$ radioactive atoms are present in the freshly prepared sample.

The mass of the sample $= 10^{19} \times 10^{-25}$ kg $= 10^{-6}$kg $= 1$ mg.

**24.** **(16.45)** Disintegration constant

$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{3.8} = 0.182$ per day

The number of particles left after time $t$

$N = N_0 e^{-\lambda t}$

or $\quad \frac{N_0}{20} = N_0 e^{-\lambda t}$

or $\quad e^{\lambda t} = 20$

or $\quad t = \frac{\ln 20}{\lambda}$

$= \frac{\ln 20}{0.182} = 16.45$ days

**25.** **($3.91 \times 10^3$)** If $A_0$ is the initial activity of radioactive sample, then activity at any time

$A = A_0 e^{-\lambda t}$

or $\quad 1 \times 10^6 = 4 \times 10^6\, e^{-\lambda \times 20}$

or $\quad e^{-20\lambda} = \frac{1}{4}$

The count rate after 100 hour is given by

$A' = A_0 e^{-\lambda \times 100} = A_0 e^{-100\lambda}$

$= A_0 [e^{-20\lambda}]^5$

$= 4 \times 10^6 \left[\frac{1}{4}\right]^5$

$= 3.91 \times 10^3$ per second

**26.** **($1.868 \times 10^9$)** Suppose x is the number of $Pb^{206}$ nulei. The number of $U^{238}$ nuclei will be 3x, Thus

$3x + x = N_0$

We know that $N = N_0 e^{-\lambda t}$

or $\quad 3x = 4x e^{-\lambda t}$

$\therefore\quad e^{\lambda t} = \frac{4}{3}$

or $\quad t = \frac{\ln 4/3}{\lambda} = \frac{\ln 4/3}{(0.693/t_{1/2})}$

$= \frac{\ln 4/3}{(0.693/4.5 \times 10^9)}$

$= 1.868 \times 10^9$ year.

**27.** **($3.8 \times 10^4$)** If $m$ kg is the required mass of the uranium, then number of nuclei

$= \frac{(m \times 1000) \times 6.02 \times 10^{23}}{235}$

Each $U_{235}$ nucleus releases energy 200 MeV,

$\therefore$ total energy released in 10 years

$E_{in} = \frac{m \times 6.02 \times 10^{26}}{235} \times 200$

Energy required in 10 years, $E_{our} = Pt$

$= (1000 \times 10^6) \times (10 \times 365 \times 24 \times 3600)$

Efficiency $\eta = \frac{E_{\text{out}}}{E_{\text{in}}}$

Substituting the values, we get

$m = 3.8 \times 10^4$ kg.

**28.** **(6.1)** We know that the rate of integration $\left|-\frac{dN}{dt}\right|$

$= A$

$\therefore\ A = A_0 e^{-\lambda t}$

or $\quad 2700 = 4750 e^{-\lambda \times 5}$

or $\quad \lambda = 0.1131$ per minute

Half life $t_{1/2} = \frac{0.693}{\lambda}$

$= \frac{0.693}{0.1131} = 6.1$ minute

**29.** **(7)** $\frac{\text{Binding energy}}{\text{Nucleon}} = \frac{0.0303 \times 931}{4} \approx 7$

**30.** **(2)** $R = R_0 (A)^{1/3}$

$\therefore\ \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{256}{4}\right)^{1/3} = 4$

$R_2 = \frac{R_1}{4} = 2$ fermi

# CHAPTER 28 Semiconductor Electronics: Materials, Devices and Simple Circuits

1. **(b)** $\lambda = \frac{hc}{E}$

$= \frac{6.62\times10^{-34}\times3\times10^{8}}{(60\times10^{-3}\times1.6\times10^{-19})} = 2.07\times10^{-5}$ m

2. **(b)** In half wave rectifier only half of the wave is rectified.
3. **(a)** Positive terminal is at higher potential (–5V) and negative terminal is at lower potential –10V.
4. **(b)** The power gain in case of CE amplifier, Power gain = $\beta^2$ × Resistance gain

$= \beta^2 \times \frac{R_o}{R_i} = (10)^2 \times 5 = 500.$

5. **(d)** Output of upper AND gate = $\overline{A}B$

Output of lower AND gate = $A\overline{B}$

$\therefore$ Output of OR gate, $Y = A\overline{B} + B\overline{A}$

This is boolean expression for XOR gate.

6. **(b)** Given : $\mu_e = 2.3\ m^2\ V^{-1}\ s^{-1}$

$\mu_h = 0.01\ m^2\ V^{-1}\ s^{-1}$, $n_e = 5\times10^{12}/cm^3$

$= 5\times10^{18}/m^3$, $n_h = 8\times10^{13}/cm^3 = 8\times10^{19}/m^3$.

Conductivity $\sigma = e[n_e\mu_e + n_h\mu_h]$

$= 1.6\times10^{-19}[5\times10^{18}\times2.3 + 8\times10^{19}\times0.01]$

$= 1.6\times10^{-1}[11.5 + 0.8]$

$= 1.6\times10^{-1}\times12.3 = 1.968\ \Omega^{-1}\ m^{-1}$.

7. **(d)** For a p-type semiconductor, the acceptor energy level, as shown in the diagram, is slightly above the top $E_v$ of the valence band. With very small supply of energy an electron from the valence band can jump to the level $E_A$ and ionise acceptor negatively

8. **(a)** $E = \frac{V}{d} = \frac{0.50}{5\times10^{-7}} V/m = 1.0\times10^{6} V/m$

9. **(c)** $\frac{I_c}{I_e} = 0.96 \Rightarrow I_c = 0.96 I_e$

But $I_e = I_c + I_b = 0.96 I_e + I_b$

$\Rightarrow I_b = 0.04 I_e$

$\therefore$ Current gain, $\beta = \frac{I_c}{I_b} = \frac{0.96 I_e}{0.04 I_e} = 24$

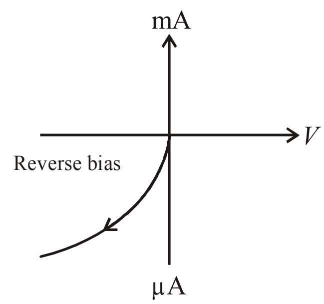
10. **(b)** $\frac{V_o}{V_{in}} = \frac{R_o}{R_{in}}\times\beta = \frac{5\times10^{3}\times62}{500}$

$= 10\times62 = 620$

$V_o = 620\times V_{in} = 620\times0.01 = 6.2\,V$

$\therefore V_o = 6.2$ volt.

11. **(d)** I-V characteristic of a photodiode is as follows :




On increasing the biasing voltage of a photodiode, the magnitude of photocurrent first increases and then attains a saturation.

12. **(d)** Given,

Wavelength of photon, $\lambda = 400$ nm

A photodiode can detect a wavelength corresponding to the energy of band gap. If the signal is having wavelength greater than this value, photodiode cannot detect it.

$\therefore$ Band gap $E_g = \frac{hc}{\lambda} = \frac{1237.5}{400} = 3.09$ eV

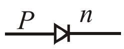
13. **(d)** Using $U_{av} = \frac{1}{2}\varepsilon_0 E_0^2$

But $U_{av} = \frac{P}{4\pi r^2 \times c}$

$$\therefore \quad \frac{P}{4\pi r^2} = \frac{1}{2}\varepsilon_0 E_0{}^2 \times c$$

$$E_0^2 = \frac{2P}{4\pi r^2 \varepsilon_0 c} = \frac{2\times 0.1\times 9\times 10^9}{1\times 3\times 10^8}$$

$$\therefore \quad E_0 = \sqrt{6} = 2.45 \text{V/m}$$

**14.** **(a)** 


For forward bias, p-side must be at higher potential than $n$-side. $\Delta V = (+)Ve$

**15.** **(c)** Relation between drift velocity and current is $I = nAeV_d$

$$\frac{I_e}{I_h} = \frac{n_e e A v_e}{n_h e A v_h}$$

$$\Rightarrow \frac{7}{4} = \frac{7}{5} \times \frac{v_e}{v_h}$$

$$\Rightarrow \frac{v_e}{v_h} = \frac{5}{4}$$

**16.** **(b)** Forward bias resistance

$$= \frac{\Delta V}{\Delta I} = \frac{0.1}{10\times 10^{-3}} = 10\,\Omega$$

Reverse bias resistance $= \dfrac{10}{10^{-6}} = 10^7\,\Omega$

Ratio of resistances

$$= \frac{\text{Forward bias resistance}}{\text{Reverse bias resistance}} = 10^{-6}$$

**17.** **(a)** In reverse biasing the width of depletion region increases, and current flowing through diode is zero. Thus, electric field is zero at middle of depletion region.

**18.** **(d)** Band gap = energy of photon of wavelength 2480 nm. So,

$$\text{Band gap, } E_g = \frac{hc}{\lambda}$$

$$= \left(\frac{6.63\times 10^{-34}\times 3\times 10^8}{2480\times 10^{-9}}\right) \times \frac{1}{1.6\times 10^{-19}} eV$$

$= 0.5$ eV

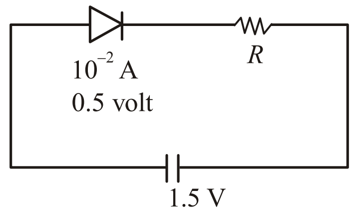
**19.** **(d)** A logic gate is reversible if we can recover input data from the output. Hence NOT gate.

**20.** **(c)** According to question, when diode is forward biased,

$V_{\text{diode}} = 0.5$ V

Safe limit of current, $I = 10$ mA $= 10^{-2}$ A

$R_{\min} = ?$




Voltage through resistance

$V_R = 1.5 - 0.5 = 1$ volt

$iR = 1\ (= V_R)$

$$\therefore R_{\min} = \frac{1}{i} = \frac{1}{10^{-2}} = 100\ \Omega$$

**21.** **(0.4)** As we know, current density,

$j = \sigma E = nev_d$

$$\sigma = ne\frac{v_d}{E} = ne\mu$$

$$\frac{1}{\sigma} = \rho = \frac{1}{n_e e \mu_e} = \text{Resistivity}$$

$$= \frac{1}{10^{19}\times 1.6\times 10^{19} - 19\times 1.6}$$

or $\quad P = 0.4\,\Omega$m

**22.** **(0.4)** Initially Ge and Si are both forward biased so current will effectivily pass through Ge diode

$\therefore V_\circ = 12 - 0.3 = 11.7$ V

And if "Ge" is revesed then current will flow through "Si" diode

$\therefore V_\circ = 12 - 0.7 = 11.3$ V

Clearly, $V_\circ$ changes by $11.7 - 11.3 = 0.4$ V

**23.** **($8.49\times 10^{26}$)** If $M$ is the molar mass and $\rho$ is the density then volume of one mole

$$V = \frac{M}{\rho}.$$

The number of atoms per unit volume

$$= \frac{N_A}{V} = \frac{N_A}{M/\rho} = \frac{N_A \rho}{M}$$

$$= \frac{(6.02 \times 10^{23}) \times (8.96)}{63.54}$$

$$= 8.49 \times 10^{22} \quad cm^{-3}$$

$$= 8.49 \times 10^{28}\ m^{-3}$$

As each copper (monovalent) atom has one electron, so number of electrons per unit volume

$$= 8.49 \times 10^{26}\ m^{-3}$$

**24.** **(650)** If $\lambda$ is the wavelength of emitted light, then

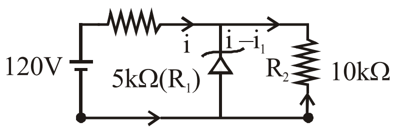
$$E_g = \frac{hc}{\lambda}$$

or $\quad \lambda = \frac{hc}{E_g}$

$$= \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{(1.9) \times (1.60 \times 10^{-19})}$$

$$= 6.5 \times 10^{-7}\ m = 650\ nm$$

**25.** **(9)** The voltage across zener diode is constant




$$i_{(R_2)} = \frac{V}{R} = \frac{50}{10 \times 10^3} = 5 \times 10^{-3}\ A$$

$$i_{(R_1)} = \frac{V}{R} = \frac{120 - 50}{5 \times 10^3} = \frac{70}{5 \times 10^3} 14 \times 10^{-3}\ A$$

$\therefore\ i_{zenerdiode} = 14 \times 10^{-3} - 5 \times 10^{-3} = 9 \times 10^{-3}$ A $= 9$ mA

**26.** **(50)** In half wave rectification, output frequency remains same as input i.e., 50Hz.

**27.** **(2.0)** The band gap

$$E_g = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{620 \times 10^{-9}}$$

$$= 3.2 \times 10^{-19}\ J$$

$$= 2.0\ eV.$$

**28.** **(50)** We know that $\beta = \frac{\Delta i_c}{\Delta i_B}$

$$= \frac{(3.5 - 1.0) \times 10^{-3}}{(80 - 30) \times 10^{-6}} = 50\ .$$

**29.** **($20 \times 10^3$)** $\beta = \frac{\Delta i_c}{\Delta i_b} = \frac{200 - 100}{10 - 5} = 20$

Voltage gain $= \beta \frac{R_2}{R_1} = \frac{20 \times 100 \times 10^3}{100} = 20 \times 10^3$

**30.** **(0.02)** As $D_2$ is reversed biased, so no current through 75Ω resistor.

now $R_{eq} = 150 + 50 + 100 = 300\ \Omega$

So, required current $I = \frac{\text{BatteryVoltage}}{300}$

$$I = \frac{6}{300} = 0.02$$

# Chapter 29 Communication Systems

1. **(b)**

$$m = \frac{\text{actual frequency deviation}}{\text{max . allowed frequency deviation}} \times 100\%$$

$$= \frac{(\Delta f)\text{actual}}{(\Delta f)\text{max}} \times 100\%$$

if $(\Delta f)$ actual = $(\Delta f)$ max

m = 100 %

2. **(c)** Modulation index

$$m_a = \frac{E_m}{E_c} = \frac{A}{A} = 1$$

Equation of modulated signal $[C_m(t)]$

$= E_{(C)} + m_a E_{(C)} \sin \omega_m t$

$= A\,(1 + \sin w_C t) \sin \omega_m t$ (As $E_{(C)} = A \sin\omega_C t$)

3. **(d)**
4. **(b)** The frequencies present in amplitude modulated wave are :

Carrier frequency = $\omega_c$

Upper side band frequency = $\omega_c + \omega_m$

Lower side band frequency = $\omega_c - \omega_m$.

5. **(c)**
6. **(b)** Modulation index $= \frac{B}{A}$

B = 25, A = 60

$$\Rightarrow \text{M.I.} = \frac{25}{60} = 0.416 \Rightarrow m\% = 41.6\%$$

7. **(d)**
8. **(a)** The critical frequency of a sky wave for reflection from a layer of atmosphere is given by

$f_c = 9(N_{max})^{1/2}$

$\Rightarrow 10 \times 10^6 = 9(N_{max})^{1/2}$

$$\Rightarrow N_{max} = \left(\frac{10 \times 10^6}{9}\right)^2 \simeq 1.2 \times 10^{12}\ m^{-3}$$

9. **(b)** From the given expression,

$V_m = 5\,(1 + 0.6 \cos 6280t) \sin(211 \times 10^4 t)$

Modulation index, $\mu = 0.6$

$\because A_m = \mu A_c$

$$\frac{A_{max} + A_{min}}{2} = A_c = 5 \qquad \text{...(i)}$$

$$\frac{A_{max} - A_{min}}{2} = A_m = 3 \qquad \text{...(ii)}$$

From equation (i) + (ii),

Maximum amplitude, $A_{max} = 8$.

From equation (i) – (ii),

Minimum amplitude $A_{min} = 2$.

10. **(d)** Modulation index, $\mu = \frac{A_m}{A_c} = \frac{2}{4} = 0.5$

Given, fe $= \frac{20000\pi}{2\pi} = 10000$ Hz.

and fm $= \frac{2000\pi}{2\pi} = 1000$ Hz.

$\therefore$ LSB $= f_e - f_m = 10000 - 1000 = 9000$ Hz.

11. **(d)**
12. **(a)** Here, $f_c = 1.5$ MHz = 1500 kHz, $f_m = 10$ kHz

$\therefore$ Low side band frequency

$= f_c - f_m = 1500$ kHz – 10 kHz = 1490 kHz

Upper side band frequency

$= f_c + f_m = 1500$ kHz + 10 kHz = 1510 kHz

13. **(b)** The frequency of AM channel is 1020 kHz whereas for the FM it is 89.5 MHz (given). For higher frequencies (MHz), space wave communication is needed. Very tall towers are used as antennas.
14. **(a)** Comparing $(x_{AM})t = 100\,[1 + 0.5\,t] \cos\omega_c t$ for $0 < t < 1$

with standard AM signal $x_{AM}$

$= E_c\,[1 + m_a \cos \omega_m t] \cos \omega_c t$

We have modulating signal t and $m_a = 0.5$.

15. **(c)** For x(t), BW = $2(\Delta\omega + \omega)$

$\Delta\omega$ is deviation and $\omega$ is the band width of modulating signal.

$\therefore$ BW = 2(90 + 5) = 190

For $x^2$(t), BW = 2 × 190 = 380

16. **(a)** Modulating signal frequency $\rightarrow$ 10kHz

Carrier signal frequency $\rightarrow$ 10 MHz

$\therefore$ Side band frequency are

USB = 10 MHz + 10 kHz = 10010 kHz

LSB = 10MHz – 10 kHz = 9990 kHz

**17.** **(b)** Comparing the given equation with standard modulated signal wave equation, $m = A_c \sin \omega_c t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_s) t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_s) t$

$$\mu \frac{A_c}{2} = 10 \Rightarrow \mu = \frac{2}{3} \text{(modulation index)}$$

$A_c = 30$

$\omega_c - \omega_s = 200\pi$

$\omega_c + \omega_s = 400\pi$

$\Rightarrow$ $f_c = 150$, $f_s = 50$ Hz.

**18.** **(b)** Frequency of radio waves for sky wave propagation is 2 MHz to 30 MHz.

**19.** **(c)** $E_{max} = (1 + m_a) E_c = (1 + 0.6) \times 10 = 16$ V

$E_{min} = (1 - m_a) E_c = (1 - 0.6) \times 10 = 4$V.

**20.** **(a)**

**21.** **(2.7)** $f_c = 9\sqrt{N_m} = 9 \times \sqrt{9 \times 10^{10}}$

$= 2.7 \times 10^6$ Hz $= 2.7$ MHz

**22.** **(10)** Comparing given expression with

$(e)_{AM} = E_c(1 + m_a \cos \omega_m t) \cos \omega_c t$

peak value of carrier wave, $E_c = 10$V.

**23.** **(104.5)** $P_t = P_c\left[1 + \frac{m_a^2}{2}\right]$

$$\Rightarrow \frac{V^2_{rms}}{2} = \frac{V^2_c}{2}\left[1 + \frac{m_a^2}{2}\right]$$

$$V^2_{rms} = V^2_c\left[1 + \frac{m_a^2}{2}\right]$$

$$\Rightarrow V_{rms} = V_c\sqrt{1 + \frac{m_a^2}{2}}$$

$$\Rightarrow V_{rms} = 100\sqrt{1 + \frac{(0.3)^2}{2}}$$

$= 104.5$ volts.

**24.** **(2.7)** $f_c = 9\sqrt{N_m} = 9 \times \sqrt{9 \times 10^{10}}$

$= 2.7 \times 10^6$ Hz $= 2.7$ MHz

**25.** **(6.25)** Average side-band power $P_{av} = \frac{m_a}{4} P_c^2$

Here $m_a = 0.5$

$P_c = 10$

$$\therefore P_{av} = \frac{0.5 \times 10 \times 10}{4} = 6.25$$

**26.** **(6)** Ratio of AM signal Bandwidths

$$= \frac{15200 - 200}{2700 - 200} = \frac{15000}{2500} = 6.$$

**27.** **(0.43)**

$$m_a = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{10 - 4}{10 + 4} = \frac{6}{14} = 0.43$$

**28.** **(9.6)** $P_c = \frac{P_t}{1 + \frac{m_a^2}{2}} = \frac{12}{1 + \frac{(0.5)^2}{2}} = \frac{12}{1.25}$

$= 9.6$ kW

**29.** **($1.28\pi \times 10^3$)** Area covered by T.V. signals

$A = 2\pi hR = 2 \times 3.14 \times 100 \times 6.4 \times 10^6 = 128\pi \times 10^8$

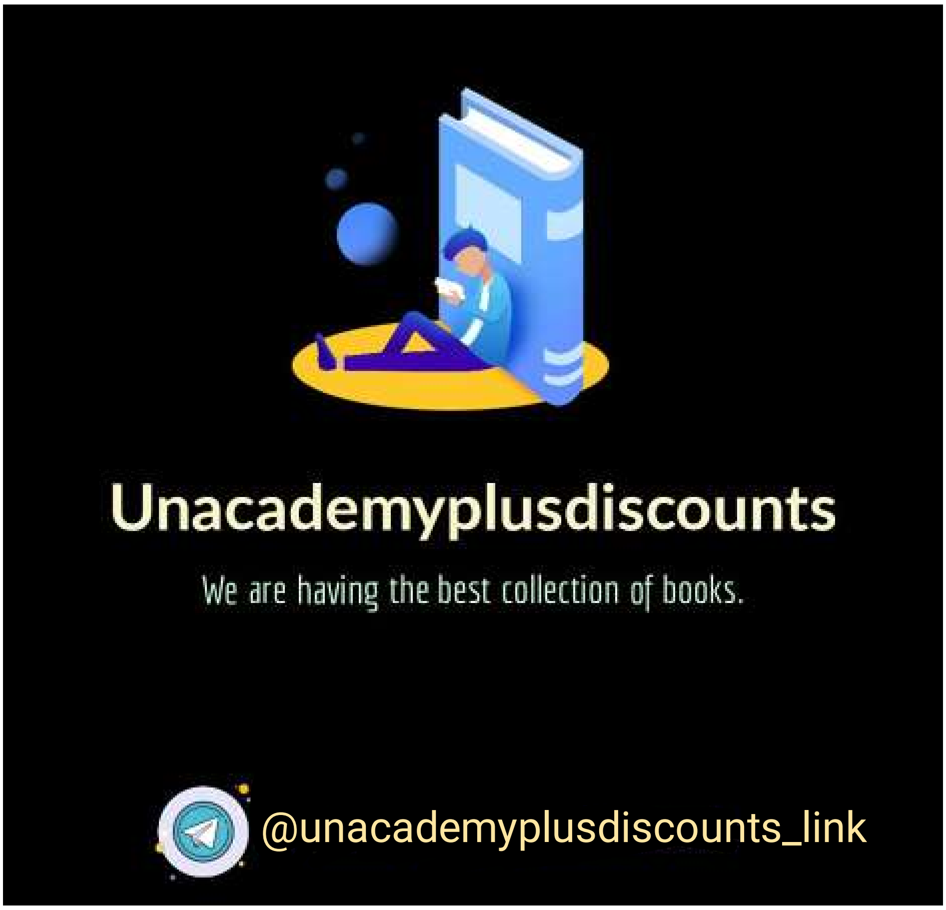
$\Rightarrow A = 1.28\pi \times 10^3 \text{ km}^2$

**30.** **(71)** Total signal B.W= $12 \times 5 = 60$ kHz

11 guard band are required between 12 signal

$\therefore$ guard bandwidth $= 11 \times 1$ kHz $= 11$ kHz

$\therefore$ total bandwidth $= 6 + 11 = 71$ kHz

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